



INTERNATIONAL CONFERENCE PROCEEDINGS ON

**INNOVATIONS AND SUSTAINABLE STRATEGIES
IN COMMERCE AND EDUCATION FOR A
GLOBALIZED ECONOMY**

ICISSCE - 2025

**CHIEF EDITOR
Dr. C. SUBATHRA**

PIONEER KUMARASWAMY COLLEGE

A Govt. Aided Institution, Reaccredited with B++ Grade by NAAC
Affiliated to Manonmaniam Sundaranar University, Tirunelveli
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DEPARTMENT OF COMMERCE

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**ZERODALE, CENTRE FOR LANGUAGE, RESEARCH, AND PUBLICATION, CANADA
UNIVERSITAS NEGERI MAKASSAR, INDONESIA**

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MESSAGE FROM THE PRINCIPAL

It is with great pride and joy that I extend my warm greetings to all the distinguished scholars, academicians, researchers, and industry professionals participating in the International Conference on Innovations and Sustainable Strategies in Commerce and Education for a Globalized Economy (ICISSCE - 2025).

In the era of rapid globalization, innovation and sustainability have become key drivers of progress in commerce and education. This conference provides an invaluable platform for intellectual exchange, fostering groundbreaking research and collaborative discussions that will shape the future of these fields. The diverse perspectives shared here will undoubtedly contribute to new insights, strategies, and solutions that address the challenges of the modern economy.

I commend the Department of Commerce, Pioneer Kumaraswamy College, for their initiative in organizing this international conference in collaboration with esteemed institutions such as ZERODALE, Canada, and Universitas Negeri Makassar, Indonesia. My heartfelt appreciation also goes to the editorial team for their diligent efforts in compiling these proceedings, which will serve as a valuable academic reference.

I encourage all participants to engage actively, exchange knowledge, and explore new avenues for academic and professional growth. May this conference be a catalyst for innovation, sustainability, and excellence.

Wishing you all a fruitful and successful conference!

Dr. C. Jeyasekaran
Principal
Pioneer Kumaraswamy College
Nagercoil

EDITOR'S MESSAGE

It is with great pleasure and enthusiasm that I present to you the proceedings of the **International Conference on Innovations and Sustainable Strategies in Commerce and Education for a Globalized Economy (ICISSCE-2025)**. This compilation of research papers, insights, and discussions reflects the academic rigor, innovative thinking, and collaborative spirit that define this conference.

In an era where technological advancements, digital transformations, and sustainable business practices are reshaping the global landscape, the role of commerce and education in fostering economic growth and societal progress has never been more significant. This conference has provided a dynamic platform for academicians, researchers, and industry experts to exchange ideas, share groundbreaking research, and propose forward-thinking strategies that address contemporary challenges in these fields.

The proceedings encapsulate a diverse range of topics, including **artificial intelligence in commerce and education, fintech innovations, digital marketing, corporate sustainability, entrepreneurship, and the future of education**. These scholarly contributions not only highlight emerging trends but also offer practical solutions that can drive meaningful change in both academia and industry.

I extend my heartfelt gratitude to all the authors, reviewers, and contributors who have enriched this volume with their valuable research. I also appreciate the efforts of the organizing committee, whose dedication and hard work have made this conference and its proceedings a great success.

May this publication serve as a valuable resource for scholars, practitioners, and policymakers, inspiring future research and fostering greater academic-industry collaboration.

Dr. SUBATHRA CHELLADURAI

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TOPIC INTEGRATING SUSTAINABILITY INTO HIGHER EDUCATION CURRICULAM

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Introduction

Sustainability has become a crucial global concern in recent decades, driven by environmental degradation, climate change, and social inequalities. Higher education institutions play a vital role in shaping future leaders, professionals, and policymakers who can address these challenges. Integrating sustainability into higher education curricula ensures that students acquire the knowledge, skills, and values necessary to contribute to sustainable development. This integration involves modifying course content, adopting interdisciplinary approaches, and promoting sustainability through research and community engagement.

This article explores the significance of embedding sustainability in higher education curricula, focusing on two primary objectives: fostering environmental and social responsibility among students and enhancing interdisciplinary learning for holistic problem-solving. By achieving these objectives, higher education institutions can contribute to building a sustainable future.

Objective 1: Fostering Environmental and Social Responsibility

One of the key objectives of integrating sustainability into higher education is to instill a sense of environmental and social responsibility among students. As future professionals and decision-makers, students need to understand the environmental impact of human activities and the importance of ethical decision-making in various fields.

1.1 Raising Awareness and Promoting Ethical Behavior

Incorporating sustainability topics such as climate change, resource conservation, and social justice into the curriculum helps students develop a deeper understanding of these issues. Courses in business, science, engineering, and humanities can integrate sustainability principles to encourage responsible thinking and ethical behavior.

1.2 Encouraging Sustainable Practices on Campus

Higher education institutions can also serve as models of sustainability by implementing green initiatives such as waste reduction, energy conservation, and sustainable food practices. Engaging students in these initiatives through coursework, projects, or extracurricular activities enhances their commitment to sustainability beyond the classroom.

1.3 Preparing Students for Green Careers

The job market increasingly demands professionals who can integrate sustainability into business strategies, engineering solutions, and public policies. By embedding sustainability into curricula, universities equip students with skills to work in renewable energy, sustainable finance, environmental management, and other green careers.

Objective 2: Enhancing Interdisciplinary Learning for Holistic Problem-Solving

Sustainability challenges are complex and require knowledge from multiple disciplines to develop effective solutions. Integrating sustainability into higher education

curricula promotes interdisciplinary learning, allowing students to approach problems from diverse perspectives.

2.1 Integrating Sustainability Across Disciplines

Traditional education often separates disciplines, limiting students' ability to see the broader implications of sustainability. By incorporating sustainability topics into business, science, law, and social science courses, universities can foster a holistic understanding of global challenges. For example, business students can learn about sustainable supply chains, engineers can explore renewable energy technologies, and social science students can study policies that promote sustainability. This cross-disciplinary approach helps students understand the interconnectedness of environmental, economic, and social issues.

2.2 Encouraging Collaborative Research and Projects

Higher education institutions can promote sustainability through interdisciplinary research projects that involve students from different academic backgrounds. Collaborative initiatives, such as sustainable urban planning, climate resilience programs, and community-based sustainability projects, provide students with hands-on experience in addressing real-world problems.

2.3 Implementing Experiential and Problem-Based Learning

Universities can adopt experiential learning approaches such as case studies, simulations, and service-learning projects that focus on sustainability challenges. These methods encourage critical thinking, creativity, and teamwork, essential skills for tackling sustainability issues.

Conclusion

Integrating sustainability into higher education curricula is essential for preparing students to address environmental, social, and economic challenges. By fostering environmental and social responsibility, universities can shape future professionals who prioritize sustainability in their careers. Additionally, enhancing interdisciplinary learning enables students to develop innovative solutions to complex sustainability problems. To successfully implement sustainability in higher education, institutions must adopt a holistic approach that includes curriculum reform, faculty training, student engagement, and collaboration with industry and government. By doing so, they can contribute to building a more sustainable and equitable future for all.

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BUSINESS ETHICS AND SUSTAINABILITY: NAVIGATING THE CHALLENGES IN GLOBAL COMMERCE

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Introduction

In today's globalized economy, businesses operate across multiple nations, cultures, and regulatory frameworks. This expansion has led to increased scrutiny on ethical practices and sustainability in commerce. Business ethics refers to the principles and standards guiding companies in conducting their operations fairly and responsibly. Sustainability, on the other hand, focuses on ensuring long-term economic growth while protecting the environment and promoting social well-being.

The integration of business ethics and sustainability has become a crucial factor in corporate success and reputation. Companies face challenges such as labor rights violations, environmental degradation, corruption, and supply chain transparency. Addressing these issues requires a balanced approach that aligns profitability with ethical responsibility. This article explores key objectives and strategies for businesses to navigate the challenges of ethics and sustainability in global commerce.

Objective 1: Promoting Ethical Business Practices in Global Markets

One of the primary objectives in global commerce is ensuring ethical business practices that adhere to universal principles such as fairness, transparency, and respect for human rights. Several multinational corporations (MNCs) have faced scandals due to unethical practices like child labor, bribery, and tax evasion. These unethical behaviors not only damage a company's reputation but also lead to legal penalties and loss of consumer trust.

To navigate these challenges, businesses must implement strict ethical guidelines and corporate governance policies. Establishing a strong code of conduct, conducting regular audits, and engaging in responsible marketing are essential strategies. Additionally, companies should embrace ethical leadership, where decision-makers prioritize integrity and corporate social responsibility (CSR) over short-term profits.

Global organizations such as the United Nations Global Compact (UNGC) provide frameworks for businesses to follow ethical guidelines, encouraging them to uphold human rights, labor standards, environmental protection, and anti-corruption measures. By integrating ethical practices into their business models, companies can foster trust, ensure compliance with international regulations, and contribute to sustainable development.

Objective 2: Ensuring Sustainability in Global Supply Chains

Sustainability in business is no longer an option but a necessity. With growing concerns over climate change, resource depletion, and pollution, companies must adopt sustainable practices, particularly in their supply chains. A significant challenge in global commerce is the environmental impact of large-scale production and distribution. From carbon emissions in transportation to excessive waste generation, unsustainable supply chains contribute to ecological harm.

To address these issues, businesses can adopt sustainable sourcing, reduce waste, and invest in renewable energy. For instance, companies like Tesla focus on eco-friendly innovations, while brands like Patagonia ensure their products are made from recycled materials. Sustainable supply chain management also involves fair labor practices, ensuring workers receive fair wages and work in safe conditions.

Technology plays a vital role in achieving sustainability. The use of blockchain for supply chain transparency, artificial intelligence for efficient resource management, and green logistics for eco-friendly transportation solutions are transforming business operations. By integrating sustainability into their global supply chains, businesses can reduce costs, enhance efficiency, and improve their corporate reputation.

Conclusion

Business ethics and sustainability are critical components of responsible global commerce. While businesses face challenges such as unethical labor practices, environmental degradation, and regulatory complexities, adopting ethical guidelines and sustainable strategies can lead to long-term success. Promoting transparency, accountability, and eco-conscious practices will not only enhance a company's reputation but also contribute to a more sustainable and equitable global economy.

As the world continues to evolve, businesses must prioritize ethical and sustainable operations to ensure a prosperous future. The integration of ethical leadership, responsible supply chains, and technological advancements will help organizations navigate global commerce challenges while making a positive impact on society and the environment.

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DIGITAL TRANSFORMATION AND ITS IMPACT ON GLOBAL BUSINESS STRATEGIES

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Introduction

In today's rapidly evolving business landscape, digital transformation has become a critical factor for organizations striving to maintain a competitive edge. It is a process of integrating digital technologies into various aspects of a business to fundamentally change how organizations operate and deliver value to customers. Digital transformation has affected virtually every industry and continues to shape the future of global business strategies. From enhanced data analytics capabilities to the advent of artificial intelligence (AI) and automation, businesses are increasingly adopting new technologies to improve efficiency, streamline operations, and foster innovation.

The digital era has given rise to new business models, disrupted traditional industries, and led to the development of new consumer expectations. Global companies are now focused on leveraging digital tools to adapt to changing market conditions and deliver innovative solutions. In this article, we will examine the concept of digital transformation and explore how it is influencing global business strategies. The article will highlight the objectives businesses aim to achieve through digital transformation and analyze the various challenges and opportunities that arise from adopting these technologies.

Objectives

The purpose of this article is to:

1. Analyze how digital transformation is reshaping business strategies on a global scale, including the opportunities and challenges it presents to organizations.
2. Examine the role of emerging technologies such as AI, cloud computing, and big data in driving digital transformation, and how businesses are leveraging these tools to stay competitive in an increasingly digital world.

Digital Transformation: The Global Context

Digital transformation is more than just adopting new technologies; it involves a cultural shift within organizations, altering how they interact with customers, employees, and partners. Organizations are adopting digital tools to improve customer experiences, streamline operations, enhance decision-making, and drive innovation. This transformation can be seen in various industries, including healthcare, manufacturing, finance, retail, and logistics.

For global businesses, digital transformation is not just a tactical move but a strategic imperative. Companies that fail to embrace digital transformation risk losing market share to more agile competitors. The integration of digital tools allows businesses to expand their reach, reduce costs, and enhance productivity. For instance, companies like Amazon and Uber have revolutionized their industries by utilizing digital platforms to provide convenient and cost-effective solutions to consumers.

The growth of cloud computing has played a key role in accelerating digital transformation. By offering scalable and flexible infrastructure, cloud solutions enable

businesses to innovate faster and provide better services to customers. Moreover, cloud computing allows companies to manage vast amounts of data and run complex applications, helping them gain insights and make more informed decisions

The Impact of Digital Transformation on Business Models

One of the most significant impacts of digital transformation is the shift in business models. Traditionally, businesses relied on physical stores, face-to-face interactions, and manual processes to conduct operations. However, the rise of digital technologies has led to the creation of new, technology-driven business models that enable organizations to reach customers in innovative ways. Some examples of these digital business models include:

1. Subscription-based models: Companies such as Netflix and Spotify have adopted subscription-based models that provide customers with continuous access to content for a recurring fee. This model ensures a steady revenue stream and offers convenience to customers.
2. Platform-based models: The platform business model, exemplified by companies like Airbnb and Uber, allows businesses to connect customers with service providers through digital platforms. These platforms typically leverage digital tools like mobile apps and cloud technologies to facilitate transactions and deliver services efficiently.
3. E-commerce models: The rise of e-commerce has fundamentally changed the retail industry. Companies like Amazon and Alibaba have revolutionized the way consumers shop, offering a wide range of products and services through online platforms. Digital transformation has enabled businesses to offer personalized experiences, target specific customer segments, and reduce operational costs.
4. Data-driven models: The use of data analytics and artificial intelligence has led to the emergence of data-driven business models. Companies are leveraging big data to personalize customer experiences, improve marketing efforts, and optimize product offerings. This data-driven approach enables businesses to understand customer preferences and predict future trends.

The Role of Emerging Technologies in Digital Transformation

Emerging technologies play a pivotal role in driving digital transformation and influencing global business strategies. Key technologies such as artificial intelligence (AI), machine learning, cloud computing, the Internet of Things (IoT), and blockchain are enabling organizations to operate more efficiently, innovate rapidly, and enhance customer experiences.

1. Artificial Intelligence (AI): AI is revolutionizing industries by automating routine tasks, enhancing decision-making, and providing personalized experiences. In sectors like finance, AI-powered algorithms are being used to detect fraudulent activities, while in healthcare, AI is helping doctors diagnose diseases more accurately and quickly.
2. Cloud Computing: Cloud computing has become the backbone of digital transformation. It enables businesses to scale their infrastructure, store vast amounts of data, and deploy applications quickly. The cloud's flexibility allows businesses to respond to changing market conditions and reduce the costs associated with maintaining on-premise IT systems.

3. Internet of Things (IoT): IoT refers to the network of physical devices connected to the internet. Businesses use IoT to gather real-time data from connected devices, enabling better decision-making and improving operational efficiency. For example, smart factories use IoT sensors to monitor production processes, detect issues early, and optimize resource allocation.
4. Blockchain: Blockchain technology is increasingly being adopted for its ability to provide secure, transparent, and immutable records of transactions. This technology is particularly valuable in industries like finance, supply chain management, and healthcare, where trust and security are critical.

Challenges and Opportunities in Digital Transformation

While digital transformation offers significant opportunities, it also presents challenges that businesses must navigate to succeed in the global marketplace.

Challenges:

1. Resistance to Change: One of the major hurdles in digital transformation is resistance to change. Employees and leaders may be reluctant to adopt new technologies due to fear of the unknown or concerns about job displacement.
2. Cybersecurity Risks: As businesses embrace digital tools and store more data online, cybersecurity becomes a major concern. Protecting sensitive data and safeguarding against cyberattacks is essential to maintaining trust and reputation.
3. Integration Complexity: Many businesses still operate with legacy systems, making it difficult to integrate new digital technologies. The cost and complexity of upgrading infrastructure can be a significant barrier.

Opportunities:

1. Improved Efficiency: Digital transformation enables businesses to streamline operations, automate routine tasks, and optimize resource allocation. This leads to significant cost savings and increased operational efficiency.
2. Enhanced Customer Experience: Digital tools enable businesses to provide personalized services, enhance customer interactions, and deliver products and services in innovative ways. This leads to higher customer satisfaction and loyalty.
3. Global Reach: Digital transformation enables businesses to expand their reach to global markets by leveraging online platforms, social media, and digital marketing tools. This provides businesses with access to new customers and markets, creating growth opportunities.

Conclusion

In conclusion, digital transformation is fundamentally changing the way businesses operate and compete in the global marketplace. It has introduced new business models, disrupted traditional industries, and empowered organizations to deliver innovative products and services to customers. The integration of emerging technologies like AI, cloud computing, IoT, and blockchain has revolutionized industries, providing companies with new tools to improve efficiency, enhance customer experiences, and drive growth.

While the journey of digital transformation presents challenges such as resistance to change, cybersecurity risks, and integration complexities, it also offers tremendous opportunities for businesses to innovate, expand globally, and achieve sustainable success. As

technology continues to evolve, the ability of businesses to adapt and integrate these advancements into their strategies will be key to their long-term success.

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THE CHALLENGES OF EDUCATIONAL EQUITY IN A GLOBALIZED WORLD

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Introduction

Education has long been recognized as a fundamental human right, essential for the development of individuals and societies alike. In a world that is increasingly interconnected through globalization, the importance of accessible and equitable education is more critical than ever. Globalization, characterized by the movement of people, ideas, goods, and technology across borders, has brought numerous benefits, including economic growth, cultural exchange, and technological advancements. However, it has also highlighted and often exacerbated the existing inequalities in educational systems worldwide.

Educational equity refers to the principle that all individuals, regardless of their socio-economic status, gender, ethnicity, or geographical location, should have access to the same quality of education. Achieving this equity in a globalized world is fraught with challenges. This article aims to explore these challenges, focusing on two key objectives: the impact of socio-economic disparities and the influence of globalization on educational systems. Understanding these issues is crucial for policymakers, educators, and activists working towards a more just and equitable global education system.

Objective 1: The Impact of Socio-Economic Disparities on Educational Equity

One of the most significant challenges to achieving educational equity in a globalized world is the persistent socio-economic divide. Across many parts of the world, access to quality education is closely tied to economic resources. In wealthier countries, educational institutions tend to have better funding, resources, and infrastructure, while schools in lower-income areas often struggle with inadequate facilities, insufficient teaching staff, and outdated materials. This disparity creates a cycle of inequality, where children from disadvantaged backgrounds receive an education that does not provide them with the same opportunities as their more affluent peers. In developing countries, the challenge is even more pronounced. For example, according to UNESCO, approximately 258 million children and youth are out of school, and many of them come from the poorest regions of the world. Factors such as child labor, early marriages, and lack of infrastructure often prevent children in these areas from accessing education altogether. Additionally, gender inequality plays a critical role, with girls in some regions facing cultural barriers and discrimination that hinder their ability to attend school.

In a globalized economy, these socio-economic disparities are magnified. As globalization increases the demand for highly skilled workers, those without access to quality education are left behind. The digital divide further exacerbates this issue, as technology becomes increasingly important in education. Students in wealthier nations or urban centers often have access to the latest digital tools and online resources, while those in rural or underdeveloped regions may lack basic internet access or the devices necessary to participate in modern educational systems.

Objective 2: The Influence of Globalization on Educational Systems

While globalization has led to remarkable technological advancements and greater global interconnectedness, it has also had a significant impact on educational systems around the world. The rapid spread of technology and information has created both opportunities and challenges in the realm of education. On the one hand, globalization has made knowledge and

educational resources more accessible than ever before. Online learning platforms, such as Coursera, edX, and Khan Academy, have democratized access to high-quality courses and learning materials, enabling individuals in remote or underserved areas to gain knowledge and skills previously unavailable to them. However, globalization also brings with it the risk of homogenization. The influence of Western educational models and practices is pervasive, and many countries feel pressure to conform to these standards, even when they may not be culturally or contextually appropriate. This trend often leads to the erosion of local knowledge systems and educational practices, as well as the marginalization of indigenous cultures and languages. In some cases, educational reforms driven by global economic pressures prioritize vocational training and technical skills over critical thinking, creativity, and the arts, which are essential for holistic human development.

Moreover, the internationalization of education, while providing opportunities for cross-cultural exchange, can also exacerbate inequalities between countries. Students from wealthier nations can afford to study abroad, access prestigious institutions, and gain international experience, while students from poorer nations face significant barriers, including financial constraints, visa restrictions, and a lack of resources. This disparity further entrenches global inequalities, as the children of affluent families benefit from educational opportunities that are out of reach for those in lower-income countries.

Conclusion

The challenges of educational equity in a globalized world are complex and multifaceted. Socio-economic disparities remain a significant barrier to achieving educational equality, as children from disadvantaged backgrounds often face systemic obstacles that prevent them from accessing quality education. At the same time, the influence of globalization on educational systems creates both opportunities and risks, as countries navigate the pressures of modernizing their educational frameworks while maintaining cultural and contextual relevance.

To address these challenges, it is crucial for policymakers, educators, and international organizations to work together to create solutions that promote educational equity on a global scale. This involves not only investing in infrastructure and resources for marginalized communities but also ensuring that education systems are adaptable to the changing demands of a globalized world. By prioritizing inclusivity, cultural sensitivity, and equal access to educational opportunities, we can build a future where education serves as a tool for empowerment and social justice for all individuals, regardless of their background or geographical location.

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THE INFLUENCE OF GLOBALIZATION ON CONSUMER PROTECTION LAWS

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Introduction

Globalization has transformed economies, trade, and consumer behavior worldwide. It has created a more interconnected world where goods, services, and information flow across borders. While this has provided consumers with diverse choices, improved quality, and competitive pricing, it has also introduced challenges such as fraud, exploitation, and unfair trade practices. To address these challenges, consumer protection laws have evolved to safeguard consumers' rights in an increasingly globalized market. This article explores the influence of globalization on consumer protection laws, focusing on its benefits, challenges, and the role of international cooperation.

Objectives of the Study

1. To Examine the Impact of Globalization on Consumer Rights and Protection Laws

Globalization has significantly altered the way businesses operate and how consumers interact with markets. With the rise of e-commerce and multinational corporations, consumer protection laws have become more complex. Issues such as product safety, misleading advertisements, and online fraud have increased, necessitating stronger legal frameworks. Countries have had to update their laws to regulate international transactions, enforce fair trade practices, and ensure that consumers are not exploited in a global market.

For instance, many nations have adopted stricter regulations to ensure product safety standards align with international norms. Organizations like the World Trade Organization (WTO) and the United Nations Conference on Trade and Development (UNCTAD) have played crucial roles in establishing global consumer protection guidelines. Additionally, regional alliances like the European Union (EU) have implemented standardized laws to protect consumers across member states.

2. To Analyse the Challenges and Solutions in Implementing Consumer Protection Laws in a Globalized Economy

While globalization has encouraged economic growth, it has also posed challenges in enforcing consumer protection laws. One of the primary issues is jurisdictional limitations. Since businesses operate across multiple countries, enforcing laws and penalizing fraudulent companies becomes difficult. Consumers may struggle to seek justice when they purchase defective or misleading products from foreign sellers.

Another major challenge is the variation in consumer protection laws across different countries. While some nations have strict regulations, others have weak enforcement mechanisms, leading to disparities in consumer rights. To address this, international cooperation has become crucial. Countries have begun collaborating through trade agreements, consumer protection networks, and global regulatory bodies to create a more uniform legal framework. For example, the International Consumer Protection and Enforcement Network (ICPEN) facilitates cooperation among consumer protection authorities worldwide.

Technological advancements also offer solutions to consumer protection challenges. Digital platforms now provide better transparency through verified reviews, secure payment

systems, and fraud detection mechanisms. Governments and consumer rights organizations are leveraging artificial intelligence and blockchain technology to track product authenticity and detect deceptive practices.

Conclusion

Globalization has had a profound impact on consumer protection laws, necessitating stronger regulations and international cooperation. While it has created new opportunities for consumers, it has also introduced risks such as fraud and exploitation. Governments and international organizations have responded by updating legal frameworks, promoting cross-border collaborations, and leveraging technology to protect consumer rights. Moving forward, a global approach to consumer protection will be essential to ensure fair trade, ethical business practices, and a safer marketplace for consumers worldwide.

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THE ROLE OF VOCATIONAL EDUCATION IN PREPARING STUDENTS FOR GLOBAL JOBS

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Introduction

In today's fast-changing job market, vocational education plays a crucial role in equipping students with practical skills, technical knowledge, and industry-specific training needed for employment in global industries. Unlike traditional academic education, vocational education focuses on hands-on learning and real-world applications, preparing students for careers in fields such as healthcare, engineering, information technology, and skilled trades. As globalization continues to shape the workforce, vocational education becomes increasingly important in helping students secure jobs that demand specialized skills and adaptability.

This article explores the role of vocational education in preparing students for global employment. It highlights two key objectives: (1) bridging the gap between education and industry needs and (2) enhancing students' adaptability and employability in a global economy.

Objective 1: Bridging the Gap Between Education and Industry Needs

One of the biggest challenges in today's job market is the mismatch between educational curricula and industry demands. Many graduates struggle to find employment because their skills do not align with what employers require. Vocational education addresses this issue by focusing on skill-based learning and industry-specific training.

Practical Skill Development

Vocational programs provide students with hands-on experience through internships, apprenticeships, and practical training sessions. This allows them to develop technical skills that are directly applicable to their chosen profession. For example, a student in a vocational automotive program will gain real-world experience in car repair and diagnostics, making them job-ready upon graduation.

Industry Collaboration

Many vocational institutions collaborate with industries to design courses that reflect the latest technological advancements and labor market trends. Companies often participate in curriculum development, ensuring that students acquire skills that are relevant to the global workforce. For example, in fields like cybersecurity and artificial intelligence, vocational training institutes partner with tech companies to provide cutting-edge knowledge and certification programs.

By aligning education with industry needs, vocational education enhances students' employability and ensures they can transition smoothly from school to work.

Objective 2: Enhancing Adaptability and Employability in a Global Economy

As globalization increases, the job market is no longer confined to a single country. Companies seek employees who can adapt to diverse work environments, handle technological advancements, and meet international standards. Vocational education plays a key role in preparing students for such challenges.

Global Competency Development

Vocational education often includes training in soft skills such as communication, teamwork, and problem-solving, which are essential in multinational workplaces. Many programs also integrate language learning and cross-cultural training to help students work in different cultural settings. For instance, a hospitality management student may receive training on how to interact with international clients and adhere to global service standards.

Technological Adaptation

With rapid advancements in technology, vocational education ensures students are equipped with the latest digital tools and technical knowledge. Programs in fields like IT, renewable energy, and healthcare teach students how to use modern software, machinery, and digital platforms. This prepares them for jobs in industries that are continuously evolving due to technological progress.

Through these initiatives, vocational education enables students to compete in the global job market by equipping them with both technical expertise and the adaptability needed to succeed in diverse work environments.

Conclusion

Vocational education plays a vital role in preparing students for the global job market by bridging the gap between academic training and industry requirements. It provides hands-on experience, aligns curricula with employer needs, and enhances students' adaptability to changing global work environments. With a strong emphasis on practical skills, technological advancements, and international competency, vocational education ensures that students are not only employable but also competitive in the ever-evolving global workforce.

As the demand for skilled professionals grows worldwide, investing in vocational education becomes essential for individuals, businesses, and governments. By fostering career readiness and professional development, vocational education contributes to economic growth and a more skilled, adaptable workforce.

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THE ROLE OF INNOVATION IN STRENGTHENING INTERNATIONAL SUPPLY CHAINS

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Introduction

Global supply chains are the backbone of international trade, connecting businesses, suppliers, and consumers across the world. However, these supply chains face numerous challenges, including disruptions caused by pandemics, geopolitical tensions, and fluctuating demand. To address these issues, innovation plays a crucial role in enhancing efficiency, resilience, and sustainability within supply chains. The integration of advanced technologies such as artificial intelligence (AI), blockchain, and the Internet of Things (IoT) has revolutionized the way supply chains operate, ensuring real-time tracking, reducing risks, and improving overall performance. This article explores how innovation strengthens international supply chains, focusing on two key objectives: improving supply chain efficiency and enhancing risk management.

Objective 1: Improving Supply Chain Efficiency Through Innovation

One of the primary roles of innovation in international supply chains is improving efficiency. Supply chain efficiency involves reducing costs, minimizing delays, and optimizing logistics operations. Several innovative solutions have been developed to achieve these goals:

1. Automation and Artificial Intelligence (AI):

Automation and AI have transformed supply chain management by streamlining processes such as inventory management, order fulfillment, and demand forecasting. AI-driven algorithms analyze vast amounts of data to predict demand trends, helping companies optimize production and avoid overstocking or shortages. Additionally, automated warehouses and robotic systems enhance order processing speed and accuracy, reducing human error and operational costs.

2. Blockchain Technology:

Blockchain innovation ensures transparency and security in supply chains by providing a decentralized and immutable ledger of transactions. This technology enables real-time tracking of goods from production to delivery, reducing fraud and errors. Companies using blockchain can verify the authenticity of products, prevent counterfeiting, and improve trust between stakeholders.

3. Internet of Things (IoT):

IoT devices, such as smart sensors and GPS trackers, enable real-time monitoring of shipments. These devices provide data on location, temperature, humidity, and other critical factors that affect product quality. For instance, in the pharmaceutical and food industries, IoT technology ensures that perishable goods are transported under optimal conditions, minimizing waste and financial losses.

Objective 2: Enhancing Risk Management in Global Supply Chains

Innovation also strengthens international supply chains by improving risk management strategies. Unforeseen disruptions, such as natural disasters, cyberattacks, and political

instability, can severely impact global trade. Innovative solutions help companies identify risks, mitigate disruptions, and ensure continuity.

1. Predictive Analytics and Big Data:

Companies leverage big data and predictive analytics to assess potential risks in supply chains. These technologies analyze historical data and external factors, such as weather patterns and economic trends, to anticipate disruptions. By identifying vulnerabilities in advance, businesses can develop contingency plans, such as alternative sourcing strategies or diversified supplier networks.

2. Resilient Supply Chain Design:

Innovations in supply chain design, such as digital twin technology, allow companies to create virtual models of their supply chains. These digital replicas simulate real-world scenarios, helping businesses test different strategies before implementing them. This proactive approach enhances resilience by ensuring that companies can quickly adapt to disruptions.

3. Cybersecurity Advancements:

With the increasing reliance on digital technologies, cybersecurity has become a critical aspect of supply chain risk management. Innovations such as AI-driven security systems and blockchain encryption protect supply chain data from cyber threats. Secure digital platforms enable safe transactions and prevent unauthorized access to sensitive information.

Conclusion

Innovation is essential for strengthening international supply chains by improving efficiency and enhancing risk management. Technologies such as AI, blockchain, IoT, and predictive analytics streamline logistics operations, increase transparency, and mitigate potential disruptions. As global trade continues to evolve, companies must embrace innovative solutions to remain competitive and resilient in an ever-changing economic landscape. By investing in technological advancements and adaptive strategies, businesses can ensure a more robust and sustainable supply chain for the future.

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SUSTAINABLE DEVELOPMENT GOALS (SDGs): BUSINESS AND EDUCATION

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Introduction

The Sustainable Development Goals (SDGs) are a global blueprint established by the United Nations (UN) in 2015 to address key challenges such as poverty, inequality, climate change, environmental degradation, peace, and justice. The SDGs consist of 17 goals and 169 targets designed to be achieved by 2030, aiming to create a more sustainable, inclusive, and prosperous world. Among the most influential sectors contributing to the realization of these goals are business and education.

Businesses play a crucial role in sustainable development by promoting ethical practices, reducing environmental footprints, and fostering innovation. Meanwhile, education is a powerful tool for equipping individuals with the knowledge and skills needed to build sustainable economies and societies. Together, business and education create a foundation for achieving SDGs by promoting economic growth, environmental responsibility, and social equity. This article explores the significance of SDGs in business and education, highlighting their objectives and impact on global sustainability.

Objective 1: The Role of Business in Achieving SDGs

Businesses are at the forefront of sustainable development because they have the resources, influence, and capabilities to drive significant change. The private sector can contribute to SDGs in various ways, including:

1. Sustainable Business Practices

Businesses can adopt sustainable practices by minimizing waste, reducing carbon emissions, and using renewable energy sources. Companies like Tesla, Unilever, and IKEA have implemented green energy solutions and sustainable supply chains, demonstrating that businesses can thrive while being environmentally responsible.

2. Corporate Social Responsibility (CSR)

CSR initiatives allow businesses to give back to society by investing in community development, education, and health programs. Many multinational corporations engage in CSR projects aligned with SDGs, such as promoting gender equality (SDG 5), providing clean water (SDG 6), and ensuring decent work conditions (SDG 8).

3. Innovation and Technology

Technological advancements play a crucial role in achieving SDGs. Businesses investing in artificial intelligence (AI), sustainable agriculture, and renewable energy can create innovative solutions for global challenges like food security, clean energy, and efficient water management.

4. Partnerships for Sustainable Development

Collaboration between businesses, governments, and non-profit organizations fosters sustainable development. Partnerships help businesses align their strategies with global sustainability goals, making significant contributions to areas like climate action (SDG 13) and sustainable cities.

By integrating sustainability into their core operations, businesses not only contribute to global progress but also improve their long-term profitability and reputation.

Objective 2: The Role of Education in Achieving SDGs

Education is a fundamental driver of sustainable development, as it empowers individuals, fosters critical thinking, and promotes awareness of global challenges. SDG 4 specifically focuses on ensuring inclusive and quality education for all, but education plays a role in achieving nearly all SDGs.

1. Education for Sustainable Development (ESD)

Education for Sustainable Development (ESD) equips learners with the knowledge, skills, and attitudes needed to build a sustainable future. Schools, universities, and training institutions integrate sustainability topics into curricula, encouraging students to adopt eco-friendly lifestyles and ethical business practices.

2. Promoting Gender Equality and Social Inclusion

Education helps eliminate gender disparities (SDG 5) and promotes equal opportunities for all. When women and marginalized groups have access to quality education, they can contribute to economic growth, innovation, and social stability.

3. Enhancing Employment and Economic Growth

Education plays a critical role in reducing unemployment and poverty (SDG 1). Vocational training, technical education, and digital literacy programs equip individuals with skills needed in industries like renewable energy, healthcare, and information technology, leading to sustainable economic growth (SDG 8).

4. Encouraging Responsible Consumption and Environmental Awareness

Sustainability-focused education raises awareness about climate change (SDG 13), responsible consumption (SDG 12), and biodiversity conservation (SDG 15). Schools and universities can promote eco-friendly practices like recycling, energy conservation, and sustainable agriculture.

By making sustainability a core component of education, future generations will be better prepared to tackle global challenges and create a more resilient and inclusive society.

Conclusion

Sustainable Development Goals (SDGs) serve as a guiding framework for building a better future. Businesses contribute to SDGs by adopting sustainable practices, investing in innovation, and engaging in social responsibility. Meanwhile, education empowers individuals with knowledge and skills to address global challenges, fostering a culture of sustainability and social equity.

The collaboration between business and education is essential for achieving SDGs. By integrating sustainability into corporate strategies and academic curricula, societies can create long-term solutions for environmental, social, and economic challenges. With continued efforts and strong partnerships, the world can progress toward a more sustainable, inclusive, and prosperous future by 2030.

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EDUCATIONAL INNOVATION IN PROMOTING FINANCIAL LITERACY WORLDWIDE

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Introduction

Financial literacy is a crucial life skill that empowers individuals to make informed financial decisions, manage their resources effectively, and secure a stable future. In an increasingly complex global economy, the need for financial literacy has grown significantly. However, many individuals lack essential knowledge about budgeting, saving, investing, and debt management. To address this gap, educational institutions, governments, and organizations worldwide are leveraging innovative approaches to promote financial literacy.

Educational innovation in financial literacy includes integrating digital tools, gamification, interactive learning, and real-life simulations into curricula. These advancements make financial education more accessible, engaging, and effective. This article explores two key objectives of educational innovation in financial literacy: enhancing accessibility through digital learning and fostering practical financial skills through experiential learning.

Objective 1: Enhancing Accessibility Through Digital Learning

One of the biggest challenges in promoting financial literacy is ensuring accessibility for diverse populations. Many individuals, particularly in developing countries, lack access to traditional financial education. Digital learning innovations have played a crucial role in bridging this gap.

1.1 Online Courses and E-Learning Platforms

Many universities, financial institutions, and non-profit organizations offer free or affordable online courses on financial literacy. Platforms such as Coursera, Khan Academy, and edX provide interactive modules on budgeting, investing, credit management, and retirement planning. These courses allow learners to acquire financial skills at their own pace, making education more flexible and inclusive.

1.2 Mobile Applications and AI-Based Tools

Mobile applications like Mint, YNAB (You Need a Budget), and MyMoneyCoach use artificial intelligence to provide personalized financial advice, budget tracking, and expense analysis. AI-driven chatbots also assist users in managing their finances, answering questions, and offering real-time recommendations based on spending patterns.

1.3 Financial Literacy Through Social Media and Digital Campaigns

Social media platforms such as YouTube, Instagram, and TikTok have become powerful tools for financial education. Influencers and financial experts create engaging content, including short videos, live Q&A sessions, and infographics, to educate people on financial topics. Digital campaigns launched by governments and financial institutions also raise awareness and provide essential financial knowledge.

Objective 2: Fostering Practical Financial Skills Through Experiential Learning

Understanding financial concepts is not enough; individuals must also develop practical skills to apply their knowledge effectively. Experiential learning methods, such as simulations,

gamification, and hands-on financial management, help learners develop real-world financial competence.

2.1 Gamification and Simulation-Based Learning

Gamification integrates financial literacy with interactive games that simulate real-life financial decision-making. Platforms like Monopoly (digital version), The Stock Market Game, and Budget Challenge allow users to experience financial planning, investment strategies, and risk management in a controlled environment. These games enhance engagement, making learning enjoyable and impactful.

2.2 School and University Financial Education Programs

Many educational institutions have incorporated financial literacy courses into their curricula. Schools teach fundamental money management skills through practical assignments, such as creating a personal budget or planning an investment portfolio. Universities offer specialized courses in personal finance, entrepreneurship, and investment, preparing students for financial independence.

2.3 Hands-On Learning Through Financial Inclusion Initiatives

Governments and financial institutions worldwide are promoting financial literacy through hands-on learning programs. Initiatives like youth savings programs, microfinance training, and financial mentorship projects enable individuals to practice financial management in real-life scenarios. For example, some schools partner with banks to allow students to open and manage savings accounts, teaching them money management skills early in life.

These experiential learning strategies ensure that financial literacy is not just theoretical but also practical, enabling individuals to develop responsible financial habits.

Conclusion

Educational innovation plays a crucial role in promoting financial literacy worldwide. By enhancing accessibility through digital learning and fostering practical financial skills through experiential learning, individuals can develop the knowledge and confidence needed to make informed financial decisions.

Governments, educational institutions, and financial organizations must continue to invest in innovative financial education strategies. By integrating technology, interactive learning, and real-world financial applications, financial literacy can be improved globally, empowering people to achieve financial stability and long-term success.

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THE IMPACT OF ONLINE LEARNING ON TRADITIONAL CLASSROOM EDUCATION

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Introduction

Education has undergone significant transformations over the past few decades, with the rapid advancement of technology playing a crucial role. One of the most notable changes is the rise of online learning, which has revolutionized how students access knowledge. Online learning provides flexibility, accessibility, and a wide range of resources that were previously unavailable in traditional classroom settings. However, this shift has also raised concerns about its impact on conventional education. This article explores how online learning affects traditional classroom education, focusing on its advantages, challenges, and long-term implications.

Objective 1: The Advantages of Online Learning Over Traditional Classrooms

Online learning has introduced several benefits that have influenced traditional classroom education. Some of these advantages include:

1. **Flexibility and Convenience** – Online learning allows students to study at their own pace and schedule. Unlike traditional classrooms that require physical attendance, online education enables learners to access materials anytime and anywhere. This flexibility is particularly beneficial for working professionals and students with other responsibilities.
2. **Cost-Effectiveness** – Online education reduces costs associated with transportation, accommodation, and physical learning materials. Many universities and platforms offer free or affordable courses, making education more accessible to a broader audience.
3. **Personalized Learning** – Unlike traditional classrooms with a one-size-fits-all approach, online learning platforms often provide customized experiences. Students can choose their preferred subjects, learning styles, and pace, which enhances their understanding and engagement.
4. **Global Access to Knowledge** – Online education connects students with educators, professionals, and peers worldwide. This exposure to diverse perspectives enriches learning and encourages collaboration across different cultures.

Objective 2: Challenges of Online Learning and Its Effects on Traditional Education

While online learning has many advantages, it also presents challenges that impact traditional classroom education. These challenges include:

1. **Lack of Face-to-Face Interaction** – Traditional classrooms promote direct interaction between students and teachers, which helps in immediate feedback and emotional connection. Online learning lacks this personal engagement, which may affect student motivation and participation.
2. **Technological Barriers** – Not all students have access to reliable internet connections or digital devices, making online education challenging for underprivileged communities. This digital divide can create inequalities in educational opportunities.
3. **Self-Discipline and Motivation** – Online learning requires strong self-discipline and time management skills. Many students struggle to stay motivated without the structured

environment of a traditional classroom. This can lead to lower completion rates in online courses.

4. Quality and Credibility of Online Education – While many reputable institutions offer online courses, some programs lack proper accreditation and standardized assessment methods. Employers and academic institutions may question the credibility of online degrees compared to traditional classroom education.

Conclusion

Online learning has significantly impacted traditional classroom education, offering flexibility, cost savings, and global access to knowledge. However, it also poses challenges such as reduced face-to-face interaction, technological barriers, and concerns about quality. While online education is a valuable supplement to traditional learning, it cannot entirely replace the benefits of in-person instruction. A balanced approach that integrates both methods—blended learning—could be the future of education, combining the best aspects of online and traditional learning environments.

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THE FUTURE OF WORK: ADAPTING EDUCATION TO PREPARE STUDENTS FOR THE GLOBAL JOB MARKET

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Introduction

The global job market is evolving rapidly due to advancements in technology, automation, artificial intelligence, and globalization. Traditional job roles are changing, and new career opportunities are emerging, requiring workers to develop new skills and adapt to a dynamic work environment. As a result, the education system must also evolve to prepare students for the future workforce. Educational institutions need to equip students with technical skills, critical thinking abilities, digital literacy, and adaptability to succeed in a competitive and technology-driven job market. This article examines how education can be restructured to meet the demands of the global workforce.

Objectives of the Study

1. To Analyze the Changing Nature of Work and Its Impact on Education

The job market is increasingly influenced by automation and artificial intelligence, reducing the demand for routine and manual labor while increasing the need for complex problem-solving and technical skills. Many traditional jobs are becoming obsolete, and new roles in data science, cybersecurity, artificial intelligence, and remote work are emerging.

To address these shifts, education systems must move beyond conventional teaching methods and embrace an interdisciplinary approach. Schools and universities should integrate subjects like coding, robotics, data analysis, and entrepreneurship into their curricula. Additionally, soft skills such as teamwork, leadership, adaptability, and emotional intelligence must be emphasized to help students thrive in a dynamic job market.

Vocational training and lifelong learning opportunities should also be prioritized. With the rise of remote work and the gig economy, students need to be prepared for flexible work arrangements, freelancing, and entrepreneurship. Education must equip learners with the ability to continuously upskill and reskill throughout their careers.

2. To Explore Innovative Educational Strategies for Preparing Students for the Global Job Market

To ensure that students are well-prepared for the global workforce, educational institutions must adopt innovative teaching strategies. Some key approaches include:

Experiential Learning: Schools should focus on project-based learning, internships, apprenticeships, and real-world problem-solving exercises. This approach allows students to develop practical skills that are directly applicable to their careers.

Digital Literacy and Technological Integration: In the digital era, proficiency in technology is essential. Education should integrate digital tools, coding, artificial intelligence, and data analysis into mainstream curricula to ensure students are technologically proficient.

Global Collaboration and Cross-Cultural Skills: Since globalization has made cross-border collaboration essential, students must develop cultural awareness, communication skills, and the ability to work in diverse teams. Virtual exchange programs and international collaborations should be encouraged.

Personalized Learning and Lifelong Education: Adaptive learning technologies and online education platforms allow students to learn at their own pace. Personalized learning ensures that students acquire relevant skills based on their career interests and strengths. Additionally, lifelong learning programs should be promoted to encourage continuous professional development.

Conclusion

The future of work demands a restructured education system that focuses on innovation, technology, and adaptability. As automation and globalization reshape industries, students must develop a combination of technical, soft, and digital skills to remain competitive in the job market. By integrating experiential learning, digital literacy, global collaboration, and personalized education, institutions can better prepare students for future careers. Continuous learning and upskilling will be essential for career success, making it crucial for education systems to evolve alongside industry trends. Adapting education for the global workforce is not just necessary—it is the key to ensuring economic growth and individual success in the 21st century.

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THE ROLE OF SOCIAL MEDIA IN ENHANCING EDUCATION IN DEVELOPING COUNTRIES

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Introduction

In the 21st century, social media has become a powerful tool for communication, collaboration, and knowledge sharing. Platforms such as Facebook, YouTube, WhatsApp, Twitter, and educational forums have transformed how people access and share information. In developing countries, where educational infrastructure may be limited, social media offers new opportunities to bridge the digital divide and provide quality education to a broader audience. With the increasing penetration of mobile phones and internet services, students and teachers can now use social media to enhance learning, improve engagement, and foster a culture of lifelong education. This article explores the role of social media in education, focusing on two key objectives: improving access to educational resources and fostering collaboration among students and teachers.

Objective 1: Improving Access to Educational Resources

One of the most significant benefits of social media in education is the ability to access a vast pool of knowledge from around the world. In many developing countries, educational institutions often struggle with inadequate libraries, outdated textbooks, and limited teaching materials. Social media helps overcome these challenges in the following ways:

1. **Free Educational Content:** Platforms such as YouTube, Facebook, and TikTok provide free access to lectures, tutorials, and study materials from experts worldwide. Students can watch educational videos, attend virtual classes, and participate in live discussions on subjects ranging from mathematics to language learning.
2. **Open Educational Resources (OERs):** Websites like Coursera, edX, and Khan Academy use social media to reach students, offering free or affordable courses. Teachers and students can share these resources on social media platforms, making quality education more accessible.
3. **Remote Learning:** In rural areas where schools are scarce, social media enables distance learning. Teachers can conduct virtual classes through Facebook Live, WhatsApp groups, or YouTube streams, ensuring students receive education regardless of location.
4. **Instant Access to Information:** Unlike traditional learning methods that rely on physical textbooks, social media provides instant access to news, research papers, and global trends in education, allowing students to stay updated with the latest developments in their fields.

Objective 2: Fostering Collaboration Among Students and Teachers

Social media also enhances collaboration and engagement among students and teachers, making learning more interactive and effective. The following points highlight how social media fosters collaboration in education:

1. **Online Study Groups:** Students can create Facebook groups, WhatsApp chats, or Telegram channels to discuss assignments, share notes, and clarify doubts. This peer-to-peer interaction strengthens understanding and improves academic performance.

2. Teacher-Student Engagement: Teachers can use social media platforms to share learning materials, provide feedback, and conduct discussions outside the classroom. This improves communication between students and educators, making learning more efficient.

3. Global Learning Communities: Social media allows students to connect with their peers and teachers from different parts of the world. Platforms like LinkedIn Learning and Twitter facilitate discussions on global educational issues, broadening students' perspectives and encouraging cultural exchange.

4. Live Q&A Sessions and Webinars: Platforms like Instagram Live, Zoom, and Facebook Live allow educators to host interactive Q&A sessions, webinars, and virtual workshops. Students can ask questions in real-time, enhancing their understanding of complex topics.

5. Encouraging Creativity and Innovation: Social media encourages students to create and share educational content through blogs, videos, and infographics. This fosters creativity and digital literacy, essential skills for the modern workforce.

Challenges and Considerations

Despite its benefits, the use of social media in education comes with challenges, including digital divide issues, misinformation, and distractions. Not all students in developing countries have access to reliable internet or digital devices. Additionally, excessive use of social media can lead to distractions and reduced concentration. To maximize the benefits of social media in education, governments and institutions must invest in digital infrastructure, promote media literacy, and encourage responsible online behavior.

Conclusion

Social media plays a crucial role in enhancing education in developing countries by improving access to educational resources and fostering collaboration among students and teachers. It has revolutionized learning by making knowledge more accessible, promoting interactive discussions, and connecting learners worldwide. However, for its full potential to be realized, governments, educational institutions, and stakeholders must address challenges such as digital inequality and online distractions. By leveraging social media effectively, developing countries can bridge educational gaps and empower future generations with quality education.

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DIGITAL LEARNING: INNOVATION IN EDUCATION FOR A GLOBALIZED ECONOMY

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Introduction

In an increasingly interconnected world, digital learning has emerged as a transformative force in education. The rapid growth of digital technologies has revolutionized traditional learning models, making education more accessible, flexible, and efficient. As economies globalize, the demand for skills that transcend borders is rising, and digital learning provides the necessary tools to meet these demands. This article explores the role of digital learning in fostering innovation, its impact on education, and its contribution to a globalized economy.

The Role of Digital Learning in Innovation

Digital learning encompasses a variety of technological tools and platforms, including online courses, virtual classrooms, artificial intelligence (AI), and augmented reality (AR). These innovations have redefined how knowledge is disseminated and absorbed. According to Bates (2019), digital learning promotes personalized education, where learners can progress at their own pace and access customized resources suited to their needs.

One of the key innovations in digital learning is adaptive learning technology, which uses AI to analyze student performance and adjust content accordingly. A study by Popenici and Kerr (2017) highlights how AI-driven education can enhance learning outcomes by providing immediate feedback and tailored lesson plans. Similarly, gamification in education, through interactive learning applications, increases engagement and motivation, making learning more enjoyable and effective (Dicheva et al., 2015).

Impact of Digital Learning on Education

Accessibility and Inclusivity

Digital learning has made education more inclusive by breaking geographical barriers. Learners from remote areas or underprivileged backgrounds can now access high-quality education through online courses. The United Nations Educational, Scientific and Cultural Organization (UNESCO) reports that digital learning tools have significantly improved educational access in developing countries, helping to bridge the knowledge gap (UNESCO, 2021).

Additionally, digital learning supports diverse learning needs, including those of individuals with disabilities. Features such as screen readers, speech-to-text software, and captioning make educational content more accessible to students with visual, auditory, or learning disabilities (Seale, 2013).

Collaboration and Global Learning Networks

The integration of digital tools facilitates collaboration among students and educators across the globe. Online discussion forums, virtual exchange programs, and cloud-based projects enable learners to engage with peers from different cultural backgrounds. According to Salmons (2019), global collaboration through digital learning fosters cross-cultural competencies, essential for success in the global workforce.

Massive Open Online Courses (MOOCs) provided by platforms like Coursera, edX, and Udacity have further expanded learning opportunities. These platforms offer courses from top universities and industry experts, allowing students to acquire skills that are in demand worldwide (Yuan & Powell, 2013).

Digital Learning and the Globalized Economy

Bridging the Skills Gap

As globalization continues to shape economies, the demand for digital literacy and technical skills is rising. Digital learning helps bridge this skills gap by offering specialized training in fields such as data science, cybersecurity, and digital marketing. A report by the World Economic Forum (WEF) (2020) emphasizes that lifelong learning through digital platforms is essential for workforce adaptability in the era of automation and artificial intelligence.

Workforce Development and Remote Learning

Organizations are increasingly adopting digital learning for employee training and professional development. E-learning platforms allow companies to upskill their workforce efficiently, reducing costs and improving productivity. Research by Bersin (2017) shows that companies investing in digital learning experience higher employee engagement and retention.

Furthermore, the rise of remote work has increased the importance of digital education. Online training programs enable professionals to acquire new skills without the need for physical presence, fostering a more flexible and resilient workforce (OECD, 2021).

Challenges and Future Prospects

Despite its numerous advantages, digital learning faces several challenges, including the digital divide, data privacy concerns, and the need for improved digital literacy among educators and students. Limited access to reliable internet and digital devices remains a barrier in many parts of the world (Van Dijk, 2020).

Looking ahead, advancements in AI, virtual reality (VR), and blockchain technology are expected to further revolutionize digital education. AI-powered tutors, immersive VR simulations, and blockchain-based credentialing systems will enhance learning experiences and provide more secure academic records (Mhlanga, 2021).

Conclusion

Digital learning is a powerful innovation that is reshaping education and preparing individuals for the demands of a globalized economy. By promoting accessibility, fostering collaboration, and bridging the skills gap, digital learning enhances educational experiences and economic opportunities. However, addressing challenges such as the digital divide and cybersecurity risks is crucial for its continued success. As technology evolves, digital learning will remain a cornerstone of modern education, equipping learners with the skills needed for the future.

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THE ROLE OF CRYPTOCURRENCIES IN MODERN FINANCIAL MARKETS

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Introduction

A Catalyst for Sustainable Economic Growth in Developing Nations

The emergence of cryptocurrencies has revolutionized the way we think about money and financial transactions. With the rise of Bitcoin, Ethereum, and other digital currencies, the financial landscape has become increasingly complex and interconnected. In this article, we will explore the role of cryptocurrencies in modern financial markets and their potential to drive sustainable economic growth in developing nations.

Objectives:

1. To examine the impact of cryptocurrencies on modern financial markets, including their benefits and drawbacks.
2. To investigate the potential of cryptocurrencies to drive sustainable economic growth in developing nations, highlighting successful case studies and innovative solutions.

Page 2: The Impact of Cryptocurrencies on Modern Financial Markets

Cryptocurrencies have introduced a new paradigm in financial transactions, offering a decentralized, secure, and transparent alternative to traditional fiat currencies. The benefits of cryptocurrencies include:

- Increased financial inclusion: Cryptocurrencies have enabled millions of people in developing nations to access financial services, promoting financial inclusion and economic empowerment.
- Improved transaction efficiency: Cryptocurrencies have reduced transaction costs and increased the speed of cross-border payments, facilitating international trade and commerce.
- Enhanced security: Cryptocurrencies have introduced advanced security features, such as blockchain technology, to protect financial transactions and prevent fraud.

However, cryptocurrencies also pose significant challenges, including:

- Market volatility: Cryptocurrency markets are highly volatile, subject to rapid price fluctuations that can result in significant losses.
- Regulatory uncertainty: The lack of clear regulations and standards has created uncertainty and risk for investors and users.
- Environmental concerns: The energy consumption required to mine cryptocurrencies has raised concerns about their environmental sustainability.

Page 3: Cryptocurrencies and Sustainable Economic Growth in Developing Nations

Despite the challenges, cryptocurrencies have the potential to drive sustainable economic growth in developing nations. Successful case studies include:

- Kenya's M-Pesa: A mobile-based financial service that has enabled millions of Kenyans to access financial services and participate in the formal economy.

- Ecuador's Dinero Electronico: A digital currency launched by the Ecuadorian government to promote financial inclusion and reduce poverty.

Innovative solutions, such as blockchain-based platforms and cryptocurrency-enabled microfinance, are also emerging. These solutions have the potential to:

- Promote financial inclusion: By providing access to financial services and promoting financial literacy.
- Foster entrepreneurship: By enabling entrepreneurs to access capital and markets.
- Enhance transparency and accountability: By utilizing blockchain technology to track transactions and ensure accountability.

Conclusion:

Cryptocurrencies have the potential to drive sustainable economic growth in developing nations by promoting financial inclusion, fostering entrepreneurship, and enhancing transparency and accountability. While challenges persist, innovative solutions and successful case studies demonstrate the potential of cryptocurrencies to transform the financial landscape and promote economic development.

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ARTIFICIAL INTELLIGENCE IN PREDICTING MARKET TRENDS AND CONSUMER PREFERENCES

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Introduction

Artificial Intelligence (AI) has revolutionized various industries, and its impact on market analysis and consumer behavior prediction is particularly significant. In an era of rapidly changing consumer preferences and dynamic market trends, businesses rely on AI-driven insights to make informed decisions, optimize marketing strategies, and enhance customer experiences.

AI enables companies to process vast amounts of data, identify patterns, and predict future trends with high accuracy. By leveraging machine learning, natural language processing (NLP), and predictive analytics, businesses can understand consumer preferences, anticipate demand, and stay ahead of competitors. This article explores two key objectives of AI in market analysis: predicting market trends and enhancing consumer preference analysis.

Objective 1: Predicting Market Trends Using AI

Market trends evolve based on economic shifts, consumer behavior, technological advancements, and global events. AI-driven analytics empower businesses to identify emerging trends and adjust their strategies accordingly.

1.1 Big Data and Machine Learning for Trend Analysis

AI systems analyze large datasets from social media, e-commerce platforms, financial reports, and industry news to identify patterns in consumer behavior. Machine learning algorithms detect correlations and predict upcoming market trends, allowing businesses to launch new products or adjust supply chains proactively.

For example, AI can analyze online search queries and social media discussions to predict fashion trends, food preferences, or travel destinations gaining popularity. By recognizing these patterns early, companies can capitalize on new opportunities before they become mainstream.

1.2 Sentiment Analysis and Social Media Monitoring

Natural Language Processing (NLP) enables AI to perform sentiment analysis on customer reviews, social media posts, and online discussions. By assessing public opinion on brands, products, or industries, AI helps businesses understand how consumers perceive their offerings and predict shifts in market sentiment.

For instance, an AI model analyzing Twitter posts may detect growing consumer interest in eco-friendly products, prompting businesses to focus on sustainable product lines. Such insights allow brands to align their marketing strategies with consumer expectations.

1.3 AI-Powered Demand Forecasting

Predictive analytics in AI enhances demand forecasting by analyzing historical sales data, economic indicators, and external factors like weather or geopolitical events. This helps businesses optimize inventory management, reduce waste, and prevent stock shortages.

Retail giants like Amazon and Walmart use AI-powered demand forecasting to maintain efficient supply chains, ensuring that popular products are stocked while minimizing overproduction. This not only reduces costs but also improves customer satisfaction.

Objective 2: Enhancing Consumer Preference Analysis Using AI

Understanding consumer preferences is essential for personalizing marketing campaigns, improving customer engagement, and boosting sales. AI provides deep insights into individual preferences and purchasing behavior.

2.1 Personalized Recommendations and Customer Insights

AI-driven recommendation engines analyze user data, including browsing history, past purchases, and demographic information, to offer personalized product suggestions. Companies like Netflix, Amazon, and Spotify use AI to recommend content, products, or music tailored to individual preferences.

For example, e-commerce platforms leverage AI to suggest relevant products based on a customer's shopping behavior. This increases conversion rates and enhances the overall shopping experience.

2.2 AI in Behavioral Analysis and Consumer Segmentation

AI helps businesses segment consumers based on their preferences, buying habits, and interactions with digital platforms. Machine learning models classify customers into different groups, allowing companies to create targeted marketing campaigns.

For instance, a financial services company can use AI to identify high-value customers who frequently invest in mutual funds and tailor financial products specifically for them. This level of personalization improves customer retention and loyalty.

2.3 Chatbots and AI-Powered Customer Support

AI-driven chatbots enhance customer service by providing instant responses, resolving queries, and offering product recommendations. These chatbots use NLP to understand customer inquiries and deliver human-like interactions.

Brands like Sephora and H&M use AI chatbots to assist shoppers with product recommendations based on their preferences and previous purchases. This not only improves customer satisfaction but also increases sales through automated yet personalized interactions.

Conclusion

Artificial Intelligence has transformed market trend prediction and consumer preference analysis, enabling businesses to make data-driven decisions with high accuracy. By leveraging AI-powered trend analysis, companies can anticipate market shifts, optimize product offerings, and stay competitive. Meanwhile, AI-driven consumer insights help businesses deliver personalized experiences, enhance customer engagement, and improve brand loyalty.

As AI technology continues to advance, businesses must invest in AI-driven analytics to gain a competitive edge. The integration of AI in market research and consumer behavior analysis is no longer a luxury but a necessity for companies aiming to succeed in the digital age.

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THE ROLE OF INNOVATION IN ACHIEVING THE UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS

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Introduction

The United Nations Sustainable Development Goals (SDGs) are a set of 17 global objectives designed to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, and peace. These goals, established in 2015 as part of the 2030 Agenda for Sustainable Development, require innovative solutions to accelerate progress and achieve tangible results.

Innovation—defined as the application of new ideas, technologies, and methods—plays a crucial role in achieving the SDGs. From renewable energy advancements to digital health solutions, innovation fosters sustainable economic growth, environmental protection, and social inclusion. This article explores how innovation contributes to the SDGs, focusing on two key objectives: advancing sustainable economic development and enhancing environmental sustainability.

Objective 1: Advancing Sustainable Economic Development

Economic growth is essential for reducing poverty and improving living standards. However, traditional development models often lead to environmental degradation and social inequalities. Innovation offers sustainable alternatives that promote inclusive and resilient economic development.

1.1 Digital Transformation and Economic Opportunities

Technological innovation, particularly in the digital sector, has revolutionized industries by creating new job opportunities and improving access to financial services. For example, mobile banking and fintech solutions have enabled financial inclusion for millions in underserved regions, aligning with SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure).

1.2 Entrepreneurship and Social Innovation

Entrepreneurs are using innovative business models to address social and environmental challenges. Social enterprises, for instance, focus on profit while also achieving positive social impact. Companies that develop affordable clean energy solutions, sustainable agriculture technologies, and eco-friendly products contribute to multiple SDGs, including SDG 1 (No Poverty) and SDG 2 (Zero Hunger).

Objective 2: Enhancing Environmental Sustainability

One of the greatest global challenges is ensuring environmental sustainability while supporting economic and social development. Innovation plays a critical role in mitigating climate change, conserving natural resources, and promoting a circular economy.

2.1 Renewable Energy and Green Technologies

The transition from fossil fuels to renewable energy sources is crucial for achieving SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action). Innovations in solar, wind, and hydroelectric power have made clean energy more accessible and cost-effective.

Energy-efficient technologies, such as smart grids and battery storage systems, further enhance sustainability efforts.

2.2 Sustainable Agriculture and Resource Management

Innovative agricultural practices, such as precision farming, vertical farming, and genetically modified crops, have improved food production efficiency while minimizing environmental impact. Water conservation technologies, such as drip irrigation and desalination, contribute to SDG 6 (Clean Water and Sanitation) and SDG 12 (Responsible Consumption and Production).

Conclusion

Innovation is a driving force behind sustainable development and plays a pivotal role in achieving the United Nations SDGs. By advancing economic development through digital transformation, entrepreneurship, and financial inclusion, innovation creates opportunities for inclusive growth. Simultaneously, sustainable solutions in energy, agriculture, and environmental management help address climate change and resource conservation challenges.

To maximize the impact of innovation on the SDGs, collaboration between governments, businesses, research institutions, and civil society is essential. Investment in research and technology, supportive policies, and knowledge-sharing initiatives will accelerate progress toward a sustainable and equitable future for all.

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SUSTAINABLE EDUCATION PRACTICES FOR A SUSTAINABLE GLOBAL ECONOMY

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Introduction

In recent years, sustainability has become an integral aspect of global development and economic policies. As the world faces challenges such as climate change, resource depletion, and environmental degradation, the need for a sustainable global economy has never been more pressing. One of the key components in achieving this vision is the role of education. Education is the cornerstone upon which societies build their future, and it holds the power to shape the behaviors, values, and decisions of future generations. Sustainable education practices are essential in fostering a mindset that prioritizes long-term environmental, social, and economic sustainability. By embedding sustainability principles into educational frameworks, we can equip individuals with the knowledge, skills, and attitudes necessary to contribute to a sustainable global economy.

This article explores the role of sustainable education practices in shaping a sustainable global economy. It highlights two main objectives: first, to examine the ways sustainable education can influence economic growth while preserving environmental integrity; second, to discuss the integration of sustainability into curricula and teaching methodologies to promote responsible global citizenship. Through these discussions, we will demonstrate how fostering a sustainable education culture can ultimately lead to a thriving, balanced, and resilient global economy.

Objective 1: Examining the Role of Sustainable Education in Economic Growth and Environmental Preservation

Sustainable education is a broad concept that encompasses various teaching practices, policies, and actions aimed at promoting sustainability. It is a form of education that focuses on equipping individuals with the knowledge and skills necessary to contribute to sustainable economic and environmental practices. One of the key areas where sustainable education can influence economic growth is through the development of green technologies and industries. By integrating sustainability into the educational system, we can cultivate a workforce that is equipped to drive innovation in areas such as renewable energy, sustainable agriculture, and green infrastructure.

For instance, programs in environmental science, renewable energy engineering, and sustainable business practices can provide students with the skills to contribute to industries that promote both economic growth and environmental preservation. By fostering a culture of innovation, sustainable education can create new markets, reduce waste, and drive resource efficiency. Additionally, educating future generations about the importance of environmental stewardship can reduce the long-term economic costs associated with environmental damage, such as the depletion of natural resources, loss of biodiversity, and climate-related disasters.

Moreover, sustainable education can have a significant impact on local and global economies by encouraging the adoption of sustainable business models. By teaching students the principles of sustainable development, such as the triple bottom line (economic, social, and environmental), educators can shape future business leaders who understand the value of incorporating social and environmental considerations into their business strategies. This

mindset can lead to the creation of sustainable enterprises that create jobs, promote social equity, and contribute to the overall well-being of society.

Objective 2: Integrating Sustainability into Curricula and Teaching Methodologies for Responsible Global Citizenship

To truly make an impact, sustainability must be integrated into educational curricula and teaching methodologies at all levels. In many regions of the world, traditional educational systems prioritize academic achievement in subjects such as mathematics, language, and science, with little attention given to sustainability. However, the need for a paradigm shift is evident. By integrating sustainability into core subjects such as economics, science, and social studies, educational institutions can ensure that sustainability is not an isolated topic but a fundamental aspect of every subject area.

Incorporating sustainability into curricula can take various forms. For example, environmental studies and sustainable development programs can be offered as standalone courses or integrated into existing subjects. This could involve teaching students about climate change, sustainable resource management, and the social and economic implications of unsustainable practices. By educating young people on the importance of reducing their ecological footprint, educators can help develop a generation that is more environmentally conscious and willing to take action.

In addition to the content of the curriculum, teaching methodologies also play a crucial role in fostering sustainability. Experiential learning, project-based learning, and community engagement initiatives are effective ways to engage students with real-world sustainability challenges. These approaches not only provide students with the opportunity to apply their learning in practical settings but also encourage critical thinking and problem-solving skills. Through hands-on experiences, students can better understand the complex interconnections between social, economic, and environmental systems, which is essential for creating sustainable solutions.

Moreover, the use of digital tools and online platforms can enhance the reach of sustainable education. By leveraging technology, educators can connect students with global sustainability initiatives, enabling them to collaborate across borders and learn from diverse perspectives. Virtual exchanges, online sustainability courses, and international projects can help nurture a global network of responsible citizens committed to addressing the pressing issues of our time.

Conclusion

Sustainable education practices are indispensable in achieving a sustainable global economy. By incorporating sustainability into educational frameworks, we can cultivate a generation of individuals equipped with the knowledge, skills, and values necessary to contribute to both economic prosperity and environmental stewardship. Through the development of green technologies, the promotion of sustainable business practices, and the integration of sustainability into curricula and teaching methodologies, sustainable education can play a transformative role in shaping a balanced and resilient global economy. It is imperative that governments, educators, and institutions work collaboratively to embed sustainability into all aspects of education to ensure that future generations are prepared to tackle the challenges of the 21st century.

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INNOVATION IN SUSTAINABLE FINANCE AND GREEN INVESTING

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Introduction

Sustainable finance and green investing have gained global prominence as businesses, governments, and investors recognize the urgent need to combat climate change and promote environmental sustainability. Traditional finance models, which focus solely on profit maximization, are evolving to incorporate environmental, social, and governance (ESG) factors. Sustainable finance involves investments that not only generate financial returns but also contribute to a healthier planet and a more equitable society.

Green investing, a crucial component of sustainable finance, directs capital towards environmentally friendly projects, such as renewable energy, energy efficiency, and climate-resilient infrastructure. Innovative financial instruments and strategies, such as green bonds, ESG-focused funds, and impact investing, are driving this transformation. This article explores two key objectives of innovation in sustainable finance: advancing financial instruments for green investing and integrating technology to enhance sustainable investment strategies.

Objective 1: Advancing Financial Instruments for Green Investing

To accelerate sustainable finance, innovative financial products and mechanisms are being developed to channel investments toward environmentally responsible projects.

1.1 Green Bonds and Sustainability-Linked Bonds

Green bonds are debt instruments issued to finance projects with environmental benefits, such as renewable energy, clean transportation, and water conservation initiatives. These bonds provide investors with a reliable way to support sustainability while earning returns. Sustainability-linked bonds, an emerging category, tie interest rates to the achievement of specific ESG goals. Companies issuing these bonds commit to measurable environmental targets, such as reducing carbon emissions, and face financial penalties if they fail to meet them.

Example: The European Investment Bank and the World Bank have issued green bonds to fund global climate projects, while corporations like Apple and Tesla have used them to finance their sustainability efforts.

1.2 ESG-Focused Investment Funds

Environmental, Social, and Governance (ESG) criteria are increasingly influencing investment decisions. ESG-focused funds screen companies based on their sustainability performance, ensuring that investments align with ethical and environmental standards.

Example: Global asset managers like BlackRock and Vanguard have introduced ESG-focused exchange-traded funds (ETFs) to cater to investors seeking sustainable portfolios.

1.3 Carbon Credit Trading and Green Derivatives

Carbon credit trading allows companies to buy and sell credits that offset their carbon emissions, encouraging businesses to reduce their environmental impact. Green derivatives, such as climate futures and sustainability-linked swaps, enable investors to hedge against climate-related financial risks.

Example: The European Union's Emissions Trading System (EU ETS) is a leading carbon market that incentivizes industries to reduce greenhouse gas emissions.

Objective 2: Integrating Technology to Enhance Sustainable Investment Strategies

Technological advancements are playing a critical role in improving transparency, efficiency, and impact measurement in sustainable finance.

2.1 Artificial Intelligence and Big Data in ESG Analysis

AI and big data are transforming ESG investing by providing real-time analysis of corporate sustainability practices. AI-driven platforms assess companies' environmental footprints, track regulatory compliance, and predict sustainability risks.

Example: AI-powered tools like MSCI ESG Ratings and Sustainalytics evaluate companies based on ESG performance, helping investors make informed decisions.

2.2 Blockchain for Transparency in Green Finance

Blockchain technology enhances transparency in sustainable finance by securely recording transactions related to green investments. It helps track the use of funds in green projects, preventing greenwashing (false claims of sustainability).

Example: The Energy Web Foundation uses blockchain to verify renewable energy transactions, ensuring the authenticity of green energy certificates.

2.3 Fintech Solutions for Sustainable Investing

Fintech innovations, such as robo-advisors and sustainable investment platforms, are making green finance more accessible to retail investors. Mobile apps and digital platforms allow individuals to invest in ESG-compliant assets effortlessly.

Example: Platforms like OpenInvest and Aspiration enable users to align their investment portfolios with sustainability goals through user-friendly digital tools.

Conclusion

Innovation in sustainable finance and green investing is reshaping the financial landscape by integrating ESG considerations into investment strategies. The development of green bonds, ESG funds, and carbon markets is directing capital toward environmentally responsible initiatives. At the same time, technological advancements such as AI, blockchain, and fintech solutions are enhancing transparency, efficiency, and accessibility in sustainable finance. As the global commitment to climate action strengthens, financial institutions, investors, and policymakers must continue driving innovation in sustainable finance. By aligning investment decisions with long-term environmental and social goals, the financial sector can play a crucial role in building a more sustainable and resilient global economy.

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INNOVATION IN CURRICULUM FOR A GLOBALIZED ECONOMY

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Introduction

The rapid expansion of globalization has transformed economies, industries, and workforce demands worldwide. In this evolving landscape, traditional education systems must adapt to equip students with the skills, knowledge, and competencies necessary for global careers. An innovative curriculum is essential to prepare learners for the interconnected and competitive nature of the modern economy.

Curriculum innovation involves integrating global perspectives, technological advancements, and interdisciplinary learning to develop students' adaptability, critical thinking, and cross-cultural competencies. This article explores two key objectives of curriculum innovation for a globalized economy: fostering global competencies and integrating technology and experiential learning.

Objective 1: Fostering Global Competencies in Education

To succeed in a globalized economy, students must develop competencies beyond traditional subject knowledge. They need cultural awareness, multilingual skills, and the ability to collaborate across borders.

1.1 Multicultural and Cross-Cultural Education

A globally relevant curriculum should include diverse perspectives, histories, and cultures to foster a sense of global citizenship. Incorporating case studies, literature, and real-world scenarios from different countries helps students understand international markets and societies.

For instance, business students learning about international trade should study case examples from Asian, European, and African economies to develop a comprehensive understanding of global business dynamics.

1.2 Language and Communication Skills

Multilingualism is a valuable asset in the global job market. Innovative curricula should emphasize learning multiple languages and effective cross-cultural communication. Institutions can integrate language immersion programs, virtual exchange programs, and collaboration with international schools to enhance linguistic skills. For example, schools can partner with global organizations to allow students to work on projects with peers from different countries, improving their cultural sensitivity and teamwork abilities.

1.3 Global Economics and Sustainable Development Focus

Understanding global economic structures and sustainability challenges is crucial for future professionals. A reformed curriculum should incorporate topics such as international trade, global supply chains, environmental sustainability, and ethical business practices. This prepares students to address global challenges, such as climate change and economic inequality, with innovative and socially responsible solutions.

Objective 2: Integrating Technology and Experiential Learning

As digital transformation reshapes industries, education must evolve to include emerging technologies and hands-on learning experiences that align with real-world demands.

2.1 Incorporating Digital Tools and Emerging Technologies

A modern curriculum should include AI, blockchain, data analytics, and other emerging technologies to prepare students for future careers. Educational institutions can use virtual reality (VR), augmented reality (AR), and artificial intelligence (AI)-driven learning platforms to enhance engagement and interactive learning. For example, business and finance students can use AI-powered simulations to analyze global market trends, while engineering students can engage in virtual labs to experiment with new technologies.

2.2 Experiential and Problem-Based Learning

Learning through real-world experiences helps students apply theoretical knowledge to practical challenges. Internships, apprenticeships, and industry collaborations should be embedded into the curriculum to give students hands-on exposure. For example, universities can establish partnerships with multinational corporations where students participate in live projects, addressing global business challenges and developing practical problem-solving skills.

2.3 Entrepreneurial and Innovation-Based Learning

A curriculum for a globalized economy should encourage creativity, entrepreneurship, and innovation. Schools and universities can integrate startup incubators, hackathons, and design-thinking workshops into their programs. For instance, students can be challenged to create business solutions for real-world economic or social issues, promoting innovation and leadership skills.

Conclusion

Innovation in curriculum design is essential to prepare students for a globalized economy. By fostering global competencies such as cultural awareness, multilingual communication, and a deep understanding of global markets, education systems can produce well-rounded professionals. Additionally, integrating technology and experiential learning ensures that students develop relevant skills for the evolving job market. To remain competitive in the global economy, educational institutions must embrace continuous curriculum reforms, leveraging digital advancements and cross-border collaborations. Only through such innovative approaches can education truly prepare students for success in an interconnected world.

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THE RISE OF E-COMMERCE: IMPLICATIONS FOR TRADITIONAL RETAIL

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Introduction

The rapid advancement of digital technology has transformed the way people shop, leading to the exponential growth of e-commerce. Online shopping platforms such as Amazon, Alibaba, and Shopify have revolutionized retail by offering convenience, variety, and competitive pricing. The rise of e-commerce has significantly impacted traditional brick-and-mortar stores, forcing them to adapt or risk obsolescence. While traditional retail has strengths such as physical customer experience and immediate product availability, e-commerce's advantages in cost-effectiveness and global reach are reshaping consumer behavior. This article explores the implications of e-commerce on traditional retail by examining its effects on business operations, consumer preferences, and the overall retail landscape.

Objective 1: The Impact of E-Commerce on Traditional Retail Businesses

E-commerce has altered the structure of the retail industry by shifting consumer preferences toward digital platforms. This transformation has affected traditional retail in several ways:

1. **Decline in Physical Store Footfall** – With the convenience of online shopping, many consumers prefer to browse and purchase products from their homes. This has led to reduced foot traffic in brick-and-mortar stores, forcing many to downsize or close. Major retailers such as Sears and Toys "R" Us have struggled to compete with e-commerce giants and ultimately shut down many of their locations.
2. **Price Competition and Cost Challenges** – Online retailers have lower operating costs compared to physical stores, allowing them to offer competitive prices. Traditional retailers must manage costs related to rent, staffing, and inventory management, making it difficult to compete with e-commerce platforms that can afford aggressive pricing strategies.
3. **Shift to Omnichannel Strategies** – To survive in the digital era, many traditional retailers have integrated e-commerce into their business models. Stores now offer online purchasing, click-and-collect services, and mobile applications to enhance customer convenience. Retailers like Walmart and Target have successfully combined their physical stores with digital operations, ensuring they remain competitive.

Objective 2: Changing Consumer Behavior and Expectations

As e-commerce continues to expand, consumer expectations have evolved significantly. Shoppers now prioritize convenience, personalized experiences, and flexible purchasing options. The following are key changes in consumer behavior due to the rise of e-commerce:

1. **Preference for Convenience and Speed** – Consumers value the ability to shop anytime, anywhere. E-commerce platforms provide 24/7 access to products with fast delivery options, making it easier for people to make purchases without visiting a store. Features like same-day delivery and subscription services (e.g., Amazon Prime) have further enhanced customer satisfaction.

2. Growth of Mobile Commerce – The widespread use of smartphones has contributed to the rise of mobile shopping. Consumers can browse products, compare prices, and make purchases with just a few clicks. This shift has prompted retailers to develop user-friendly mobile applications and optimize their websites for mobile users.

3. Increased Demand for Personalization – Online retailers use artificial intelligence and data analytics to offer personalized recommendations based on consumer behavior. Traditional retailers, on the other hand, must leverage customer relationship management (CRM) systems and loyalty programs to provide a similar level of personalization and engagement.

4. Social Media Influence on Shopping – Platforms like Instagram, Facebook, and TikTok have become powerful tools for influencing purchasing decisions. E-commerce brands leverage social media marketing, influencers, and targeted ads to reach potential customers. Traditional retailers must also embrace digital marketing strategies to remain relevant.

Conclusion

The rise of e-commerce has fundamentally transformed the retail industry, posing significant challenges to traditional retail stores. While e-commerce provides consumers with convenience, affordability, and a wider selection of products, physical stores still offer tangible experiences that digital platforms cannot replicate. To stay competitive, traditional retailers must adapt by integrating digital solutions, enhancing customer experiences, and leveraging data-driven insights. The future of retail lies in a hybrid model, where both e-commerce and physical stores coexist to meet diverse consumer needs. By embracing technological advancements and evolving with market trends, traditional retailers can continue to thrive in the digital age.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE ON EDUCATION AND LEARNING METHODS

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Introduction

Artificial Intelligence (AI) has emerged as a revolutionary force across various industries, and education is no exception. AI-driven technologies are reshaping traditional learning methods, enhancing personalized learning experiences, automating administrative tasks, and improving overall efficiency in the education sector. With the rise of intelligent tutoring systems, AI-driven assessment tools, and adaptive learning platforms, students and educators alike are experiencing a paradigm shift in how knowledge is imparted and acquired.

This article explores the impact of AI on education and learning methods by examining its role in personalizing education, automating administrative tasks, and improving learning outcomes. Two key objectives will be discussed: enhancing personalized learning experiences and streamlining administrative processes for educators.

Objective 1 : Enhancing Personalized Learning Experiences

One of the most significant impacts of AI in education is its ability to personalize learning based on individual student needs, learning styles, and performance. Traditional education systems often follow a "one-size-fits-all" approach, but AI introduces adaptive learning platforms that adjust content and pace based on a learner's strengths and weaknesses.

1.1 Adaptive Learning Technologies

AI-driven platforms such as Coursera, Khan Academy, and Duolingo use machine learning algorithms to analyze a student's progress and adapt the curriculum accordingly. These platforms can provide additional resources, suggest exercises, or modify the difficulty level of assessments to ensure that students fully grasp the concepts.

1.2 Intelligent Tutoring Systems (ITS)

AI-powered intelligent tutoring systems (ITS) simulate human tutors by providing real-time feedback, answering queries, and guiding students through problem-solving steps. Examples include IBM Watson Tutor and Carnegie Learning's AI-driven software, which offer interactive learning experiences tailored to individual students.

1.3 Chatbots and Virtual Assistants

AI-powered chatbots and virtual assistants provide 24/7 assistance to students by answering frequently asked questions, helping with assignments, and offering explanations of complex topics. These tools reduce dependency on human educators for routine queries, allowing teachers to focus on more critical aspects of teaching.

1.4 Speech and Image Recognition for Accessibility

AI technologies, such as speech-to-text and image recognition, have enhanced accessibility for students with disabilities. For instance, Microsoft's AI-powered Immersive Reader helps dyslexic students by reading text aloud, while real-time translation tools assist non-native language speakers in understanding course materials.

With these advancements, AI ensures that learning is not only personalized but also more engaging and inclusive, helping students of all backgrounds maximize their educational potential.

Objective 2: Streamlining Administrative Processes for Educators

Beyond personalized learning, AI plays a crucial role in reducing the administrative burden on educators, allowing them to focus more on teaching and mentoring students. Automating repetitive tasks enhances efficiency and improves the overall educational experience.

2.1 Automated Grading and Assessment

AI-powered grading systems, such as Gradescope and Turnitin, automate the evaluation of assignments, quizzes, and exams. These tools use natural language processing (NLP) and machine learning to assess student responses, provide instant feedback, and detect plagiarism. This significantly reduces the time educators spend on manual grading.

2.2 AI-Driven Curriculum Development

AI can analyze educational trends, student performance data, and industry requirements to assist educators in developing relevant curricula. By assessing vast amounts of data, AI helps institutions design courses that align with job market demands and future skills requirements.

2.3 Predictive Analytics for Student Performance

Machine learning models can analyze students' historical performance, attendance records, and engagement levels to predict their future success or risk of dropping out. Institutions can use this data to offer personalized interventions, ensuring that struggling students receive timely support.

2.4 Automated Scheduling and Resource Management

AI-powered scheduling systems help schools and universities optimize class schedules, assign faculty members efficiently, and manage resource allocation. This ensures that institutions operate smoothly and students receive a seamless learning experience.

By streamlining these administrative tasks, AI empowers educators to dedicate more time to mentoring students, fostering creativity, and improving the overall quality of education.

Conclusion

Artificial Intelligence is transforming the education sector by personalizing learning experiences and automating administrative tasks to enhance efficiency. With adaptive learning platforms, intelligent tutoring systems, and AI-driven chatbots, students receive customized educational experiences tailored to their needs. At the same time, educators benefit from automated grading, predictive analytics, and AI-assisted curriculum development, reducing their workload and enabling them to focus on high-impact teaching activities.

While AI offers numerous advantages, challenges such as data privacy concerns, the digital divide, and ethical considerations in AI-driven decision-making must be addressed. However, with responsible implementation and continuous advancements in AI technology, the future of education is set to become more inclusive, efficient, and accessible for learners worldwide.

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SMART CITIES AND SUSTAINABILITY: INNOVATIONS FOR THE FUTURE

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Introduction

As urban populations continue to grow, cities face numerous challenges, including traffic congestion, pollution, resource depletion, and energy consumption. In response to these issues, the concept of smart cities has emerged as a transformative solution. Smart cities leverage digital technologies, data-driven strategies, and innovative infrastructures to enhance urban living while promoting sustainability. By integrating intelligent systems into transportation, energy management, waste disposal, and public services, smart cities aim to create efficient, eco-friendly, and livable environments for residents. This article explores the objectives of smart city innovations, their role in sustainability, and the potential impact on future urban development.

Objective 1: Enhancing Energy Efficiency and Environmental Sustainability

One of the core objectives of smart cities is to improve energy efficiency and reduce environmental impact. By adopting innovative solutions, cities can effectively manage energy consumption, reduce carbon emissions, and promote renewable energy sources.

Key Innovations in Energy Management

Smart Grids: These advanced electrical grids use real-time data to optimize energy distribution and consumption. They efficiently manage power demand, reduce energy wastage, and enhance grid stability.

Renewable Energy Integration: Solar panels, wind turbines, and geothermal systems are integrated into smart city infrastructures to reduce reliance on fossil fuels. Energy storage solutions like battery systems also support sustainable energy use.

Energy-Efficient Buildings: Smart buildings utilize automated systems that control lighting, heating, and cooling based on occupancy patterns. These innovations significantly lower energy consumption and operational costs.

Green Infrastructure: Smart cities invest in eco-friendly designs such as green roofs, vertical gardens, and urban forests to improve air quality, reduce heat islands, and promote biodiversity.

By integrating these innovations, smart cities aim to minimize their carbon footprint while improving resource efficiency.

Objective 2: Improving Mobility and Transportation Systems

Another significant goal of smart cities is to develop efficient, safe, and sustainable transportation systems. Traffic congestion and excessive vehicle emissions pose major environmental and economic concerns in urban areas.

Key Innovations in Transportation

Intelligent Traffic Management: Advanced traffic systems use sensors, AI algorithms, and real-time data to optimize traffic flow, reduce congestion, and improve road safety.

Smart Public Transit Systems: Cities are adopting digital payment systems, GPS tracking, and mobile apps to enhance public transportation services. These technologies improve route planning, reduce wait times, and encourage the use of eco-friendly transit options.

Electric Vehicles (EVs) and Charging Stations: Encouraging EV adoption reduces greenhouse gas emissions. Smart charging stations powered by renewable energy sources further support sustainable transportation.

Shared Mobility Solutions: Bike-sharing programs, e-scooters, and ride-sharing services provide alternative transport options, promoting reduced vehicle dependency and lower emissions.

By investing in these transportation innovations, smart cities can significantly enhance mobility while improving air quality and reducing environmental harm.

Conclusion

Smart cities represent a forward-thinking approach to urban development that integrates technology, innovation, and sustainability. By adopting energy-efficient systems, intelligent infrastructure, and sustainable transportation solutions, smart cities aim to improve the quality of life for residents while minimizing environmental impact. As technological advancements continue to evolve, smart cities will play a crucial role in shaping a greener, more sustainable future. Governments, urban planners, and private sectors must collaborate to implement smart city initiatives that prioritize both innovation and environmental responsibility. Through these efforts, smart cities can become global models for sustainable urban living.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN TRANSFORMING GLOBAL COMMERCE

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Introduction

Artificial Intelligence (AI) is revolutionizing global commerce by enhancing efficiency, automating processes, and improving customer experiences. Businesses across industries, from retail and finance to logistics and manufacturing, are leveraging AI to gain a competitive edge. AI-powered technologies such as machine learning, predictive analytics, chatbots, and robotic process automation (RPA) are streamlining operations, reducing costs, and providing data-driven insights to improve decision-making.

As globalization accelerates, AI is playing a crucial role in breaking barriers, enabling personalized marketing, optimizing supply chains, and ensuring seamless digital transactions. This article explores two key objectives of AI in global commerce: improving operational efficiency through automation and enhancing customer experience with intelligent solutions.

Objective 1: Improving Operational Efficiency through Automation

AI is transforming commerce by automating repetitive tasks, improving supply chain management, and optimizing business processes to increase productivity and reduce operational costs.

1.1 AI in Supply Chain and Logistics

AI-powered supply chain solutions help businesses predict demand, manage inventory, and optimize delivery routes. Machine learning algorithms analyze historical data to anticipate market fluctuations and improve supply chain efficiency.

Example: Companies like Amazon and Walmart use AI-driven inventory management systems to ensure products are stocked efficiently, reducing waste and improving delivery speed. AI-powered logistics platforms like Flexport optimize global shipping by analyzing weather patterns and traffic conditions.

1.2 Robotic Process Automation (RPA) in Business Operations

RPA uses AI to automate routine tasks such as data entry, invoice processing, and compliance reporting. This reduces human error, increases accuracy, and allows employees to focus on strategic tasks.

Example: Banks and financial institutions use AI-powered automation for fraud detection, risk assessment, and customer onboarding, improving efficiency while reducing operational risks.

1.3 AI-Driven Financial Transactions and Fraud Prevention

AI algorithms detect suspicious activities and prevent fraudulent transactions by analyzing spending patterns and identifying anomalies. This ensures secure and efficient global transactions.

Example: PayPal and Mastercard use AI-driven fraud detection systems that analyze transaction behaviors in real-time, preventing cybercrime and enhancing customer security.

Objective 2: Enhancing Customer Experience with Intelligent Solutions

AI-driven innovations are revolutionizing customer interactions by providing personalized shopping experiences, improving support services, and enhancing engagement through predictive analytics.

2.1 Personalized Marketing and AI-Powered Recommendations

AI analyzes customer data to understand preferences and predict future buying behaviors. AI-driven recommendation engines help businesses offer customized product suggestions, improving customer satisfaction and increasing sales.

Example: Netflix, Amazon, and Spotify use AI algorithms to recommend content and products tailored to individual preferences, boosting customer engagement.

2.2 AI Chatbots and Virtual Assistants

AI-powered chatbots and virtual assistants handle customer inquiries, provide instant support, and enhance customer service efficiency. These AI tools operate 24/7, reducing wait times and improving response accuracy.

Example: Companies like Sephora and H&M use AI chatbots for personalized shopping assistance, while banks like HSBC deploy AI-powered virtual assistants for customer queries.

2.3 AI in E-Commerce and Smart Retail

AI-driven visual search technology allows consumers to find products using images, while voice-enabled assistants like Alexa and Google Assistant facilitate hands-free shopping experiences.

Example: Fashion retailers such as ASOS and Zara use AI for virtual try-ons, helping customers visualize products before purchasing. Smart retail stores, like Amazon Go, use AI to enable checkout-free shopping experiences.

Conclusion

AI is fundamentally reshaping global commerce by automating business operations, optimizing supply chains, and delivering personalized customer experiences. By integrating AI-driven technologies, businesses can enhance efficiency, reduce costs, and improve consumer satisfaction.

As AI continues to evolve, its role in commerce will expand, driving innovations in predictive analytics, fraud prevention, and autonomous decision-making. Companies that embrace AI-driven strategies will remain competitive in the rapidly changing global marketplace, ensuring sustained growth and improved customer engagement.

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HOW AI AND BIG DATA ARE REVOLUTIONIZING BUSINESS ANALYTICS

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Introduction

Artificial Intelligence (AI) and Big Data are transforming the way businesses analyze information, make strategic decisions, and optimize operations. The massive volume of structured and unstructured data generated daily, combined with advanced AI-powered analytics, enables organizations to gain valuable insights, predict market trends, and enhance customer experiences.

Traditional data analysis methods relied on manual processes and limited computing power, making it difficult to process vast amounts of information efficiently. However, with the emergence of AI and machine learning algorithms, businesses can now analyze data in real-time, detect patterns, and make data-driven decisions with precision.

This article explores two key objectives of AI and Big Data in business analytics: enhancing decision-making with predictive analytics and optimizing operational efficiency through automation.

Objective 1: Enhancing Decision-Making with Predictive Analytics

AI and Big Data analytics are empowering businesses to make informed decisions by predicting future trends, customer behavior, and market fluctuations.

1.1 Predictive Modeling for Market Trends

AI-driven predictive analytics helps businesses anticipate changes in consumer demand, stock market trends, and economic fluctuations. By analyzing historical data and market indicators, companies can make proactive decisions to stay ahead of competitors.

Example: Retail giants like Walmart and Target use AI-powered forecasting tools to predict demand spikes and optimize inventory management, reducing stock shortages and excess inventory.

1.2 Customer Behavior Analysis and Personalization

AI analyzes vast amounts of customer data, including browsing history, purchase patterns, and social media interactions, to predict consumer preferences. Businesses can then offer personalized recommendations and targeted marketing campaigns.

Example: Streaming services like Netflix and music platforms like Spotify use AI-based recommendation engines to analyze user preferences and suggest relevant content, enhancing customer satisfaction and engagement.

1.3 Fraud Detection and Risk Management

AI and Big Data enable businesses to identify fraudulent activities by analyzing transaction patterns and detecting anomalies. This enhances security and reduces financial losses.

Example: Banks and financial institutions like JPMorgan Chase and PayPal use AI-powered fraud detection systems to monitor real-time transactions and flag suspicious activities before they escalate.

Objective 2: Optimizing Operational Efficiency Through Automation

AI and Big Data analytics streamline business operations, reduce costs, and improve productivity by automating repetitive tasks and optimizing resource allocation.

2.1 AI-Powered Process Automation

Machine learning algorithms and robotic process automation (RPA) reduce human intervention in data entry, report generation, and administrative processes, allowing employees to focus on strategic tasks.

Example: Companies like IBM and UiPath provide AI-powered automation tools that help businesses handle complex workflows, reducing operational costs and improving efficiency.

2.2 Real-Time Data Processing for Business Insights

Big Data platforms process real-time information, allowing companies to react quickly to market changes and customer demands. AI-driven dashboards and analytics tools help executives make data-driven decisions on the spot.

Example: Amazon Web Services (AWS) and Google Cloud provide AI-powered analytics solutions that enable businesses to track performance metrics, sales patterns, and operational efficiency in real-time.

2.3 AI in Supply Chain Optimization

AI-driven analytics improve supply chain management by optimizing logistics, predicting demand fluctuations, and reducing delivery delays.

Example: Companies like DHL and FedEx use AI-powered route optimization and demand forecasting to enhance delivery speed, reduce fuel costs, and minimize supply chain disruptions.

Conclusion

AI and Big Data are revolutionizing business analytics by providing predictive insights, automating operations, and optimizing decision-making. Companies that leverage these technologies gain a competitive advantage by improving efficiency, reducing costs, and delivering personalized customer experiences.

As AI and Big Data continue to evolve, their role in business analytics will expand, enabling smarter decision-making and deeper insights into market dynamics. Organizations that invest in these technologies will stay ahead in an increasingly data-driven world.

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FINTECH AND THE FUTURE OF BANKING: INNOVATIONS AND CHALLENGES

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Introduction

Over the past decade, the financial services industry has been significantly disrupted by the rapid rise of Financial Technology (FinTech). These innovations are not merely enhancing existing banking processes but are transforming the entire landscape of financial services. FinTech leverages technology to improve and automate the delivery of financial services, offering consumers and businesses new ways to manage, invest, and save money. By integrating advanced technologies such as blockchain, artificial intelligence (AI), big data analytics, and mobile applications, FinTech is reshaping how banks operate, engage with customers, and deliver products and services.

As these technological innovations continue to evolve, they present both exciting opportunities and formidable challenges. The future of banking is undeniably intertwined with FinTech, but its widespread adoption requires the resolution of several challenges related to regulation, cybersecurity, financial inclusion, and adaptation by traditional financial institutions. This article will explore the innovative strides made by FinTech and delve into the challenges that need to be addressed to ensure a smooth transition toward a digital-first financial ecosystem.

Objective 1: Understanding FinTech Innovations in Banking

The rapid evolution of FinTech has introduced numerous innovations that are transforming banking operations and the customer experience. One of the most profound changes has been the digitization of banking services. Traditional banks are increasingly offering mobile banking apps, digital wallets, and online banking solutions that provide customers with round-the-clock access to their accounts and financial services. These innovations make banking more convenient, especially for underbanked populations who lack access to physical bank branches. With mobile money services, consumers can transfer funds, pay bills, and even receive loans via their smartphones, thus reducing financial barriers and contributing to greater financial inclusion.

Blockchain technology has also been a significant innovation in the FinTech sector. Initially popularized by cryptocurrencies such as Bitcoin, blockchain offers a decentralized ledger system that ensures secure, transparent, and efficient transactions. In banking, blockchain eliminates the need for intermediaries in various financial processes, thus reducing transaction costs, increasing speed, and enhancing security. The rise of decentralized finance (DeFi) further emphasizes the potential of blockchain by enabling peer-to-peer transactions that bypass traditional banking channels.

Artificial intelligence (AI) and machine learning are increasingly being used to automate banking services. AI-powered chatbots, for instance, are revolutionizing customer service by providing real-time support and addressing customer queries around the clock. AI also plays a critical role in fraud detection, as algorithms can analyze vast amounts of data to spot patterns and detect suspicious activities. In lending, AI helps assess creditworthiness by considering a broader range of data, including alternative data sources, rather than relying solely on credit scores. This enables a more inclusive and personalized approach to lending.

Another significant innovation is open banking, which is gaining momentum globally. Open banking allows third-party developers to create financial products and services by securely

accessing a customer's financial data from various banks, with their consent. This has led to the development of more tailored financial products, better competition in the market, and improved customer experiences. With open banking, consumers are empowered to make more informed financial decisions, while also benefiting from the enhanced services offered by new entrants to the financial sector.

Objective 2: Addressing the Challenges of FinTech in Banking

While FinTech innovations offer numerous benefits, they also present several challenges that need to be addressed for the banking sector to fully harness their potential. One of the primary concerns is cybersecurity. As more banking services move online, the risk of cyberattacks increases. Financial institutions hold vast amounts of sensitive data, including personal and financial information, which makes them prime targets for hackers. Ensuring the protection of this data is paramount. Financial institutions must invest heavily in robust cybersecurity measures and technologies to prevent data breaches, identity theft, and other forms of cybercrime. Additionally, as the use of cryptocurrencies and blockchain technologies grows, new cybersecurity challenges are emerging, as these systems are not yet fully regulated or immune to hacking attempts.

Another challenge is regulatory uncertainty. FinTech operates in a highly complex regulatory environment that varies by region, with some jurisdictions adopting clear guidelines, while others remain hesitant or unclear. As FinTech evolves rapidly, existing financial regulations often fail to adequately address the unique risks posed by new technologies such as blockchain and AI. There is a need for global regulatory cooperation to create clear standards that protect consumers, ensure market stability, and foster innovation. However, balancing innovation with regulation remains a challenge, as excessive regulation can stifle creativity, while insufficient oversight can lead to risks such as fraud, money laundering, or market manipulation.

Traditional banks also face challenges in adapting to the digital-first world driven by FinTech. Many legacy financial institutions are burdened by outdated infrastructure and are slow to embrace digital transformation. FinTech startups, on the other hand, are agile and can quickly adapt to technological advancements, enabling them to deliver better and more efficient services. This places immense pressure on traditional banks to invest in new technologies, streamline their operations, and rethink their customer engagement strategies. Partnerships between FinTech startups and traditional banks are increasingly seen as a potential solution, allowing banks to leverage the innovative capabilities of FinTech while maintaining their trusted reputation.

Another critical issue is the digital divide. While FinTech promises to improve financial inclusion by making banking more accessible, there are concerns that it may inadvertently exclude individuals who lack the necessary technology or digital literacy. Older adults, people in rural areas, and low-income groups may not have access to smartphones, reliable internet, or the knowledge required to navigate digital financial platforms. It is crucial for financial institutions to ensure that FinTech solutions are designed to be inclusive and accessible to all members of society.

Conclusion

FinTech has undeniably transformed the banking sector, introducing a wide range of innovative solutions that are changing how financial services are delivered. Digital banking, blockchain technology, artificial intelligence, and open banking are reshaping the way consumers access and interact with financial services, creating more efficient, secure, and

personalized banking experiences. However, these innovations also present several challenges that must be addressed to ensure the continued growth and stability of the banking sector.

Cybersecurity, regulatory uncertainty, the adaptation of traditional financial institutions, and the digital divide are key challenges that need to be addressed. By focusing on robust security, developing clear regulatory frameworks, investing in digital infrastructure, and ensuring financial inclusion, the banking sector can leverage the full potential of FinTech while minimizing its risks.

The future of banking is undoubtedly digital, and with the right measures in place, FinTech has the power to create a more inclusive, efficient, and secure financial system for all.

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ENTREPRENEURSHIP IN A DIGITAL ECONOMY: OPPORTUNITIES AND RISKS

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Introduction

The digital economy has transformed the way businesses operate, creating new opportunities for entrepreneurs while also presenting significant risks. With advancements in technology, the internet, and digital platforms, entrepreneurs can reach a global audience, leverage innovative business models, and operate with lower overhead costs. However, the digital landscape also comes with challenges such as cybersecurity threats, data privacy concerns, and market competition.

This article explores the opportunities and risks associated with entrepreneurship in the digital economy, highlighting key aspects that modern entrepreneurs must consider to thrive in this evolving business environment.

Objective 1: Opportunities in the Digital Economy

The digital economy offers several benefits that empower entrepreneurs to launch and scale their businesses efficiently. Some of the key opportunities include:

1. Global Market Access

Digital platforms such as e-commerce websites, social media, and online marketplaces allow entrepreneurs to reach customers worldwide. Businesses are no longer limited to local markets, enabling small startups to compete with established brands on a global scale.

2. Lower Business Costs

Starting a business in the digital economy requires less capital compared to traditional businesses. Online tools, cloud computing, and digital marketing reduce operational costs, making entrepreneurship more accessible.

3. Flexible Business Models

Entrepreneurs can adopt innovative business models such as freelancing, subscription-based services, dropshipping, and software-as-a-service (SaaS). These models allow businesses to scale rapidly and generate recurring revenue.

4. Data-Driven Decision Making

The digital economy provides access to vast amounts of data, helping entrepreneurs understand customer behavior, optimize marketing strategies, and make informed business decisions. Analytics tools and artificial intelligence (AI) further enhance business efficiency.

5. Automation and AI Integration

Technological advancements enable entrepreneurs to automate various aspects of their business, such as customer service (chatbots), sales processes (CRM systems), and inventory management. This increases efficiency and reduces manual workload.

Objective 2: Risks and Challenges in the Digital Economy

Despite the numerous opportunities, entrepreneurs must navigate several risks when operating in the digital economy. Some of the key challenges include:

1. Cybersecurity Threats

With increased online transactions, businesses are vulnerable to cyberattacks, data breaches, and hacking attempts. Entrepreneurs must invest in cybersecurity measures to protect customer information and business data.

2. Intense Market Competition

The digital space is highly competitive, with businesses from around the world competing for customer attention. Entrepreneurs must differentiate their products, build strong brand identities, and continuously innovate to stay ahead.

3. Data Privacy and Regulatory Compliance

Governments worldwide have introduced strict regulations to protect user data, such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). Entrepreneurs must ensure compliance to avoid legal penalties.

4. Dependence on Digital Infrastructure

A business in the digital economy relies heavily on technology, internet connectivity, and cloud services. Any technical failure, system outage, or cyberattack can disrupt operations, affecting business continuity.

5. Rapid Technological Changes

The digital economy is constantly evolving, with new technologies emerging frequently. Entrepreneurs must stay updated with trends such as blockchain, AI, and the Internet of Things (IoT) to remain competitive and adapt to market changes.

Conclusion

Entrepreneurship in the digital economy presents a wealth of opportunities for individuals looking to start and grow businesses with minimal resources. The ability to access global markets, leverage data-driven insights, and implement innovative business models has transformed traditional entrepreneurship. However, digital entrepreneurs must also be aware of the risks, including cybersecurity threats, regulatory challenges, and intense competition.

To succeed in the digital economy, entrepreneurs must embrace technology while ensuring they implement robust security measures, comply with regulations, and continuously innovate to remain relevant in a rapidly changing business environment. By balancing opportunities and risks, digital entrepreneurs can build sustainable businesses and thrive in the modern economy.

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THE FUTURE OF FINTECH IN A GLOBALIZED ECONOMY

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Introduction

Financial technology (FinTech) has become a driving force in reshaping the global financial ecosystem, leveraging digital innovation to enhance efficiency, security, and accessibility. In a globalized economy, where financial transactions transcend borders, FinTech plays a crucial role in fostering economic integration and providing cutting-edge solutions for businesses and consumers. The convergence of blockchain, artificial intelligence (AI), and big data analytics has accelerated digital transformation, challenging traditional financial institutions while unlocking new growth opportunities. This article explores the future of FinTech in a globalized economy, focusing on two key objectives: enhancing financial inclusion and strengthening cross-border transactions.

Objective 1: Enhancing Financial Inclusion

One of the primary goals of FinTech is to promote financial inclusion, ensuring that individuals and businesses, especially in developing economies, have access to financial services. Traditional banking systems have often excluded low-income populations due to high costs, geographic barriers, and complex requirements. FinTech bridges this gap by offering alternative financial solutions, such as mobile banking, digital wallets, and peer-to-peer (P2P) lending platforms.

Innovative technologies like decentralized finance (DeFi) are redefining accessibility by enabling users to engage in financial transactions without intermediaries. AI-driven credit scoring models analyze alternative data sources, including mobile phone usage and social behavior, to provide credit access to individuals with no formal financial history. Furthermore, the adoption of biometric authentication and blockchain-based digital identity systems enhances security and trust, making financial services more inclusive and efficient. As FinTech continues to evolve, its role in expanding financial access will be pivotal in fostering global economic growth.

Objective 2: Strengthening Cross-Border Transactions and Security

In an interconnected global economy, efficient and secure cross-border transactions are essential for international trade, remittances, and investment flows. Traditional payment systems, such as SWIFT, are often slow, costly, and subject to regulatory constraints. FinTech disrupts these inefficiencies by introducing blockchain-based solutions, digital currencies, and AI-driven fraud detection mechanisms.

Blockchain technology enhances transparency and security by providing decentralized and tamper-proof ledgers for financial transactions. Stablecoins—cryptocurrencies pegged to fiat currencies—are gaining traction as reliable mediums for international payments, reducing transaction costs and settlement times. Additionally, AI-powered risk assessment tools analyze transaction patterns in real-time, helping to detect fraudulent activities and improve cybersecurity.

Regulatory harmonization is critical to the widespread adoption of FinTech in cross-border finance. Governments and financial institutions must collaborate to establish standardized regulations that foster innovation while ensuring compliance with anti-money laundering (AML) and know-your-customer (KYC) policies. As digital identity verification methods

become more advanced, cross-border transactions will become more seamless, reducing financial friction and promoting economic connectivity.

Conclusion

The future of FinTech in a globalized economy is marked by rapid technological advancements and increased financial integration. By enhancing financial inclusion and streamlining cross-border transactions, FinTech will continue to reshape the financial landscape, making services more accessible, efficient, and secure. However, challenges such as regulatory compliance, cybersecurity threats, and technological adoption barriers must be addressed through strategic collaborations and policy frameworks. As FinTech continues to evolve, it will play an instrumental role in fostering global economic development, bridging financial gaps, and driving innovation in the digital age.

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THE ROLE OF E-LEARNING PLATFORMS IN INTERNATIONAL EDUCATION

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Introduction

The global education landscape has been significantly transformed by the rise of e-learning platforms. These digital tools have revolutionized the way students and educators interact, breaking down geographical barriers and providing access to quality education worldwide. With advancements in technology, international education is no longer confined to physical classrooms; instead, it has expanded to virtual environments where learners can access courses from prestigious institutions, collaborate with peers from different countries, and gain skills that prepare them for a globalized workforce.

E-learning platforms offer various benefits, including flexibility, cost-effectiveness, and accessibility. They provide a bridge for students who may lack opportunities due to economic, political, or geographical constraints. This article explores the role of e-learning platforms in international education, focusing on two key objectives: improving access to quality education and promoting cross-cultural learning experiences.

Objective 1: Improving Access to Quality Education

One of the primary advantages of e-learning platforms in international education is their ability to provide access to high-quality learning materials regardless of location. Many prestigious institutions, such as Harvard University, MIT, and Oxford University, offer online courses through platforms like Coursera, edX, and Udemy. These courses allow students from developing countries or remote areas to gain knowledge from top educators without the need to travel abroad.

Additionally, e-learning platforms cater to different learning styles by offering a variety of content formats, including video lectures, interactive quizzes, and live discussions. This approach enhances engagement and knowledge retention. The affordability of online courses compared to traditional education also makes higher learning more inclusive, enabling students who may not afford tuition fees or living expenses in foreign countries to pursue international education online.

Furthermore, e-learning platforms support lifelong learning, allowing individuals to update their skills and stay competitive in the job market. Many professionals use these platforms to earn certifications that improve their career prospects in global industries such as IT, business management, and healthcare.

Objective 2: Promoting Cross-Cultural Learning Experiences

Another crucial role of e-learning platforms in international education is fostering cross-cultural understanding. These platforms connect students from diverse backgrounds, creating a global learning environment where learners exchange ideas, perspectives, and experiences. Through online forums, group projects, and live webinars, students can interact

with peers from different countries, enhancing their cultural awareness and communication skills.

Moreover, universities and organizations offer specialized courses that focus on international relations, global business practices, and multicultural studies. These courses equip students with knowledge about different cultures, preparing them to work in multinational organizations or engage in diplomatic and trade-related activities.

Language learning is another significant aspect of cross-cultural education. E-learning platforms such as Duolingo, Rosetta Stone, and Babbel provide opportunities for students to learn new languages, making it easier for them to integrate into international academic and professional environments.

By fostering cross-cultural exchanges, e-learning platforms help break down stereotypes, encourage diversity, and prepare students for a more interconnected world.

Conclusion

E-learning platforms have become a vital tool in international education, bridging gaps in accessibility and cultural exchange. They provide learners with opportunities to access world-class education, regardless of their location or financial status, and promote cross-cultural understanding through global collaboration. The integration of technology in education continues to evolve, making international learning experiences more dynamic and inclusive.

As more institutions and educators embrace e-learning, the future of international education looks promising. Digital learning tools will continue to break barriers, foster global connections, and empower students to acquire knowledge and skills necessary for the 21st-century workforce.

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THE ROLE OF DIGITAL MARKETING IN A GLOBALIZED ECONOMY

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Introduction

The advent of digital marketing has radically transformed the way businesses operate in today's globalized economy. The integration of the internet and digital tools into marketing strategies has shifted the traditional dynamics of business communication and customer engagement. In a globalized world, where businesses have the ability to operate across borders and connect with customers from different cultures, the significance of digital marketing cannot be overstated. Digital marketing enables organizations to reach larger audiences, create personalized customer experiences, and drive growth in an increasingly competitive environment. This article explores the critical role digital marketing plays in shaping business success within the globalized economy.

Objectives

1. To Understand How Digital Marketing Facilitates Business Expansion in a Globalized Economy Digital marketing has enabled businesses, both large and small, to expand their reach beyond local borders and tap into international markets. This objective explores the strategies that digital marketing provides for businesses to effectively expand their footprint globally, from targeting international consumers to building brand awareness worldwide.
2. To Examine the Role of Digital Marketing in Enhancing Customer Engagement and Experience Across the Globe In a globalized economy, consumer expectations are influenced by diverse cultural, social, and technological factors. Digital marketing allows businesses to create personalized and localized experiences for customers from various regions. This objective delves into how digital marketing helps in creating engaging and customer-centric experiences that are tailored to different global markets.

The Role of Digital Marketing in Business Expansion

The primary role of digital marketing in a globalized economy is its ability to allow businesses to expand their presence internationally without the substantial costs traditionally associated with expansion. Online platforms, including websites, social media channels, search engines, and email marketing, offer businesses a direct line to international consumers. Businesses can leverage these platforms to advertise their products and services to a wide audience, regardless of geographic location.

For instance, social media channels like Facebook, Instagram, LinkedIn, and Twitter provide companies with the ability to connect with users across the world. This is achieved through targeted advertising campaigns that can be customized for different regions, languages, and cultures. Similarly, search engine optimization (SEO) and content marketing enable companies to improve their online visibility, helping them rank higher in search results and attract global traffic.

Furthermore, digital marketing tools such as analytics software help businesses track the success of their campaigns in real time, allowing for quicker adjustments and better-informed decisions. These tools also enable businesses to measure the effectiveness of their efforts and tailor their strategies for specific international markets. The ability to gain insights into customer behavior on a global scale is a significant advantage in navigating the complexities of global markets.

Digital Marketing and Customer Engagement

In the modern business landscape, customer engagement has become more complex due to the wide range of platforms, devices, and communication channels available to consumers. Digital marketing plays a vital role in addressing these challenges by enabling businesses to engage with customers on a personalized level. Through data-driven marketing, companies can identify consumer preferences and deliver tailored experiences.

One of the most effective tools for customer engagement is social media. Brands use social platforms to not only promote their products but also engage with customers in real-time. For example, companies use platforms like Twitter and Instagram to answer customer queries, address concerns, and gather feedback, which fosters a sense of connection and loyalty.

Moreover, digital marketing allows businesses to personalize their marketing efforts based on customer data. By analyzing browsing history, past purchases, and social media interactions, companies can create highly targeted advertisements and product recommendations. This personalization increases the likelihood of conversion and enhances the customer experience.

For global brands, it is essential to customize marketing strategies to suit the cultural and regional differences of their diverse customer base. Localizing digital content, language, and promotions ensures that the brand resonates with different markets. For example, multinational companies often create region-specific campaigns that appeal to local tastes, preferences, and customs, thereby enhancing their brand's relevance and engagement with international consumers.

Challenges and Considerations

While digital marketing offers numerous benefits, it also comes with challenges that businesses must address to succeed in a globalized economy. One of the most significant hurdles is the need to understand and respect cultural differences. Marketing campaigns that work well in one country may not be as effective in another due to differences in values, customs, and consumer behavior. Therefore, businesses must invest in market research and localization strategies to ensure that their digital marketing campaigns resonate with diverse audiences.

Another challenge is the increasing complexity of digital advertising platforms. With so many platforms and tools available, businesses may struggle to identify the most effective ones for their needs. Moreover, digital marketing is subject to changing algorithms, privacy regulations, and competitive pressures, requiring businesses to stay updated on the latest trends and adjust their strategies accordingly.

Conclusion

In conclusion, digital marketing plays a pivotal role in the success of businesses operating in a globalized economy. It enables companies to reach international audiences, drive engagement, and foster customer loyalty, all while overcoming geographic and cultural barriers. However, for businesses to truly succeed, they must embrace the complexities of digital marketing, from data analysis to personalization, while also recognizing and respecting the diverse needs of global consumers. As the digital landscape continues to evolve, the role of digital marketing will only become more integral to business success in the global marketplace.

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ENHANCING COLLABORATION IN INTERNATIONAL EDUCATION THROUGH TECHNOLOGY

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Introduction

International education has increasingly become a vital part of the global learning ecosystem. It fosters cross-cultural understanding, facilitates knowledge exchange, and promotes global citizenship among students and educators. Technology plays a pivotal role in enhancing these collaborative efforts by bridging geographical, cultural, and language barriers. Through digital platforms, virtual classrooms, and communication tools, technology enables seamless interactions between students and educators from different parts of the world, creating opportunities for collaborative learning and academic partnerships. This paper explores the potential of technology to enhance collaboration in international education, discussing how digital tools can transform educational experiences by promoting inclusivity, increasing accessibility, and fostering dynamic, multicultural learning environments.

Objectives: Enhancing collaboration in international education through technology

1. **Fostering Global Collaboration:** To examine how technological tools can create global learning communities, enabling students and educators from different cultural and academic backgrounds to collaborate effectively.
2. **Facilitating Cross-Border Partnerships:** To explore the role of technology in enhancing academic partnerships between institutions across the globe, allowing for joint research, collaborative programs, and shared academic resources.
3. **Enhancing Communication and Interaction:** To investigate how virtual platforms and communication tools improve interaction between students, faculty, and institutions, enabling more effective global learning experiences.
4. **Increasing Accessibility in Education:** To analyze how technology can make international education more accessible to students from diverse socioeconomic backgrounds and remote locations, overcoming traditional barriers to participation.
5. **Improving Educational Quality:** To assess how digital learning tools and platforms can enhance the quality of education by providing more personalized learning experiences, real-time feedback, and access to diverse resources.
6. **Supporting Cross-Cultural Exchange:** To highlight the role of technology in promoting cross-cultural understanding and exchange, allowing students to experience different educational methodologies and perspectives.

Conclusion

In conclusion, technology has the potential to significantly enhance collaboration in international education by breaking down geographical, cultural, and institutional barriers. Through the use of digital platforms, online resources, and virtual communication tools, students and educators from around the world can engage in meaningful collaborations that enhance their educational experiences. Technology not only facilitates greater inclusivity and accessibility but also fosters a deeper understanding of global issues, cultures, and perspectives. As international education continues to evolve, technology will remain central to its growth, offering new opportunities for academic partnerships, cross-border

collaborations, and a more interconnected global learning community. The effective integration of technology in education will continue to pave the way for a future where collaboration in international education is more dynamic, inclusive, and accessible to all.

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E-COMMERCE IN EMERGING MARKETS: CHALLENGES AND OPPORTUNITIES

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Introduction

E-commerce has transformed the global retail landscape, offering businesses new ways to reach customers and expand their markets. While developed economies have seen a rapid rise in digital commerce, emerging markets present both significant opportunities and unique challenges for e-commerce growth. These markets, including countries in Asia, Africa, and Latin America, offer vast consumer bases, increasing internet penetration, and growing digital payment adoption.

However, despite their potential, emerging markets face barriers such as inadequate infrastructure, regulatory hurdles, and consumer trust issues. Businesses looking to establish or expand their e-commerce presence in these regions must navigate these challenges while capitalizing on the available opportunities. This article explores two key objectives of e-commerce in emerging markets: overcoming challenges and leveraging growth opportunities.

Objective 1: Overcoming Challenges in Emerging E-Commerce Markets

While emerging markets present great potential, several obstacles hinder the seamless expansion of e-commerce. Addressing these challenges is crucial for sustainable growth.

1.1 Infrastructure and Logistics Barriers

One of the primary challenges in emerging markets is the lack of advanced infrastructure for transportation, logistics, and delivery services. Poor road networks, inefficient postal systems, and unreliable supply chains can delay product deliveries, affecting customer satisfaction.

Solution: Companies are investing in localized logistics solutions, such as micro-fulfillment centers and partnerships with local courier services. For example, Amazon and Alibaba have adapted their delivery networks to cater to the unique needs of emerging markets by working with regional logistics providers.

1.2 Digital Payment and Financial Inclusion

Many consumers in emerging markets lack access to traditional banking services, making online transactions difficult. Cash-on-delivery (COD) remains a dominant payment method in many regions, increasing risks for e-commerce businesses.

Solution: The rise of mobile wallets and fintech solutions, such as M-Pesa in Africa and Paytm in India, has enabled secure digital transactions. Governments and financial institutions are also working to improve digital financial inclusion by promoting online banking and mobile payment adoption.

1.3 Consumer Trust and Cybersecurity Issues

In emerging markets, many consumers are skeptical about online shopping due to concerns about fraud, counterfeit products, and poor customer service. Cybersecurity risks, including data breaches and identity theft, further hinder trust in e-commerce.

Solution: Businesses must invest in secure payment gateways, transparent return policies, and strong customer support. Implementing trust-building measures such as verified customer reviews, product authenticity checks, and secure website encryption can encourage consumer confidence.

Objective 2: Leveraging Growth Opportunities in Emerging Markets

Despite these challenges, emerging markets present vast opportunities for e-commerce businesses willing to adapt to local conditions.

2.1 Rapid Internet and Smartphone Penetration

The increasing affordability of smartphones and mobile internet access has driven e-commerce growth in emerging markets. Millions of new consumers are coming online every year, expanding the digital shopping audience.

Opportunity: Companies can adopt a mobile-first approach, developing user-friendly mobile apps and websites optimized for low-bandwidth connections. Platforms like Jumia in Africa and Flipkart in India have successfully leveraged mobile commerce to reach customers in remote areas.

2.2 Social Commerce and Influencer Marketing

Social media plays a significant role in shaping consumer behavior in emerging markets. Many consumers rely on platforms like Facebook, Instagram, and TikTok for product recommendations and shopping.

Opportunity: E-commerce businesses can use social commerce strategies, collaborating with local influencers to build brand trust and engage with potential customers. Live-stream shopping events and interactive product promotions can further boost online sales.

2.3 Localized Product Offerings and Cultural Adaptation

Understanding consumer preferences and local culture is essential for success in emerging markets. A one-size-fits-all approach does not work due to differences in shopping habits, language, and economic conditions.

Opportunity: Businesses can customize their product catalogs, pricing strategies, and marketing campaigns to align with local consumer needs. For example, e-commerce platforms in India offer regional language support and festival-specific discounts to attract diverse customer segments.

Conclusion

E-commerce in emerging markets presents both challenges and opportunities for businesses seeking to expand their digital footprint. Addressing logistical, financial, and trust-related barriers is essential for building a sustainable e-commerce ecosystem. At the same time, leveraging mobile commerce, social media influence, and localized strategies can unlock significant growth potential.

As digital infrastructure continues to improve and consumer confidence in online shopping grows, emerging markets are poised to become major drivers of global e-commerce expansion. Businesses that adapt to these dynamic environments and invest in innovative solutions will thrive in the evolving digital economy.

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AI-POWERED EDUCATIONAL TOOLS: TRANSFORMING TEACHING AND LEARNING

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Introduction

Artificial Intelligence (AI) is revolutionizing multiple sectors worldwide, including education. AI-powered educational tools have emerged as game-changers, significantly improving teaching and learning processes. In developing nations, where educational challenges such as teacher shortages, lack of resources, and accessibility gaps persist, AI-driven innovations offer a transformative solution. These technologies provide personalized learning, automated assessments, and enhanced educational content delivery, making education more effective and inclusive.

Beyond improving education, AI-driven learning tools contribute to sustainable economic growth by fostering a skilled workforce. With enhanced digital literacy, students and professionals in developing countries can access better job opportunities, drive innovation, and contribute to national economic development. This article explores how AI-powered educational tools are transforming education and how these innovations can lead to sustainable economic growth in developing nations.

Objective 1: Enhancing Teaching and Learning Efficiency Through AI-Powered Tools

1. **Personalized Learning:** AI-driven learning platforms, such as adaptive learning software, analyze students' progress and tailor educational content to their needs. This ensures that students receive customized instruction, improving comprehension and retention.
2. **Automated Assessments and Feedback:** AI tools can grade assignments, analyze student performance, and provide instant feedback. This reduces teachers' workload and allows them to focus on interactive teaching.
3. **Virtual Tutors and Chatbots:** AI-powered virtual tutors offer round-the-clock assistance to students, helping them with queries and guiding them through complex subjects. This is especially beneficial in regions with a shortage of qualified educators.
4. **Language Translation and Accessibility:** AI-powered translation tools help overcome language barriers, making education accessible to diverse populations. AI-driven speech-to-text tools also support students with disabilities by converting lectures into readable text.
5. **Smart Content Creation:** AI helps generate educational materials such as interactive simulations, video lessons, and gamified content that enhance learning experiences. This makes education more engaging, especially for students in rural areas with limited resources.

Objective 2: AI-Driven Education as a Catalyst for Economic Growth

1. **Developing a Skilled Workforce:** AI tools facilitate skill-based learning in fields such as coding, data science, digital marketing, and other emerging sectors. This prepares students and workers for high-demand jobs, reducing unemployment rates.
2. **Bridging the Digital Divide:** AI-powered online learning platforms provide access to quality education in remote and underserved areas. This promotes digital inclusion and ensures that individuals from all backgrounds can participate in the global economy.

3. Encouraging Entrepreneurship and Innovation: AI-driven educational platforms equip learners with entrepreneurial skills, enabling them to create startups and innovative solutions. This fosters business growth, job creation, and overall economic development.

4. Boosting Productivity in Various Sectors: AI-trained professionals contribute to industries such as healthcare, agriculture, finance, and manufacturing, increasing productivity and innovation. Developing nations can benefit from AI-driven automation and efficiency in these sectors.

5. Supporting Government Policies and Sustainable Development: AI-powered data analytics can help governments make informed decisions about education policies, workforce development, and economic planning. By aligning education with national economic goals, AI enhances long-term sustainability.

Conclusion

AI-powered educational tools are transforming teaching and learning, addressing key challenges in education systems across developing nations. By enhancing learning efficiency, personalizing education, and automating assessments, AI empowers both students and educators. Moreover, AI-driven education contributes to economic growth by developing a skilled workforce, reducing unemployment, and fostering entrepreneurship.

For AI-powered education to have a lasting impact, governments, private sectors, and educational institutions must invest in digital infrastructure, training, and policy development. By embracing AI-driven learning tools, developing nations can achieve sustainable economic growth and bridge the global digital divide, ensuring a brighter future for generations to come.

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CIRCULAR ECONOMY MODELS IN INTERNATIONAL BUSINESS

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Introduction

In the current global economy, sustainability has become a key area of focus, especially in developing nations where economic growth is often hindered by resource depletion, environmental degradation, and increasing waste. One of the most promising solutions to address these challenges is the adoption of Circular Economy (CE) models. Circular economy models, as opposed to the traditional linear "take-make-dispose" model, promote a regenerative approach that emphasizes reducing waste, reusing resources, and recycling materials. The innovative adoption of these models in developing nations offers an opportunity to not only improve environmental sustainability but also spur economic growth by enhancing resource efficiency, fostering green jobs, and driving technological innovation.

This article explores how circular economy models in international business can facilitate sustainable economic growth in developing nations. Through the integration of innovation in circular business practices, these nations can leverage sustainability as a competitive advantage in the global market. By adopting circular economy principles, developing nations can achieve economic resilience, promote social inclusion, and contribute to global sustainability goals, such as those outlined in the United Nations Sustainable Development Goals (SDGs).

Objectives

To explore how circular economy models can be integrated into international business strategies to promote sustainable economic growth in developing nations.

To analyze how innovations in circular economy practices can foster sustainable development by creating new business opportunities, improving resource efficiency, and enhancing environmental outcomes.

1. The Role of Circular Economy Models in International Business

A circular economy model differs significantly from the traditional linear economic model, where products are designed, manufactured, used, and discarded. Circular models, instead, emphasize the reduction of waste and the continual use of resources through practices such as recycling, remanufacturing, and reuse. In international business, this model encourages the design of products with longer lifecycles, which can be repaired or reused multiple times, thus reducing the need for raw material extraction and minimizing environmental impact.

For developing nations, adopting circular economy principles can have profound implications for sustainable growth. These nations, often rich in natural resources, can use the principles of CE to manage their resources more efficiently and reduce environmental degradation. For instance, instead of exporting raw materials, developing nations can add value by processing these materials within their borders, extending their life cycles through innovative reuse strategies, and developing new markets for recycled products. This can lead to the creation of green jobs and industries centered around recycling, remanufacturing, and sustainable product design.

Additionally, international business strategies involving circular economy models offer new opportunities for developing nations to integrate into global supply chains that prioritize sustainability. Companies and consumers in developed countries are increasingly prioritizing environmental responsibility, creating new markets for sustainably produced goods. By implementing circular models, developing nations can become key suppliers of sustainably sourced products, thus fostering economic development while aligning with global sustainability goals.

2. Innovations Driving Sustainable Economic Growth

Innovation is at the heart of circular economy practices. Through technological advancements and creative business solutions, companies can optimize resource use, reduce waste, and create closed-loop systems that support long-term sustainability. For developing nations, innovation in circular economy practices can lead to new industries and job opportunities, facilitating economic growth.

One of the primary innovations in the circular economy is the concept of product life extension. Technologies such as 3D printing, remanufacturing, and repair services enable businesses to extend the lifespan of products, reducing the need for raw materials and energy consumption. In developing nations, this innovation can be particularly beneficial for industries such as electronics, textiles, and automotive, where high-value products can be refurbished and resold, creating new revenue streams and employment opportunities.

Another innovation with significant potential for developing nations is the rise of digital technologies such as the Internet of Things (IoT), artificial intelligence (AI), and blockchain. These technologies enable the tracking and management of resources throughout the supply chain, making it easier for businesses to optimize resource use, reduce waste, and monitor product life cycles. By incorporating IoT into manufacturing processes, businesses can create real-time data systems that help monitor resource consumption and waste production, driving more efficient use of materials and reducing environmental impact.

Blockchain technology, on the other hand, can enhance transparency and traceability in the supply chain, ensuring that products are sourced responsibly and that waste is minimized. By enabling a more transparent supply chain, blockchain can help businesses in developing nations participate in global markets that demand higher standards of sustainability and ethical production.

3. Addressing Challenges and Harnessing Opportunities

While the potential benefits of adopting circular economy models in developing nations are clear, challenges remain. These nations often face barriers such as limited access to capital, insufficient infrastructure, and a lack of technical expertise. Additionally, many businesses in developing nations operate under cost constraints, which can make it difficult to invest in innovative technologies or sustainable practices.

However, these challenges also present opportunities for innovation. By embracing partnerships between governments, businesses, and international organizations, developing nations can leverage resources to create the infrastructure needed for a circular economy. Public-private partnerships can provide the necessary capital for technological investments and infrastructure development, while international cooperation can help share best practices and knowledge related to circular economy implementation.

Developing nations can also benefit from regional collaborations, where countries with similar economic conditions can share resources and develop joint circular economy strategies. These collaborations could involve creating recycling facilities, pooling resources for research and development, and developing local circular business networks that foster entrepreneurship and innovation.

Conclusion

Circular economy models offer a compelling pathway for developing nations to achieve sustainable economic growth. By adopting innovations in product design, resource management, and digital technologies, these nations can optimize their use of resources, reduce waste, and create new business opportunities. Moreover, circular economy practices can help improve environmental sustainability, generate green jobs, and contribute to meeting global sustainability goals.

To fully realize the benefits of the circular economy, developing nations must overcome challenges related to infrastructure, access to capital, and technical expertise. By fostering innovation, encouraging public-private partnerships, and collaborating regionally, these nations can transform their economies into resilient, sustainable systems that thrive in the global marketplace. Circular economy models offer a unique opportunity for developing nations to not only contribute to global sustainability but also to position themselves as leaders in a rapidly evolving international business landscape.

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BLOCKCHAIN TECHNOLOGY: INNOVATIONS AND APPLICATIONS IN INTERNATIONAL TRADE

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Introduction

In the digital era, where businesses and economies are increasingly becoming interconnected, the need for secure, efficient, and transparent methods of conducting international trade has become more important than ever. Traditional trade mechanisms have long been plagued by inefficiencies such as delays, fraud, high costs, and lack of transparency. However, the advent of blockchain technology is poised to revolutionize the way international trade operates, offering innovative solutions to these challenges.

Blockchain, initially known for supporting cryptocurrencies like Bitcoin, has evolved far beyond its original purpose. It is a decentralized, distributed ledger technology that ensures transparency, immutability, and security in transactions. These features make blockchain a valuable tool for transforming industries, especially international trade. Through blockchain, businesses can enhance the efficiency, security, and traceability of cross-border transactions, thereby improving trust among trade partners, reducing paperwork, and streamlining processes.

This article aims to explore the innovations brought by blockchain technology in international trade and examine the various applications that have emerged, addressing both the opportunities and challenges of adopting blockchain for global commerce.

Objective 1: To Examine the Innovations Blockchain Brings to International Trade

Blockchain technology has introduced several key innovations that have made it an attractive option for international trade. The fundamental principles of blockchain—decentralization, immutability, and transparency—have allowed for the development of groundbreaking solutions in trade processes.

1. **Enhanced Transparency and Security:** Traditional international trade transactions often involve multiple intermediaries, each introducing potential delays and risks. Blockchain removes the need for these intermediaries by providing a single source of truth accessible to all stakeholders. The decentralized nature of blockchain ensures that no single entity controls the data, which reduces the likelihood of fraud and increases transparency. All transactions are recorded on the blockchain in an immutable ledger, ensuring that once data is entered, it cannot be altered. This makes blockchain particularly useful for verifying the authenticity of goods and preventing the circulation of counterfeit products.

2. **Smart Contracts:** One of the most innovative aspects of blockchain technology is the use of smart contracts. These are self-executing contracts with the terms of the agreement directly written into code. In international trade, smart contracts can automatically execute transactions and enforce contractual agreements when predefined conditions are met. For example, a smart contract in a supply chain might release payment to a supplier once the goods are delivered and verified. This reduces the need for intermediaries like banks or legal professionals, minimizes human error, and increases the efficiency of trade processes.

3. Supply Chain Traceability: Blockchain enhances the traceability of products as they move across borders. Each transaction and movement of goods can be logged on the blockchain, providing a transparent record that stakeholders can access in real time. This is particularly useful in industries such as food, pharmaceuticals, and electronics, where the provenance and safety of products are of utmost importance. Blockchain ensures that each step in the supply chain is verified and traceable, making it easier to recall defective products, reduce fraud, and enhance consumer confidence.

Objective 2: To Analyze the Applications of Blockchain in International Trade

Beyond the innovations that blockchain has introduced, it is important to explore the various practical applications of this technology in international trade.

1. Cross-Border Payments and Remittances: Traditional cross-border payments are slow and expensive, often taking several days and involving multiple banks, which can result in high fees and currency conversion costs. Blockchain enables faster and cheaper cross-border payments by removing intermediaries. Cryptocurrencies like Bitcoin and Ethereum, as well as stablecoins pegged to fiat currencies, allow businesses and individuals to send payments across borders almost instantaneously and at a fraction of the cost of traditional methods. Companies like Ripple and Stellar are already utilizing blockchain to enable efficient international remittance systems.

2. Trade Finance: Blockchain is transforming trade finance by simplifying and automating the process of securing and verifying transactions in international trade. Letters of credit, one of the most commonly used financial instruments in global trade, can be digitized and placed on the blockchain. This eliminates the need for paper-based documentation, reducing the time and cost involved in the trade finance process. Additionally, blockchain enhances the security of these financial instruments by reducing the risk of fraud and ensuring that the terms of the trade are met.

3. Customs and Regulatory Compliance: One of the biggest challenges in international trade is ensuring compliance with the various regulations and customs procedures in different countries. Blockchain can streamline the customs process by providing a secure and immutable record of transactions, which customs authorities can use to verify compliance in real time. By reducing the need for paperwork and automating verification processes, blockchain can significantly reduce delays and increase the efficiency of customs procedures.

4. Digital Identity and KYC (Know Your Customer) Processes: Blockchain technology can also be used to improve digital identity verification and the KYC process, which is essential for reducing fraud and ensuring compliance with anti-money laundering (AML) regulations. A decentralized, blockchain-based digital identity can enable businesses to securely verify the identity of their trading partners without relying on centralized authorities. This increases security and reduces the risk of identity theft or fraudulent activities in international trade.

Conclusion

Blockchain technology holds significant promise for the future of international trade, offering innovative solutions to long-standing challenges such as inefficiency, fraud, and lack of transparency. Its applications, ranging from enhancing cross-border payments to streamlining trade finance and improving regulatory compliance, are already transforming the way global commerce operates.

While the adoption of blockchain in international trade is still in its early stages, the potential benefits are clear. Businesses and governments around the world are increasingly exploring

blockchain solutions to improve the security, speed, and cost-effectiveness of their international trade operations. However, challenges such as regulatory uncertainty, technological integration, and the need for widespread adoption must be addressed before blockchain can be fully integrated into global trade systems.

As blockchain continues to evolve, it is likely that its impact on international trade will only grow, enabling more secure, efficient, and transparent global commerce.

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GLOBAL COMPETENCY AND INTERNATIONALIZATION OF HIGHER EDUCATION

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Introduction

In the 21st century, globalization has transformed various sectors, including education. The concept of global competency has gained importance as higher education institutions strive to prepare students for an interconnected world. Internationalization of higher education refers to integrating global perspectives into teaching, research, and service, enhancing students' ability to engage with international challenges. Institutions worldwide are incorporating global competency skills, such as cultural awareness, multilingual proficiency, and cross-border collaboration, to ensure graduates are competitive in a global job market.

This article explores the significance of global competency, the role of internationalization in higher education, and how these elements contribute to academic excellence, employability, and cultural exchange. It also discusses the objectives of internationalization and its long-term impact on students and institutions.

Objectives of Internationalization in Higher Education

1. Enhancing Global Competency among Students

One of the primary objectives of internationalization is to equip students with global competency skills, enabling them to thrive in multicultural environments. This includes:

Cross-Cultural Understanding – Exposure to diverse cultures through exchange programs, international curricula, and global collaboration helps students develop intercultural communication skills.

Language Proficiency – Learning new languages fosters communication skills and allows graduates to work in international settings.

Problem-Solving in a Global Context – Addressing international challenges, such as climate change, economic crises, and technological advancements, requires a well-rounded global perspective.

2. Promoting Research Collaboration and Institutional Partnerships

Another key objective is to foster collaboration between universities across borders, leading to:

Joint Research Initiatives – International partnerships encourage research in areas such as science, technology, social sciences, and humanities.

Student and Faculty Exchange Programs – These programs allow students and educators to experience different academic environments, improving knowledge-sharing and innovation.

Global Recognition and Institutional Growth – Universities that participate in global collaborations enhance their reputation, attract international students, and receive funding for academic development.

Conclusion

The internationalization of higher education plays a crucial role in preparing students for a rapidly changing global landscape. By focusing on global competency, institutions ensure that graduates possess the necessary skills to work in diverse environments, contribute to international research, and foster cross-cultural understanding. The collaboration between universities, the integration of global perspectives into curricula, and the promotion of international partnerships enhance both student learning experiences and institutional growth. Moving forward, higher education institutions must continue to adapt and innovate to keep pace with global trends and meet the evolving expectations of students and employers.

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THE ROLE OF TECHNOLOGY IN PROMOTING SUSTAINABLE AGRICULTURE

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Introduction:

Sustainable agriculture is a practice that meets current food and fiber needs without compromising the ability of future generations to meet their own needs. It emphasizes the importance of ecological balance, resource conservation, and social and economic viability in agricultural practices. With a rapidly growing global population, increasing climate variability, and depletion of natural resources, the need for sustainable agricultural practices has become more urgent than ever. Technology plays a pivotal role in making agriculture more sustainable by improving productivity, resource use efficiency, and reducing the environmental impact of farming. From precision farming to artificial intelligence (AI) and drone technology, innovations are transforming the way farmers approach crop production, pest management, irrigation, and soil health. This article explores the role of technology in promoting sustainable agriculture by focusing on its impact on resource efficiency, environmental conservation, and enhanced productivity. The objective is to understand how technological advancements contribute to improving agricultural practices and securing food security while protecting the environment.

Objectives

1. To examine how technological innovations in agriculture contribute to improving resource efficiency and minimizing waste, ensuring that resources like water, energy, and fertilizers are used more efficiently.
2. To explore the environmental benefits of technology in agriculture, including reduced carbon footprints, enhanced soil health, and the reduction of harmful agricultural practices that lead to land degradation and loss of biodiversity.

Technological Innovations in Sustainable Agriculture

1. Precision Agriculture

Precision agriculture is a technology-driven approach that uses data from various sources, including GPS, sensors, and satellite imagery, to monitor and manage farm activities. By precisely measuring soil conditions, crop health, weather patterns, and other variables, farmers can make more informed decisions about planting, fertilization, irrigation, and pest management. This technology helps optimize resource usage, reduce waste, and increase yields. For example, precision irrigation systems can deliver water only where and when it is needed, reducing water usage and preventing over-irrigation. Similarly, precision fertilization ensures that fertilizers are applied in the correct amounts at the right time, which can improve soil health and reduce nutrient runoff, thereby preventing water pollution.

2. Drones and Remote Sensing

Drones equipped with cameras and sensors are increasingly used to monitor crop health, assess soil conditions, and identify pest or disease outbreaks. By flying over fields, drones capture high-resolution images that are processed using machine learning algorithms to identify areas of concern, such as nutrient deficiencies, pest infestations, or water stress. This data enables farmers to take targeted actions, such as applying pesticides or adjusting irrigation, only where necessary.

Remote sensing technology, often used in conjunction with drones, can monitor large agricultural areas and provide real-time information on crop health. This allows farmers to implement timely interventions and avoid the overuse of pesticides and fertilizers, promoting environmental sustainability.

3. Artificial Intelligence and Machine Learning

Artificial intelligence (AI) and machine learning are revolutionizing agriculture by enabling predictive analytics, decision-making, and automation. Machine learning algorithms can analyze large datasets generated by sensors, satellites, and drones to predict crop yields, optimize planting schedules, and even forecast pest invasions or disease outbreaks.

AI-based systems can also assist in precision farming by automating tasks such as irrigation and fertilization. This reduces the need for human intervention and ensures that resources are used more efficiently. Additionally, AI can help farmers assess the impact of various environmental factors on crops, thus improving decision-making and reducing reliance on chemical inputs.

4. Biotechnology and Genetic Engineering

Biotechnology and genetic engineering are playing a critical role in developing crops that are more resistant to diseases, pests, and environmental stress. Genetically modified organisms (GMOs) have been developed to withstand harsh weather conditions such as drought, reduce the need for chemical pesticides, and increase crop yields.

For example, drought-resistant crops can help conserve water, which is a critical resource for sustainable agriculture. By reducing the need for irrigation, these crops can help farmers in water-scarce regions produce food with minimal environmental impact. Additionally, crops engineered to resist pests or diseases reduce the dependency on harmful chemical pesticides, which can harm ecosystems and biodiversity.

5. Sustainable Livestock Management Technologies

Sustainable livestock farming is equally important for promoting overall agricultural sustainability. Technology in this area includes smart sensors that monitor animal health, behavior, and environmental conditions. For instance, sensors placed on animals can track their vital signs and alert farmers to potential health issues before they become serious, minimizing the need for antibiotics and other chemical treatments.

Moreover, innovations like automated feeding systems ensure that livestock are fed the right amount of nutrition, reducing waste and improving the efficiency of feed usage. This technology also helps reduce the environmental impact of livestock farming by minimizing methane emissions, a significant contributor to greenhouse gas emissions from agriculture.

Environmental Benefits of Agricultural Technology

1. Water Conservation

Water scarcity is one of the biggest challenges facing agriculture, especially in regions that experience prolonged droughts or erratic rainfall patterns. Technological solutions, such as drip irrigation systems and smart irrigation technologies, allow farmers to use water more efficiently, ensuring that crops receive the right amount of moisture without wastage.

By integrating weather forecasting and soil moisture sensors, farmers can adjust irrigation schedules in real time, minimizing water usage while maintaining optimal growing conditions

for crops. This approach not only conserves water but also helps to prevent waterlogging and salinization, which can degrade soil quality.

2. Reduction in Chemical Usage

Traditional farming practices often rely heavily on the use of synthetic fertilizers, pesticides, and herbicides, which can have harmful effects on soil health, water quality, and biodiversity. Technology-driven solutions like precision agriculture and remote sensing help farmers apply chemicals only where and when needed, reducing overall usage and minimizing environmental damage.

The integration of AI and machine learning also allows farmers to predict pest and disease outbreaks more accurately, reducing the need for broad-spectrum pesticide applications. This results in healthier ecosystems and less contamination of soil and water resources.

3. Soil Health Management

Healthy soil is essential for sustainable agriculture, as it provides essential nutrients for plant growth and plays a key role in carbon sequestration. Technologies like soil sensors, which measure parameters such as pH, temperature, moisture, and nutrient levels, allow farmers to monitor and manage soil health more effectively. These tools enable farmers to take corrective measures when soil quality deteriorates, reducing soil erosion and degradation.

Additionally, no-till farming technologies, which reduce soil disturbance, help preserve soil structure and organic matter. By reducing the need for plowing, these technologies prevent soil erosion, improve water retention, and promote biodiversity.

Conclusion

In conclusion, technology is playing a crucial role in promoting sustainable agriculture by enhancing productivity, reducing resource usage, and minimizing the environmental impact of farming practices. Precision farming, drones, artificial intelligence, biotechnology, and sustainable livestock management technologies are transforming the agricultural landscape, helping farmers to meet the growing global demand for food while protecting the planet's resources. The adoption of these technologies is essential for ensuring food security, mitigating climate change, and promoting long-term agricultural sustainability. As technology continues to evolve, its role in fostering a more sustainable and resilient agricultural system will only become more significant.

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THE ROLE OF WOMEN ENTREPRENEURS IN SUSTAINABLE DEVELOPMENT

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Introduction

Women entrepreneurs are a driving force in the global economy, contributing to innovation, job creation, and sustainable development. In recent years, the number of women engaging in entrepreneurship has grown significantly, breaking gender stereotypes and fostering economic empowerment. Women-led businesses have not only contributed to economic growth but have also played a vital role in promoting social equity and environmental sustainability.

Sustainable development, as defined by the United Nations, focuses on meeting present needs without compromising the ability of future generations to meet their own. Women entrepreneurs contribute to this vision by adopting eco-friendly business models, promoting inclusive growth, and fostering social responsibility. However, despite their growing influence, they continue to face challenges such as financial constraints, limited access to resources, and societal biases.

This article explores the role of women entrepreneurs in sustainable development, focusing on two key objectives: economic empowerment and environmental sustainability. It also highlights the barriers they face and the need for supportive policies to enhance their contributions.

Objective 1: Economic Empowerment Through Women's Entrepreneurship

Economic empowerment is one of the most critical aspects of sustainable development. Women entrepreneurs play a significant role in reducing poverty, increasing financial independence, and strengthening national economies.

1.1 Job Creation and Poverty Reduction

Women-led businesses generate employment opportunities, particularly for marginalized groups, helping to reduce poverty. In many developing countries, female entrepreneurs create job opportunities for other women, promoting gender equality in the workforce.

For example, women in small and medium-sized enterprises (SMEs) often engage in local industries such as textiles, agriculture, and handicrafts, providing jobs and improving livelihoods in their communities. By fostering economic stability, they help families afford healthcare, education, and better living conditions.

1.2 Financial Independence and Social Progress

Entrepreneurship provides women with financial independence, reducing their dependence on traditional male-dominated economic structures. This independence allows women to make decisions that benefit their families and communities. Studies have shown that when women control income, they reinvest a significant portion into the well-being of their families, leading to better education, healthcare, and nutrition.

Furthermore, economic empowerment through entrepreneurship helps close the gender wage gap. Women who own businesses have greater control over their income and professional growth compared to those in traditional employment.

1.3 Driving Innovation and Business Resilience

Women entrepreneurs bring fresh perspectives to business and innovation. Their unique problem-solving approaches often result in the development of products and services that address social needs.

For instance, female entrepreneurs have pioneered advancements in sectors such as education technology, healthcare solutions, and social enterprises. Many women-led businesses prioritize ethical sourcing, fair wages, and community welfare, making them more sustainable in the long run.

Despite these contributions, women entrepreneurs face financial and institutional barriers, such as limited access to credit and investment opportunities. Governments and private sectors must implement gender-inclusive policies to ensure women have equal access to funding, training, and mentorship programs.

Objective 2: Promoting Environmental Sustainability

Women entrepreneurs are at the forefront of environmental sustainability, integrating eco-friendly practices into their businesses. From renewable energy startups to waste reduction initiatives, female-led enterprises play a crucial role in addressing environmental challenges.

2.1 Sustainable Business Models

Many women entrepreneurs focus on sustainable business models that reduce environmental harm. Some key areas of contribution include:

Eco-friendly Products: Women-led businesses often produce organic goods, biodegradable packaging, and sustainable fashion products, reducing carbon footprints.

Renewable Energy Solutions: Female entrepreneurs in developing countries have pioneered solar energy initiatives, providing clean electricity to rural communities.

Waste Management and Recycling: Women entrepreneurs engage in businesses that promote recycling, waste reduction, and circular economy principles.

2.2 Sustainable Agriculture and Food Security

Women play a crucial role in agriculture, particularly in small-scale farming and organic food production. Many women entrepreneurs in the agricultural sector use sustainable farming techniques such as:

Organic Farming: Reducing pesticide use and promoting healthier soil and water conservation.

Agroforestry: Combining agriculture with tree planting to enhance biodiversity and combat deforestation.

Water Conservation: Implementing irrigation techniques that reduce water wastage and improve efficiency.

By engaging in sustainable agriculture, women entrepreneurs contribute to food security while preserving the environment for future generations.

2.3 Advocacy and Environmental Awareness

Women entrepreneurs are not just business leaders; many also serve as environmental advocates. Through social enterprises and educational initiatives, they promote sustainability awareness in local communities.

For example, female business leaders often lead campaigns on climate change, pollution reduction, and sustainable consumer habits. Their ability to influence social change makes them powerful agents of environmental progress.

However, to enhance their impact, women entrepreneurs need better access to green financing, government incentives for sustainable businesses, and technological support for eco-friendly innovations.

Challenges Faced by Women Entrepreneurs

Despite their growing presence in entrepreneurship, women still encounter significant challenges that hinder their full potential. Some of the key barriers include:

1. **Limited Access to Capital:** Women entrepreneurs often struggle to secure funding from banks and investors due to gender biases and societal stereotypes.
2. **Cultural and Societal Barriers:** In many regions, traditional gender roles discourage women from pursuing entrepreneurship, limiting their opportunities.
3. **Lack of Mentorship and Training:** Women entrepreneurs frequently lack access to mentorship programs and business education, affecting their ability to scale their businesses.
4. **Legal and Institutional Challenges:** Policies and regulations in some countries do not support women-owned businesses, making it harder for them to compete in the market.

Addressing these barriers requires a multi-faceted approach, including policy reforms, financial inclusion initiatives, and gender-sensitive support systems.

Conclusion

Women entrepreneurs are key contributors to sustainable development, driving economic empowerment, social progress, and environmental sustainability. Their businesses create jobs, foster financial independence, and promote innovative solutions to pressing global challenges. Additionally, female entrepreneurs play a vital role in advocating for and implementing sustainable practices, from eco-friendly products to renewable energy initiatives.

However, despite their growing influence, women entrepreneurs still face numerous obstacles, including financial limitations, societal biases, and lack of institutional support. To fully unlock their potential, governments, financial institutions, and organizations must implement policies that provide equal opportunities, access to resources, and training programs for women entrepreneurs.

By empowering women in business, societies can move toward a more inclusive and sustainable future, where economic growth and environmental responsibility go hand in hand.

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DIGITAL LITERACY: THE NEW ESSENTIAL SKILL FOR THE GLOBAL WORKFORCE

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Introduction

In today's digital age, technology is deeply integrated into every aspect of life, including the workplace. Digital literacy—the ability to use digital tools, access information online, and communicate effectively in a digital environment—has become an essential skill for the global workforce. The increasing reliance on digital platforms for work, education, and communication has made it crucial for employees to develop these skills to remain competitive.

The rise of automation, artificial intelligence (AI), and remote work has transformed traditional job roles, making digital proficiency a necessity rather than an option. Workers who lack digital literacy may struggle with basic tasks such as using email, managing online documents, or participating in virtual meetings. Moreover, digital skills are not only required for technical professions but are also essential for roles in administration, customer service, marketing, and healthcare.

This paper explores the importance of digital literacy, highlighting two key objectives: improving workplace efficiency and enhancing global collaboration. It also discusses how organizations and individuals can work towards developing digital skills to thrive in an increasingly digitalized world.

Objectives

1. Improving Workplace Efficiency

Digital literacy enhances workplace productivity by enabling employees to complete tasks more efficiently using modern technology. Digital tools such as project management software, cloud storage, and data analysis applications streamline processes and reduce manual effort. For example, cloud-based platforms like Google Drive and Microsoft OneDrive allow employees to store, share, and collaborate on documents in real time, eliminating delays caused by traditional paper-based workflows.

Furthermore, automation and artificial intelligence have simplified repetitive tasks such as data entry and customer inquiries. Employees with digital literacy can leverage these technologies to focus on more strategic and creative aspects of their work. For instance, customer service representatives use AI-powered chatbots to handle routine queries, allowing human agents to concentrate on complex customer issues.

Digital literacy also improves time management. Employees who can navigate digital calendars, task management tools, and virtual collaboration platforms can organize their workload efficiently, reducing the risk of missed deadlines or miscommunication. As businesses increasingly adopt digital workflows, employees who lack these skills may face difficulties adapting to new technologies, leading to reduced productivity and job insecurity.

2. Enhancing Global Collaboration

The modern workforce is no longer confined to physical office spaces; digital connectivity has enabled seamless collaboration across borders. Organizations now operate in

a globalized environment where employees from different countries work together on projects using digital communication tools. Digital literacy ensures that employees can effectively communicate and collaborate regardless of geographical location.

Video conferencing platforms like Zoom, Microsoft Teams, and Google Meet have become essential for remote work and virtual meetings. Employees must be proficient in using these platforms to participate in discussions, share screens, and present ideas effectively. Additionally, email communication, instant messaging apps, and digital document-sharing tools have revolutionized teamwork, allowing employees to work together efficiently without being physically present in the same location.

Digital literacy also promotes cultural awareness and inclusivity in global teams. By using digital platforms, employees can engage with colleagues from diverse backgrounds, fostering a more inclusive and dynamic work environment. Understanding digital etiquette, such as respecting different time zones and communication styles, is crucial for maintaining professionalism in virtual collaborations.

Furthermore, businesses that operate internationally require employees to understand cybersecurity best practices. Cyber threats such as phishing, data breaches, and online fraud pose significant risks to organizations. Employees with strong digital literacy skills can identify and mitigate cybersecurity risks, ensuring data protection and business continuity.

Conclusion

Digital literacy has become a fundamental skill for the global workforce, influencing productivity, collaboration, and career growth. In an era where technology continues to evolve, employees must continuously update their digital skills to remain competitive in the job market. Companies that invest in digital literacy training empower their workforce to adapt to technological advancements, enhance efficiency, and contribute to global collaboration.

Governments, educational institutions, and businesses must work together to promote digital literacy through training programs and accessible learning resources. By recognizing digital literacy as a critical skill, individuals and organizations can harness the power of technology to drive innovation and success in the global economy.

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THE ROLE OF ONLINE EDUCATION IN BRIDGING GLOBAL KNOWLEDGE GAPS

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Introduction

Education has always been the foundation of human development, yet access to quality education remains uneven worldwide. Many people in developing countries, rural areas, and marginalized communities struggle to obtain the same level of education as those in more developed regions. The rise of online education has revolutionized learning, offering a solution to bridge global knowledge gaps by making education more accessible, affordable, and flexible.

Online education, facilitated by digital platforms and the internet, enables students across the world to access quality learning resources. Universities, institutions, and independent platforms offer courses covering various disciplines, ensuring that knowledge is no longer restricted by geographical, financial, or social barriers. This digital revolution in education has played a crucial role in reducing educational inequalities, promoting lifelong learning, and fostering global collaboration.

Objectives

1. Enhancing Access to Quality Education for All

One of the primary objectives of online education is to provide universal access to quality learning resources. Traditional education systems often face challenges such as limited infrastructure, a shortage of qualified teachers, and high tuition fees. Online education eliminates these barriers by offering:

Affordable Learning Opportunities: Many online courses are free or available at lower costs compared to traditional education. Platforms like Coursera, edX, and Khan Academy provide free courses from top universities.

Flexible Learning Options: Learners can study at their own pace, making education accessible to working professionals, parents, and individuals with disabilities.

Global Reach: Students from rural and underprivileged regions can access the same courses as those in developed countries, reducing the disparity in educational opportunities.

2. Promoting Knowledge Exchange and Skill Development

Another crucial objective of online education is to foster global knowledge exchange and skill development. With rapid advancements in technology and industry requirements, traditional education often struggles to keep up with evolving skills. Online education addresses this by:

Providing Up-to-Date Knowledge: Digital learning platforms frequently update their courses to align with industry trends, ensuring that learners acquire relevant skills.

Encouraging Cross-Cultural Learning: Online education enables students from different countries to engage in discussions, collaborate on projects, and share diverse perspectives. This cultural exchange enriches learning experiences and fosters global understanding.

Enhancing Professional Development: Many online courses focus on skill-based training, such as coding, data analysis, and digital marketing, helping individuals improve their employability. Certifications from reputed institutions also add value to job applications.

Conclusion

Online education has emerged as a transformative force in bridging global knowledge gaps by making education more accessible, flexible, and relevant to modern needs. It has broken down geographical and economic barriers, ensuring that knowledge reaches even the most remote parts of the world. By fostering skill development and global collaboration, online education contributes to a more inclusive and knowledge-driven society.

However, challenges such as digital literacy, internet accessibility, and the need for self-discipline among learners must be addressed to maximize the potential of online education. Governments, educational institutions, and technology providers must work together to ensure that digital learning continues to evolve as an effective tool for global education equity.

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GREEN FINANCE AND SUSTAINABILITY IN BUSINESS PRACTICES: A NEW PARADIGM

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Introduction

The world is facing unprecedented environmental challenges, from climate change to biodiversity loss. As businesses continue to grow and expand, their impact on the environment has become a pressing concern. In response, the concept of green finance and sustainability has emerged as a critical aspect of business practices. This article explores the importance of green finance and sustainability in business practices, highlighting its objectives, benefits, and future directions.

What is Green Finance?

Green finance refers to the use of financial instruments and mechanisms to support environmentally friendly projects and businesses. It involves the allocation of financial resources to projects that promote sustainable development, reduce carbon emissions, and mitigate environmental risks.

Why is Sustainability Important in Business Practices?

Sustainability is essential in business practices as it enables companies to reduce their environmental footprint, improve their social responsibility, and enhance their economic performance. Sustainable business practices not only contribute to a healthier environment but also improve brand reputation, increase customer loyalty, and reduce operational costs.

Objectives:

The objectives of this article are:

1. To examine the role of green finance in promoting sustainable business practices.
2. To discuss the benefits of sustainability in business practices, including environmental, social, and economic benefits.

Objectives

Objective 1: The Role of Green Finance in Promoting Sustainable Business Practices

Green finance plays a critical role in promoting sustainable business practices by providing financial support to environmentally friendly projects and businesses. Green finance instruments, such as green bonds, green loans, and impact investing, enable companies to access funding for sustainable projects, reducing their reliance on fossil fuels and promoting renewable energy.

Case Study: Green Bond Issuance

In 2020, the Indian Renewable Energy Development Agency (IREDA) issued a green bond worth ₹10 billion to finance renewable energy projects. The bond was oversubscribed, demonstrating the growing demand for green finance instruments.

Objective 2: Benefits of Sustainability in Business Practices

Sustainability in business practices offers numerous benefits, including:

- Environmental Benefits: Reduced carbon emissions, water conservation, and waste reduction.
- Social Benefits: Improved brand reputation, increased customer loyalty, and enhanced social responsibility.
- Economic Benefits: Reduced operational costs, improved resource efficiency, and increased competitiveness.

Case Study: Sustainable Supply Chain Management

Unilever, a multinational consumer goods company, has implemented a sustainable supply chain management system, reducing its carbon emissions by 40% and water usage by 30%. The company's sustainable practices have not only improved its environmental performance but also enhanced its brand reputation and reduced operational costs.

Conclusion:

Green finance and sustainability are critical components of business practices in the 21st century. As companies continue to grow and expand, their impact on the environment has become a pressing concern. Green finance instruments and sustainable business practices offer numerous benefits, including environmental, social, and economic benefits. As the world transitions to a low-carbon economy, businesses must adopt sustainable practices and leverage green finance instruments to reduce their environmental footprint and improve their competitiveness.

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IMPACT OF GLOBAL TRADE POLICIES ON SMALL AND MEDIUM ENTERPRISES (SMES)

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Introduction

Global trade policies shape the international business landscape, influencing how companies operate, compete, and grow. These policies include tariffs, trade agreements, import/export regulations, and international standards that can either facilitate or hinder cross-border business activities. While large corporations often have the resources to navigate complex trade regulations, Small and Medium Enterprises (SMEs) face significant challenges due to limited financial and administrative capabilities. SMEs play a crucial role in national economies, contributing to job creation, innovation, and economic diversification. However, changes in global trade policies—such as trade wars, tariff escalations, and regulatory shifts—can disrupt their operations. This article explores the impact of these policies on SMEs, focusing on two key objectives: understanding the challenges and identifying strategies to mitigate negative effects.

Objective 1: Understanding the Challenges Faced by SMEs Due to Global Trade Policies

1. Tariffs and Trade Barriers

Trade policies often involve tariffs, quotas, and non-tariff barriers that increase the cost of importing and exporting goods. For SMEs, which operate on tighter profit margins than large corporations, these additional costs can be burdensome. Higher tariffs on raw materials, for example, can lead to increased production costs, making it difficult for SMEs to remain competitive in international markets.

2. Complex Regulatory Compliance

Global trade regulations require businesses to comply with a wide range of legal and procedural requirements, such as certification standards, labeling laws, and customs documentation. Many SMEs lack the resources to navigate these regulations efficiently, leading to delays, penalties, and restricted market access.

3. Currency Fluctuations and Financial Risks

Trade policies influence currency values, impacting SMEs engaged in international transactions. Currency depreciation in key markets can lead to higher costs for imported goods and raw materials. Additionally, restrictive trade policies can cause economic instability, leading to unpredictable financial conditions that negatively affect SME investments and profitability.

4. Trade Wars and Political Uncertainty

Geopolitical tensions and trade wars—such as those between the U.S. and China—can create an unpredictable business environment. SMEs that depend on international supply chains or export markets may face sudden policy changes that disrupt their business operations. For

instance, retaliatory tariffs between nations can make certain markets unviable for SME exports.

Objective 2: Strategies to Mitigate the Negative Impact of Trade Policies on SMEs

1. Diversification of Markets and Supply Chains

SMEs can reduce dependence on a single market or supplier by diversifying their business operations. Expanding into multiple international markets and sourcing from various suppliers can mitigate risks associated with sudden trade restrictions. Free Trade Agreements (FTAs) can also provide opportunities to enter new markets with lower trade barriers.

2. Leveraging Digital Trade and E-Commerce

The digital economy presents an opportunity for SMEs to bypass traditional trade barriers. By leveraging e-commerce platforms, cloud-based supply chain solutions, and digital payment systems, SMEs can engage in international trade with reduced costs and regulatory hurdles. Governments and trade organizations are increasingly promoting digital trade agreements to support SME participation in global markets.

3. Enhancing Compliance Capabilities

Investing in legal and regulatory expertise can help SMEs navigate complex trade regulations. Partnering with trade consultants, joining industry associations, and utilizing government support programs can assist SMEs in meeting compliance requirements efficiently. Automation tools and trade management software can also streamline the documentation and compliance process.

4. Accessing Financial Support and Trade Facilitation Programs

Many governments and international organizations provide financial assistance, trade credit insurance, and export incentives to help SMEs manage trade-related risks. Programs from entities such as the World Trade Organization (WTO), the International Trade Centre (ITC), and regional trade bodies can offer valuable support in overcoming financial and logistical barriers.

Conclusion

Global trade policies significantly impact SMEs, affecting their cost structures, market access, and financial stability. While trade barriers, regulatory complexities, and geopolitical uncertainties pose challenges, SMEs can adopt strategic approaches to mitigate these risks. By diversifying markets, leveraging digital trade, enhancing compliance capabilities, and accessing financial support, SMEs can strengthen their resilience in the evolving global trade environment. Governments and international organizations must also work toward creating SME-friendly trade policies to ensure sustainable economic growth and global trade inclusivity.

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SUSTAINABLE BUSINESS MODELS IN THE AGE OF DIGITAL TRANSFORMATION

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Introduction

In today's rapidly evolving world, businesses are experiencing significant changes driven by technological advancements and societal expectations. The digital transformation, characterized by the adoption of digital technologies in all areas of business, has opened new opportunities for innovation, efficiency, and customer engagement. However, it has also increased the need for sustainability, urging businesses to balance profitability with environmental responsibility and social equity. Sustainable business models (SBMs) are frameworks that help companies create, deliver, and capture value in ways that are economically viable, socially responsible, and environmentally friendly. As the global community faces challenges such as climate change, resource depletion, and social inequality, the role of SBMs becomes increasingly critical. Digital transformation accelerates this shift by providing tools and technologies that enable more sustainable practices, from reducing waste in supply chains to enhancing energy efficiency through smart technologies. This paper explores the concept of sustainable business models in the digital era, focusing on their importance, key components, and how digital technologies support their implementation.

Objectives

Objective 1: To Analyze the Role of Digital Transformation in Promoting Sustainable Business Models

Digital transformation enables businesses to reimagine their operations, products, and services through innovative technologies such as artificial intelligence (AI), big data analytics, the Internet of Things (IoT), blockchain, and cloud computing. These technologies offer new pathways for integrating sustainability into core business strategies.

For example, IoT devices can monitor energy consumption in real-time, helping companies optimize their operations and reduce carbon emissions. Blockchain technology enhances transparency in supply chains, ensuring ethical sourcing and responsible production. AI-driven data analytics allow businesses to predict market trends and customer preferences, minimizing waste and overproduction.

By adopting digital solutions, businesses can develop sustainable products, improve resource efficiency, and create more transparent and ethical operations. This not only benefits the environment and society but also builds consumer trust and brand loyalty, offering long-term competitive advantages.

Objective 2: To Identify the Key Components of Sustainable Business Models in the Digital Age

A sustainable business model in the digital age typically includes several key components:

1. **Value Proposition for Sustainability:** Businesses focus on providing products and services that create environmental and social value, in addition to economic returns. This includes eco-friendly products, fair trade practices, and services that promote well-being.

2. **Stakeholder Integration:** Sustainable businesses engage with a broad range of stakeholders, including customers, employees, suppliers, communities, and regulators. Digital platforms enable more effective communication and collaboration with stakeholders, facilitating co-creation and shared value.
3. **Circular Economy Practices:** Many SBMs are adopting circular economy principles, where products are designed for reuse, refurbishment, and recycling. Digital technologies like 3D printing and predictive maintenance support these practices.
4. **Sustainability-Driven Revenue Streams:** Businesses explore new revenue models, such as product-as-a-service or sharing platforms, which emphasize access over ownership and reduce resource consumption.
5. **Measurement and Reporting:** Digital tools help businesses measure and report their environmental and social impacts more accurately. Sustainability metrics and reporting frameworks such as ESG (Environmental, Social, and Governance) are increasingly supported by data analytics and AI technologies.

Conclusion

Sustainable business models have become an essential strategy for companies striving to remain competitive and responsible in the age of digital transformation. As digital technologies continue to evolve, they offer new ways to enhance sustainability, enabling businesses to create long-term value for all stakeholders. By leveraging innovations such as AI, IoT, and blockchain, companies can design operations and products that minimize environmental impact, promote ethical practices, and foster inclusive growth. Moreover, these sustainable practices align with growing consumer demand for responsible business conduct and can lead to increased brand loyalty and market differentiation. The future of business lies in integrating sustainability with digital transformation, ensuring that economic success does not come at the expense of people and the planet. Businesses that embrace sustainable models today are more likely to thrive in tomorrow's digital and sustainable economy.

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GAMIFICATION AND ITS IMPACT ON MODERN EDUCATION

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Introduction

Education has evolved significantly in recent years, with digital technologies playing a crucial role in shaping new learning methods. One of the most innovative approaches is gamification, which involves incorporating game-like elements such as points, badges, leaderboards, and interactive challenges into the learning process. Gamification makes learning more engaging, interactive, and enjoyable, helping students stay motivated and improving knowledge retention.

In modern education, gamification is used in classrooms, online learning platforms, and corporate training programs. Teachers and educators use it to make complex subjects more accessible, encourage healthy competition, and enhance student participation. This article explores the impact of gamification on modern education, focusing on its role in increasing student engagement and improving learning outcomes.

Objective 1: Increasing Student Engagement

One of the most significant impacts of gamification in modern education is its ability to increase student engagement. Traditional learning methods, such as lectures and textbooks, often fail to capture students' interest. Gamification changes this by making learning fun, interactive, and rewarding.

How Gamification Engages Students

1. Use of Rewards and Recognition – Students earn points, badges, or certificates for completing tasks, encouraging them to stay motivated.
2. Healthy Competition – Leaderboards and challenges inspire students to perform better and improve their skills.
3. Interactive Learning – Gamification includes quizzes, role-playing, and storytelling, making lessons more engaging.
4. Instant Feedback – Students receive immediate feedback on their progress, helping them understand their mistakes and improve.

Research shows that when students enjoy learning, they participate more actively, retain information better, and develop a positive attitude toward education. Platforms like Kahoot!, Duolingo, and Classcraft use gamification techniques to make subjects like math, science, and language learning more exciting.

By transforming the learning experience, gamification ensures that students are not just passive learners but active participants who enjoy and look forward to learning.

Objective 2: Improving Learning Outcomes

Beyond engagement, gamification plays a crucial role in enhancing learning outcomes by improving memory, comprehension, and problem-solving skills.

How Gamification Enhances Learning

1. Better Knowledge Retention – Studies show that students remember information more effectively when learning through games and simulations rather than traditional methods.
2. Encourages Critical Thinking – Problem-solving challenges and strategy-based games help students develop logical thinking and decision-making skills.
3. Personalized Learning – Gamification allows students to learn at their own pace, making education more inclusive and effective.
4. Boosts Confidence – As students progress through levels and earn rewards, they gain confidence in their abilities and feel encouraged to tackle more challenging tasks.

For example, Minecraft: Education Edition helps students learn subjects like coding, history, and environmental science through interactive storytelling and problem-solving. Similarly, Duolingo gamifies language learning by awarding points for completing lessons and providing instant feedback to improve grammar and vocabulary.

By making learning more interactive and goal-oriented, gamification ensures that students develop practical skills and retain knowledge for the long term.

Conclusion

Gamification is transforming modern education by making learning more engaging, interactive, and effective. By incorporating game-like elements into lessons, educators can increase student motivation, improve participation, and enhance learning outcomes. Whether through points, rewards, or interactive challenges, gamification encourages students to enjoy the learning process and develop essential skills for the future.

Despite its many advantages, gamification also presents challenges such as the risk of distraction, over-competitiveness, and the need for well-designed educational games. However, with continuous advancements in technology and educational research, gamification will continue to shape the future of learning in schools, universities, and online education platforms worldwide.

By integrating gamification into modern education, we can create a more engaging, fun, and effective learning experience for students of all ages.

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THE INFLUENCE OF MOOCS (MASSIVE OPEN ONLINE COURSES) ON GLOBAL LEARNING

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Introduction

Education has undergone a digital transformation with the rise of Massive Open Online Courses (MOOCs). These online platforms provide free or affordable access to high-quality educational resources, allowing learners from all over the world to gain knowledge and skills without traditional barriers like location, cost, or time constraints. MOOCs are offered by prestigious universities and institutions, making education more accessible and flexible. This article explores the impact of MOOCs on global learning, focusing on their role in expanding educational opportunities and improving skill development worldwide.

Objective 1: Expanding Educational Opportunities

One of the key influences of MOOCs on global learning is their ability to provide access to education for millions of people, including those in remote or underserved regions. Traditional education systems often face challenges such as high costs, limited infrastructure, and geographic constraints. MOOCs help to bridge these gaps by offering free or low-cost courses in various fields, from science and technology to arts and humanities.

Many top universities, such as Harvard, MIT, and Stanford, provide MOOCs through platforms like Coursera, edX, and Udemy. These courses include video lectures, reading materials, quizzes, and discussion forums, allowing students to learn at their own pace. This accessibility empowers learners who may not have the opportunity to attend university due to financial, social, or geographical barriers.

Additionally, MOOCs have been instrumental in educating refugees, underprivileged students, and working professionals who seek to improve their knowledge without leaving their jobs or homes. As a result, MOOCs contribute significantly to the democratization of education, ensuring that learning is not limited to privileged individuals but available to everyone worldwide.

Objective 2: Enhancing Skill Development and Career Growth

Another major influence of MOOCs is their role in skill development and career advancement. In today's fast-changing job market, continuous learning is essential for staying competitive. MOOCs provide professionals with opportunities to acquire new skills, update their knowledge, and earn certifications recognized by employers globally.

Many MOOCs offer specialized courses in fields such as programming, business management, data science, artificial intelligence, and digital marketing. Some platforms also provide certification programs in collaboration with major companies like Google, Microsoft, and IBM, helping learners gain industry-relevant expertise.

Moreover, MOOCs promote lifelong learning by enabling individuals to upskill or reskill as needed. Employees can take short courses to enhance their job performance, while job seekers can gain new qualifications that make them more employable. Companies and organizations also benefit from MOOCs by using them to train their workforce efficiently and affordably.

With interactive assignments, peer collaboration, and real-world projects, MOOCs ensure that learners not only gain theoretical knowledge but also practical experience. This hands-on approach increases their confidence and prepares them for real-life challenges in the workplace.

Conclusion

MOOCs have revolutionized education by making learning accessible, flexible, and affordable for people across the globe. They have expanded educational opportunities, allowing individuals in remote and underprivileged areas to access quality learning resources. Additionally, MOOCs have played a crucial role in skill development and career advancement, enabling professionals to stay competitive in an evolving job market.

Despite their many benefits, MOOCs also face challenges such as low completion rates, the need for internet access, and the lack of direct interaction with instructors. However, ongoing improvements in technology and instructional design are addressing these challenges, making MOOCs an increasingly effective tool for global learning. As digital education continues to grow, MOOCs will remain a powerful force in shaping the future of education worldwide.

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THE EVOLUTION OF CONSUMER BEHAVIOR IN THE DIGITAL ERA

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Introduction

The digital revolution has significantly transformed consumer behavior, reshaping how people discover, evaluate, and purchase products and services. With the rise of e-commerce, social media, artificial intelligence (AI), and mobile technology, consumers now have more access to information, personalized recommendations, and convenient shopping experiences than ever before.

Traditional buying patterns, once influenced mainly by word-of-mouth and in-store interactions, have shifted to digital platforms where reviews, online influencers, and data-driven marketing play a crucial role. Businesses must understand these evolving behaviors to adapt their strategies and remain competitive in the market.

This article explores two key objectives in understanding the evolution of consumer behavior: the impact of digitalization on purchasing decisions and the role of personalization and data-driven marketing in shaping consumer choices.

Objective 1: The Impact of Digitalization on Purchasing Decisions

The digital era has revolutionized the way consumers make purchasing decisions. Several factors influence their behavior, including the accessibility of online shopping, mobile commerce, and the power of online reviews.

1.1 Shift from Physical to Online Shopping

The convenience of e-commerce has led to a significant shift from traditional brick-and-mortar shopping to online retail platforms. Consumers can browse and purchase products from anywhere, reducing the need for in-person shopping.

Example: Companies like Amazon and Alibaba have built global e-commerce ecosystems that cater to consumer preferences for fast delivery, seamless checkout, and a wide range of product choices.

1.2 The Role of Social Media and Influencers

Social media platforms such as Instagram, TikTok, and YouTube have become key drivers of consumer behavior. Influencers and content creators shape opinions and purchasing decisions by showcasing products through engaging content.

Example: Fashion brands partner with influencers who demonstrate the latest trends, leading to increased consumer trust and brand engagement. Studies show that consumers are more likely to buy products recommended by influencers they follow.

1.3 The Power of Online Reviews and Ratings

Consumers rely heavily on online reviews and ratings to evaluate products and services. Platforms like Google Reviews, Trustpilot, and Yelp help users make informed decisions before purchasing.

Example: A restaurant with positive online reviews attracts more customers, while a product with low ratings on an e-commerce platform may struggle to gain consumer trust.

Objective 2: The Role of Personalization and Data-Driven Marketing

Advancements in data analytics and AI have allowed businesses to understand consumer preferences better, leading to highly personalized shopping experiences.

2.1 AI and Predictive Analytics in Consumer Behavior

Artificial intelligence helps businesses analyze consumer behavior patterns and predict future purchasing decisions. AI-powered recommendation engines suggest products based on browsing history, past purchases, and user preferences.

Example: Netflix recommends movies based on users' viewing history, and Amazon suggests products tailored to individual shopping habits, enhancing customer satisfaction.

2.2 Personalized Advertising and Targeted Marketing

Digital marketing has evolved from mass advertising to targeted campaigns that focus on individual consumer preferences. Businesses use data-driven marketing to deliver relevant content and advertisements based on demographics, interests, and online activity.

Example: Google Ads and Facebook Ads enable businesses to reach specific audiences through personalized campaigns, ensuring higher engagement and conversion rates.

2.3 The Growth of Subscription-Based and On-Demand Services

Consumers increasingly prefer subscription-based services that offer convenience and continuous access to products. The rise of on-demand streaming, meal delivery, and personalized beauty subscription boxes reflects this trend.

Example: Companies like Spotify, Netflix, and Birchbox provide personalized subscription services, allowing users to customize their experiences based on preferences.

Conclusion

Consumer behavior has undergone a remarkable transformation in the digital era, influenced by the accessibility of online shopping, social media influence, and the power of data-driven marketing. The shift from traditional to digital purchasing habits has reshaped business strategies, emphasizing convenience, personalization, and customer engagement.

As technology continues to evolve, businesses must adapt to these changing behaviors by leveraging AI, data analytics, and personalized marketing techniques. Understanding digital consumer behavior is crucial for companies to enhance customer satisfaction, drive sales, and remain competitive in the ever-evolving digital marketplace.

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ENTREPRENEURSHIP EDUCATION IN THE GLOBALIZED ECONOMY

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Introduction

In an increasingly interconnected world, entrepreneurship has emerged as a driving force behind economic growth, innovation, and job creation. Globalization, characterized by the movement of goods, services, capital, and people across borders, has drastically reshaped business landscapes and presented both challenges and opportunities for entrepreneurs. As a result, entrepreneurship education has gained significant importance in preparing individuals to navigate and thrive in this dynamic global economy.

Entrepreneurship education refers to the process of equipping individuals with the knowledge, skills, and mindset needed to identify business opportunities, take risks, and launch and manage successful ventures. In the context of a globalized economy, entrepreneurship education not only focuses on traditional business skills but also incorporates global awareness, cross-cultural understanding, and an ability to leverage technological advancements. Given the complexities of global markets, educational institutions and governments are increasingly emphasizing entrepreneurship as a means to foster economic growth, reduce unemployment, and drive innovation on a global scale.

The role of entrepreneurship education in the globalized economy is multifaceted. It enables individuals to contribute to their local economies while also preparing them to compete and collaborate in a global marketplace. Through effective entrepreneurship education, individuals are encouraged to think creatively, adopt innovative practices, and utilize technology to overcome geographical and cultural barriers. This article explores the significance of entrepreneurship education, its objectives, and its role in shaping future leaders capable of thriving in an interconnected global economy.

Objectives of Entrepreneurship Education in the Globalized Economy

1. To Equip Learners with the Necessary Skills for Global Competitiveness In a globalized world, entrepreneurs must possess a diverse skill set to remain competitive. Traditional entrepreneurship education focuses on foundational skills such as financial management, marketing, and operations. However, in the context of a global economy, students must also develop competencies in international business practices, digital literacy, and intercultural communication. The first objective of entrepreneurship education is, therefore, to provide learners with the knowledge and tools needed to understand global markets and compete on an international scale. This includes fostering skills in international marketing, cross-border trade, negotiation, and adaptation to varying regulatory environments.

Additionally, entrepreneurs need to be adept in leveraging technology, digital platforms, and social media to connect with global customers and partners. As businesses become increasingly digital and interconnected, knowledge of emerging technologies like artificial intelligence, blockchain, and e-commerce is essential for sustaining growth in the global economy. Entrepreneurship education programs must, therefore, integrate these elements into their curricula to ensure that students are well-prepared for the challenges and opportunities of a global marketplace.

2. To Foster Innovation and Sustainability in Business Practices Another critical objective of entrepreneurship education in a globalized economy is to encourage innovation and sustainable business practices. With the rapid pace of technological advancement and the growing awareness of environmental and social issues, entrepreneurs today are expected not only to create profitable ventures but also to contribute positively to society and the environment. Education systems must foster a culture of innovation, where students learn to think creatively and identify solutions to global challenges, such as climate change, poverty, and inequality.

Sustainability has become a key driver of global business, with consumers, investors, and governments increasingly prioritizing eco-friendly and socially responsible businesses. Entrepreneurship education must, therefore, teach students how to incorporate sustainable practices into their business models, whether through green technologies, ethical supply chains, or socially conscious products and services. By emphasizing sustainability alongside innovation, entrepreneurship education can contribute to the creation of businesses that are not only economically viable but also socially and environmentally responsible.

Conclusion

In conclusion, entrepreneurship education plays a pivotal role in preparing individuals for the complexities and opportunities of a globalized economy. As businesses become increasingly interconnected, entrepreneurs must possess a diverse set of skills, including an understanding of global markets, technological proficiency, and an ability to innovate in sustainable ways. The objectives of entrepreneurship education in this context are twofold: to equip learners with the necessary skills to compete globally and to foster a mindset of innovation and sustainability. By achieving these objectives, educational institutions can help shape the next generation of entrepreneurs who are capable of navigating and thriving in an ever-changing global landscape. Entrepreneurship education is not only essential for individual success but also for the overall economic development of nations. As globalization continues to shape the business environment, it is crucial for educational systems worldwide to adapt and equip students with the skills and knowledge needed to succeed in a global economy. The future of entrepreneurship lies in the ability to innovate, collaborate across borders, and contribute to a sustainable and equitable global economy.

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VERBAL COMMUNICATION USED BY TEACHERS IN CLASSROOM INTERACTION: A STUDY BASED ON FLANDER'S INTERACTION ANALYSIS CATEGORIES SYSTEM (FIACS)

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ABSTRACT

This research was to find out the kinds of verbal communication usually used by teachers in classroom interaction. It employed qualitative research. The participants of this research were the students of English Education Department in Universitas Negeri Makassar. It consisted of three classes; those were class A, B, and advanced class. Two teachers were taken as the subject of this research. The data were obtained from the classroom observation and observation checklist. The data from observation were analysed by using interactive model of data analysis procedure consisting of data collection, data display, data condensation, and conclusion-drawing/verifying.

The findings showed that all kinds of teachers' verbal communication based on Flander Interaction Analysis Categories System (FIACS) appeared in teaching and learning process. Those were accepts of feelings, praise or encouragement, accepts or uses ideas of students, asking question, lecture or lecturing, giving direction, and criticizing or justifying. The most dominant kind usually used was asking question and the rare kind was accepts of feelings.

Keywords: verbal, communication, classroom interaction, FIACS.

INTRODUCTION

Teaching and learning process is a planned interaction that involves teacher and students in the classroom. The interaction between teacher and students is the main condition for on-going learning. The teacher should interact to the students to make the students get the benefit of learning and to make the teacher knows whether or not the students understand what he/she said. However, it is a quite challenging for teachers to build good interaction in the classroom. He/she needs to have attitudes, abilities, efforts and skills to support the learning process (Susanti, et al, 2014: 248). One of the skills that teacher should have is communication. Communication is essential for connecting, sharing, and transferring information about knowledge, attitudes, and skills from student to teacher and teacher to student (Harisnawati, et al, 2022: 67). Moreover, communication is the process of changing the others' behavior. Through communication, the interaction can be done.

Communication requires a medium in which the communicated information is understood by both sender and receiver. One primary medium is verbal communication, which involves spoken or written language to exchange messages effectively. Verbal is the use of word which can be spoken or written.

In classroom setting, teachers should be aware of the use of verbal and nonverbal communication. Communication in EFL classroom, however, is complicated phenomenon which has central part to classroom activities. Teacher should use the words carefully to be successful in teaching. Each word provokes a feeling, specific emotions, and distinct function to the students. If the words are applied in proper place, it will affect the soul and body of students immediately. Xiao-yan (2006:6) states that in English classrooms, teachers' language

is not only the object of the course, but also the medium to achieve the teaching objective. The learning will not be encountered by the students without language or verbal communication.

Flander in Flander Interaction Analysis Categories System (FIACS) divided two categories of verbal communication, which are: (1) Indirect teachertalk which consists of four subcategories like accepts students' feelings, praise or encourage students, accepts students' ideas, asking question, and (2) Direct teacher talk which consists of three subcategories like lecturing, giving direction, and criticizing or justifying students.

However, many teachers are ignorant and apathetic to verbal communication during teaching and learning, and the reality in the field is that teachers' verbal communication is not maximized (Yunius& Joseph, 2022). They may not be completely aware of how their verbal communication affects students' participation, motivation, and understanding. Some teachers may struggle with effectively using verbal strategies to encourage student engagement, the rise of online learning, for example has led to a decline in students' attention spans and verbal responsiveness, making it more challenging for teachers to maintain meaningful dialogue. Miscommunication or ineffective verbal expressions, such as unclear instructions, lack of encouragement, or monotonous lecturing can contribute to student disengagement, decreased motivation, and difficulty in grasping key concepts.

Hence, based on the illustration above, this research is to explore the kinds of verbal communication usually used by teachers in classroom interaction based on Flander's Interaction Analysis Categories System (FIACS).

RESEARCH METHOD

This research employed qualitative research design to describe the kinds of teachers' verbal communication in EFL classroom. The subject of this research was English lecturers in English Education Department, UniversitasNegeri Makassar. The researcher chose two lecturers as the representative lecturers where the observation was held in two meetings for each. They were professional and experienced lecturers who taught English more than many years. It also found that they were active and had good performance in the classroom. Moreover, for the students, the researcher took the classes taught by the lecturers. Two classes for teacher 1 and two classes for teacher 2, but because there was one class in common taught by them, it consisted of three classes but in four meetings.

To get the data, the researcher applied some instruments namely observation which consisted of observation checklist and field note. Overall, there were 4 recordings taken from this study which represented 4 meetings. The data were analyzed by using interactive model of data analysis consisting of data collection, data display, data condensation, and conclusion-drawing/verifying (Miles and Huberman, 2013). The data were then interpreted and analyzed in the forms of conversation extracts. In the extracts, relevant data were identified and discussed. The results were processed based on the framework of verbal communication by Flander Interaction Analysis Categories System (FIACS).

RESULT

After analyzing the recording, the data about teachers verbal communication in EFL classroom were classified and identified based on Flander Interaction Analysis Categories System (FIACS). The data shows that two teachers performed verbal communication in their teaching and learning activities. The kinds of verbal communicationwere presented as follows:

a) Accepts of Feelings

The first kind of verbal communication is accepts of feelings. In this category, teacher accepts the feelings of the students. This category was rarely used by the teacher. It can be proved when it only occurred 3 times in four meetings.

Extract 1: Asking students' condition

T : **how are you today?**

SSS : fine and you?

T : good thank you

In extract 1 above, it presents the conversation between teacher and students in the classroom where teacher accepted the students' feeling. At the beginning of the lesson and before starting the material, teacher asked the students' condition by saying **“how are you today?”** by smiling. The teacher seemed try to build relationship with the students by asking them this question and wanted to get the students' attention because at the time some students were still busy by themselves. They then answered by saying “fine and you”. At last, the teacher replied it by saying “good thank you”. The first utterance shows that the teacher accepted the feeling of the student by using verbal communication.

b) Praise or Encouragement

The second kind of verbal communication is praise or encouragement. It occurred 17 times in four meetings. This type of verbal communication was used to give positive reinforcement to the students after the students gave an answer to the question asked by teacher or after the students participated in classroom interaction.

Extract 2: Pronouncing 'p' sound

T : sorry?

SSS : /pi/

T : /pi/ is the letter, this is sound

S1 : /pe/

T : **good.**

Extract 2 shows that the teacher praised the student. It occurred when the teacher commanded the students to pronounce 'p' sound which was shown in the slide. After hearing the students' pronunciation, the teacher realized that their pronunciation was not correct. She then commanded them to repeat it by saying “sorry?” But, the students pronounced it with “/pi/” which was still wrong. The teacher then said “/pi/ is the letter, this is sound”. Suddenly, S1 said “/pe/” with higher voice. Hearing it, the teacher then replied him by saying **“good”** followed by rising her thumb. It can indicate that the teacher's verbal communication is used to praise the student followed by hand gesture.

c) Accepts or uses ideas of students

This category is indicated when the teacher clarifies, uses, interprets, and summarizes the ideas of the students. Furthermore, the ideas must be rephrased by the teacher but it is still recognized as being student's contribution. It occurred in 14 times in four meetings.

Extract 3: Asking the meaning of a phrase

T : if we erase the word step? Standing on the door?... without using the doorstep, why not using standing on the door?

S1 : maybe it's like we stand on the door

T : **yah, you standing on me. Yah,if on the door it's like you are standing on me. Yah, that's standing on the door.**

From the extract above, it shows that the teacher accepted the student's ideas. The teacher asked the students about the meaning of doorstep but no one answered it. To make them realized, he asked them again by saying "if we erase the word step? Standing on the door?" and stopped talking for a while to make the students think. He then continued his question "why not standing on the door?". S1 then answered it by saying "maybe it's like we stand on the door".Hearing the student's answer, the teacher said "**yah, you standing on me**"then he added his answer"**Yah if on the door it's like you are standing on me. Yah, that's standing on the door**"with an enthusiastic face.He also used his hands to demonstrate the meaning of the phrase. This teacher's verbal sign then indicates that the teacher accepted the student's ideas by paraphrasing the answer of the student.

d) Asking question

This category occurred 169 times in four meetings. It can be identified when the teacher asked a question to the student based on his/her own ideas and expects an answer from the students.

Extract 4: Asking information

T : iya, it might be voiceless or voiced we will see later which one is voiceless which one is voiced. **The first one?**

SSS : places of articulation

Extract 4 indicates that the teacher was asking a question to the students. At the time, she had informed the students about three factors of English consonant. To convince herself that the students paid attention to what she had explained, she asked the students by saying "**the first one?**"by raising her forefinger to make sure that the students that she wanted them to mention the first factors of English consonant. So, by hearing the teacher's utterance and seeing her gesture, the students answered the question by saying "places of articulation". The utterance and the gesture of the teacher are included as asking question with using hand gesture.

e) Lecture/lecturing

This category is indicated when the teacher gives facts or opinions about content or procedure expressed by his own ideas, gives explanation, cites an authority other than students, or asks rhetorical questions. It occurred 129 times in four meetings.

Extract 5: Explaining how to pronounce p consonant

T : **the lower lip is functioning here, it is to the front teeth, so these teeth don't move.**
/əf/

SSS: /əf/

From extract 5, it indicates that lecturing occurred in this time. It showed that the teacher was explaining how to pronounce p sound in order the students could pronounce it correctly. In this time the teacher told the students about the position of mouth when pronouncing p sound by saying "**the lower lip is functioning here, it is to the front teeth, so these teeth don't move**". In addition, when the teacher said the utterance, she pointed her teeth using her finger, or in another word, she used hand gesture while explaining. She did it to inform the students how to pronounce the consonant f correctly. Hearing the teacher's explanation, the students pronounced it correctly.

f) Giving Directions

This category involves commanding, ordering or initiating in which the student is expected to do. It occurred 84 times in four meetings.

Extract 6: Directing how to pronounce ‘f’ consonant

T : **pay attention, upper teeth don’t move but the lower lip moves**, /əf/
 S16 : /əp/

Extract above shows that the teacher was directing one of the students to pronounce the consonant f. At the time, the teacher saw that S13 did not pronounce the consonant f correctly. She then came close to him and told him **“pay attention, upper teeth don’t move but the lower lip moves”**. When she said it, she also used her hand by pointing her lower lip to make sure that the students know what she said. Here, the teacher gave direction to the students by following her nonverbal behaviour: hand gesture. It is done in order that the student could catch up what she directed. At last, the students still did not pronounce the sound correctly.

g) Criticizing or Justifying Authority

Criticizing or justifying authority is the last kind of verbal communication which speaker does it to make the listener change his/her behavior. It occurred 24 times in four meetings.

Extract 7: Pronouncing the consonants in English

T : once again
 SSS : /pə/, /bə/
 T : **no, your lips like this, no not also like this, no, don’t press your all lips**, try
 S14 : /pə/
 S9 : /pə/

Extract 7 above shows that the teacher was criticizing the students because they did not pronounce the sound correctly. At this situation, the teacher directed the students to pronounce the first line of English consonant which was shown in the slide by saying “once again”. The students then pronounced it, but before the students finished their pronunciation the teacher interrupted them by saying **“no”** using her two hands. This indicated that the students were wrong in pronouncing the sounds. She later said **“your lips like this”** by pressing her lips firmly to show the students how to pronounce the sound before. She then strengthened her utterance by saying **“not also like this, no, don’t press your all lips”**. She told the students not to press all their lips. This utterance showed that the teacher criticized the students’ pronunciation.

DISCUSSION

Seven extracts of conversations were presented as the samples of interaction between teacher and students in the classroom. Overall, the researcher found that two teachers of English Education Department applied and performed all kinds of verbal communication based on Flander Interaction Analysis Categories System (1970) in their teaching activity. The verbal communication used by teacher in the classroom includes accepts of feelings, praise or encouragement, accepts of ideas, asking question, lecture or lecturing, giving direction, and criticizing or justifying authority.

The most dominant kind of verbal communication used by two teachers in classroom interaction was asking question with 169 utterances in four meetings. This finding is supported by Sofyan and Mahmud (2014) who found that asking questions in classroom interaction dominates the teacher talk, and it was very positive for the students. Based on the observation, the researcher found that most asking question is used by teacher 1 to review or to seek specific information from the students, while teacher 2 used it to increase or stimulate the students' participation in learning.

The second dominant kind of verbal communication used by two teachers was lecturing. It occurred 128 times in four meetings. It mainly involved explanation and giving information to the students. This often occurred when the teacher explained the task or the activities that the students should do. This was in line with Pujiastuti (2013) who stated that giving direction and lecturing were affected by the way the teacher led the students to accomplish the learning task.

The next kind was giving direction. It occurred 84 times in four meetings. This category was mainly used by the teachers to maintain the interaction with the students and made them do the activity in the classroom. Herman (2009) stated that the lecturer could give the students a good opportunity to practice their English language by giving them direction to do the activities.

Another kind of verbal communication which appeared in this observation was criticizing. It occurred 24 times in four meetings. Based on the observation, only teacher 1 used this category, teacher two never did it. It was caused by the lesson she taught. At the time, the teacher taught Phonology in different classes. So, the students were required to pronounce the words correctly, and to correct their pronunciation. It was usually done by criticizing them. Pujiastuti (2013) stated that criticizing happened when the teacher criticized and corrected the student's unacceptable performance and behavior.

The next category was praising or encouraging. This category occurred 17 times in four meetings. It was used after the students gave an answer to the question asked by teacher or after the students participated in classroom interaction. This is in line with Herman (2009) who found that most praises occurred after student's response to teacher questions. So, praise or encouragement was used as the positive reinforcement for the students.

The next kind of verbal communication appeared in this observation was accepting the students' ideas. It only occurred 14 times in four classroom meetings. This category was found when the student answered the question and the teacher clarified it by rephrasing the idea of the student but it was still recognized as being student's contribution. By doing it, the students could involve in classroom activity. It is supported by Sofyan and Mahmud (2014) who stated that using ideas of students was one way for encouraging the students to be confident and to show them that their ideas were valuable.

While accepts of feeling was the last and the rare category which occurred in this observation. It was proved when this category just happened 3 times in four meetings. This is supported by Herman (2009) who found that accepted students' feelings were used less than one percent of the time (.10%) for combined groups. The percentage meant that this category rarely happened in his observation.

CONCLUSION

All kinds of verbal communication which are based on Flander in Flander Interaction Analysis Categories System (FIACS) appeared in this observation. Those are divided into two categories which are indirect teacher talk and direct teacher talk. Indirect teacher talk includes accepts of feelings, praise or encouragement, accepts or uses ideas of students, and asking question, while direct teacher talk involves lecture or lecturing, giving direction, and criticizing or justifying. The dominant kind of verbal communication used by two teachers in

classroom interaction is asking question with 169 utterances in four meetings. While the rare kind of verbal communication is accepts of feelings. It only occurs 3 times in four meetings.

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EFFECTIVENESS OF EXPERIMENTAL MEDIA FOR HEAT UPTAKE IN BLACK BODY RADIATION ASSISTED BY LIGHT MEASUREMENT APPLICATIONS (LUX METER) ON LEARNING OUTCOMES AND SCIENCE PROCESS SKILLS

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ABSTRACT

This study aims to determine the learning outcomes of students' cognitive aspects after the application of experimental media for heat uptake in black body radiation assisted by light measurement applications (Lux Meter) is effectively used in learning black body radiation material and to determine the science process skills of students when conducting practicum with experimental media for heat uptake of black body radiation assisted by light measurement applications (Lux Meter) which includes six indicators namely designing experiments, formulating problems, hypothesizing, classifying, observing and interpreting.

The results of the descriptive analysis obtained that the average learning outcomes of students show the population in the experimental class reached a completeness value exceeding the KKM standard > 70, namely with an average learning outcome of 80 with a percentage of 81%, it can be stated that the experimental media of black body radiation heat uptake assisted by light measurement applications (Lux Meter) is effective on learning outcomes. In comparison, science process skills refer to 6 indicators with an average value of science process skills of 76% in a high category and only one KPS indicator in the low category, namely in hypothesizing skills. So, it can be concluded that simple radiation heat uptake experimental media can be used to train students' skills and improve cognitive learning outcomes.

INTRODUCTION

The world of education is very important in human life. An educated person's life will always develop for the better. In every era, there will always be changes that lead to better educational progress. In addition, the world of education also requires various innovations. This is important for the advancement of the quality of education, which not only emphasizes theory but must also be directed to practical matters. According to (Laseinde & Dada, 2023), adopting and utilizing the progressing educational change is a personal decision and development that requires long-term efforts. Therefore, learning innovations are needed so that students become excited, have the motivation and interest to learn, and enthusiastically welcome learning at school. According to (Muflihini, 2016), In the era of globalization and the progress of science and technology today, changes in all aspects of life cannot be avoided. Increasing the quality of higher education is expected to increase the nation's competitiveness in the international arena at regional and global levels.

The issue of 21st-century skills has been the main subject in recent years, both among educational researchers and educational policy-making authorities worldwide. These skills are not developed through conventional learning but are shifted through ICT-based learning. This means that 21st century skills must develop simultaneously with digital literacy. Digital literacy is attitudes and skills using digital technology and communication tools to access, manage, integrate, analyze and evaluate information, build new knowledge, and create and communicate with others. Therefore, the core skills of the 21st century (creativity, critical

thinking, collaboration and communication) are skills that are integrated with digital literacy (Novitra et al., 2024).

The implementation of learning assessment in Indonesia is a joint effort of educators and students to share and process information, with the hope that the knowledge provided is useful for students and becomes a sustainable foundation; the assessment of learning outcomes by education units aims to assess the achievement of graduate competency standards for all subjects. In the 2013 curriculum, the implementation of the evaluation in the 2013 curriculum explicitly requests that educators in schools be balanced in conducting assessments in three domains, namely cognitive, affective, and psychomotor, in accordance with the objectives to be measured. The emphasis on a comprehensive assessment of the three aspects provides a major change compared to the previous curriculum (Almutairi et al., 2020). Assessment in learning, commonly called Integrated Cumulative Grade Pointing (Integrated GPA), aims to encourage constructive alignment practices to ensure curriculum design curriculum implementation. Integrated Cumulative Grade Point Average (IPK) is a system or mechanism to evaluate and report on learners' improvement and development in an integrated manner in the aspects of ethics (ethics and noble values), declarative and functional knowledge and technical skills in disciplines. This mechanism aims to assist stakeholders in planning and determining the most aligned improvement measures (Hashim et al., 2019).

Based on regulations on Indonesian education, according to Article 3 of Law No. 20 of 2003 concerning the national education system, three aspects of educational goals are achieved: cognitive, affective, and psychomotor (). The mental aspect becomes the benchmark for assessing the development of students, the affective element refers to the response of students in learning, while the psychomotor aspect includes movement behaviour and coordination; in measuring the achievement of teaching materials only emphasized results and only on cognitive elements so that it often ignores other aspects, so that the results are usually ineffective because to be able to achieve a certain level of knowledge that is needed is precisely a process and understanding of the concepts that can be achieved by also paying attention to the affective and psychomotor aspects of students, the psychomotor aspect aims so that students can imitate and practice the stimulus that we have provided, with an emphasis on the process aspects of students will master the material that educators have provided will support the effective learning outcomes to be much better. In the world of education, studying is very important and even a necessity that each individual requires.

Physics is a field of science that discusses a lot about nature and its symptoms, from the real (visible in real terms) to the abstract. In learning physics in SMA / MA, there are two things that can not be separated from physics, namely physics as a product (in the form of facts, concepts, principles, laws and theories) and physics as a process (scientific work). Therefore, educators are required to be able to deliver physics in a form that is easier to understand and more interesting so that it can reduce the image that physics is scary (Maiyena et al., 2018).

Based on the results of observations made by researchers with one of the physics educators on April 10, 2021, at SMAN 06 Makassar, it is known that many students respond well to science learning, but on the other hand, the learning outcomes, especially in physics subjects, are below the KKM (Minimum Completeness Criteria) of 70 set by the school due to the lack of motivation to learn from students in the learning process in class.

Based on the facts that have been described, the researcher feels the need to conduct research using props in the form of experimental media because it will be easier to understand physics material when doing experiments while proving the theory taught. By giving him

experimental media in the form of props that can support his learning outcomes by emphasizing the process aspect of experimenting in order to stimulate his motivation in learning because motivation is needed to support the achievement of maximum learning outcomes, learning outcomes will be optimal if there is motivation. The more appropriate the motivation given, the more successful the lesson will be. Motivation will always determine the intensity of students' learning efforts.

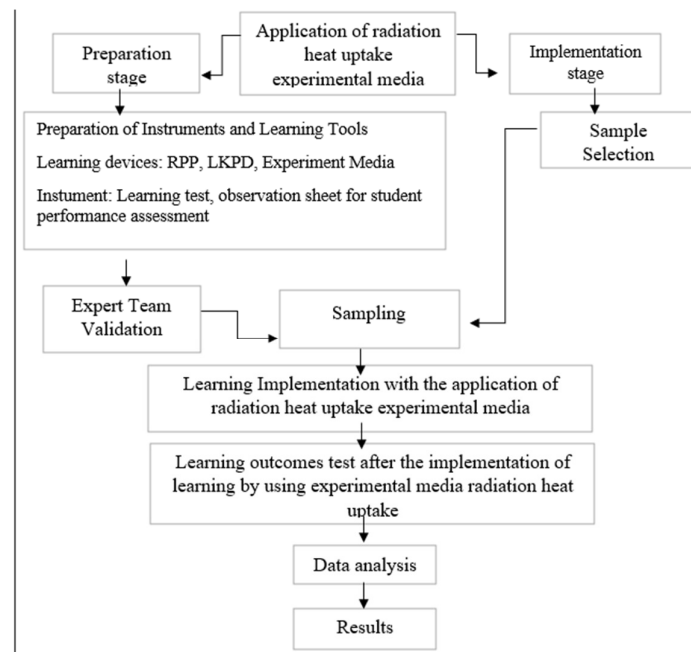
The advantages of this experimental media are that it is easy to use by students because of its portable design, does not use AC currents that endanger students, and can be utilized and used anywhere as long as there is light from sunlight that can produce electric current, meaning that this tool does not require electric current and can even produce electric current from nature. The tools and materials in this experimental media have advantages that follow the development of technology. Materials on this media, such as solar panels, solar panels are materials that have many benefits for life because solar panels can deliver electric current using only direct sunlight, and the drawback is that these solar panels cannot be used at night when there is no light. The application (lux meter) is only a measuring tool for the intensity of sunlight.

METHOD

The type of research used was pre-experiment research with a one-shot case study research design. This research was conducted at SMAN 06 Makassar. This study uses a population that is all students of class XII IPA SMAN 06 Makassar. The sample used was 1 class from 3 classes, namely class XII IPA 2, with a total of 31 students. The sampling technique that will be used in this study is purposive sampling. According to (Sugiyono, 2017), purposive sampling is a sampling technique for specific purposes only. The research procedure consists of the preparation stage, the implementation stage, and the data collection stage. In addition, there are also instruments used in this study, namely learning outcomes tests and student skill assessment observation sheets. The learning tools in this study are semester learning plans, experimental media for heat uptake of black body radiation and student worksheets.

In this case, the data collection technique of this study was carried out by taking data obtained through learning outcome tests and skill observation sheets on students. The following is a diagram of data collection techniques during the research.

RESULTS DISCUSSION



AND

1. Learning Outcomes of Cognitive Aspects of Learners

The results of descriptive analysis of the learning outcomes of students in class XII IPA SMA Negeri 6 Makassar after the application of black body radiation heat uptake media. Based on the results of the learning test conducted in class XII IPA SMA Negeri 6 Makassar with the application of black body radiation heat uptake media, the data on student learning outcomes are obtained in the frequency distribution table 1:

Table 1. Frequency Distribution of Learning Outcomes of Cognitive Aspects of Learners

No.	X_i	F_i
1	60	1
2	67	3
3	80	9
4	86	5
5	93	3
Total	1.695	21

The data in Table 1 above shows the physics learning outcomes of students after applying the experimental media obtained data. Three students got the highest score of 93, five students got a score of 86, nine students got a score of 80, three students got a score of 67, and one student got a score of 60. So, from the distribution table, descriptive statistical data are obtained, which are shown in the following table

Table 2. Data on Student Learning Outcomes

Descriptive Statistics	statistical value
N	21
Mean	80
Std. Deviation	9
Variance	82
Range	33
Minimum	60
Maximum	93
Sum	1.695

Based on Table 2 above, the maximum value is the highest learning outcome value obtained by students in the experimental class after the test with a score of 93, while the minimum value is the lowest learning outcome value obtained by students in the experimental class after the test with a score of 60.

The number of samples in the experimental class was 21 students. Mean or average is the sum of all values in a distribution divided by the number of cases, where the average or mean value obtained is 80. In addition, the value of standard deviation, variance, and range can also be seen. Standard deviation is a measure that describes the level of data distribution from the average value of 9; the range obtained is 33, and the variance is 82. In addition, the sum value obtained is 1,695.

The following is presented a table and a picture of the frequency diagram of the student learning outcomes test regarding the level of effectiveness of the black body radiation heat absorption experimental media as follows:

Table 3. Analysis of Learning Outcome Completeness

Classical Completeness	KKM	F	%	Category
$\geq 75 \%$	≥ 70	17	81 %	Completed
$< 75 \%$	< 70	4	19 %	Not Completed

Based on this, it can be concluded that learning black body radiation material using simple black body radiation heat uptake learning media obtained an average student learning outcome value of 80. The indicator of learning effectiveness is learning completeness; learning completeness can be seen from learning outcomes that have achieved individual completeness, namely, students who have met the minimum completeness criteria (KKM) determined by the relevant school. This learning completeness can be measured if a student can achieve 65% absorption individually and 75% classically with an adequate level of competency achievement, so if the criteria for the effectiveness of the learning completeness value is at least 75% of the total population of students who have obtained a score equal to 70 (KKM standard), then the experimental media used is effective on the learning outcomes of students. Based on the results of the analysis, the value of the minimum completeness criteria (KKM) of students is greater than equal to 70, which is in the complete category with a percentage of 81% and the value of the minimum completeness criteria (KKM) of students more than 70 which is in the category of incomplete with a rate of 19%. So, it can be concluded that the learning outcomes of students have reached effective indicators, so the experimental media used is effective in improving student learning outcomes.

Based on the results of the analysis, the value of the minimum completeness criteria (KKM) of students is more significant than ≥ 70 , which is in the complete category with a percentage of 81% and the value of the minimum completeness criteria (KKM) of students who are less than < 70 (KKM) which is in the incomplete category with a percentage of 19% and the high average value of student learning outcomes is 80 which is also supported by the process of learning activities by doing practical activities directly, where students can understand the theory of black body radiation well and apply the theory during practical activities. Apart from that, the confidence and enthusiasm of students in learning activities, supported by experimental media for solar panel radiation heat absorption with a simple physical form and easy for students to understand in applying the concept of black body radiation. The number of students who get low learning outcomes is due to students being less cooperative with their group members during the learning process and doing practicum while it is going on. The learning outcomes test is measured by the number of correct answers divided by the number of questions so that students' learning outcomes can be obtained to measure the impact received by students after learning using experimental media.

The advantages of experimental media for heat absorption of black body radiation are that it can support the learning process by directly proving the theory taught, and when learning takes place, students are enthusiastic about seeking their knowledge; the learning process takes place not passively with educators, students are ready and confident in the learning process, the enthusiasm of students in the learning process by using experimental media makes students easily remember the material studied so that students can understand the material presented.

2. Learners' sciences process skills

In addition to test instruments to measure student learning outcomes, this study also used observation sheets to assess students' science process skills. The main purpose of analyzing the observation of the assessment of students' science process skills is to determine the effect of experimental media on skills in conducting experiments when the learning process takes place according to each indicator of science process skills. The results of the observation analysis of students' skills assessment for each indicator can be described in Table 4:

Table 4. Percentage Results of Science Process Skills of Learners..

No	KPS Indicator	(%)	Category
1.	Designing Experiments	87 %	Very high
2.	Formulating the Problem	75 %	High
3.	Hypothesizing	51 %	Low
4.	Classifying	75 %	High
5.	Observing (Observation)	86 %	Very high
6.	Interpreting (Interpreting)	85 %	Very high
Average		76 %	High

Based on the data above, the results of science process skills can be said to be in the high category. The details of the data on each indicator of science process skills are the indicator of designing experiments by 87%, formulating problems by 75%, hypothesizing by 51%, classifying by 75%, observing by 86% and interpreting by 85%. At the same time, only one indicator of science process skills is categorized as low, namely 51% hypothesizing skills. According to Purwanto, science process skills are classified as low, if less than 54%. The average value of science process skills reached 76%, which is a high category. After observation, there was one aspect that was classified as low, namely observing, so it can be concluded that the experimental media for radiation heat absorption assisted by lux meter applications can develop students' science process skills and can expand students' physics knowledge, especially on black body radiation material.

It is concluded that the skills teaching approach contributes to understanding abstract concepts in learning, which would remain implicit if taught theoretically. In addition, the skills teaching approach can foster learners' scientific knowledge presented through practical activities and dancing.

Based on the analysis of observation data, it can be concluded that the problem of the lack of hypothesizing skills is caused by several factors, namely, lack of facilities and infrastructure, such as laboratories, books and so on, then because the model, strategy and learning approach still do not explore learning activities that can train students' science process skills optimally, so experimental media is needed in learning that can improve students' science process skills.

Science process skills are all directed scientific skills used to discover facts, concepts or theories. This ability will give students direct experience of a physical phenomenon around them and change their perception of essential things. Process skills involve cognitive or intellectual, manual, and social skills. Mental or intellectual skills are involved because by performing process skills, learners use their minds. Manual skills involve process skills, including using tools and materials, measuring, building or assembling tools. Social skills mean interacting with each other in teaching and learning activities with process skills, such as discussing the results of observations. Based on the view of physics as a process, process

skills are currently used in learning physics. Science process skills can be interpreted as an insight or role model for the development of intellectual, social, and physical skills derived from the fundamental abilities that, in principle, are in the learners who provide opportunities for students to be able to find facts, build concepts, through activities and or experiences such as scientists (Yusuf & Wulan, 2015).

The skills-teaching approach contributes to understanding abstract concepts in learning, which would remain implicit if taught theoretically. In addition, the skills teaching approach can foster learners' scientific knowledge presented through practical and exciting activities.

3. Test of the Effectiveness of the Blackbody Radiation Heat Uptake Experiment Media on the Learning Outcomes of the Cognitive Aspects and Skills of Learners

The effectiveness test of the black body radiation heat uptake experiment media on the learning outcomes and skills of students was analyzed using the sample T-test statistical analysis assisted by the IBM SPSS Statistics 25 application as follows:

a. T-test of Cognitive Aspect Learning Outcomes

The T-test of learning outcomes was analyzed using the IBM SPSS Statistics 25 application. The results can be seen in the following table:

One-Sample Test						
	Test Value = 70					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Learning outcomes	5.408	20	.000	10.714	6.58	14.85

Table 5. T-test I sample of students' learning outcomes

The experimental media of radiation heat uptake is called effective on learning outcomes if the score of the significant level of the T-test analysis result 1 is smaller than the value of 0.05 or Thitung is greater than the value of Ttabel. Based on the analysis, the Thitung obtained is 5.408 while the Ttabel obtained is 2.045. From the comparison of significant scores between Thitung and Ttabel, where $Thitung > Ttabel$, the use of experimental media heat absorption of black body radiation is called effective in student learning outcomes.

The effectiveness of using experimental media on the heat uptake of black body radiation on learning outcomes is supported by the analysis of the significant level in the 1-sample T-test (table 4.8), which is 0.000. The substantial score obtained is smaller than 0.05 ($sig. < 0.05$). Therefore, it can be said that the experimental media of heat uptake of black body radiation is significantly effective in improving learning outcomes.

The experimental media of radiation heat uptake is called effective if the score of the significant level of the T-test analysis 1 is smaller than the value of 0.05 or the Thitung is greater than the Ttabel value. Based on the analysis, the Thitung obtained is 62.048 while the Ttabel obtained is 2.045. From the comparison of significant scores between Thitung and

$T_{hitung} > T_{tabel}$, the use of experimental media heat uptake of black body radiation is called effective on students' skills.

The effectiveness of the use of experimental media on the heat uptake of black body radiation on the skills of students is supported by the analysis of the significant level in the T-test I sample (table 6), which is 0.000. The substantial score obtained is smaller than 0.05 ($\text{sig.} < 0.05$). Therefore, it can be said that the experimental media of heat uptake of black body radiation is significantly effective in improving students' skills.

Based on the analysis results, T_{hitung} obtained the media's effectiveness on learning outcomes at 5.408, while T_{tabel} obtained 2.045. The comparison of significant scores between T_{hitung} and T_{tabel} where $T_{hitung} > T_{tabel}$, which means the use of experimental media heat uptake of black body radiation is effective on student learning outcomes, shows that experimental media heat uptake of black body radiation is effective on student learning outcomes.

The relationship between skills and student learning outcomes, namely by conducting experiments in the learning process can increase student knowledge when students conduct experiments while directly proving the theory of the material presented can streamline the learning process by performing experiments as well as honing student knowledge and the use of experimental media in the learning process can support the learning process.

Based on the analysis, the T_{hitung} obtained is 62.048, while the T_{table} obtained is 2.045. The comparison of significant scores between T_{hitung} and T_{tabel} , where $T_{hitung} > T_{tabel}$, which means the use of experimental media heat uptake of black body radiation is called effective on students' skills, shows that the experimental media heat uptake of black body radiation is effective on students' skills.

Based on the description above, it can be concluded that students' skills must continue to be honed in supporting the acquisition of learning outcomes and also need to be improved. The role of educators is to create a more interesting and effective atmosphere in the physics learning process by using simple props that can activate skill abilities so that students can understand the concept as well as the theory of physics material.

Students' science process skills must continue to be honed and improved, and teachers must play a role in creating a more interesting and effective atmosphere for the physics learning process by using simple props that can activate psychomotor abilities (skills) so that students can understand the concept and theory of physics material.

CONCLUSIONS

Based on the data that has been obtained in this study, the following conclusions can be drawn:

1. The level of learning outcomes of students after taking part in learning black body radiation material using simple black body radiation heat uptake experimental media with an average value of 80 with a percentage of the completeness of learning outcomes is 81% of the total population.
2. The average level of science process skills of students who use simple black body radiation heat uptake experimental media with an average value of 76%, with a high category covering six aspects of science process skills and only one low category, namely the science process skills indicator hypothesizing.
3. The experimental media of black body radiation heat uptake is effective on the learning outcomes and skills of students based on the results of the analysis of the T-

Test I sample of students' learning outcomes value Thitung 5.408 greater than Ttable 2.045, while the T-Test I sample of students' skills value Thitung 62.048 greater than Ttable 2.045, then the radiation heat uptake experiment media is effectively used in supporting the learning process.

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TRANSFORMING LEARNING: INNOVATIONS RESHAPING KNOWLEDGE TRANSFER

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Abstract

Education has a vital role to play in developing the knowledge, skills, attitudes and values that enable people to contribute to and benefit from an inclusive and sustainable future. Examining the historical background and the circumstances that led to the emergence of language teaching highlights the importance of reflecting on our current context. We must consider how political, economic, cultural, and social changes are influencing classroom practices, often in ways we may not immediately recognize. This reflection is crucial, as understanding past and present trends in language education helps shape its future trajectory. Changes in pedagogy and curriculum may originate from within the profession itself—driven by educators, administrators, researchers, and theorists. However, in light of globalization, technological advancements, and shifts in employment, it is essential for TESOL professionals to recognize that English is evolving beyond being merely a subject of foreign language study. Instead, it is increasingly becoming an additional language that L2 speakers use actively to engage with and influence the world around them.

Introduction

English language education is constantly evolving, with new trends and emerging issues shaping the way we teach and learn. In recent years, there has been a growing focus on the psycho-emotional traits of students in language education. This shift towards understanding the psychological and affective well-being of teachers and learners is crucial in creating effective and engaging language learning environments. In today's fast-paced world, students are under immense pressure to perform academically, leading to high levels of stress and anxiety. Understanding and addressing these psycho-emotional traits (i.e., positive and negative affect as well as individual's emotional health and overall functioning) is essential in promoting a positive learning experience. As we look towards the future of language education, it is clear that a more holistic approach is needed to address the psycho-emotional needs of students. Futurology in language studies involves predicting future trends and developments in the field which seeks to explore ways in which teachers and students can work towards sustainable development and think creatively and strategically about long-term outcomes. The field of language studies has the potential to explore a diverse array of possibilities when it comes to predicting the future. Notably, it can cover a wide range of scopes in education policy, development and management, reforms and technology in education, teaching and learning methods, strategies, assessment, sustainable education, inclusion, equity, lifelong learning, and etc. The central focus of this research topic revolves around the exploration of emerging themes in the field of language studies, particularly those related to psycho-emotional traits. By experiencing and incorporating positive affect, mindfulness practices, emotional intelligence training, and stress management techniques into language education, teachers can support students' well-being and enhance their language learning abilities. These characteristics encompass a wide range of factors that play a crucial role in shaping individuals' ability to navigate through stressful and difficult circumstances (Heidarzadi et al., 2022).

Due to the invaluable role of psycho-emotional factors in educational setting, a variety of studies (e.g., Dewaele et al., 2019; Heidarzadi et al., 2022; MacIntyre et al., 2022; Shirzad et al., 2022; Solhi et al., 2023; Wang et al., 2022; Xie & Derakhshan, 2021) have looked into the impact of these factors on teaching and learning including foreign language anxiety, enjoyment, and resilience, academic buoyancy, boredom, emotion regulation strategies, identity, L2 grit, coping strategies, self-esteem, self-efficacy, self-regulation, involvement and engagement, well-being, mindfulness, boredom. The intersection of psycho-emotional traits and language education is a critical area that requires further exploration and research. Looking towards the future, a focus on futurology in language studies will be essential in preparing teachers and students for the challenges and opportunities of tomorrow. Research has shown that the emotional and psychological traits of language learners can influence the future of learning environment and teachers' professional development as well as their level of engagement (Derakhshan, 2022; MacIntyre, 2021; Xuehua et al., 2023). The majority of studies in this field have used traditional, retrospective approaches to explore the interplay between psychological and emotional factors and the process of teaching and learning. Therefore, this research topic adopts a prospective approach to assist educators and students to conscientize responsibility and to map their preferred futures with the objective to act for sustainability now and in the future. The impact of psycho-emotional characteristics on the quality of teaching and effectiveness of learning in language studies is a topic that remains unresolved. The question of whether strategic planning in this field can be influenced by these factors is still up for debate. Accordingly, this research topic takes an emerging issue analysis (EIA) of teachers' and learners' psycho-emotional constructs that might be influential in the future direction of education.

Language teaching introduced itself as a profession in the twentieth century. Central to it was the emergence of the concept of "methods" of language teaching. The method concept in language teaching - the notion of a systematic set of teaching practices based on a particular theory of language and language learning (Richards & Rodgers, 2001, p. 20) - is a powerful one, and the search for better methods was a preoccupation of both teachers and applied linguists throughout the 20th century (Rodgers, 2001). During its rather short life span, the profession has witnessed tremendous changes including shifts in underlying theories of language and language learning, and pertinent consequential changes in stated objectives, syllabus specifications, types of activities, roles of teachers, learners, materials, and so forth.

How do we feel the language teaching profession will move ahead in the near, or even in a more distant future? Discussing the future of anything is always challenging, and that of the language teaching is a particular one. We may be conservatively tempted to assume that things will carry on much as they have in the past and that the future will be recognizable from clues in the present (Rodgers, 2001). Alternatively, we may be tempted to predict a science-fiction future in which the future is nothing like the present (Rodgers, 2001).

A brief look at our past, and the context in which language teaching arose, establishes the necessity for us to look at our context today and ask ourselves how similar forces of political, economic, cultural, and social changes may be finding reflection in our classroom practices, something which we may not otherwise be aware of. The question is an important one, because it will be important for us to see what was and is happening in language education is paving the way for what will be happening.

Futurology in language Education

We are facing unprecedented challenges – social, economic and environmental – driven by accelerating globalization and a faster rate of technological developments. The future is

uncertain and we cannot predict it; but we need to be open and ready for it. Futurology in language studies involves predicting future trends and developments in the field which seeks to explore ways in which teachers and students can work towards sustainable development and think creatively and strategically about long-term outcomes. The field of language studies has the potential to explore a diverse array of possibilities when it comes to predicting the future. Futurology in language studies involves predicting future trends and developments in the field which seeks to explore ways in which teachers and students can work towards sustainable development and think creatively and strategically about long-term outcomes. The field of language studies has the potential to explore a diverse array of possibilities when it comes to predicting the future. We are living in a time when education, like every other facet of society, is experiencing a seismic shift. The way we approach knowledge, the way we learn, and the tools we use to understand the world are evolving faster than ever before. At the center of this transformation is AI—a technology with the power to shape, enhance, and redefine learning on a global scale. As we look ahead to the future of education in the third millennium, one thing is crystal clear: Artificial Intelligence is not just a tool—it is rapidly becoming the heart of how we will learn, teach, and grow in the years to come.

Prophesying Next

Richards and Rodgers (2001) have identified some of the factors that have had influenced language teaching trends in the past and that can be expected to continue to do so in the future. These factors are:

1-Government policy directives: Increased demands for accountability on the part of both funding agencies and governments have paved the way for educational changes in the past and will be doing so in the future (Richards & Rodgers, 2001, p. 253).

2-Trends in the profession: Professional certification for teachers and endorsement of special trends or approaches by professional organizations can have an important impact on the teaching profession (Richards & Rodgers, 2001, p. 253).

3-Guru-led innovations: As teaching has been described as artistry rather science and is shaped by the influence of powerful individual practitioners such as Gattegno, Lozanov, and the like of them in the past, so new gurus will shape teaching practices in the future (Richards & Rodgers, 2001, p. 253).

4-Responses to technology: The potential of the Internet, the computer interfaces and other technological innovations such as the booming cell phone industry is likely to shape the future of the teaching profession and influence both its content and the form of the instructional deliver (Richards & Rodgers, 2001, p. 253).

5-Influence from academic disciplines: As new theories and insights emerge in disciplines such as linguistics, psychology, sociolinguistics, psycholinguistics and other interdisciplinary branches of science, they are likely to have their related impacts on future theories of teaching (Richards & Rodgers, 2001, p. 253).

6-Research influences: Second language teaching and learning is increasingly a field for intensive research and theorizing and has provided the researchers with the impetus for further research (Richards & Rodgers, 2001, p. 253).

7-Learner-based innovations: Learner-based focuses such as individualized instruction, the learner-centered curriculum, learner training, learner strategies, and Multiple Intelligences

recur in language teaching and the trend is anticipated to be continued (Richards & Rodgers, 2001, p. 254).

8-Crossover educational trends: Crossover movements from general education into second language teaching such as Cooperative Learning, the Whole Language Approach, Neurolinguistic Programming, and Multiple Intelligences continue since the field of language teaching does not exercise monopoly over theories of teaching and learning (Richards & Rodgers, 2001, p. 254).

9-Crossovers from other disciplines: Crossover movements from other disciplines such as cognitive psychology, psychotherapy, communication science, to name a few, have left their imprint on language pedagogy and seems to continue to do so in the future (Richards & Rodgers, 2001, p. 254).

Next Phases in Language Teaching Methodology

The future is almost never certain, and this is true in foreseeing methodological directions in second language teaching as well. Rodgers (2001) outlined 10 scenarios which he thinks are likely to shape the teaching of second languages in the next decades of the new millennium:

1. Teacher/Learner Collaborates

Matchmaking techniques will be developed which will link learners and teachers with similar styles and approaches to language learning. Considering various Teacher and Learner roles, one can anticipate development of a system in which the preferential ways in which teachers teach and learners learn can be matched in instructional settings, perhaps via on-line computer networks or other technological resources (Rodgers, 2001). If such matchmaking becomes theoretically viable, a major challenge for the future will be how to put such information into practice in ELT classes. This problem challenges other notions of how individual differences in learning and teaching can be analyzed and accommodated (Rodgers, 2001).

2. Method Synergistics

Crossbreeding elements from various methods into a common program of instruction seems an appropriate way to find those practices which best support effective learning. It seems reasonable to combine practices from different approaches and methods where the philosophical foundations are similar through an approach called "Disciplined Eclecticism" (Rodgers, 2001).

3. Curriculum Developmentalism

Language teaching has not profited much from more general views of educational design. The curriculum perspective comes from general education and views successful instruction as an interweaving of Knowledge, Instructional, Learner, and Administrative considerations as follows:

- **Knowledge Considerations:** In language education, knowledge considerations involve the input/output assumptions about what language is, as well as specification of the content - the topical range - of the instructional language examples or texts presented and the student responses anticipated (Rodgers, 2001).
- **Instructional Considerations:** Instructional considerations reflect the input of teachers and other staff involved with instruction. They also include methods, materials, programs, technologies, and educational environments, as well as

time and scheduling techniques and plans for reporting on learning progress to all stakeholders (Rodgers, 2001).

- **Learner Considerations:** Learner considerations involve the ages, proficiency levels, and developmental stages of the learner or learners. Considerations include societal expectations and learners' self-perceptions, prior learning experiences and preferred learning styles, strategies, environments, and groupings (Rodgers, 2001).
- **Administrative Considerations:** Administrative considerations comprise the choice of instructional models and the scale, pace, and style of educational delivery. Plans for and execution of teacher and learner selection, evaluation, and promotion, as well as environmental development and institutional image, are also administrative considerations (Rodgers, 2001).

Successful educational program design and delivery demands successful integration of all four sets of considerations rather than a dominance by any one set. From this perspective, methodology is viewed as only one of several instructional considerations that are realized in conjunction with all other curricular considerations (Rodgers, 2001).

4. Content-Basics

Content-based instruction assumes that language learning is a by-product of focus on meaning and that content topics to support language learning should be chosen to best match learner needs and interests and to promote optimal development of second language competence (Rodgers, 2001). A critical question for language educators is "what content" and "how much content" best supports language learning. The natural content for language educators is literature and language itself, and we are beginning to see a resurgence of interest in literature and in the topic of "language: the basic human technology" as sources of content in language teaching (Rodgers, 2001).

5. Multitelligencia

The notion of Multiple Intelligences view of human talents proposed by Howard Gardner (1983, as cited in Christison, 1996). This model is one of a variety of learning style models that proposed eight native intelligences and indicates classroom language-rich task types that play to each of these particular intelligences. The challenge here is to identify these intelligences in individual learners and then to determine appropriate and realistic instructional tasks in response (Rodgers, 2001).

6. TotalFunctionalResponse

As Communicative Language Teaching was founded on earlier notional/functional proposals for the description of languages, new leads in discourse and genre analysis, schema theory, pragmatics, and systemic/functional grammar are rekindling an interest in functionally based approaches to language teaching as well (Rodgers, 2001). Refinement of functional models will lead to increased attention to genre and text types in both first and second language instruction (Rodgers, 2001).

7. Strategopedia

"Learning to Learn" is the key theme in an instructional focus on language learning strategies. Research findings suggest that strategies can indeed be taught to language learners, that learners will apply these strategies in language learning tasks, and that such application does produce significant gains in language learning. Simple and yet highly effective strategies will attract considerable instructional interest in Strategopedia (Rodgers, 2001).

8. LexicalPhraseology

The lexical phraseology view holds that only "a minority of spoken clauses are entirely novel creations" and that "memorized clauses and clause-sequences form a high proportion of the fluent stretches of speech heard in every day conversation" (Rodgers, 2001). For language teachers, the results of such views have led to conclusions that language teaching should center on these memorized lexical patterns and collocations and the ways they can be pieced together, along with the ways they vary and the situations in which they occur (Rodgers, 2001).

9. O-zoneWholeLanguage

Renewed interest in some type of "Focus on Form" proposals in second language acquisition (SLA) research, variously labeled as consciousness-raising, noticing, attending, and enhancing input, are founded on the assumption that students will learn only what they are aware of. Whole Language proponents have claimed that one way to increase learner awareness of how language works is through a course of study that incorporates broader engagement with language, including literary study, process writing, authentic content, and learner collaboration (Rodgers, 2001).

10. Full-FrontalCommunicativity

We know that the linguistic part of human communication represents only a small fraction of total meaning. Language teaching should not restrict its attention to the linguistic component of human communication. The methodological proposal is to provide instructional focus on the non-linguistic aspects of communication, including rhythm, speed, pitch, intonation, tone, and hesitation phenomena in speech and gesture, facial expression, posture, and distance in non-verbal messaging so that instruction can improve other aspects of communicative competence as well (Rodgers, 2001) .

Towards a Transformative Shift

With rapid technological advancements and changing societal needs, the landscape of higher education is set to undergo a major transformation. Universities of the future will embrace digitalization, smart technologies, and personalized learning models to provide a more accessible, flexible, and skill-driven educational experience. Below, we explore the most significant changes that will define higher education in the coming decades.

1. Personalized Learning with Artificial Intelligence (AI)

Traditional one-size-fits-all curricula will give way to adaptive, AI-powered education systems that tailor learning experiences to individual students. AI-driven platforms will analyze students' learning patterns, strengths, and interests to create customized coursework, helping learners progress at their own pace. This shift will ensure that education becomes more efficient, engaging, and aligned with students' career aspirations.

2. Hybrid and Remote Learning: Beyond Traditional Classrooms

Future universities will not be confined to physical campuses. Instead, students will access classes from anywhere in the world through hybrid learning models that blend online and in-person education. Advanced tools like Virtual Reality (VR) and Augmented Reality (AR) will create immersive learning environments, allowing students to conduct virtual experiments, attend simulated field trips, and collaborate in digital classrooms.

3. The Rise of the Metaverse in Higher Education

The concept of virtual universities will expand through the metaverse, offering fully digital campuses where students can interact, attend lectures, and conduct experiments in a 3D simulated environment. This shift will eliminate geographical barriers, allowing students worldwide to enroll in prestigious institutions without physically relocating. Virtual university tours and AI-powered career counseling will further enhance the student experience.

4. Skill-Based Learning Replacing Traditional Degrees

The future of education will focus on skill acquisition rather than degree accumulation. Employers are shifting their priorities toward competencies and practical abilities rather than formal diplomas. As a result, universities will increasingly offer short-term courses, industry certifications, and micro-credentials that align with workforce demands. This change will empower students to gain specialized knowledge and adapt quickly to evolving job markets.

5. Stronger Industry Collaboration and Real-World Learning

Universities will forge closer partnerships with industries, ensuring that students gain hands-on experience through internships, apprenticeships, and project-based learning. Applied learning will become a core component of higher education, allowing students to work on real-world challenges from the outset. This model will bridge the gap between academia and employment, making graduates more job-ready.

6. Lifelong Learning for All Age Groups

Higher education will no longer be limited to young students. Instead, universities will become hubs for continuous learning, offering courses for professionals, managers, and lifelong learners seeking to upgrade their skills. This shift will make education more fluid and adaptable, integrating seamlessly with professional growth and career transitions.

7. Data-Driven Education and Advanced Analytics

Universities will leverage big data and AI-powered analytics to track student progress, identify learning gaps, and offer personalized recommendations. These data-driven insights will enhance teaching strategies, optimize course content, and predict future career paths based on students' evolving interests and job market trends.

8. Decentralized and Globalized Universities

Physical campuses will become less relevant as decentralized and global institutions rise. International online universities will offer degrees that are recognized worldwide, allowing students from different countries to study together in collaborative programs. Blockchain-based universities will also emerge, providing secure, verifiable academic credentials, reducing fraud, and making credentialing more transparent.

9. Increased Accessibility and Reduced Tuition Costs

The transition to online learning and open educational resources (OER) will significantly lower the cost of higher education. Universities will adopt innovative financial models, such as “pay-after-employment” schemes, making education more accessible to students in developing regions. This democratization of education will help bridge the global knowledge gap.

10. A Revolution in Assessments and Exams

The traditional exam model will be replaced by dynamic, competency-based assessments. AI-powered systems will evaluate students' real-world problem-solving skills rather than memorization. Performance-based evaluations, group projects, and AI-driven plagiarism detection will ensure fairer, more comprehensive assessments that reflect students' true abilities.

Informationalism, Multiliteracies and Future Pedagogy

With the fast-paced changes brought about by globalization and technological development, TESOL professionals need to understand current socioeconomic factors and their influence on English language teaching (ELT) (Warschauer, 2000). The industrial societies of the past are giving way to a new postindustrial economic order based on globalized manufacturing and distribution which sees the application of science, technology, and information management as the key elements of productivity and economy growth (Warschauer, 2000). This new global economic order, termed Informationalism by Castells (1996, as cited in Warschauer, 2000), first emerged in the 1970s following advances in computing technology and telecommunications.

Although informationalism is still in its infancy, it has already had an important impact on the field of TESOL. As this new stage of global capitalism expands and develops, the ELT profession will face new challenges. In the 21st century, three consequences of informationalism are likely to affect ELT: (a) the growth of global Englishes, (b) changing employment patterns, and (c) the development and spread of technology (Warschauer, 2000).

Just as businesses and media have experienced globalization and relocation, so has the English language. The past few decades have seen a growth in the role of English around the world as the lingua franca for economic and scientific exchange (Warschauer, 2000). But the very growth of English has shifted the balance of forces within it, with L2 speakers by some accounts now outnumbering L1 speakers (Crystal, 1997, as cited in Warschauer, 2000). This explains in part the shift to a trend in which it would be rather odd to insist that all learners adapt to a British or North American model when L2 speakers increasingly use English to speak to other L2 speakers rather than to native speakers of the language (Warschauer, 2000).

A second major way that informationalism will affect ELT in the 21st century relates to trends in employment. Given ongoing discrimination against speakers of other languages, including stigmatized varieties of English, the ability to speak English is associated in many settings with economic advantage (Tollefson, 2007). Simply put, the jobs that existed in the industrial era are disappearing and are being replaced by new types of job and work requirements, such as software engineers, management consultants, strategic planners, lawyers, real estate developers, and research scientists, which are applied in English language contexts (Warschauer, 2000).

The final consequence of informationalism, and the one that underlies all the other changes of global Englishes and changing employment patterns, is the development and spread of information and communications technology (ICT). The rapid development and diffusion of ICT is both a contributor to and a result of the broader socioeconomic changes and it affects the entire context and ecology of language teaching today and tomorrow (Warschauer, 2000).

The spread of world Englishes, changes in employment patterns, and the emergence of new technological literacies are mutually enforcing trends of the global informational economy, and a key pedagogical concept that responds is multiliteracies, put forth by a group

of specialists in education, critical literacy, and discourse analysis (New London Group, 1996; Cope & Kalantzis, 2000, as cited in Warschauer, 2000).

The multiliteracies concept recognizes the inadequacy of educational approaches that limit themselves to “page-bound, official, standards forms of the national language” (New London Group, 1996, as cited in Warschauer, 2000) and suggests instead that students should learn to negotiate a multiplicity of media and discourses.

To cope with the demands of informationalism, we need a framework going far beyond the linguistic syllabi that are most common today, based on collections of syntactic or functional items. It also goes far beyond the notion of task-based learning if such learning is interpreted as consisting of a progression of narrow tasks designed principally to assist learners in grasping particular grammatical forms. A better framework for a new pedagogy is project-based learning (e.g., Stoller, 1997, as cited in Warschauer, 2000). Projects themselves may include many individual tasks, but the umbrella of the project allows opportunities to critique and transform practice in ways that individual tasks do not.

Projects can take many guises and should be based in large measure on students’ backgrounds, needs, and interests. When possible, they may involve electronic communication and collaboration to increase students’ on-line literacy skills. They may also provide opportunities to grapple with cultural and identity issues emerging in the new global era (Beckett & Slater, 2005; Beckett, 2002). Projects might include long-distance exchanges in which students debate and discuss issues related to cultural identity, service learning projects in which students use their knowledge of English and technology to assist their local communities, or multimedia creation and publishing projects in which students collaboratively experiment with new genres (Sokolik, 1999, as cited in Warschauer, 2000).

Potential Components of Future ELT Curriculum

Littlejohn (2001) tries to enumerate some characteristics of future ELT curriculum as follows:

- 1- **Coherence:** The use of themes, topics, projects to bind lessons together and provide coherence and a deeper focus and understanding (Littlejohn, 2001).
- 2- **Significant content:** The selection of content that is worth learning and thinking about, dealt with in appropriate ways, which does not, on the one hand trivialise significant issues or, on the other hand, make trivial things seem important. A key topic could itself be “the future” - attempting to raise students awareness of future developments and discuss their own hopes, aspirations, worries and personal action (Littlejohn, 2001).
- 3- **Decision-making in the classroom:** A structured plan for actively involving students in making decisions in the classroom, taking on more responsibility for what happens in their lessons (Littlejohn, 2001).
- 4- **Use of students’ intelligence:** The use of types of exercises which require thinking, beyond memory retrieval or repetition, for examples, and involving students in hypothesising, negotiating, planning, and evaluating (Littlejohn, 2001).
- 5- **Cultural understanding:** Tasks and texts which require students to look through the eyes of others, to learn the relative nature of values, to understand why people in different contexts think and do different things (Littlejohn, 2001).
- 6- **Critical language awareness:** To view all language use critically - that is, to look beyond the surface meaning and ask oneself questions such as “Why are they saying that?” “What is

not being said?” and “Who benefits from what is being said?” We might for example ask students to think about deeper reasons for why the passive voice is used in a newspaper headline or why particular adjectives are used to describe a consumer product (Littlejohn, 2001).

Possible Characteristics of Future ELT teachers

According to Breen (2007), the question is: “what may be the characteristics of future language teacher development that can be grounded in localized communities of practice while also generating strategies for engaging with regional and global issues that impinge upon teachers’ work?”

Teacher development needs to be holistic in addressing practitioners who enact imagination, values, alignments, intuitions, and diverse knowledge systems during the teaching-learning process. Teachers need to integrate alternative ways of working into current practice; they need to make their own links between what has been personally meaningful in their work so far and what can be seen to be meaningful in future ways of acting (Breen, 2007). Teacher development must therefore address all the attributes of professional identity and self-esteem grounded in ongoing achievements rather than merely the attainment of external imperatives: it must start where teachers have come to and why, assuming that their personal experience and knowledge are evolving rather than things to be superseded (Breen, 2007).

The gap between these activities and curriculum or materials planning appears to be greater than previously because of the current centralization of language policies. Clearly, future teacher development needs to restore dynamic interchange between all these activities and the people engaged in them, thereby reducing unnecessary spaces between them while also recognizing that each of the activities has its own integrity of purposes and procedures (Breen, 2007).

The future development of English language teachers requires their explicit recognition of the pivotal role they play at the heart of current educational, cultural, and political change and the responsibilities that such a role involves (Breen, 2007).

Conclusion

The universities of the future will be smarter, more flexible, and focused on skill-based education rather than traditional degree pathways. Learning will take place in dynamic digital environments, driven by AI, immersive technologies, and industry collaboration. As education evolves, the emphasis will shift toward lifelong learning, making higher education a continuous journey rather than a one-time experience. In light of the profound transformations brought about by globalization, shifting employment landscapes, and rapid technological advancements, the role of English language teaching in the 21st century is evolving. For TESOL professionals, the key shift in perspective is recognizing that English is no longer merely an object of foreign language study for L2 learners; rather, it is becoming an additional language that empowers individuals to engage with and shape the world around them (Warschauer, 2000). In this new paradigm, English functions as a tool for communication, self-expression, and global participation, rather than just an academic subject confined to classrooms. Learners will increasingly use English alongside digital technologies to articulate their identities, advocate for their perspectives, and contribute meaningfully to international discourse. Through social media, digital storytelling, and online collaboration, they can connect across cultures, challenge narratives, and assert their presence on a global stage.

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FOREIGN LANGUAGE LEARNING AND SUSTAINABILITY: TOWARDS INCLUSIVE LITERACY

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Abstract

Foreign language literacy is crucial in fostering global communication, cultural exchange, and economic growth. However, its development must be examined through the lens of sustainable development to ensure equitable access, inclusive learning, and long-term educational impact. This paper highlights the need for sustainable language education policies that promote lifelong learning, environmental consciousness, and global citizenship. Educators can create more adaptable, inclusive, and ethically responsible learning environments by integrating sustainability principles into language education. Ultimately, this study argues that foreign language literacy should focus on linguistic proficiency and align with the broader goals of sustainable development to contribute to a more interconnected and equitable world. This paper also deals with language learning in relation to ESL literacy learner and the relationship between L1 and L2 literacy. Subsequently, some instructional strategies and methodologies for practicing ESL literacy development will be discussed in order to come up with a better understanding of how literacy goals and objectives can be implemented in language classrooms. Next, the issue of language literacy and the relevant future directions for English language teaching, in general, and FL literacy practice, in particular, will be discussed in detail.

Keywords: Critical literacy; Critical thinking, Education; Sustainable development

Introduction

Sustainable language learning integrates equitable education policies, digital literacy, and cultural preservation to support diverse learners and promote lifelong skills. Education for Sustainable Development is an education that aims to help people to develop the attitudes, skills and knowledge to make informed decisions for the benefit of themselves and others, now and in the future, and to act upon these decisions. ESD is hoped to meet the needs of the present without compromising those for future generations. It is a vision of development that encompasses populations, animals and plant species, ecosystems, natural resources and that integrates concerns such as the fight against poverty, gender [in] equality and human and animal rights are observed (Radushinsky, et al., 2024; Vasilev, et al., 2024). By adopting sustainable teaching methodologies and addressing disparities in language education, we can create a more inclusive and resilient learning environment that empowers individuals to actively participate in a globally connected world. Language literacy is the ability to effectively read, write, speak, and comprehend a language for communication, learning, and critical thinking. It extends beyond basic proficiency to include cultural understanding, digital literacy, and the ability to navigate complex social and professional contexts. When integrated with sustainable development in education, language literacy becomes a powerful tool for fostering equitable access to knowledge, promoting lifelong learning, and supporting inclusive societies. Sustainable language education ensures that linguistic skills contribute to global citizenship, environmental awareness, and economic opportunities. It emphasizes:

1. Equitable Access – Ensuring language education is available to all, reducing barriers related to socio-economic status, gender, and geographic location.
2. Cultural and Linguistic Diversity – Promoting multilingualism and preserving indigenous languages to foster inclusive and sustainable communities.

3. Digital and Media Literacy – Incorporating technology to enhance language learning while ensuring responsible digital citizenship.
4. Lifelong Learning – Encouraging continuous skill development for personal growth, career advancement, and social engagement.
5. Global Communication and Collaboration – Equipping learners with language skills to engage in international cooperation and address global challenges. By aligning language literacy with sustainable development, education systems can empower individuals to participate actively in society, contribute to economic growth, and support a more inclusive and environmentally conscious world.

According to the Center for Literacy (2008), to date literacy is a concept with many definitions and categories, and there is no consensus on its definition. Yet, as McKay (1993) states the prevalent ways of viewing literacy can be categorized into two groups, namely, literacy as an individual skill and literacy as a social practice.

For the advocates of the former group, literacy is viewed as a skill that

is acquired by an individual, generally within an educational context, utilizing oral language as a basis and ultimately, affecting cognitive development. Viewed from this perspective, discussions of literacy often involve a delineation of skill level and an examination of the relationship between oral and written language as well as between literacy and cognitive development. (McKay, 1993, p. 8)

Thus, it can be assumed that in this definition the focus is on the individual rather than on the larger social context in which the individual functions. This definition can be related to what Norton (2007) refers in mentioning that "the dominant conception of literacy among governments, policy-makers, and many members of the general public is that literacy references the ability, on the part of individuals, to read and write (p. 6)."

Back to McKay (1993), the second prevailing view of literacy is that of literacy as a social practice. According to her, for those who view literacy as a social practice, what is meant by literacy depends on the historical, economical, political, and sociocultural context in which the learner operates. Thus, from this perspective, what literacy is cannot be distinct from how literacy is applied by individuals within their community and how it is valued.

But what is meant by these historical, economical, political, and sociocultural contexts?

A historical perspective: According to McKay (1993), those who approach literacy from a historical perspective point out that both the value a society puts on literacy and also the definition of literacy differ and change over time. This fact can account for one of the reasons why it is not easy to find one single definition for literacy.

An economic perspective: According to Williams (2004), there has long been a belief that investment in education would have an effect on developing countries, similar to that claimed for developed countries. Yet, "the failure of the Experimental World Literacy program, organized by UNESCO from 1967 to 1972, provided evidence that literacy alone cannot be a causal factor in development" (p. 595). He goes on to state that "reflecting upon the failure, UNESCO observed that if development is to occur, then literacy, economic and social reforms must be integrated" (p. 596). Thus, literacy may be a necessary condition for economic development, but is not the sufficient condition. Nevertheless, as McKay (1993) maintains when literacy is highly valued by a society, there may be economic advantages to becoming literate for the individual and for society. Individuals who try to gain second language literacy, as well as those who are involved in teaching literacy, can also view literacy as an economic asset.

Political Perspective: This view of literacy refers to "how literacy both positions people in society and can provide a means for transforming the social order" (McKay, 1993, p. 22). According to McKay, literacy positions people in the sense that common social labels can marginalize individuals. Also, those who support a political perspective of literacy such as Freire (1970, cited in McKay, 1993) maintain that literacy can empower people to change in the sense that for him "learners are sociohistorical, creative and transformative beings, and literacy is the process through which these learners can come to critically reflect on reality and take actions to change oppressive conditions" (p. 18). The ultimate goal of literacy is, therefore, empowerment and social transformation, and this approach to literacy depends upon critically examining the existing social order. This Freireian view has resulted in what is termed a critical approach to literacy. According to Norton (2007) educators who are interested in critical literacy are interested in investigating written text, or any other kind of representation of meaning as a site of struggle, negotiation, and change. Mentioning Luke (1997), Norton states that:

While earlier psychological perspectives perceived literacy as the acquisition of particular behaviors, cognitive strategies, and linguistic processing skills, more recent insights from ethnography, cultural studies, and feminist theory have led to increasing recognition that literacy is not only a skill to be learned, but a practice that is socially constructed and locally negotiated. (p. 6)

According to Clark and Ivanic (1997, p. 217, cited in Pennycook, 2004), critical literacy is linked to language awareness in that its aim is to empower learners by providing them with a critical analytical framework "to help them reflect on their own language experiences and practices and on the language practices of others in the institutions of which they are a part and in the wider society within which they live" (p. 786). One way to develop this awareness is through questioning or what Freire refers to in the "problem-posing" education (Shor, 1992).

Sociocultural perspective: Those who approach literacy from a sociocultural perspective focus on "literacy as a social and cultural phenomenon, something that exists between people and something that connects individuals to a range of experiences and to different points in time" (Schieffelin & Cochran-Smith, 1984, cited in McKay, 1993, p. 1991). One of the main proponents of such a view is Street (1991, cited in Lotherington, 2007) who was among the first to challenge the notion of a singular literacy and proposes what he calls an ideological model of literacy which recognizes a multiplicity of literacies in which the meaning and uses of literacy are related to specific cultural contexts. He contrasted this model with the autonomous model, which represents much of the historical study of literacy, approaches literacy as cognitive advancement: a skill or set of skills developed by the individual, detached from social context. This view of literacy accords with the first prevailing view of literacy proposed by McKay that was already discussed. Street (1991) claims that the ideological model, which is oriented to the future of literacy, is a critical approach to literacy rooted in social agency that sees the individual as embedded in a social and cultural context within which literacy practices have meaning (Lotherington, 2007). Lam (2000) also refers to this fact by stating that "as a socially situated practice, literacy appears in multiple forms that have political and ideological significance, hence it is more appropriate to refer to literacies in multiple manifestations that bear no universal consequences" (p. 458). In short, what literacy means for a particular individual or community depends both on the demands made of them by the cultural context and on the social needs within that context (Center for Literacy, 2008). Based on what has been argued so far it can be assumed that the concept of literacy and the different forms it can take, by any definition, are quite context-sensitive and

culture-specific, and each given circumstance calls for its own framework both in terms of definition and practice.

English language literacy learner

There is no commonly accepted answer to the question of who is an English language literacy learner (Center for Literacy, 2008). "ESL literacy" is a concept that is even more difficult to define than "literacy" (p. 3). As it was already mentioned, what is meant by literacy depends on the historical, economical, political, and sociocultural context in which the learner operates. Likewise, the definition of EL literacy learner depends on the context to which the ESL learner belongs. In the literature, individuals are commonly categorized as ESL literacy learners based on level of education and native language alphabet, nonetheless, this term is defined differently by various organizations or researchers (Center for Literacy, 2008). For example, Florenz and Terrill (2003) define a literacy-level learner as a person with six or fewer years of education in their native country who needs focused instruction on learning to read and write English. They also place people into six categories, depending on their native tongue:

- Non-literate (The native language has a written form but the learner cannot read the native language, for instance, a single mother from rural Vietnam who grew up without access to education)
- Semi-literate (the learner's reading abilities are minimal, for example, a person who attended a rural school in El Salvador for 3 years, but could not continue for the family's economic purposes)
- Pre-literate (the learner's native language has no writing system, for instance a Bantu from Somalia, knows her native Af-Maay only orally, because a written form of the language is just now being developed.)
- Non-alphabet literate (the learner can read a non-alphabetic language, for example, a retired minor bureaucrat from China who is highly literate in the Mandarin script, but he is unfamiliar with any alphabet, including Roman)
- Non-Roman alphabet language literate (the learner can read a language that has a non-Roman alphabet writing system, for instance, an Arab literate in Arabic)
- Roman alphabet language literate (for instance, a senior from Russia who also knows French)

In another attempt to define the EL literacy learner, The Centre for Canadian Language Benchmarks (2000) include in their definition people with up to eight years of schooling who have not acquired "study skills" and may have "preconceived notions of reading and writing that may hinder progress in the class" and anyone who comes from a country with a non-Roman alphabet whether or not they are literate in their first language (p. ii). By contrast, the U.S. organizations such as The National Institute for Literacy (2000) exclude those who are literate in a non-Roman alphabet, based on the premise that such learners know that written language represents speech, and as it will be discussed in the following section, this is a necessary piece of prior knowledge for English language acquisition, and these learners are able to transfer their skills and adapt more quickly.

Thus, as it was already pointed out, like literacy, the definition of ESL literacy learner is quite context-sensitive and based on the organization that defines it, its designation may change.

The interplay between L1 literacy and L2 literacy

As one of the effects of L1 literacy on L2 literacy McKay (1993, p.6) refers to Bell and Burnaby (1984) who point out that adults who are already literate in their native language, particularly if it is an alphabetic language, have the following advantages:

They are trained in the visual discrimination of significant features in letter discrimination. They understand the concept of a particular sound being represented by a particular symbol. They have expectations of a certain content being presented in certain formats. Most important of all, they expect print to yield meaning. (p. 14)

But just like there is a great body of research suggesting that there is a positive transfer of literacy skills learned in the first language to the second language, there is evidence that this sort of transfer is not always beneficial. This aspect is well discussed by Bell (1995) in her paper "The relationship between L1 and L2 literacy: Some complicating factors" and her autobiographical study of Chinese literacy acquisition. While learning to be literate in Chinese, Bell found that Chinese literacy meant something different from English literacy. Her Chinese literacy teacher emphasized on the importance of calligraphy, the need to proceed very slowly (perfecting one character before moving to the next). Bell also observed a difference in learning styles: learning Chinese, she was expected to observe and digest, rather than analyze and comment. Thus, as in the case of Bell, when references are made to the issue of culture the negative effect of L1 literacy on L2 is more highlighted. As she puts it her "L1 literacy skills often seemed to be a stumbling block rather than an advantage" (p. 692). This issue is also reflected in the studies that investigated the differences in organizing discourse under the concept of Contrastive Rhetoric (Kaplan, 1966). Moreover, to Bell, this experience suggested that literacy is not 'neutral', but affected by class, gender, culture, ideology and ethnicity. Auerbach (1999, cited in McKay, 1993, p. 8) also draws attention to this fact that no view of literacy is neutral. He notes:

There can be no disinterested, objective, and value-free definition of literacy: The way literacy is viewed and taught is always and inevitably ideological. All theories of literacy and all literacy pedagogies are framed in systems of values and beliefs which imply particularly reviews of the social order and use literacy to position people socially. (p. 71)

Then it can be assumed that it is not easy to conclude firmly whether first language literacy helps or hinders the ESL Literacy learner. Perhaps one possible answer can be that it depends on what the first language is. Barton and Pitt (2003) note both the advantage and the difficulties. They refer to a study (Auerbach 1996) which was done in the United States, where learners were first taught literacy in their native tongue (either Haitian Creole or Spanish), and after six months to a year these learners were judged by the teachers to be ready to move on to transitional bilingual ESL classes. Although no details of the pedagogic practice are given, the report recommended that students gained initial literacy in their native language.

Barton and Pitt (2003) refer to helpful outcomes when teaching first language literacy, but they also quote Bell's paper where, based on the author's experience of learning Chinese, she challenges the belief that learners who are literate in their native language generally make better progress than those without native language literacy. As Hornberger (1989, cited in Bell, 1995) points out in her discussion of the research literature on biliteracy "the relationship between first and second literacies is highly complex, so that not all aspects of the L1 will necessarily aid the development of L2" (p. 688). Thus, as Bell argues literacy in

one language or culture cannot necessarily be assumed to be helpful in developing literacy in another. In fact, the cultural differences add a layer of complexity to the learning experience.

Instructional Strategies and Methodology for ESL literacy development

As it is pointed out by the Center for Literacy (2008, p. 5) "the discussions of effective pedagogical practices that do occur emphasize the complex and multifaceted nature of ESL learning processes. They suggest that a multifaceted teaching approach is therefore required to accommodate different learner backgrounds, interests, learning styles and literacy levels". In the literature, references are usually made to What Works Study (Condelli & Wrigley, 2003, p. 27) which concluded that three instructional strategies were instrumental in improving literacy and language development:

- Connection to the outside world, Using materials from everyday life

One of the key findings of the study carried out by Condelli and Wrigley (2003, p. 27) was that connecting literacy teaching to everyday life made a significant difference in reading basic skills development. To implement this strategy, teachers applied materials from daily life that held information that students wanted to know about or with which they had some experience. For instance, a teacher might bring in grocery flyers from different stores and ask students to compare prices or use phone and electricity bills, letters from schools or immigration authorities, and other items that appear in students' mailboxes to highlight literacy for adult contexts. According to these researchers:

Activities of this sort might foster literacy development by linking new information to what learners already know and by engaging the learner in topics of interest. By starting with familiar materials that are of interest to learners and by creating situations for cognitive involvement, teachers can create interest, maintain high levels of motivation, engage students' minds and through this process build literacy skills that have importance in the lives of adults. (p. 27)

In this regard, Florez and Terrill (2003) also have suggestions for literacy practices in the literacy class, particularly in dealing with adults. In this regard, they refer to Malcolm Knowles' (1973) principles of adult learning that are applicable to planning instruction for adult English language learners' literacy maintaining that adults are self-directed, practical, and problem solving. They have reservoirs of experience to help them learn new things, and they want to know why something needs to be learned and how it will be applicable to their lives.

By the same token, Florez and Terrill (2003) suggest the integration of the four language classrooms. According to them, skills in literacy-level classes vary because of program type (general, family, workplace, or corrections); intensity; and learner needs and goals. Accordingly, in real life and regardless of the context, language tasks involve integrating the four skills of reading, writing, listening, and speaking. For example, a trip to the health clinic includes reading and filling out health forms, explaining symptoms, and understanding the doctor's response. Likewise, The Centre for Canadian Language Benchmarks (2000) refers to the fact that many ESL Literacy learners prefer "experiential" learning and so the connection to real life in both content and skills developed in the classroom is important. Thus, they suggest the teachers to think about the real-life applications of any skills they plan to cover to the extent it is feasible (p. xii).

Florez and Terrill (2003) also highlight the need for incorporating the learners' needs and wants. In this regard they refer to the point that since learners have many purposes for

developing English literacy, needs assessment assures learners a voice in their instruction and keeps content relevant to their lives and goals. It also gives the teacher an opportunity to learn what skills learners bring to class and which ones they feel they need to strengthen (Brod, 1999; Shank & Terrill, 1997, cited in Florez and Terrill, 2003).

- Use of native language for clarification

Condeli and Wrigley (2003) in their study conclude that if instructions are given to students in their native tongue, they are able to focus on the task, and not be stressed over whether or not they have understood the directions correctly and thus exhibit faster growth in both reading comprehension and oral communication skills. The idea of using first language is also confirmed in Cook's (2007) discussions of the idea of multicompetence. Cook argues that since multicompetence means that the first language is always present in the users' minds, it would be artificial and sometimes inefficient to avoid its use.

- Varied practice and interaction

This third strategy refers to using multiple modes of teaching. As Condeli and Wrigley (2003) discuss:

Language and literacy development encompass complex linguistic and cognitive processes that are not yet fully understood by current science. Since learning a second language also has psychological dimensions, such as motivation to learn, and sociocultural dimensions, it is not surprising that the relationship between teaching processes in the classroom and learning outcomes is not often linear. No single teaching strategy invariably leads to success in language learning. Multiple modes of learning and teaching tend to be more successful in advancing language skills, particularly oral communication skills, as demonstrated through our analyses. (p. 30)

- Not making assumptions about what a learner knows or does not know

One other effective literacy classroom practice comes from the Centre for Canadian Language Benchmarks' (2000) suggestions for ESL literacy methodology and it is: based on what this center calls for ESL Literacy instruction is based on building a body of knowledge and skills that includes but is not limited to reading and writing. Learners may be unfamiliar with concepts the instructor considers common knowledge, or may not have a sub-skill required to complete a larger task. For example, a learner cannot acquire the language skills related to telling time if he has not yet learned to read a clock. It is important that we check the learner's understanding of what we teach and not make quick assumptions about why someone is not learning.

Conclusion

In general, one significant point to keep in mind while working with ESL/EFL literacy learners can be summarized in Brod's (1999, p. 5) quotation that "literacy-level learners may be beginning learners, but they are not beginning thinkers" (cited in Florez and Terrill, 2003). Like all learners, they bring varied strengths and needs to the adult ESL classroom. Teachers need to provide instruction that addresses these strengths and needs, engages learners in demanding and relevant topics, and provides them with tools they can use to meet their responsibilities and goals (Florez and Terrill, 2003).

As van Lier (2004) puts it:

Learners approach a second language with a given history and ongoing construction of their social self and identity. In the new language, various aspects of the self must be renegotiated and reconstructed, and this often entails clashes and struggles. This is true in both second and foreign language learning, though the processes and problems can differ in different contexts. Teachers can encourage students to develop their own 'voice' in the new language...by embedding language in meaningful activity. Focusing on the here and now brings the benefit that indicational uses of language are established first, allowing the learners to develop ecological and interpersonal perceptions in the language, and on the basis of these they can construct translingual and transcultural selves. (p. 130)

Taking into account the diverse and changing definitions and viewpoints about literacy Lotherington (2007, p. 901) believes that "the evolving reconceptualization of literacy as multiliteracies reconstructs the educational goals and outcomes of ELT to be more linguistically and culturally sensitive and inclusive, socially and linguistically repositioning the English language". He goes on to assert that:

ELT teachers cannot morally ignore the ecological debate about the relative status of English and its encroachment on other language domains. The teacher must ensure that her learners are learning English, at the same time complementing and maintaining the many literacies learners will need to engage in their complex, multicultural, and multilingual lives. (p. 901)

Such changes in perspectives demand responsive ways of learning and teaching one of which is that of nurturing multilingualism, multiculturalism, and in turn, one way to achieve this multimodalism can be through the use of new literacies such as digital computer literacies (Lotherington, 2007).

Considering what has been covered in this paper about the notions and issues about the concept of literacy and reflecting upon the current situation in our own country, Iran, certain questions can be raised:

- To what extent have the historical, economical, political and sociocultural issues been taken into account when defining and setting the objectives and goals of foreign language literacy in our educational programs?
- To what degree has the educational system been successful in touching the FL literacy issue critically?
- How far has the educational system been successful in identifying the real needs and requirements of the language learners in Iran and how well has it tried to incorporate their background knowledge and previous experience into the learning process?
- What role has needs analysis and needs assessment played in providing the learning materials?
- How many times have we, as language learners, felt we have been rendered "voiceless" (Giroux, 1999) or "silenced" (Fine, 1987, both cited in Cummins, 1999) in the language classrooms? On the other hand, how well, have we, as teachers, tried to listen to our learners' voices or take into account their identities?
- To what extent have we been successful to proceed with the changing definition of literacy to multiliteracies and multimodalism and incorporating the new literacies including the use of the Internet as "a potential solution for establishing a viable postmodern multilingualism" (Lotherington, 2007, p. 901) into our educational system?
- Strain (1971) investigated the English language instruction in Iran and came up with these shortcomings: teachers are inadequately prepared, classrooms are overcrowded,

materials are not properly used and basic writing skills are not taught. How far have we made real progress toward overcoming these problems over the last forty years?

To the author's best knowledge, little, if any, attempt has been made to answer these questions in the available literature but a probe into our own educational experience and what we observe in general English or even ESP classes at the university level reveals to me that these issues have rarely been addressed in our FL instructional programs. Even if there have been attempts, they have been so unproductive that it makes it hard to trace their effect in the learners' knowledge of English as the outcome. Therefore, given the ever-increasing need for FL literacy and the potentials and capabilities of EFL learners in Iran, reconsideration and reconstruction of EFL literacy issues seem to be essential for reaching our English language learning goals in a more effective, reasonable and even economical way.

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RETHINKING ENGLISH LANGUAGE EDUCATION FOR A SUSTAINABLE FUTURE: BRIDGING INNOVATION AND EDUCATION

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Abstract

Sustainability education is an approach that equips learners with the knowledge, skills, values, and attitudes needed to address interconnected global challenges. It goes beyond environmental awareness to include social justice, economic equality, and cultural perspectives. This paper explores the integration of sustainability into English language education, emphasizing (a) Innovative practices that enhance learning and sustainability, (b) Collaborative networks that support educators and institutions, and (c) The role of teachers in promoting sustainable development. Besides, the paper explores how integrating sustainable development principles into English language teaching (ELT) can enhance learning outcomes while addressing global educational needs. By leveraging digital tools, interdisciplinary approaches, and eco-friendly teaching practices, ELT can promote linguistic competence alongside environmental and social responsibility. The study highlights the role of technology, project-based learning, and culturally responsive pedagogy in preparing learners for a future where communication, adaptability, and sustainability are key. The paper highlights how Education for Sustainable Development (ESD) can be effectively applied in schools, especially in remote and rural areas, and discusses strategies to better equip teachers for sustainability education.

Keywords: English Language Education; Innovation; Sustainability; Sustainable Development

Introduction

The third millennium is essentially interwoven with significant progress in different aspects of human life, such as economic, social, cultural, and political. In an era of rapid globalization and environmental challenges, English language education must evolve to foster sustainability and innovation. Education, in general, and applied linguistics, in particular, follow the same trend. Recently, some applied linguists (Canagarajah, 1999; Crooks, 2009; Pennycook, 1994; Phillipson, 1992), inside the English language teaching (ELT) community, interrogate the lack of critical perspective in the teaching and the role of English internationally. More precisely, they have strived to intensify the notion of the sociopolitical and sociocultural issues in ELT. Their concern is that ELT is neither neutral nor always culturally appropriate. The general dissatisfaction with the efficiency of the mainstream English language teaching (ELT) in teaching English as a foreign language context (TEFL) has been discussed in expanding and outer circle countries (e.g., Bangladesh, Ghana, the Philippines, China, Egypt, Korea, Iran, Japan, Russia, to name but a few) and at different national and international conferences and symposiums. The main dissatisfaction concerns the centralized, top-down educational management and policy. Most of the teaching models and materials fail to meet the needs of learners. Notably, they do not link learners' content and real-life concerns (i.e., society). Thus, this dissatisfaction gives rise to the adoption of different internationally marketed materials in the classroom. The problem emerges because internationally marketed materials disseminate western cultural values by providing a utopian environment to the learners. Accordingly, the need is felt to accommodate ELT materials in the EFL classrooms in the TEFL contexts to serve the local

needs. With the development of critical approaches in education and their application to the field of ELT and the hidden curriculum (HC), the local perspectives towards the ELT industry are emerging worldwide. Considering the notion that ELT has been regarded as a political activity, it is crystal transparent that language policymakers should not contemplate ELT as a value-free or a mere educational concern. Notably, they should approach education, in general, and teaching methods as well as materials development, in particular, from the sociocultural and sociopolitical dimension. These perspectives realize the notion of a hidden aspect in education. This aspect is the hidden curriculum and value that is inherent in any educational material. Probably, one approach to meet such a demand for domesticating ELT and helping students think about their learning can be critical thinking (CT) and critical language pedagogy (CLP) with a hope to transform the knowledge in the real-world situation.

Current Trends in Language Teaching

The emergence of different methods in a second language (L2) acquisition aims to facilitate and promote the learning process. Traditionally, teachers are supposed to be unquestioned authorities, decision-makers, and knowledgeable individuals in a class who transmit information to the students without a questioning manner, and students are assumed to be passive recipients and consumers of given information (Freire, 1970). Current trends in language teaching seem to experience a critical turn toward the role of language, learning, teachers, and learners. In Kumaravadivelu's (2006) term, this critical shift recognizes language as an ideology than a system. It realizes language teaching and learning more than learning and teaching a language. In other words, it extends an educational setting to the social, political, and cultural dynamics of learning. While the main tenets of conventional education are to bring designated information to the mind of passive learners, a number of critical approaches as the progeny of post-method in language teaching, deeply rooted in Marxist approach and Frankfurt School, considered a paradigm shift in the assumed role for teachers and students (Giroux, 1988). The critical theories in language teaching gave prominence to learners' empowerment, critical consciousness, conscientization, and dialogism (Giroux, 1988; Kincheloe, 2005; Kumaravadivelu, 2006). From a critical perspective, the term authority implies a shared power between teachers and students. Teachers are likely to be what Giroux and McLaren (1996) described as transformative intellectuals who combine "scholarly reflection and practice in the service of educating students to be thoughtful, active citizens" (Giroux, 1988, p. 122). Students are active agents whose viewpoints are underscored through dialogue and discussion (McLaren, 2003). The upholders of critical theory (Freire, 1970; Kincheloe, 2005; Morgan, 1998; Shor, 2012) encourage learners to act in a questioning manner, to construct their own understanding, to be independent, and to develop their full potentialities in classroom milieu.

Informed by the tenets of critical theorists, knowledge is no more realized as passive information. In fact, it is gradually constructed in interaction through a dynamic nature (Mortimer & Scott, 2003). Cazden and Beck (2003) posit that dynamic nature helps students collaboratively construct meaning in critical discourses and dialogical interactions. The dialogic teaching approach rooted in Bakhtin's (1981) concept of dialogism is based on such teacher-student communication, in which higher forms of cognitive processes are dominant on the students' part. Students in this kind of teaching are actively engaged, endowed with high levels of autonomy, and empowered to influence the development of the classroom discussion to a certain degree. Discourse is a matter of the oral use of language in an instructional setting that can encourage interaction or what Bakhtin (1981) called a "responsive understanding" (p. 279). Bakhtin described instructional discourse patterns in two terms of dialogic and monologic discourse. From this perspective, a classroom discourse

is monologic when the main speaker, typically the teacher, performs a prior script. It is often controlled by one individual, albeit two or more persons participate. By contrast, a dialogic discourse encourages the participants to develop or change the contributions of the peers as one voice “refracts” another.

For Bakhtin (1981), dialogic teaching encourages learners’ voices, values, and perspectives. Bakhtin postulates that knowledge is not in an individual mind, but it is built by engaging participants in a critical interaction. From a Bakhtinian perspective, an interaction is dialogic when both teachers and students have the authority and the autonomy to voice their ideas. This perspective was supported by Freire (1970) ,who addressed that the dialogic discourse can be created by discussing learners’ real-life concerns to raise their critical awareness. Woods (2014) outlined L2 learners’ role in a dialogic classroom. Woods maintained that learners initiate a conversation and engage in a discussion to pose reflective questions. Accordingly, within the framework of dialogic teaching, a teacher expects to work as a director, to control learners’ discussion, and to authorize students for sharing their knowledge through interaction. Likewise, Shor (2012) believed that a teacher in a dialogic classroom makes use of learners’ knowledge to commence the discussion and introduce deeper levels of knowledge.

Drawing on Bakhtin’s (1984) conceptualization of classroom discourse, monologic and dialogic patterns are considered the opposing poles of teachers’ discourse continuum. Analyzing discourse content in language classrooms and examining the nature of teacher and student interactions can be promising on how language is used, and what kind of input and interaction affect language learning process (Chappell, 2014; Cullen, 2002, Nystrand, Lawrence, Gamoran, Zeiser, Long, 2010). Recently, numerous studies have been conducted into the quality of classroom interactions which delve into a number of issues comprising teacher talk (Thornbury, 1996), conversation analysis (Seedhouse, 2005), turn allocation patterns (Xie, 2010), reflective discourse analysis (Anderson, 2017), L2 interactional competence (Hall, 2018) to name but a few. A sizable body of researches (Alexander, 2008; Nystrand, 1997; Scott, Mortimer & Aguiar, 2006) suggests that the acquisition of useful knowledge is linked to the quality of classroom interaction because students are provided with various inputs, discourses, and interactions in the classrooms (Maftoon& Rezai, 2013). Investigating the nature of discourse content from a critical perspective can yield promising results; particularly, the application of Bakhtin’s dialogic discourse pattern (DDP) in EFL classrooms can uncover how EFL students actively construct new knowledge. To date, only a few studies (Ahmadi, 2017; Cazden, 2001: Hemati & Valadi, 2017; Sedova, 2017) have been carried out to appraise the discourse contents of EFL classrooms. What is particularly novel in this study is appraising 12 classroom interactions adopting Bakhtin’s instructional discourse pattern. To address this gap in research, both teachers’ and students’ naturally occurring interactions in EFL classrooms were observed and audiotaped with a hope to explore if the interactions follow or violate the principles of DDP. To undertake the study, a qualitative research method was adopted. More specifically, a Non-experimental observational approach was employed to probe the classroom interactions. In educational research, one of the most common uses of direct observation is the study of classroom observation to determine the extent to which a particular behavior(s) is present (Ary, Jacobs, Sorensen & Walker, 2013).

Given the pivotal role of teachers and students in classrooms and the importance of classroom spoken discourse features in the learning process, analyzing teachers’ and students’ interactions can provide insightful information on what type of interaction promotes learning outcome and how learners construct new knowledge (Seedhouse, 2005; Walsh, 2011). One pedagogical advantage of investigating classroom spoken discourse features would be

determining whether or not the current classroom discourse provides students with critical awareness to transform knowledge and to engage students in a cooperative dialogue.

Kumaravadivelu argues that a need to look beyond the notion of the method has emerged out of the inherent contradictions between method as conceptualized by theorists and method as actualized by practitioners. This claim is again far from being scientific. He argues in such a way as to pinpoint the existence of a taken-for-granted sort of hostility between theorizers and practitioners. It seems to be more reasonable to try to encourage both theorizers and practitioners to compromise in more favorable ways. Given the chance of reasonable discussion on points of major controversy, this compromise does not seem to be out of reach. Teacher autonomy is another pedestal upon which the post method era stands. The crucial problem with the traditional notion of method, according to the proponents of post-method condition, is an ethical one in the sense that method, as outlined by theorizers, keeps practitioners away from the practice of their potentials. "The post-method condition, however, recognizes the teacher's potentials: teachers know not only how to teach but also know how to act autonomously within the academic and administrative constraints imposed by institutions, curricula, and textbooks" (Kumaravadivelu, 1994, p. 30).

Principled pragmatism reveals itself in the form of a teacher's sense of plausibility (i.e., teacher's subjective understanding of the teaching he does). Teachers' plausibility connotes the involvement of both teachers and students in the learning activity. As such, it is quite reasonable to emphasize the importance of teacher plausibility in language teaching pedagogy. Teacher plausibility should not be interpreted in such a way as to empower the teacher to change language teaching or learning experience to unidirectional flow of information from the teacher to the learner. It should, on the contrary, entail the teacher's endeavor to assess learner needs, and his attempt to involve learners in learning activities. Principled pragmatism is based on Widdowson's (1990) notion of 'pragmatics of pedagogy', which construes the immediate activity of teaching as the medium through which the relationship between theory and practice can be realized.

Kumaravadivelu (1994) takes the characteristics outlined above as a point of departure to propose a strategic framework for L2 teaching. The framework, Kumaravadivelu claims, is offered not as a dogma for uncritical acceptance but as an option for "critical appraisal in light of new and demanding experience and experimentation in L2 learning and teaching" (p. 32). As Kumaravadivelu delineates it, the post method condition is a descriptive, open-ended set of options, and an interim plan to be continually modified, expanded and enriched by classroom teachers. The paradigm of the post-method condition was later enriched by Kumaravadivelu's (2003) attempt to characterize language teaching in a post-method era and to provide the fundamentals of the post-method pedagogy as a three-dimensional system consisting of the pedagogic parameters of particularity, practicality, and possibility. The parameter of particularity facilitates context-sensitive language teaching with a true understanding of local linguistic, sociocultural, and political particularities. Practicality ends the conventional role relationship between theorists and practitioners by empowering teachers to construct their own practice theory. A Possibility is the parameter that allows learners, teachers, and teacher educators to be socio-politically conscious and to search for identity formation and social transformation.

The Politics of Language

The field of language planning and policy (LPP) is concerned with the explicit and implicit policies that influence what languages are spoken when, how, and by whom, as well as the values and rights associated with those languages. Four major types of language planning are status planning (about the social standing of a language), corpus planning (the structure of a language), language-in-education planning (learning), and

prestige planning (image). Language planning may occur at the macro-level (the state) or the micro-level (the community). Different key terms are connected to language policy and the politics of language: Linguistic Imperialism, Cultural Imperialism, and World Englishes.

Linguistic Imperialism, Linguiscism, is the theory that languages may be seen as occupying a dominant or dominated role in society. It is argued that English plays a dominant role internationally and plays a role in maintaining the economic and political dominance of some societies over others. Because of the role of English as the dominant international language, many other languages have been prevented from going through processes of development and expansion. The spread of English is viewed as imposing aspects of Anglo-Saxon Judeo-Christian culture and causing a threat to the cultures and languages of non-English speaking countries.

Much has been written about the colonality of the English language. Now more than ever, one sense among a section of the applied linguistics community a critical awareness of the fact that the English language, in its long march to its current global status, was aided and abetted by colonialist and imperialist projects that trampled upon the political, cultural and linguistic heritage of millions of people across the globe. The nature of English linguistic imperialism (Phillipson, 1992), the colonial character that still adheres to it (Pennycook, 1998), and the subaltern attempts to resist its dominance (Canagarajah, 1999) have all been documented in detail. Equally, well-documented is the globality of the English language. “A language achieves a genuinely global status,” observes Crystal (1997), “when it develops a special role that is recognized in every country.” (p. 2). Clearly, English has achieved such a role. In that global role, English has spawned a large number of local varieties, aptly called world Englishes. An interesting aspect of the emergence of world Englishes is that it is not confined to former British colonies alone. In addition to English in India, Singlish in Singapore, etc., one now hears about Franglais in France, Denglish in Germany, and so on. The emergence of world Englishes with their amazing form, function, and spread has been the result of what Kachru has called nativization (Kachru, 1982, 1983). Varieties such as Indian English, Nigerian English, and Singaporean English represent the extent to which a foreign language can be profitably reconstructed into a vehicle for expressing sociocultural norms and networks that are typically local. Creative writers such as Agnihotri and Khanna, Tollefson, Chinua Achebe, and others have shown how the Western language can be used for communicating sociocultural nuances that are completely alien to the Western culture. Common people who speak English as an additional language see it more as a language of communicational necessity than as a symbol of cultural identity. They use English according to their individual and institutional needs and keep it separate from their local cultural beliefs and practices.

Education for Sustainable Development

Education for Sustainable Development (ESD) is a holistic and transformative learning approach that equips learners with the knowledge, skills, values, and attitudes necessary to shape a more sustainable future. ESD emphasizes participatory and action-oriented learning, empowering students to become proactive agents of change. By integrating sustainability into education, we prepare individuals to understand sustainability and actively contribute to a more just, resilient, and livable world. Sustainability education has emerged as a critical pathway to creating a more equitable and environmentally sound future in a world facing unprecedented environmental challenges. By empowering young minds with the knowledge, skills, and values needed to address complex global issues, we can foster a generation of change-makers committed to sustainability. Sustainability education is an approach that equips learners with the knowledge, skills, values, and attitudes needed to address

interconnected global challenges. It goes beyond environmental awareness to include social justice, economic equality, and cultural perspectives. Sustainability simply refers to a concept that is not only applicable in environmental issues but also in fields such as education, including English language education. Under the context of English language education, the term so called sustainability operationally refers to an approach that ensures the long-term success of language learning processes. It is all about creating and maintaining conditions that foster continuous improvement, adaptability, and inclusiveness, and ensuring that future generations of learners have access to quality education. The goal of promoting sustainable English language education involves addressing the needs of both students and educators while considering the broader societal and global contexts in which learning takes place. True sustainability education balances the three pillars: environmental stewardship (protecting ecosystems), social responsibility (ensuring equity and human rights), and economic viability (meeting present needs without compromising future generations). UNESCO's Education for Sustainable Development (ESD) framework emphasizes transformative learning experiences that enable learners to take informed decisions and responsible actions for environmental integrity, economic viability, and a just society for present and future generations. Education for Sustainable Development (ESD) is a comprehensive and transformative educational methodology that prepares students with the essential knowledge, abilities, principles, and perspectives to contribute to a more sustainable world. ESD transcends the basic introduction of sustainability concepts—it motivates learners to:

1. Cultivate critical thinking skills for understanding and addressing multifaceted global challenges.
2. Participate actively in civic life by committing to the well-being of both the environment and society.
3. Promote cross-disciplinary learning by integrating sustainability into diverse subject areas.
4. Strengthen problem-solving abilities to devise inventive solutions for sustainability-related concerns.

ESD empowers students to become environmentally conscious and socially responsible citizens who can drive change and create a more sustainable future. Key Factors Influencing Sustainability (a) Personal Factors (knowledge, skills, openness to change); (b) Technological Factors (ease of use, compatibility); (c) Organizational Factors (administrative support, infrastructure); (d) Social Practices (collaboration, peer learning); (e) Perceived Value (tangible benefits in teaching). Educational networks are instrumental in promoting Education for Sustainable Development (ESD) by encouraging cooperation, pooling resources, and spurring creativity. A notable case in point is ENSI (Environment and School Initiatives), which has actively contributed to ESD (figure 1) for more than three decades by:

Figure 1. Environment and School Initiatives and ESD

Bridging Gaps	Connecting schools, communities, and policymakers to integrate sustainability into education.
Platform for Best Practices	Providing spaces to share innovative teaching strategies.
Resource Development	Creating materials to support educators in sustainability education.
Knowledge Exchange	Facilitating communication among educators to help schools adapt teaching methods.

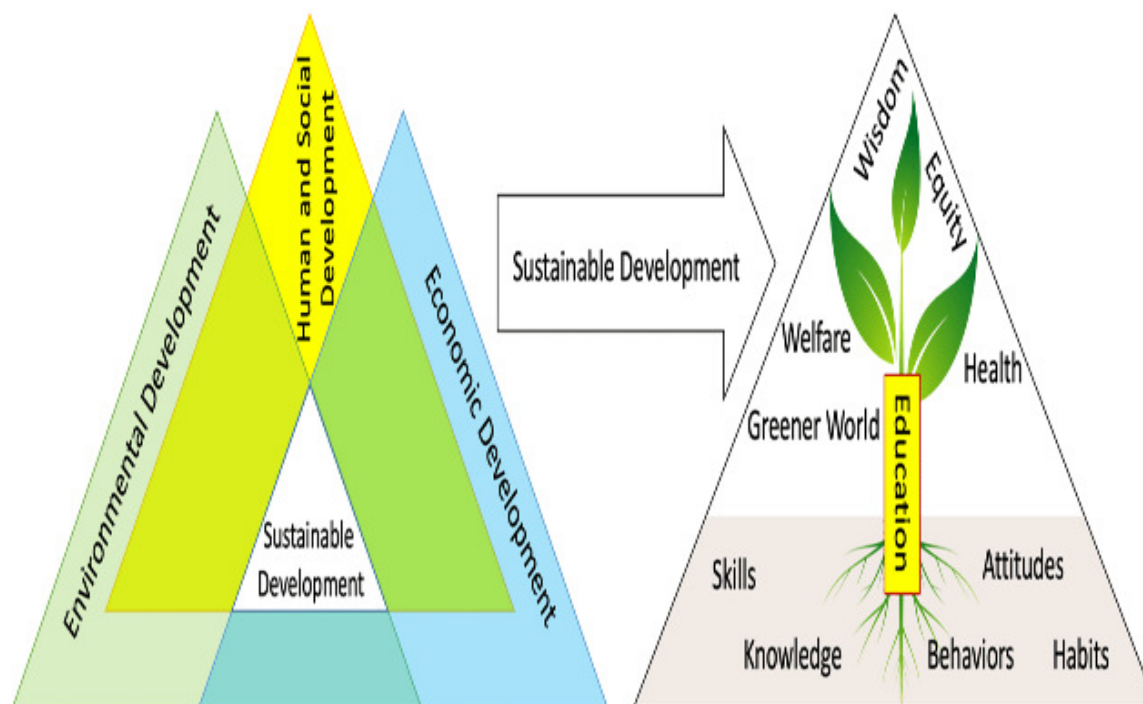
Utilizing these networks allows educational institutions to develop lively and adaptable learning spaces that empower students to tackle sustainability challenges more effectively. Navigating Conceptual Ambiguity and Evolving Sustainability Challenges in ESD Although there is no shortage of initiatives related to Education for Sustainable Development (ESD), a significant challenge arises from the myriad definitions and interpretations of sustainability. This ambiguity can impede educators' ability to integrate ESD into their teaching effectively. Additionally, the constant evolution of sustainability issues necessitates ongoing adaptation and innovation in education. Educators often encounter obstacles like resource limitations, inadequate training, and opposition to change. Overcoming these hurdles demands concerted action from all parties involved—from policymakers and educators to the broader community.



Challenges of implementing education for sustainable development in K-12 education.

Several factors significantly influence the successful incorporation of new educational approaches, including:

1. Individual Attributes: Educators' knowledge, skillset, and adaptability to change.
2. Technological Aspects: The user-friendliness and compatibility of new tools and systems.
3. Institutional Context: Availability of administrative backing and necessary infrastructure within organizations.
4. Social Dynamics: Promotion of cooperation and peer learning among educators.
5. Recognized Worth: Clear, practical advantages for day-to-day teaching practices.



Sustainable English language education is not just about what happens in individual classrooms; it also requires considering the broader societal and global contexts in which learning takes place. English is a global language, used in a wide range of contexts, from business to diplomacy to popular culture. As such, English language education must prepare students to engage with the world in a meaningful way. This involves promoting intercultural competence, which refers to the ability to communicate effectively and appropriately with people from different cultural backgrounds. Intercultural competence is essential in a globalized world, where English is often used as a lingua franca between speakers from different countries.

Key Competencies in Sustainability Education

Self-awareness

The ability to reflect on one's own role in the local community and global society; to evaluate one's actions and motivations; and to deal with one's feelings and desires. Pennycook (2004) identified three critical moments in the teacher-learner's classroom, arising from a) the actions of a disruptive male student, b) the use of practice dialogs, and c) the recognition of non-standard English in the classroom. Each of these critical moments raises larger questions of power and authority in society and provides an opportunity for critical discussion and reflection. He suggests that trying to be a critical educator is more than seeking and seizing small moments to open the door on a more critical perspective. So his account locates praxis in the emerging critical awareness of teacher-learners and its potential for social transformation.

Anticipatory Thinking

The ability to understand and evaluate multiple futures – possible, probable and desirable; to create one's own visions for the future; and to assess the consequences of actions. Critical thinking involves the formation of logical inferences (Simon &

Kaplan, 1989). Some scholars and educators erroneously assume critical thinking to be higher order thinking or cognitive processing (Paul, 2012). According to Paul and Elder, "Critical thinking is best understood as the ability of thinkers to take charge of their own thinking. This requires that they develop sound criteria and standards for analyzing and assessing their own thinking and routinely use those criteria and standards to improve its quality."

According to Chafee (1988 as cited in Fahim & Barjesteh) critical thinking is "our active, purposeful, and organized efforts to make sense of our world by carefully examining our thinking, and the thinking of others, in order to clarify and improve our understanding" (p.29). According to Halpern (1989) critical thinking is "thinking that is purposeful, reasoned and goal directed. It is the kind of thinking involved, in solving problems, formulating inferences, calculating likelihoods, and making decisions" (p. 5).

Critical thinking can be set apart from problem solving (Hedges, 1991) in that problem solving is a linear process of evaluation, while critical thinking is a comprehensive set of abilities allowing the inquirer to properly facilitate each stage of the linear problem-solving process. Thus, based on the several definitions above, the researcher believes that critical thinking is a complex process, and it is generally higher order thinking or cognitive processing. A critical thinker is able to solve problems, make decisions, evaluating information and formulating inferences. This means that critical thinking involve the ability to use our minds to achieve our goals.

Critical Thinking

The ability to question norms, practices and opinions; to reflect on own values, perceptions and actions; and to take a position in the sustainability discourse. Critical Consciousness is one of the essential components of critical language pedagogy and the key to people's emancipation. Barjesteh (2020) conceptualizes CLP as a philosophy of education and a method of teaching that pursues to raise teachers' and learners' power that enable them to critique and challenge the problems by helping them to build critical consciousness. Coming to critical consciousness, Barjesteh maintains that it is a process whereby students foster a capacity to think critically. It is defined as a process by which students achieve an awareness of the social realities which discover their abilities to recreate them. Freire introduces the notion of 'conscientization' and defines it primarily as "the effort to enlighten men about the obstacles preventing them from a clear perception of reality. In this role, conscientization effects the ejection of cultural myths that confuse people's awareness and make them ambiguous beings." (1985, p. 89). According to him, conscientization envisages people's gaining critical consciousness.

The Urgency of Sustainability Education

Global Challenges: We face unprecedented environmental challenges including accelerating climate change, rapid biodiversity loss, and widespread ecosystem degradation. These environmental issues intersect with growing social inequality and economic instability, creating complex sustainability problems that require informed action.

SDG: The UN Sustainable Development Goal 4.7 explicitly calls for ensuring all learners acquire knowledge and skills needed to promote sustainable development through education for sustainability, human rights, gender equality, and global citizenship.

Future Readiness: Today's students will navigate a world of resource constraints, climate impacts, and social transformation. Sustainability education equips them with adaptive thinking and problem-solving skills necessary to thrive in uncertain future scenarios and contribute to positive change.

Critical Approaches to Pedagogy

The Critical aspect in pedagogy is a relatively new paradigm in thinking about education in general and applied linguistics, materials development which takes a new dimension as praxis oriented movement. As Giroux (2003 as cited in Barjesteh&Birjandi, 2015) asserts critical perspective is an educational movement guided by passion and principle that aims to help students develop consciousness of freedom, to disclose the hidden cultural values of an educational setting, to make both teacher and students transformative intellectuals, to recognize authoritarian tendencies, and to connect knowledge to power and the ability in order to take constructive actions. Over the last two decades, transformative-based education, critical education, and humanized ELT have been faced with more demand than transmission based pedagogy. Transformative-based education deeply rooted in the constructivist approach.

With the advent of the constructivist approach to education, instruction takes a critical and socio-political dimension. More precisely, the constructivist schooling, challenges teachers and students to rethink the purpose of schooling, and the role that they might play as "cultural workers"- a term borrowed from Freire (1998, p. 30). These cultural workers can think critically and act transformatively (Freire, 2005; Moreno-Lopez, 2005). Kumaravadivelu (2003) states that teachers in constructivist orientation can theorize about their practices and perform their personal theories. To him, personal theories are those that teachers can employ in the classrooms based on professional assumptions, i.e., those that are generated by experts and are transmitted from centers of higher learning. From this standpoint, teachers and students take a new identity in a classroom. Teachers as unquestioned authorities turn to be facilitators and act by posing problems wherein both teachers and students engage with different aspects of the class procedures, such as the selection of the materials, assessment, evaluation, and their role in the classroom (Freire, 1970; Giroux, 1988; McLaren, 1989). Teachers go further as "reflective practitioners" and embrace the position of transformative intellectual (Giroux, 1988, p. 125) who can engage in a critical thinking activity where both teachers and students teach each other, and no one possesses authority over another. This reflects the constructivist notion in which learners are not viewed as empty vessels to be filled with their teachers' knowledge (Freeman & Johnson, 1998).

Based on the different ways in which critical approaches looked at education, critical pedagogy is a philosophy of education and a method of teaching that pursues to raise teachers' and learners' power that enable them to critique and challenge the problems by helping them to build critical consciousness. Accordingly, CLP can be operationally defined as a dynamic framework of education. that encompasses and encourages the ways of learning and teaching that are of adequate relevance to the cultural and sociopolitical realities (Giroux, 1997). This orientation rejects the banking education, a term coined by Freire (1970), as the process of transferring information, and it embraces a view of education as consisting of the acts of cognition that occur through dialogue. Students and teachers become critical co-investigators in which no one teaches one another, nor is anyone self-taught. CLP considers learning as a social event where teaching is mediated by the world and cognizable objects, which in banking education are manipulated by the teacher (Keesing-Styles, 2003).

Conclusion

The future of Education for Sustainable Development (ESD) depends on the seamless integration of sustainability into all levels of education. This includes fostering global collaboration, empowering communities, and adapting to emerging sustainability challenges. Innovation and continuous research are pivotal for advancing ESD, as they allow us to stay ahead of the rapidly changing global landscape. By uniting all stakeholders—policymakers, educators, and communities—we can ensure a sustainable future for all. This collective commitment will lead to transformative change, where sustainability becomes a central pillar in educational frameworks worldwide. Innovation and sustainability in education require a holistic approach that integrates the principles of ESD, promotes collaboration, and empowers learners to take action. By embracing place-based learning and leveraging ICT, we can create a more sustainable future. Educators play a crucial role in this transformation, as they shape the next generation of sustainability leaders. Providing students with the skills and knowledge to tackle the challenges of tomorrow will lead to a more sustainable and equitable world for all. Rethinking English language education through the lens of sustainability and innovation is essential for preparing learners to navigate an increasingly interconnected and environmentally conscious world. By integrating sustainable development principles, leveraging technology, and adopting interdisciplinary and culturally responsive pedagogies, ELT can go beyond linguistic proficiency to foster critical thinking, global awareness, and social responsibility. Emphasizing eco-friendly teaching practices, digital tools, and project-based learning can create dynamic and future-ready educational environments. Ultimately, bridging innovation and education in ELT not only enhances language learning outcomes but also contributes to a more sustainable and equitable future, empowering learners to be active participants in global change.

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INNOVATIVE LEARNING: MAPPING THE FUTURE POSSIBILITIES FOR A TRANSFORMATIVE EDUCATION

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Abstract

In the 1970s, the talk was about the leisure society and how the decline of work would leave people aimless and dissatisfied. Education would need to move from a concentration on the development of workplace skills to those of leisure and voluntary pastimes. A discourse of postponed retirement has replaced that discourse. It is always dangerous to make predictions. However, present social changes have implications for future educational policy and practices, and the future depends on decisions made now. The future is not determined. Emerging trends suggest a shift towards hybrid and digital-first learning environments, integrating artificial intelligence (AI), virtual reality (VR), and adaptive learning platforms to enhance student engagement and comprehension. An exploration of some of the issues raised by change can reveal trends which, once exposed, can be interrogated and challenged. As education continues to evolve, the focus will likely remain on fostering critical thinking, creativity, and adaptability—skills necessary for success in an increasingly interconnected and automated world. This paper highlights the possibility of changes for the future and the tentative action educators need to manage to face the future.

Keywords: Futurology; education; critical thinking; Individualization; dichotomization of Education

Introduction

Throughout history, events have swung one way and then there has been a rebalancing, as in the swings between classical and romantic periods in the eighteenth and nineteenth centuries, or those towards collectivism and away from individualism in the latter part of the nineteenth century. But this is too simple – one movement always undercuts and interpenetrates another. For example, since the 1980s there has been a definite swing in education towards instrumentalism and the knowledge economy, but this is starting to be countered by a re-emergence of liberal values in education (Jarvis and Preece 2001: 222–3).

However, profound transformations have occurred as a result of the interactions between three emerging social phenomena; those of ‘individualization’, ‘networks and ‘globalization’. Individuals may now have their most meaningful experiences in cyberspace, within which they can construct new forms of individual identity by means of ‘avatars’ (on-screen representations of themselves), engage in cyber relationships with others, and alter the ‘nature’ of human relationships (Ahlstrom 2007). According to the social geographer Miguel Castells, body bio-rhythms, physical activity, seasonal change and global location become irrelevant because they are not necessary to the network. A culture is created of ‘real virtuality’ (Castells 1996/1999).

Individualization, networks and globalization are interconnected in a dynamic relationship, each forming and being formed by the other. We explore the nature of these phenomena and, although certainty is impossible, their future has educational implications.

Dichotomization of Education

In the traditional model of teaching, the teacher is conceptualized as the knower in the class, and students were considered as having nothing to offer. That is why teaching in the traditional perspective is that of transferring the knowledge from the mind of one who knows

everything to the minds of learners as the receiver of knowledge. Teachers in this model are the pillars of knowledge who disseminate knowledge in students' minds. Freire (1970) refers to this view of education as the 'banking' model of education wherein students are viewed as "empty vessels waiting to be filled by the teacher" (p.72). In this model, as the name implies, education "becomes an act of depositing, in which the students are the depositories, and the teacher is the depositor" (p. 72). This model considers knowledge as a package that can easily be transferrable.

Thus, in the banking model of education, the teacher is the only source of information and decision-making in the class. Kumaravadivelu (2003) comments on the teachers' role in the banking model by arguing that their "primary role in the classroom is to function as a conduit, channeling the flow of information from one end of the educational spectrum (i.e., the expert) to the other (i.e., the learner) without significantly altering the content of information" (p. 8). Correspondingly, students are passive in the traditional model of education, in that students are only expected to receive the information and memorize it. This information doesn't have anything to do with learner real-life situations and thus, don't foster their awareness. Instead, they reflect the world as a fixed reality that cannot be critiqued or changed (Freire, 1970). He maintains that students, this way, become passive obedient receivers of the knowledge that they are not encouraged to think about or allowed to question. They are hindered from becoming either conscious or critical. The more students get the passive role of deposit receivers, the more they will adapt themselves to the world and reality as given.

Shor (1999) criticizes this notion for its monolithic discourse that establishes educational programs in a top-down manner. The prevalence of this top-down discourse stems from the common belief that there is only one right and legitimate method of doing things, and this method is transmitted by certain experts who can theorize to practitioners and consumers who are believed to lack the ability to theorize and, therefore, must practice experts' theories (Clarke, 1994). To Freire, this transmission-oriented pedagogy leads to the abrogation of critical consciousness on the part of men and women who will not be able to rethink their original reading of the world (Freire, 2005). Freire maintains that this model hinders creative power, which causes the submersion of consciousness.

Freire (1970) contends that "the dominant elites utilize the banking concept to encourage passivity in the oppressed" (p. 95). If students get used to accepting passively whatever decision is made for them in the class, they can never dare to question the discriminations and injustices in the real world in the future. Freire offers problem-posing education as an alternative to the banking model of education. He highlights learners' real-life problems as the main issue in a curriculum instead of prepared, static, and dissociated knowledge. He believes that this model aims to raise learners' critical awareness, critical consciousness and make them aware of the injustices in their society. He advocates a liberatory education by posing problems (Barjesteh&Niknezhad, 2020). McLaren (1995) believed that the overall mission of CP is to encourage students to critically analyze the relation between their quotidian experiences, the knowledge they produce, and the social, political, cultural, and economic problems of their society. Duncan-Andrade and Morrell (2007) summarize the main philosophy of this model as:

"an approach to education that is rooted in the experiences of marginalized people; that is centered in a critique of structural, economic, and racial oppression; that is focused on dialogue instead of one-way transmission of knowledge, and that is structured to empower individuals and collectives as agents of social change." (p. 183)

In problem-posing education, the reality is not presented as fixed and given, rather as an ongoing process that an individual's action can intervene. Given such a perspective to the

world, students learn to hope for possibility, change, and improvement. According to Freire (1970), in problem-posing education, people develop their power to perceive critically the way they exist in the world with which and in which they find themselves; they come to see the world not as a static reality, but as a reality in process, in transformation. Students in this model are encouraged to reflect critically on the world and their society. They learn that they have the right to speak and that they can change and improve their living conditions.

Individualization

According to the British sociologist, Antony Giddens, the Industrial Revolution in the eighteenth century helped develop individualism through the separation of work from private life. New institutional structures such as the workplace and schools developed outside the home, leaving it as a more private space. Within this space, private subjective feeling became possible and developed. This last phase of modernity has furthered the development of individualism, as the decline of traditional social values has created a social disembeddedness. There is now a compelling need for people to construct their own identities and find new means for its expression, rather than being thrown back on types of categorizations that no longer fit. Individuals also require the skill of reflexivity, which entails a reappraisal of the past, self-awareness in the present, and projects for the future in order to give themselves a strong sense of self-identity (Giddens 1991).

Individualization and education

Traditionally, education was seen to have a socializing function, inducting individuals into the roles expected of them by society (Durkheim 1956). But as society is no longer homogeneous, this suggests a different focus for education – that of the recognition and celebration of individualism as part of the strengthening of identity. There are many indications of the promotion of individualization as a modern aim of education. These include:

- an emphasis on self-expression in learner's writing and literature;
- the promotion of personal projects in education;
- an emphasis on the learner's individual psychology;
- an emphasis on the individual learner's strengths and styles;
- a concern for the development of self-esteem in learners;
- a promotion of learner reflection and reflexivity and the development of reflexive tools.

Not all of these emphases on individualism arise from responses to social fragmentation. The recognition and celebration of learner difference may also arise from equality and diversity agendas in a pluralist society.

The network society

The 'network' represents a new form of social organization that has emerged since the mid-1980s. There are many reasons for the growth of networks, among them a decline in state power, the spread of capitalism and the growth of digital communication systems. We talk commonly about computer, family, social, political, terrorist or media networks (Kelly 1998: 3–5). Such networks operate across distance, their structures are not obvious, and they have the power to make connections in any number of different ways, as hyperlinks do. One feature of networks is that they are inclusive in the way that information flows between members and exclusive in ensuring that information and knowledge stays within the network. It is not surprising to find that the metaphor of the network is used as a form of knowledge representation in a number of disciplines, replacing an older computer hardware model. In cognitive science, connectionist networks are widely used to represent and explain cognitive activity (Bruning et al. 2004: 60).

The network society is underpinned by the revolution in information communication technology. This began with the rise of the personal computer and continued through the

development of the Internet, the World Wide Web and increasingly sophisticated software applications for communication and information exchange. The future is likely to include the experiences of multiple realities, sentient robots or computers, and the translation of vicarious into immediate experience.

Networks and education

Educational networks have always existed, connecting groups of scholars who communicate together in scholarly activity and publication across institutions and countries. Such scholarly groups may share their research and expert knowledge without the necessity for physical contact.

Modern ideas of educational networks are shown in ‘communities of practice’ in which practitioners can actively create and share knowledge. In communities of practice, knowledge is always ‘situated’, not necessarily within a specific location, but always within a culture and a background. Knowledge is not pre-existing, but constructed from the practices of the group. The community confers identity on its members; to be excluded from the community is to be excluded from the knowledge the network constructs, and to possess no community identity. Power is dispersed or shared between the group members, rather than in a hierarchical or top-down fashion (Wenger 1998: 4).

There are many implications of seeing education as comprising communities of practice, not least for the traditional power of the teacher.

An unforeseen type of educational networking is shown by Internet sites such as ‘Rate my teacher’ which are constructed by school children and students to provide their own evaluations of schools and teaching. Such sites, which have sprung up in many countries, are almost impossible to control and show the effective ways in which networks can operate to subvert formal systems in the interests of the members.

The value of networks lies in their power to:

- connect groups of learners and practitioners separated by distance;
- disseminate power throughout the network to create more equal relationships between learners and teachers;
- confer identities on learners;
- create instantaneous communication-flows;
- actively construct and disseminate knowledge.

The dangers of educational networking lie in:

- the ability of the network to exclude outsiders from knowledge;
- the possibilities of illusion and deception created by virtuality;
- the difficulty of monitoring and regulating networks;
- privacy issues in relation to membership of networks.

Globalization

The other major and most obvious social phenomenon is that of globalization. The global economy, which acts as another form of network, this time at a macro-level, is self-expanding, unregulated and uncontrollable. It began as an economic network reflecting a form of advanced capitalism, but now stretches beyond economics to affect all aspects of social life (Giddens 1998: 30–1). As in other networks, power is dispersed between competing markets, often in the form of multinational corporations that operate globally, and now have more power than nation states (Beck 2000).

Globalization and education

The influx of market forces and the colonization of education by business practices are shown in many ways. Educational institutions are beginning to act more like corporations as they adopt the discourse of ‘customers’, ‘markets’ and ‘efficiency’. Conversely, corporations

are themselves beginning to take over educational enterprises. For example, large multinational companies like Motorola are now creating their own universities which can be used to disseminate precisely the training and knowledge that the corporation requires (Jarvis and Preece 2001: 220–1). In 2008 the MacDonalds fast-food chain gained British government approval to offer its employees awards equivalent to an A level or advanced diploma level (BBC 2008).

The characteristics of the corporate educational institution are that it:

- identifies internal and external markets:
 - internal markets: units or academic departments;
 - external markets: learners, parents, communities, government agencies;
- packages knowledge, educational programmes and services as commodities;
- distributes resources competitively;
- measures educational effectiveness by the achievement of targets;
- operates bureaucratic top-down rather than collegial management systems;
- sells a share of the educational enterprise to external investors.

The educational corporation:

- may operate virtually, on a global basis, with no physical location;
- combines work-based, professional training with education;
- cooperates with educational institutions on equal terms;
- devises its own curricula and programmes of study;
- may admit external learners as customers;
- awards certification and credits to its learners;
- blurs the boundaries between education and business.

In the future there is likely to be a further merging of corporations and educational institutions as the commodification of education continues and spreads to all sectors of education. The use of online and blended learning processes will facilitate this process.

The other aspect of globalization which has already had a marked effect on education is that of the discourse of lifelong learning. According to this discourse, workers will need to upskill on a continuous basis, as knowledge is transformed, in order that companies, regions and states can maintain their competitive advantage. Lifelong learning is not simply concerned with the workplace and workplace training – it affects all educational providers and all types of provision (Kogan 2000: 341–59), though the extent to which it acts as a catchphrase to justify economic imperatives is debateable.

Knowledge itself has become the main commodity for exchange – the World Trade Organization (WTO) has recently categorized education as a tradable commodity – which has far-reaching implications for learners, or ‘customers’ as they are increasingly known. If education is a commodity, it must be like other products and be accountable to the shareholders who own it. Education as a commodity therefore has had to standardize its products in the interests of the market and its major shareholders, the consumers and providers. In higher education, a proliferation of private providers has led in many countries to greater student choice, but also to the potential for lowered educational standards.

The commodification of education has affected all levels of the educational system. In school systems, market forces have led to consumer choice in relation to educational provision. For example, the British school system under the ‘New Right’ Conservative government in the 1980s responded to the demand for parental school choice and an open educational market. This led to the need for educational transparency in the form of school league tables and the assessment of school performance on the basis of Standard Attainment Testing (SAT) of students. This publication of performance indicators meant that schools wanted to attract only the clever or the academically motivated. The net result was, ‘It is not

what the school can do for the child that counts, but what the child can do for the school' (Gerwitz 2001: 176).

Glocalization: the global, the local, and the hybrid

Every discussion of modern life, no doubt, incorporates the controversial term of globalization. Globalization, according to Giddens (as cited in Block, 2004), is defined as "the intensification of worldwide social relations which link distant localities in such a way that local happenings are shaped by events occurring many miles away and vice versa" (p. 75). In language teaching, this has manifested itself in the way English is widely used as the international language or "the shared linguistic code" (Block & Cameron, 2002, p. 1) in international communication.

Richards (2008) points out "English is no longer viewed as the property of the English-speaking world but is an international commodity sometimes referred to as world Englishes, a deterritorialized English detached from its geographical and cultural origins" (p. 2). Although there is no doubt that we are living in a globalized world, thanks to technological innovations that are in fact "evaporating borders and distances" (Daly, 2009, p. 7), there is little agreement on the various aspects of the issue. In fact, as Sifakis and Sougari (2004) maintain, "there seem to be different interpretations, or versions, or realizations of it, depending on the observer" (p. 60). Blok (2004) believes that some see globalization phenomenon as "a done deal"; others view it as "a work in progress" which is developed in different parts of the world. Some see it "as both progress and progressive, benign and good," others see it as "the steamroller of late modernity taking away all that is authentic and meaningful in our lives" and finally some see it as "hegemonically Western" or "an extension of American imperialism", others see the process as more "egalitarian" (p. 75). While many primarily acknowledged the advantages of globalization, the so-called problems it could cause have given rise to lively debates over its legitimacy.

Perhaps the view that has gained much attention and debate over the last decades is the one which considers globalization synonymous with "westernization," "McDonaldization," or "Europeanisation" (Sifakis & Sougari, 2004) and which is accused of practicing a western hidden curriculum imposing "sets of social and cultural values which are inherent in their make-up" (Hurst, 2007). Rogers (1982) maintains that one of the ethical issues of the ELT industry is the cultural imposition of non-neutral values by ELT professionals in host countries. Similarly, Vongalis-Macrow (2005) believes that "this phenomenon is particularly important to education because it impacts on how a national system, such as education, can be governed by supranational bodies rather than remaining a national concern" (p. 2).

The discussion about the merits and demerits of universalizing and the degree it needs to be considered as a homogenizing procedure goes on the third millennium; however, the arrival of more individual and student-centered techniques to learning, and the post-method refusal of one-size-fits-all system, has risen to the need to trace the teaching materials. This is accomplished to ideal match the specific language gains and desires of every particular context. McDonough and Shaw (2003) stated that localizing considers the global geography of English language teaching and awareness that what might work in one specific setting might not inevitably do so in another context. It is probably because of the same conflict among internationalize and positioning in modern life that the balancing phrase of "internationalize"; which has been presented to the literature to offer the saying "think globally, act locally."

As defined by Weber (as cited in Gurko, 2011), glocalization is a blend of universal systems and local training. It highly grows the possibility of autonomous learning by combining global knowledge and domestic knowledge. Still, according to Moss (2008), glocalization does not demonstrate the intermediate or temporary plan or a period between

the local and the global level, instead of applying medium standards to explain the objectives and thus make local plans to obtain the universal models. Khondker (2004, p. 3) enumerates the following points as the main propositions of glocalization:

1. Diversity is the essence of social life;
2. Globalization does not erase all differences;
3. The autonomy of history and culture give a sense of uniqueness to the experiences of groups of people whether we define them as cultures, societies or nations;
4. Glocalization is the notion that removes the fear from many that globalization is like a tidal wave erasing all the differences.
5. Glocalization does not promise a world free from conflicts and tensions but a more historically grounded understanding of the complicated – yet, pragmatic view of the world.

As far as language curriculum development is concerned, Daly (2009) believes that in this post-methods age, non-native speaker teachers are in the best position to act as mediators in the EFL profession to combine local knowledge and teaching strategies with CLT or other principles from second language acquisition (SLA) or Applied Linguistics. In this way, a more systematic (and hopefully less teacher-responsible) approach to foreign language (FL) teaching for exam-oriented contexts can be devised to better prepare EFL students for the needs of a globalized world (p. 14). Further, Daly refers to this globalizing activity as "a post-methods principled pragmatics," which is currently "cautious, tentative, and hazy" and which "places a terrific burden on teachers" (p. 15). Hence, glocalization is not an easy straightforward path to take and some requirements must be met to pave the way towards achieving it. In what follows, some of the necessities and requirements for a globalized language curriculum are presented.

The Critical Shift: Hidden Curriculum in the ELT context

In the 1990s, the ELT discipline decides to turn into a critical perspective. Kumaravadivelu (2006) attributes the evolution of this vital position in ELT to Phillipson's (1992) linguistic imperialism. Such a necessary opinion is defined as the linguistic imperialism theory (Ghaffar Samar & Davari, 2011). As stated by Aghagolzadeh and Davari (2012), the aspect of a critical intellectual change in applied linguistics has considerably disputed the mainstream ELT. It has also dramatically introduced CLP as a substitute approach to the general ELT, particularly in the Periphery.

Within this framework, curriculum and materials development would be based on the assumption that no one program fits all populations and a set of the predetermined curriculum would ever be imposed in a program. It is assumed that foreordained program content contradicts the ideology of transformative teaching (Richards, 2013). Therefore, all curriculum decisions would rely on learners' needs in a program and the choices of what would be studied through negotiation or dialogical interaction (Giroux, 1997; Shor, 1996). Accordingly, creating and adopting materials within the framework of TP could be vital for EFL teachers. While materials are the essential elements in ELT programs (Richards, 2010), few studies have been conducted on the transformative L2 materials preparation.

Akbari (2005) also introduces linguistic imperialism as a recent development in foreign language teaching and as a concept that tries to sensitize language teachers to the political complications of what they do in the classroom. Reviewing the premises, mechanisms and outcomes of linguistic imperialism, Akbari notes the fact that since ELT scholars had no real knowledge of what was actually taking place in other countries in which English was taught as a foreign language, they had to resort to the invention and promotion of theoretical knowledge that is basically of a linguistic nature. In his words, through the creation and application of theories that are thousands of miles away from the realities of the EFL classroom, center scholars could secure leadership in the profession, making big publishers their economic allies. Akbari maintains that while there is no overt resistance to the

dominance of the Center in the Iranian academic setting, some modifications in the syllabus or textbook structures can be viewed as measures to counter this silent hegemony. Trying to introduce linguistic imperialism and its impacts in Iran, Aghaei (2009) aims at presenting the ideologies or, in his word, the false ideas currently exist in the field of ELT, especially with their probable impacts on learners and teachers. In his critical view, these impacts have made the Iranian English language teachers, and learners have native-based opinions and attitudes about English language teaching and learning. Reviewing the works indicates some significant and new issues. Firstly, as new attempts, they have taken a look beyond the current state of ELT in the Iranian context and have not confined themselves to some purely pedagogical issues. Referring to cultural, ideological, political, and religious aspects of the context, these works mostly intend to shed light on the cultural politics of English language, its role, and function both locally and globally. Secondly, avoiding an extremist view to reject English as well as insisting on the importance of English and its instruction in the global context. What follows will provide some practical suggestions towards developing critical materials.

Transformative nature for education

CLP has been used as an umbrella term in L2 professional literature, which assumes different identities for teachers and students. Transformative Pedagogy (TP) has been used, a term by Giroux (1988), as a version of CLP. It is an activist pedagogy that combines the elements of constructivism and critical pedagogy by empowering students to examine their beliefs, values, and knowledge critically. The goal is to develop a reflective knowledge and to challenge the status quo to promote a democratic and emancipatory learning context (Freire, 2005; Moreno-Lopez, 2005). In TP, teachers, and students engage in dialogic relationships to empower each other for transformative intellectualism and offer students a perspective to voice their ideas (Clark, 2018; Kim & Pollard, 2017; McLaren, 1989; Shor, 1996). Teachers are seen as transformative intellectuals who should develop critical attitudes and skills among students (Kareepadath, 2018).

TP advocates a constructivist dialogical perspective of knowledge. It aims to disclose the connections between knowledge and the cultural norms of the society. This perspective opposes the traditional theory of education that serves to reinforce certainty, conformity, and technical control of knowledge and power (Darder, Baltodano, & Torres, 2009). Dialectical constructivism posits that knowledge forms interactions among people. From this perspective, Kincheloe (2000) asserts that knowledge is socially constructed through the dialogue between “the world and human consciousness” (p. 107). Thus, TP seeks to support a dynamic interactive element instead of absolute dichotomies of thought or practice. This point of view is rooted in the dialectical view of knowledge. That is why Darder et al. (2009) maintain that TP endorses a flexible and flowing “view of human and nature that is relational; objectivity and subjectivity that is interconnected and a coexistent understanding of theory and practice” (p. 11).

A plethora of works (Atai & Moradi, 2016; Barjesteh, 2017; 2019; Bellis, 2018; Kincheloe, 2008; Shor, 1996) have been done in TP indicate that teachers can empower their students by incorporating the principles of CLP in the classroom context. Crooks (2009) notes the word empowerment has turned to be a password among critical pedagogists. In brief, TP is mainly concerned with the critique and elimination of human sufferings, the creation of consciousness, voice, hope, possibility, and the improvement of life through education. It results in the reconceptualization of the notion of education and transforms it into a means of justice. Giroux (1988) posits that empowering students to become critical and active citizens rests on teachers who have the potential to “combine scholarly reflection and practice in the service of educating students to be thoughtful, active citizens” (p. 122).

Critical Language Teacher Education Practices

Critical theory mainly deals with elements of power in any community and is meant to serve justice by providing people with the means they need to free themselves from the unjust or cruel exercise of authority. Critical pedagogy, unlike critical theory, recognizes education as the instrument that can emancipate people from these forms of discrimination. Critical theory is mainly concerned with power relations in society and seeks to bring about justice by empowering people to emancipate themselves from oppression. CP differs from critical theory in that it views education as the way to emancipation of people from these forms of discrimination.

McDonald and Zeichner (2008, as cited in Hawkins & Norton) discuss the current move from multicultural education to social justice teacher education. Social justice teacher education seeks to address institutionalized as well as individual power differentials, intending to promote teacher's recognition and ownership of their roles as social activities. There is an increasing body of literature within the fields of TESOL that addresses theoretical stances around language teaching and language planning (Pennycook, 2001; and Phillipson, 1992), but there are few accounts of critical language teaching practices. These accounts can be categorized as critical awareness, critical self-reflection, and critical pedagogical relations, Critical Discourse Analysis, Critical Literacy (ies, Critical Literacy Pedagogy, and Literacy in the Third Millennium. Barjesteh and Mozafari (2016) support that incorporating critical approaches in the classrooms can foster learners' consciousness and self-seeking activities. In each of the critical accounts, the notion of praxis is hidden.

Critical awareness

The main emphasis of critical teacher educators is on encouraging critical awareness through raising the realization of how power relations are built and act in society and the degree to which historical, social, and political practices structure educational discrimination. This refers to a case in which teacher educators attempt to make visible to teacher-learners inequitable relations of power in their communities and how these affect the language learners they teach. Pennycook (2004, p. 341) offers an insightful reflection on what it means to be a critical teacher educator, coining the term *praxicum* to capture how theory and practice come together to create a new understanding of the TESOL practicum. Three critical aspects in the teachers-learners classroom were identified by Pennycook, taken from a) the activities of a disturbing male student, b) the use of rehearsal conversations, and c) the acknowledgment of non-standard English in the classroom. Each of these critical aspects increases greater questions of power in the community and offers a chance for critical argument and reflection. He recommends that trying to be a critical teacher is more than looking for and grasping small moments to open the door to a more critical viewpoint. So his account finds praxis in the developing critical awareness of teacher-learners for social revolution.

Critical Consciousness

Critical Consciousness is one of the essential components of critical language pedagogy and the key to people's emancipation. Barjesteh (2020) conceptualizes CLP as a philosophy of education and a method of teaching that pursues to raise teachers' and learners' power that enable them to critique and challenge the problems by helping them to build critical consciousness. Coming to critical consciousness, Barjesteh maintains that it is a process whereby students foster a capacity to think critically. It is defined as a process by which students achieve an awareness of the social realities which discover their abilities to recreate them. Freire introduces the notion of 'conscientization' and defines it primarily as "the effort to enlighten men about the obstacles preventing them from a clear perception of

reality. In this role, conscientization affects the ejection of cultural myths that confuse people's awareness and make them ambiguous beings" (1985, p. 89). According to him, conscientization envisages people's gaining critical consciousness.

Freire (1970b) assumes different modes or levels of consciousness (i.e., semi-intransitive, naive transitive, and critical). At the lowest level or 'semi-intransitive' consciousness, people almost accept their lives as they are. This mode of consciousness does not have enough distance from reality to see reality concretely or critically. This consciousness has a distorted portrait of the challenges and problems of reality. Therefore, individuals with this type of consciousness attribute problems and misfortunes to super-natural powers or to their incapability and consequently, they will not attempt to change their lives and transform reality. The next level is 'naive-transitive consciousness,' which moves away to some extent from blind myths toward awareness. People with naive consciousness perceive a little more accurately the source of problems in their lives and begin to take action (Freire, 1970). The third level is 'critical consciousness' therewith individuals can contextualize and analyze their problems deeply and view reality and their lives as changeable entities. Thus they can engage in both reflection and action, i.e., praxis. According to Freire (1985), gaining critical consciousness cannot be achieved only by reflection, but through praxis.

Heaney (1995) distinguishes consciousness-raising and conscientization. He points out that the former is the result of banking education. Thus, it is influenced by the transmission of knowledge. The latter implies achieving new levels of awareness as the subject in the real world rather than an object. He explains:

"In this movement from naivety to critical pedagogy, individuals grasp the social, political, economic, and cultural contradictions that subvert learning. Teachers and students with a critical consciousness conceptually pull back from their lived reality to gain a new vantage point on who they are and how they came to be this way." (p. 73)

Critical self-reflection

Critical language educators inspire teacher-learners to reflect on their identities and situations in society analytically. Self-reflection opens a window into the link between the learner and the social world, by the emphasis on the limitations and probabilities of changes in society. Pavlenko (2003) presented a self-reflective report that showed how she used the theory to supply authorization alternatives in a graduate language learning class in the United States. She discovered that many instructors engaged in learning interiorized traditional indigenous and non-indigenous speech. She offered the concept of multiple competencies to strengthen her reflection, which her students consider as legal customers of the target language, instead of the "unsuccessful native speakers" (Cook, 1999). Ikuku's idea, a Japanese female student, demonstrates the capability of theory to supply a wide range of identity alternatives for teachers who are engaged in learning language: I learn new perspicacities into the English and socio-cultural aspects of the language day by day, authorizing our knowledge. For example, until recently, I did not consider myself as a bilingual person and constantly thought that my English knowledge was not good enough, and in the end, I must be able to use the English language like a native until I learn the notion of multiple competencies which is introduced by Cook (p. 262). In this sample, praxis can characterize the development of perception (by teacher-learners) in a way in which social

discourses have framed self-perception and as well as their capability to perform on the world.

Critical Pedagogical Relations

If the purpose of critical education is to give power to students, the educational relationship between teacher educators and teacher-learners must be built on fair terms. Teacher educators should strive to rebuild power relations between their teacher-learners and themselves, not only to model important teaching methods but also to inspire teacher-learners to assume approaches in which their pedagogy can provide opportunities for language learners during the classrooms. As an account, Crookes and Leher (1998) refer to a language teacher education course taught in Hawaii in which they intended to interrupt what Freire has referred to as the banking model of education in support of one in which all members are equally in charge of planning and taking part in learning. By starting the discussion of the educational program, they tried to revision the structure and environment of the class. They describe in detail their attempt to accomplish an important educational program in a long-established setting. This is, to a large extent, a reflection of the anxieties between their positions of authority as instructors and their willingness to adopt participatory dialogue. Thus, praxis requires a fundamental review and changes in the foundation of the educational programs and classes.

Conclusion

The principal goal in education is to promote the academic success of students and prepare them to raise their real life. The goals are fulfilled only if teachers train students to become critical thinkers and take charge of their learning (Milner, 2003). Most of the traditional approaches and methods in education acknowledge pre-established patterns to transmit knowledge to the learners. These approaches consider learners as passive receivers of knowledge and teachers as the authority in the classrooms who deposit knowledge in learners' minds (Hetherington & Wegerif, 2018). However, the current trend in English language teaching (ELT) deals with creating and developing critical skills, dialogic interaction, and reflective practices (e.g., inquiring, doing, imagining, and negating) at schools (Garcia et al., 2020). This trend criticized the traditional ELT approach due to its monologic and lecture-based instruction (Li, 2019). Zhang (2018) postulates that monologic instruction promotes passive talk and keep students reticent. L2 professional literature proposed three skills categories (i.e., technical, behavioral, creative) to promote learners' CT mode. These skills are to help students act in a questioning manner, construct their understanding, and to be the agent of their learning process (LaGarde & Hudgins, 2018). Both teachers and learners have a bilateral role in which the learners are active, practice exploratory talk, think reasonably, organize their learning practices. Teachers help learners voice their ideas, and share their authority with the students (Kissling-Styles, 2003). Bakhtin (1981) believes that they learn together. Using Freire's (1970) words, "no one teaches another, nor is anyone self-taught, men teach each other, mediated by the teacher" (p.67). One of the recommendations evolved to incorporate talk effectively for students' learning process and to involve teachers and students for transforming and constructing knowledge is DT (García-Carrión, López de Aguilera, Padrós, & Ramis Salas, 2020). The future of education is being shaped by rapid technological advancements, evolving societal needs, and a growing emphasis on personalized learning. Emerging trends suggest a shift towards hybrid and digital-first learning environments, integrating artificial intelligence (AI), virtual reality (VR), and adaptive learning platforms to enhance student engagement and comprehension. Competency-based education, which prioritizes skill mastery over seat time, is gaining traction, aligning curricula with real-world demands. Additionally, lifelong learning is becoming essential, driven by the fast-paced evolution of industries and the gig economy.

Equity and accessibility remain critical, with efforts to bridge the digital divide and provide inclusive educational opportunities worldwide. As education continues to evolve, the focus will likely remain on fostering critical thinking, creativity, and adaptability—skills necessary for success in an increasingly interconnected and automated world

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INNOVATION IN HR MANAGEMENT AND WORKPLACE CULTURE

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ABSTRACT

Human Resource Management (HRM) revolves around this core matter of managing relations at work place. In general Human Resource Management is a management function concerned with hiring, training, motivating, developing and maintaining workforce in an organization. Human resource management (HRM) department can be summarized as the effective management of teams and individuals in an organization aimed at competitive advantage and performance success. Automation and artificial intelligence (AI) assimilation into human resources is a critical part of the future of employment. In a positive workplace, employees work together efficiently to complete tasks and overcome hurdles. Human resource management bridges the gap by incorporating goal-setting into their performance management processes, connecting employees' individual and team goals to overall organizational goals. Effective leadership ensures that the culture is consistently reflected in daily operations, keeping employees motivated and engaged throughout. The most common use case for AI in human resources is using software to identify ideal candidates for specific positions. AI Chatbots are one of the most impactful innovations in HR. They answer common questions quickly and help resolve simple issues when HR teams are unavailable.

Keywords: Increase competitiveness, Recruitment and selection, Job satisfaction, AI recruitment tool, Chatbot

INTRODUCTION

Human beings are social beings and hardly ever live and work in isolation. We always plan, develop and manage our relations both consciously and unconsciously. The relations are the outcome of our actions and depend to a great extent upon our ability to manage our actions. From childhood each and every individual acquire knowledge and experience on understanding others and how to behave in each and every situations in life. Later we carry forward this learning and understanding in carrying and managing relations at our workplace. The whole context of Human Resource Management revolves around this core matter of managing relations at work place.

There is no best way to manage people and no manager has formulated how people can be managed effectively, because people are complex beings with complex needs. Effective HRM depends very much on the causes and conditions that an organizational setting would provide. Any Organization has three basic components, People, Purpose, and Structure

That predicted future is today's reality. Most managers in public- and private sector firms of all sizes would agree that people truly are the organization's most important asset. Having competent staff on the payroll does not guarantee that a firm's human resources will be a source of competitive advantage. However in order to remain competitive, to grow, and diversify an organization must ensure that its employees are qualified, placed in appropriate positions, properly trained, managed effectively, and committed to the firm's success. The goal of HRM is to maximize employees' contributions in order to achieve optimal productivity and effectiveness, while simultaneously attaining individual objectives (such as having a challenging job and obtaining recognition), and societal objectives (such as legal compliance and demonstrating social responsibility).

Definition

In general Human Resource Management is a management function concerned with hiring, training, motivating, developing and maintaining workforce in an organization. Human resource management ensures satisfaction of employees so as to get maximum contribution of employees for the achievement of organizational objectives. According to Armstrong (1997), Human Resource Management can be defined as —a strategic approach to acquiring, developing, managing, motivating and gaining the commitment of the organization's key resource – the people who work in and for it.

REVIEW OF LITERATURE

Potgieter and Mokomane (2020) argue that the strategic emphasis of a human resource management (HRM) department can be summarized as the effective management of teams and individuals in an organization aimed at competitive advantage and performance success. Thus, there is growing interest in investigating the role of HRM departments and practices in supporting companies' capacity for innovation (Engelsberger et al., 2021). Due to the recent transformation (such as digitization) of most organizations, HRM's role in strategic management has become more important (Zhou et al., 2020), as these practices can provide tools for change and innovation and support strategic decision-making in organizations (Sheehan et al., 2016).

The management of talent also consists of the ongoing evaluation of the performance of the employees. This includes setting up goals and providing regular feedback and also conducting performance reviews. That is how an organization makes sure that the employees contributions are going side by side with the objectives of the organization (Zel, & Kongar, 2020).

Automation, Artificial Intelligence, and the Changing Workforce: Automation and artificial intelligence (AI) assimilation into human resources is a critical part of the future of employment. To increase productivity, businesses are gradually automating routine processes. Amazon is a famous example, since it deploys robots in its factories to increase efficiency while decreasing labour expenses (Vrontis et al., 2022). The emergence of technology reshapes employment responsibilities and involves the acquisition of fresh abilities. Employees at organizations like Tesla must adapt to AI-driven production procedures, highlighting the importance of continual learning

STATEMENT OF THE PROBLEM

Human resource management maintains a positive workplace culture by tending to employee needs and offering support where they can. In a positive workplace, employees work together efficiently to complete tasks and overcome hurdles. They feel secure in the workplace and can express their thoughts, ideas, and emotions. This can help company operations and day-to-day tasks run more smoothly. Despite of the employee engagement and retention, many organizations struggles to create a workplace culture that fosters innovation, creativity and collaboration, leading to decreased productivity. Furthermore, many organizations struggle to create a workplace culture that promotes inclusivity, collaboration, and work-life balance, all which are critical to employee satisfaction and organizational success

OBJECTIVES OF THE STUDY

- To evaluate the impact of workplace culture in organizational performance and innovation.
- To identify the key challenges and opportunities for innovation in HR.

IMPACT OF WORKPLACE CULTURE

Job Satisfaction

A company culture that spreads positivity and is supportive can create an environment where employees feel respected, valued, and heard. As a result, the employees showcase satisfaction with their jobs, avoiding chances of seeking opportunities elsewhere.

Employee Motivation and Productivity

A company culture is deemed powerful when it communicates expectations and goals transparently to its employees. This clarity makes them derive a sense of meaning and purpose from their work, boosting their motivation and empowering them to maximize productivity. This is because employees who understand their roles are motivated to go above and beyond. They enthusiastically contribute to the organizational goals which leave no scope for any shortcomings.

Innovation and Adaptability

A culture that embraces creativity and innovation enables employees to experiment and be innovative in their approach. They are inspired to think outside the box and adapt to changing trends. This encouragement directly leads to a rise in engagement levels, where employees produce fresh ideas and ultimately benefit the organization.

Collaboration and Teamwork

A strong company culture encourages teamwork and collaboration, which binds employees with trust and support for each other. This unity helps break down silos which result in better communication, innovation, and effective problem-solving among employees.

Leadership Influence

Leaders, as true representatives of company culture, have the power to influence and promote it effectively. By embodying the culture, they can inspire their teams and reinforce the values that drive engagement.

Effective leadership ensures that the culture is consistently reflected in daily operations, keeping employees motivated and engaged throughout.

BENEFITS OF HUMAN RESOURCE MANAGEMENT

Increased competitiveness

When HR managers work toward identifying issues within the workforce and implementing solutions to correct them, organizations can better adapt to the market's changing needs and create a competitive edge for themselves.

Additionally, creating a culture where HR leaders continually encourage employees to come up with new solutions can lead to a more engaged workforce and increased organizational efficiency.

Greater Focus on Company Goals

Human resource management bridges the gap by incorporating goal-setting into their performance management processes, connecting employees' individual and team goals to overall organizational goals. This helps everyone see how their work contributes to larger goals, making it easier for employees to understand not only their impact but also that of their coworkers. Goals are an essential part of creating a culture of transparency and accountability where employees are united in their efforts to help the company achieve long-term success.

Better recruitment and retention

Innovative HR solutions, where workplace cultures align with the organization's mission and values, can help a workplace find and keep diverse, talented employees.

Hiring is one of the most impactful areas of diversity, equity, and inclusion progress. Finding new ways to broaden the reach of a job search and helping team members check biases in the interview process can help you build a team for the future of work.

Increased Employee Engagement

The power of employee engagement is hard to understate - and it's been top-of-mind for HR professionals for the past few years. Companies have been facing difficulties hiring and retaining employees and navigating trends that indicate a lack of engagement, like quiet quitting and the Great Resignation. Luckily, engagement comes much more naturally with an HRM strategy.

The increased connections between employees and their work, motivating factors like career advancement and regular recognition, and clear objectives established through HRM all contribute to a more engaged workforce.

CHALLENGES OF HUMAN RESOURCE MANAGEMENT

Recruitment and Selection

One of the major challenges faced by human resource managers is recruitment and selection. As technology advances, new HR management tools and techniques are developed. As a result, HR professionals find it challenging to find a suitable candidate who fits job requirements and adapt to technological changes. They will have to make some changes in their recruitment methods, including offering new training services to ensure quality results.

Managing a Diverse Workforce

Hiring a diverse workforce means that you bring onboard employees with a broader range of skills and different experiences that can maximize productivity. However, managing such workers is one of the HR challenges in many organizations. For instance, handling frequent disagreements and encouraging effective communication between a diverse workforce with different backgrounds can leave other employees feeling neglected. In this regard, HR professionals must develop an organizational culture through various team-building activities.

Retaining the Best Talent

This is one of the most common challenges of HRM faced by organizations. With employers competing for the best employees, retaining them requires a structured onboarding plan. Remember that employee turnover can be costly and have an adverse effect on your business

growth. To maintain the best talent, you should also create a conducive work environment, provide room for career development, and motivate them to perform better.

Engaging Employees

Besides talent acquisition challenges, HR departments struggle to keep their workforce engaged and maximize productivity. You see, as your business grows, you will also alter its structure, work plan, and even internal processes. Getting your workers to adapt to this change require effective communication. Let them know the benefits of the changes and why their input is crucial in taking the business to a higher level. Consider having frequent meetings and let them know about recent developments. Regular feedback is also essential to identify areas that need extra attention.

HR INNOVATIONS

AI-based recruitment tools

The most common use case for AI in human resources is using software to identify ideal candidates for specific positions. AI-driven software has powered recruitment digital transformation with new automation such as screening resumes, cover letters, and other application materials and then selecting the most qualified candidates based on job requirements. It can also weed out candidates that don't meet the minimum qualifications for a job.

Employee self-support portals

Employee self-support portals provide employees with direct access to HR resources, such as benefits information, payroll information, and company policies. These portals allow companies to take the burden off the HR teams while enabling employees to quickly find the information they need without having to reach out to the HR staff.

The digital workplace

A digital workplace is a cloud-based virtual environment that enables employees to work, collaborate, and access information and tools digitally rather than relying solely on physical office space.

In many cases, employees use a mix of physical and digital workplaces. For instance, you're already part of the digital workplace transformation if you are using tools like Slack, Zoom, or Microsoft Teams.

Digital onboarding

HR technology now streamlines the onboarding process, often starting when a candidate signs their contract. These tools ensure new hires stay engaged and don't feel overlooked. Digital onboarding uses technology to guide employees through their initial days, automating tasks like paperwork, emails, training, and other repetitive activities.

Employee wellness tools

One of the innovative approaches to supporting well-being at work is adopting employee wellness tools that help employees maintain their physical and mental health. Online wellness portals such as wellspaceoffer accessible well-being support for organizations looking to

incorporate health initiatives in the workplace. These tools usually integrate with HR systems and benefit management platforms.

Chatbots

AI Chatbots are one of the most impactful innovations in HR. They answer common questions quickly and help resolve simple issues when HR teams are unavailable.

Chatbots can address queries about policies, benefits, and processes and even assist with onboarding new employees or managing annual leave.

CONCLUSION

The innovation in HR management and workplace culture has provided valuable insight into the importance of adopting innovative HR practices and fostering a positive workplace culture in driving organizational success. The study suggest that arrangements, employee recognition programs, and professional development opportunities and significantly enhance employee engagement, retention and productivity.

Foster a culture of feedback to encourage open communication, regular feedback, and recognition to create a culture of transparency and trust. Utilize HR technology, such as AI – powered chatbots, virtual reality training and predictive analytics to streamline process enhance employee experience and drive business outcomes. To promote a culture of inclusivity, respect, empathy and celebrate diverse backgrounds, perspectives and experience. To implement recognition and reward programs that acknowledges the employee.

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AI AND DATA ANALYTICS IN DECISION MAKING

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ABSTRACT

In modern management, Artificial Intelligence (AI) and Data Analytics play a transformative role in decision-making by providing actionable insights, predictive capabilities, and automation. AI-driven algorithms analyze vast datasets to identify patterns, trends, and correlations, enabling managers to make informed, data-driven decisions with greater accuracy and efficiency. Machine learning models, natural language processing, and big data analytics enhance strategic planning, risk assessment, and operational optimization. Additionally, AI-powered decision support systems improve real-time decision-making, reducing uncertainty and human bias. This paper explores the integration of AI and Data Analytics in management decision-making, highlighting their impact on productivity, competitiveness, and innovation while addressing challenges such as data privacy, ethical considerations, and implementation complexities.

Keywords: Intelligence, Data Analytics, Decision-making, Management, Predictive Analytics

INTRODUCTION

AI technologies, such as machine learning, natural language processing, and computer vision, play a crucial role in decision-making. These technologies can quickly process and analyze large datasets, identifying insights that humans might miss. AI-powered predictive models can forecast future events, enabling proactive decision-making. Additionally, AI can automate routine, repetitive tasks, freeing up human resources for more strategic decision-making. By leveraging AI, businesses can make faster, more accurate decisions, driving growth and competitiveness.

Data analytics is also a critical component of decision-making, providing businesses with a solid foundation for strategic decision-making. Data analytics helps businesses understand their strengths, weaknesses, opportunities, and threats. By analyzing data, businesses can identify areas of inefficiency, optimize operations, and inform strategic decisions. Data analytics also enables businesses to measure the effectiveness of their decisions, making adjustments as needed. By combining data analytics with AI, businesses can unlock new insights, drive innovation, and achieve their goals.

The future of decision-making with AI and data analytics holds much promise. As these technologies continue to evolve, we can expect to see increased automation, improved predictive analytics, and enhanced collaboration between humans and machines. AI-powered tools will facilitate decision-making, providing businesses with the insights and expertise needed to drive growth and success. By embracing AI and data analytics, businesses can stay ahead of the curve, making informed, strategic decisions that drive competitiveness and achievement.

STATEMENT OF THE PROBLEM

The integration of Artificial Intelligence (AI) and data analytics in decision-making has become a critical imperative for organizations seeking to stay competitive in today's fast-paced, data-driven business environment. However, despite the vast potential of AI and data analytics, many organizations continue to struggle with harnessing these technologies to

support informed decision-making. The sheer volume and velocity of data, coupled with the complexity of business environments, have created an information overload that overwhelms decision-makers, making it difficult to extract actionable insights. Furthermore, traditional decision-making approaches often rely on intuition, anecdotal evidence, and limited data analysis, leading to suboptimal decisions. As a result, organizations are failing to capitalize on the full potential of AI and data analytics, hindering their ability to make timely, accurate, and informed decisions. This research aims to investigate the challenges and opportunities of integrating AI and data analytics into decision-making processes, exploring the benefits and limitations of these technologies in supporting informed decision-making.

OBJECTIVES OF THE STUDY

1. To evaluate the influence of AI and data analytics on decision-making speed and accuracy in management.
2. To identify the key AI technologies and data analytics methods used in management decision-making.
3. To examine the challenges faced by organizations in adopting AI and data analytics for decision-making.

CONTENTS OF “AI AND DATA ANALYTICS IN DECISION MAKING”

Introduction

The integration of Artificial Intelligence (AI) and data analytics has transformed the decision-making landscape, enabling businesses to make informed, strategic decisions. In today's fast-paced, data-driven world, organizations are constantly seeking ways to unlock hidden insights, predict future outcomes, and optimize operations. AI-powered data analytics has emerged as a game-changer, providing businesses with the ability to analyze vast amounts of data, identify patterns, and make predictions.

Importance of AI and Data analytics in Decision Making

In today's data-driven world, Artificial Intelligence (AI) and Data Analytics play a vital role in enhancing decision-making processes in management. These technologies help businesses analyze vast amounts of data, identify trends, and make informed decisions that improve efficiency, reduce risks, and drive growth.

- 1. Data-Driven Decision-Making** – AI eliminates guesswork by analyzing vast amounts of data, ensuring accurate and objective decisions.
- 2. Faster and Real-Time Insights** – AI processes data instantly, enabling managers to respond quickly to market changes and business challenges.
- 3. Predictive and Strategic Planning** – AI forecasts trends, risks, and opportunities, allowing businesses to plan proactively.
- 4. Operational Efficiency and Cost Reduction** – AI automates tasks, optimizes resources, and minimizes waste, improving overall efficiency.
- 5. Enhanced Risk Management** – AI detects fraud, cybersecurity threats, and financial risks, helping businesses take preventive actions.

How AI and Data Analytics in Decision-Making hep Management Transformation

AI and Data Analytics are revolutionizing management decision-making by shifting organizations from traditional, intuition-based strategies to data-driven, intelligent, and automated decision-making processes. These technologies help leaders make faster, more accurate, and predictive decisions, transforming business operations and strategic management.

1. Shift from Reactive to Proactive Decision-Making

- Traditional management relied on historical data and intuition, often leading to reactive decision-making.
- AI-powered analytics enables predictive decision-making, allowing managers to anticipate market trends, risks, and opportunities before they arise.

2. Faster and More Accurate Decision-Making

- AI automates data collection, analysis, and reporting, enabling real-time decision-making.
- Machine learning algorithms improve accuracy by identifying complex patterns that humans might overlook.

4. Enhanced Risk Management and Fraud Detection

- AI detects anomalies in financial transactions, cybersecurity threats, and operational risks, allowing businesses to take preventive actions.
- AI-driven risk assessments provide more accurate credit scoring, investment predictions, and policy enforcement.

5. Data-Driven Decision-Making for Competitive Advantage

- AI provides real-time insights into market conditions, customer behavior, and business performance.
- Businesses gain a competitive edge by making informed, data-backed strategic decisions faster than competitors.

6. Personalized Customer Experiences and Engagement

- AI analyzes customer data to provide personalized recommendations, targeted marketing campaigns, and improved customer service.
- Companies can tailor products and services to meet customer needs, increasing engagement and loyalty.

7. Cost Reduction and Resource Optimization

- AI-powered analytics optimize supply chain management, workforce planning, and budget allocation, reducing waste and costs.
- Predictive maintenance prevents equipment failures and downtime, improving efficiency.

8. Real-Time Performance Monitoring and Decision Adjustments

- AI enables real-time tracking of key performance indicators (KPIs), helping businesses quickly adjust strategies for better outcomes.
- AI-powered dashboards provide interactive visualizations for data-driven decision-making.

9. Smarter Human Resource Management

- AI enhances hiring, performance evaluation, and workforce planning by analyzing employee data and industry trends.
- Predictive analytics helps businesses retain top talent and improve employee engagement.

10. Driving Innovation and Business Transformation

- AI enables businesses to explore new business models, digital transformation strategies, and product innovations.
- Organizations using AI for continuous improvement stay ahead in dynamic industries.

BENEFITS OF AI AND DATA ANALYTICS IN DECISION MAKING

- AI and data analytics help managers make data-driven decisions by analyzing vast amounts of structured and unstructured data. Predictive analytics can forecast market trends, financial risks, and customer behaviors, reducing uncertainties and improving decision accuracy. This minimizes the chances of errors and helps in risk management.
- AI-powered analytics tools process data in real time, enabling quick and informed decision-making. Business Intelligence (BI) systems provide dashboards and reports that allow managers to react promptly to changes in market conditions, customer demands, or operational challenges. This agility gives companies a competitive advantage.
- AI automates routine decision-making tasks, reducing the need for manual analysis. Machine learning models can optimize supply chains, resource allocation, and financial planning, leading to reduced operational costs and improved efficiency. Organizations can allocate resources more effectively, maximizing profitability.
- AI-driven analytics enable organizations to personalize customer experiences and optimize marketing strategies. By analyzing consumer preferences, purchase behaviors, and engagement patterns, businesses can tailor products and services to specific customer segments, increasing customer satisfaction and revenue.
- Companies leveraging AI and data analytics gain a strategic edge over competitors by identifying emerging trends, market opportunities, and innovative business models. AI helps organizations experiment with new strategies, simulate different business scenarios, and drive continuous innovation, keeping them ahead in dynamic industries.

CHALLENGES OF AI AND DATA ANALYTICS IN DECISION MAKING

1. Data Quality and Integration Issues

- AI and data analytics rely on high-quality data for accurate decision-making. However, many organizations struggle with inconsistent, incomplete, or inaccurate data from multiple sources. Integrating data from different departments, systems, and formats can be complex, leading to errors that affect decision-making reliability.

2. Ethical Concerns and Bias in AI Algorithms

- AI models can inherit biases from the data they are trained on, leading to unfair or discriminatory decisions. For example, biased hiring algorithms may favor certain candidates based on historical hiring patterns. Ethical concerns related to data privacy, transparency, and accountability make it challenging for organizations to ensure fair decision-making.

3. Lack of Skilled Talent and AI Literacy

- Many managers and employees lack the necessary AI and data literacy to interpret analytical insights effectively. Organizations often face a skills gap in hiring professionals with expertise in machine learning, data science, and AI-driven decision-making. Without proper understanding, AI recommendations may be misinterpreted or misused.

4. High Implementation Costs and Resource Requirements

- Deploying AI and advanced analytics requires significant investment in infrastructure, software, and expertise. Small and medium-sized enterprises (SMEs) may struggle with the high costs of acquiring AI tools, maintaining data storage systems, and training employees, making adoption difficult.

5. Resistance to Change and Organizational Culture

- Adopting AI-driven decision-making requires a shift in organizational culture, which can be met with resistance from employees and managers. Decision-makers who are used to traditional methods may be hesitant to rely on AI insights, fearing loss of control or job displacement. Successful adoption requires strong leadership and change management strategies.

Conclusion

In this study highlights the AI and data analytics have revolutionized decision-making in management by enabling data-driven strategies, reducing uncertainty, and improving efficiency. By leveraging AI-powered predictive analytics, business intelligence tools, and big data processing, organizations can make more informed, accurate, and strategic decisions.

AI automates routine tasks, identifies patterns, and provides real-time insights, allowing managers to focus on innovation and long-term planning. Data analytics, on the other hand, ensures that decisions are based on facts rather than intuition, reducing risks and improving business outcomes. While AI and data analytics enhance decision-making, human expertise remains essential in interpreting insights, addressing ethical considerations, and ensuring responsible AI adoption. The integration of AI and analytics into management will continue to evolve, driving competitive advantage and sustainable growth in the digital era.

ANALYZE THE DIGITAL LEARNING AND AI IN EDUCATION

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ABSTRACT

The Changes in the demands of education require innovation and creativity in the learning process. With the development of Artificial Intelligence (AI) in the field of education to help process daily activities including teaching and learning. The objective of this study is to investigate Artificial Intelligence (AI) in education, especially in the teaching and learning process. This research uses library research. The result analysis shows that AI has been widely applied to various educational technology platforms such as virtual Aided Instruction (ICAI). Artificial Intelligence (AI) is the process of modeling human thinking and designing a machine so it can behave like humans. In the future progress of science and technology, teachers' work such as correcting, student attendance, giving daily tests and exams, explaining knowledge, making administrative reports and other systemic work can be submitted to be completed by technology devices. Teachers can save more energy and can focus more on non-systemic work to create a golden generation with more character and quality with natural intelligence where robots cannot do it. Technology only runs systemically and is automated based on human commands, while the human mind, especially teachers deliver new knowledge. Therefore, the teacher's intelligence will be unmatched. AI that emerged as the industrial revolution is also the result of the creative minds of human natural intelligence. So when compared, between the two will never have an equal position.

Keywords : Digital Learning, Artificial Intelligence (AI) in Education, E-learning, Personalized Learning, Adaptive Learning Systems.

INTRODUCTION

Today's technology has become an unavoidable part of the passage of time. Technology has not only changed people's lifestyles but has also changed how we work, learn, and interact. Various kinds of innovations appear all the time, making our activities and work more practical and effective. A more recent technological development is the emergence of the term artificial intelligence which is abbreviated as AI (artificial intelligence) which is currently starting to steal attention as a tool to act like humans. In its development, artificial intelligence has also penetrated the world of education. AI systems allow people to learn with the help of education assistants such as bots. The development of the times requires the world of education to adapt to technological developments to improve the quality of education, especially the adjustment of information and communication technology.

Digital learning content that is developing today can be presented thanks to the application of AI. Thick textbooks can now be presented into content that is more concise, easier to read and understand by students, such as study guides, material summaries, or short notes. AI as a pillar of the industrial revolution 4.0 plays a central role in facilitating the learning process mediated by technology. Artificial Intelligence (AI) is the process of modeling human thinking and designing a machine so that it can behave like humans or other terms called cognitive tasks, namely how machines can learn automatically from programmed data and information. Artificial intelligence can also be interpreted Artificial intelligence or AI is one part of computer science that makes machines (computers) able to do work as and as well as humans do. The use of Artificial Intelligence consciously or not we have applied it in everyday life.

Many applications have implemented artificial intelligence as an advantage of these applications. Artificial intelligence technology referred to here refers to machines that can think, weigh the actions to be taken, and can make decisions as humans do. Artificial intelligence (AI) is currently being developed on a large scale so that this technology will imitate and even take over the work normally done by humans. Based on the definition of AI, it was created to be able to act like humans in the form of programs and robots. Of course to facilitate human work. Even various digital platforms have used AI as a part of it. AI is used to make things easier for humans to do. Several technology companies have implemented AI including Amazon, Facebook, Microsoft, and Google.

Artificial intelligence or (AI) is a technology where machines can learn and understand logic like humans. This technology is said to be able to help simplify human life which is very complex (Fitria, 2021a). AI itself works by combining the presence of several data, iterative processing, and intelligent algorithms. This allows the software to learn automatically from patterns or features in the data. AI can also be said to be a very broad field of study. The scope of theories, methods, technologies, and subfields that exist in AI is very wide, including machine learning, neural networks, cognitive computing, computer vision, and scientific language processing. The role of Artificial Intelligence (AI) technology is increasingly evident in various sectors, including the education sector.

The presence of AI technology has transformed the educational curriculum, especially in the fields of technology, science, mathematics, and engineering. But AI will also change the face of the world of education as a whole. One technology that has recently received attention is Artificial Intelligence (AI). This technology has an important role in facilitating various job functions, including in the field of education. AI can also be implemented in the world of education. Teachers/lecturers can understand student needs more easily and more deeply (Fitria, 2021b). The students also can learn according to their needs without encountering difficulties.

Artificial Intelligence (AI) is believed to be able to help humans learn better and achieve educational goals more effectively. So it's not surprising that currently many AI-based innovations and breakthroughs are being and will be applied to support the learning process to make it more practical and effective. So when AI is present in the education sector, which then raises concerns by teachers, it is a challenge that must be faced so that the existence of education continues.

Some of the strong beliefs that teachers cannot be replaced by AI are teacher collaboration with AI in the implementation of learning. Teachers need to have skills in utilizing science and technology (Science and Technology). So that teachers can take advantage of AI in terms of completing school administration such as making lesson plans, student attendances reporting student learning outcomes, making learning media and learning resources. The researcher is interested to investigate Artificial Intelligence (AI).

Therefore, the objective of this study is to investigate Artificial Intelligence (AI) in education, especially in the teaching and learning process.

Education has evolved from traditional classroom-based teaching to digital and AI-driven learning environments. With the rise of online platforms, smart classrooms, and AI-powered tools, the education sector is experiencing a paradigm shift. This study aims to analyze how digital learning and AI contribute to better learning outcomes, student engagement, and education accessibility.

AI can be defined as “automation based on associations.” When computers automate reasoning Based on associations in data (or associations deduced from expert knowledge), two shifts Fundamental to AI occur and shift computing beyond conventional edtech: (1) from capturing Data to detecting patterns in data and (2) from providing access to instructional resources to Automating decisions about instruction and other educational processes.

Detecting patterns and Automating decisions are leaps in the level of responsibilities that can be delegated to a computer System. The process of developing an AI system may lead to bias in how patterns are detected And unfairness in how decisions are automated. Thus, educational systems must govern their use Of AI systems. This report describes opportunities for using AI to improve education, recognizes Challenges that will arise, and develops recommendations to guide further policy development.

The U.S. Department of Education (Department) is committed to supporting the use of technology to improve teaching and learning and to support innovation throughout educational Systems. This report addresses the clear need for sharing knowledge and developing policies for “Artificial Intelligence,” a rapidly advancing class of foundational capabilities which are Increasingly embedded in all types of educational technology systems and are also available to the public.

Key aspects of digital learning:

1.Accessibility:

Enables learning anytime, anywhere, across diverse devices, breaking geographical barriers.

2.Variety of formats:

Offers diverse content like videos, interactive simulations, quizzes, and virtual labs, catering to different learning styles.

3.Collaboration tools:

Facilitates online interaction between students and teachers through forums, chat rooms, and video conferencing.

4.Adaptive learning platforms:

Adjusts content difficulty and pace based on individual student progress.

How AI enhances digital learning:

5.Personalized learning:

Analyze student data to identify strengths and weaknesses, creating customized learning pathways and providing targeted feedback.

6.Intelligent tutoring systems:

Mimic human tutors by providing real-time guidance, explanations, and practice questions.

7.Automated grading:

Quickly assesses assignments, freeing up teacher time for more focused interactions.

8.Predictive analytics:

Identifies students at risk of falling behind, allowing for early intervention strategies.

9.Chatbots and virtual assistants:

Provide immediate support and answer student queries.

REVIEW OF LITERATURE

1.Digital learning, encompassing online platforms, learning management systems (LMS), and mobile applications, has redefined traditional learning environments. According to Siemens (2005), the theory of connectivism emphasizes how digital technology facilitates

knowledge acquisition through interconnected networks. Research by Means et al. (2013) found that students in blended learning environments performed better than those in traditional classrooms.

2.AI in Education: Personalization & Automation AI has introduced intelligent tutoring systems, adaptive learning platforms, and automated grading, significantly enhancing education. Woolf(2010)emphasizes that AI tutors can offer personalized feedback, allowing students to learn at their own pace.

3.Benefits of AI and Digital LearnpacPersonalized Learning: AI tailors content based on student needs (Luck in et al., 2016).Increased Accessibility Digital tools bridge the educational gap for students with disabilities (Rose et al., 2005).Efficiency in Education AI reduces teacher workload by automating administrative tasks (Selwyn, 2019).

4.Despite the benefits, AI and digital learning pose challenges. Selwyn (2019) and Holmes et al. (2021). Data Privacy & Security AI relies on student data, raising ethical concerns. Digital Divide Unequal access to technology affects students from low-income backgrounds.Teacher Adaptation Educators require training to effectively integrate AI in tools.

STATEMENT OF THE PROBLEM

Highlight the challenges of effectively integrating AI technologies into existing educational systems, including concerns around equitable access to technology, potential for algorithmic bias, privacy issues with student data, the need for teacher training to effectively utilize AI tools, and the potential for AI to replace human interaction, potentially impacting student engagement and holistic developments.

OBJECTIVES OF THE STUDY

□ To analyze the impact of digital learning on education understand how technology enhances teaching and learning experiences.

CONTENTS OF DIGITAL LEARNING AND AI EDUCATION

Introduction

Digital learning has transformed traditional education by integrating technology into teaching and learning processes. It encompasses online courses, e-learning platforms, virtual classrooms, and digital resources that make education more accessible, engaging, and personalized. With the rise of artificial intelligence (AI), education has seen further advancements, enabling smart tutoring systems, automated assessments, and adaptive learning experiences tailored to individual students' needs.

What is Digital Learning?

Digital learning refers to the use of digital tools, the internet, and multimedia resources to enhance learning experiences. It includes:

☐ Online Courses & E-Learning

Platforms like Coursera, Udemy, and Khan Academy offer interactive courses.

☐ Virtual Classrooms

Video conferencing tools like Zoom and Google Meet facilitate remote learning.

☐ Educational Apps & Gamification

Apps such as Duolingo and Byju's make learning engaging.

☐ Digital Content & E-Books

Digital libraries and resources replace traditional textbooks.

AI IN EDUCATION

AI enhances digital learning by offering:

☐ Personalized Learning

AI analyzes students' progress and customizes lessons accordingly.

☐ Smart Tutoring Systems

AI tutors provide real-time feedback and assistance.

☐ Automated Assessments

AI-powered grading systems reduce workload for educators.

☐ Chatbots & Virtual Assistants

AI-driven bots answer student queries instantly.

☐ Data-Driven Insights

AI helps institutions track student performance and improve curriculum design.

CHALLENGES OF AI IN DIGITAL LEARNING

Although AI has generally been a positive force within digital learning, there is still a lot unknown about how effective its application is for digital learning. As with many novel technologies, it does come with some challenges and has not escaped some myths being created about its impact. We'll look at three of the main challenges of AI in digital learning.

☐ Ethics

One challenge of AI in digital learning revolves around the ethical and transparent use of data. As AI-driven machine learning algorithms operate by using learner data, this has raised concerns around data storage and protection. Another ethical issue associated with AI in digital learning is the fallibility. Since machine learning intelligence is based on pattern recognition, it lacks the discernment of human intelligence. Therefore, there is a risk that AI could unfairly discriminate or make inappropriate recommendations. For this reason, the ethics around accountability and liability are another challenge that must be addressed.

☐ Skills gaps

Although many AI-powered digital learning tools are relatively user-friendly and don't require specialised skills, they are nevertheless a new type of technology that may require training and onboarding to use. Therefore, one challenge of AI in digital learning is equipping teachers, course facilitators, and L&D teams with the knowledge to leverage the most benefit from AI.

As some professionals may be apprehensive of these novel types of technologies, an effective approach to change management is helping them to understand the time-saving benefits that can come from tools, such as chatbots and task automation.

☐ Data system capabilities

L&D teams within corporations may already have access to sufficient employee data for AI processes. However, when it comes to gathering data for public education purposes, this can be much more challenging. Since you're dealing with larger data sets and inconsistent procurement, this makes it difficult for AI technologies that rely on data to function. So when it comes to reaping the most benefit for decentralised education systems, this is still an ongoing challenge.

BENEFITS OF AI IN DIGITAL LEARNING:

☐ Accessibility

One of the most profound benefits of AI in digital learning is the impact that it has on making learning more accessible. With translation tools, chatbots and tools that provide descriptions of visual content, AI is making learning more accessible for neurodivergent learners, and those with hearing or visual impairments. However, AI isn't only making learning more accessible for learners with differing needs, but also learners in different locations. AI makes it possible to learn from wherever you are in the world, without having to enter a classroom.

☐ Improved learner experience

It's well known that emotions play an important role in the ability to learn and retain knowledge. Therefore, it's vital to ensure that learners are motivated, engaged, and

overall have a positive learning experience. And AI-powered tools can contribute greatly to the enhancement of the learning environment. With content personalised to learners' needs and the ability to get assessment scores and answers to questions quicker than ever, learners benefit from an overall improved experience.

☐ Personalised learning

One of the main advantages of using AI in digital learning is having the ability to more effectively and efficiently personalise learning experiences. AI-powered technology can create tailored learning experiences in a way that an instructor would never have the time to be able to. This technology can create personalised learning paths, curate and recommend content and address knowledge gaps or difficulties. In turn, this personalisation helps create a better learner experience and also improve the learning outcomes.

☐ Save time

A further benefit of AI in digital learning is the amount of time it can save for instructors and other digital learning professionals. In a traditional learning environment, instructors previously had to spend hours arduously grading exams and answering student queries. But with AI, these tedious tasks are able to be automated. AI-powered tools can grade assessments, recognise and intervene if learners are struggling, and answer common learner queries with chatbots.

□ Greater insights

We're currently living in the Age of Big Data. And this increased access to extensive data has touched nearly every industry – including digital learning. As we previously mentioned, machine learning has created even further opportunities to benefit from data. This AI-driven data analysis method involves computers essentially learning from data patterns. Because of the ability to make predictions about future learning outcomes and learners' needs, machine learning has become an invaluable tool within digital learning.

HOW IS ARTIFICIAL INTELLIGENCE CHANGING THE FUTURE OF EDUCATION?

Artificial Intelligence (AI) is transforming the future of education by making learning more personalized, accessible, and efficient. Here's how AI is shaping education:

1. Personalized Learning – AI adapts lessons based on individual student needs, helping learners progress at their own pace.
2. Automated Assessment – AI-driven tools provide instant feedback and grading, reducing teacher workload.
3. AI-powered Tutoring – Virtual tutors and chatbots offer 24/7 assistance, making learning more interactive.
4. Enhanced Accessibility – AI supports students with disabilities through speech recognition, text-to-speech, and other assistive technologies.
5. Data-Driven Insights – AI analyzes student performance, predicting learning gaps and suggesting improvements.
6. Immersive Learning with AR/VR – AI enhances virtual and augmented reality experiences, making education more engaging.
7. AI in Administrative Tasks – Automating scheduling, attendance, and other tasks allows educators to focus on teaching.
8. Smart Content Creation – AI generates study materials, quizzes, and summaries, improving learning efficiency.
9. AI-powered Career Guidance – AI helps students choose careers based on skills, interests, and job market trends.
10. Global and Inclusive Education – AI-powered translation and speech tools break language barriers, making education accessible worldwide. AI is not replacing teachers but enhancing their capabilities, ensuring adaptive, inclusive, and data-driven learning experiences for students worldwide.

CONCLUSION

The existence of artificial intelligence may be able to provide knowledge to students, but developing character cannot be done. That is an educator's job. How to inspire, motivate, make students become good students." So the role of the teacher in providing motivation, inspiration, and developing character are what AI cannot replace because AI is not given feelings and emotions like humans in general. In the end, if we look at technological developments, we must be able to adapt as technology advances. If we do not adjust, we are an educator (teacher/lecturer) may be replaced by technology.

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DIGITAL LEARNING AND AI IN EDUCATION

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ABSTRACT

This study explores the role of Artificial Intelligence (AI) in digital learning, highlighting its benefits, challenges, and future implications. AI-driven educational tools enhance personalization, engagement, and accessibility, allowing students to learn at their own pace with real-time feedback. The study, based on secondary data sources, examines how AI improves learning outcomes while addressing challenges such as data privacy, high implementation costs, and reduced human interaction. Findings suggest that AI can revolutionize education if integrated effectively with traditional teaching methods. The study concludes that strategic implementation, ethical considerations, and proper training for educators are essential for maximizing AI's potential in education.

Keywords: Artificial Intelligence, Personalized Education, Student engagement, Education technology, Data privacy.

INTRODUCTION

The rapid advancement of technology has transformed education, with Artificial Intelligence (AI) playing a crucial role in digital learning. AI-powered tools enhance personalized learning, automate administrative tasks, and provide data-driven insights to improve student outcomes. Digital learning platforms integrated with AI offer adaptive learning experiences, ensuring students receive content tailored to their needs and progress. Through intelligent tutoring systems, chatbots, and AI-driven assessments, students can receive instant feedback, helping them grasp complex concepts more effectively.

In addition to enhancing student learning, AI supports educators by streamlining tasks such as grading, lesson planning, and tracking student performance. AI-powered analytics help teachers identify learning gaps and provide targeted interventions, making education more efficient. However, while AI has the potential to revolutionize education, challenges such as data privacy, ethical concerns, and the digital divide must be addressed to ensure equitable access for all learners.

Despite its advantages, AI in education also presents challenges, including concerns over data privacy, ethical use, and accessibility disparities. While AI enhances engagement and efficiency, reliance on technology raises questions about the diminishing role of human interaction in education. This study explores the impact of AI on digital learning, examining both its benefits and challenges. By analyzing secondary data sources, the research aims to provide insights into AI's role in shaping the future of education and offer recommendations for its effective implementation.

STATEMENT OF THE PROBLEM

The rapid integration of digital learning and Artificial Intelligence (AI) in education is transforming traditional teaching methods, making learning more accessible, personalized, and efficient. However, challenges such as unequal access to technology, data privacy concerns, teacher adaptation, and the risk of reduced human interaction pose significant barriers to its sustainability. Ensuring the effectiveness of AI-driven education requires balanced integration, ethical considerations, and continuous innovation. This study explores

the challenges and opportunities of digital learning and AI in education, aiming to enhance its impact while maintaining educational quality and inclusivity.

OBJECTIVES OF THE STUDY

1. To assess the effectiveness of AI-driven digital learning in improving student engagement and academic performance.
2. To identify challenges related to accessibility, teacher preparedness, and ethical concerns in AI-based education.
3. To evaluate the impact of the digital divide on students' learning opportunities and outcomes.

CONTENTS OF “DIGITAL LEARNING AND AI IN EDUCATION”

Introduction

Digital learning and Artificial Intelligence (AI) are transforming education by making learning more personalized, efficient and accessible. AI-powered tools, such as virtual tutors, adaptive learning platforms and automated grading systems, enhance student engagement and academic performance. Digital learning enables flexible, self-paced education through online courses, interactive simulations and multimedia content.

Opportunities for Digital Learning and AI in Education

1. **Personalized Learning Experiences** – AI can tailor lessons based on individual student needs, helping them learn at their own pace.
2. **Global Access to Education** – Digital learning removes geographical barriers, making quality education available to students worldwide.
3. **AI-Powered Tutoring and Support** – Virtual tutors and chatbots provide instant assistance, improving student understanding.
4. **Enhanced Teacher Support** – AI automates grading, attendance, and administrative tasks, allowing teachers to focus on instruction.
5. **Data-Driven Insights** – AI analyzes student performance to identify learning gaps and suggest targeted interventions.
6. **Immersive Learning with AR/VR** – AI-powered simulations, virtual reality (VR), and augmented reality (AR) make complex subjects more interactive.
7. **Inclusion and Accessibility** – AI-driven assistive technologies help students with disabilities by providing speech-to-text, text-to-speech, and real-time translations.
8. **Lifelong Learning and Skill Development** – AI-based courses help professionals and students upskill and reskill continuously.
9. **Cost-Effective Education** – Digital learning reduces the need for physical materials, making education more affordable.
10. **AI in Assessments and Examinations** – Automated testing and proctoring ensure fair and efficient evaluation of student progress.

Tools for Digital Learning and AI in Education

1. Learning Management Systems (LMS)
Examples: Moodle, Blackboard, Google Classroom.
Helps educators manage courses, assignments, and student progress online.
2. AI-Powered Adaptive Learning Platforms
Examples: DreamBox, Knewton, Squirrel AI
Provides personalized learning experiences based on student performance and learning pace.

3. Virtual Tutors and Chatbots
Examples: ScribeSense, Querium, Watson Tutor
Offers instant support and explanations to students through AI-powered tutoring.
4. Automated Grading and Assessment Tools
Examples: Gradescope, Turnitin, EvalAI
Uses AI to assess assignments, detect plagiarism, and provide feedback.
5. Speech-to-Text and Text-to-Speech Assistive Tools
Examples: Google Speech-to-Text, Microsoft Immersive Reader
Helps students with disabilities by converting speech to text and vice versa.
6. AI-Powered Language Learning Apps
Examples: Duolingo, Rosetta Stone, Babbel
Uses AI to personalize language learning based on user progress.
7. Virtual and Augmented Reality (VR/AR) Tools
Examples: Google Expeditions, zSpace, EON Reality
Creates immersive learning experiences for subjects like science, history, and medicine.

Benefits of Digital Learning and AI in Education

1. Personalized Learning

- AI adapts to students' learning styles and pace, providing customized lessons.
- Helps students focus on areas where they need improvement.

2. Enhanced Student Engagement

- Digital learning platforms use interactive content like videos, quizzes, and gamification.
- AI-powered tools such as virtual tutors and chatbots keep students engaged.

3. Real-Time Feedback and Assessment

- AI provides instant feedback on assignments, quizzes, and exams.
- Automated grading saves teachers time and ensures accurate evaluations.

4. Accessibility and Inclusivity

- Digital learning supports students with disabilities through AI-driven assistive technologies (e.g., speech-to-text, text-to-speech).
- AI-enabled translation tools help students learn in different languages.

5. Flexible and Remote Learning

- Students can access learning materials anytime, anywhere, promoting self-paced learning.
- Online courses and AI-driven platforms make education more accessible.

6. Data-Driven Decision Making

- AI analyzes student performance data to identify learning gaps.
- Helps educators create targeted interventions to improve student outcomes.

7. Efficient Administrative Support

- AI automates administrative tasks like attendance tracking, scheduling, and record-keeping.
- Reduces the workload for educators, allowing them to focus on teaching.

8. Cost-Effective Education

- Digital learning reduces the need for physical materials, lowering costs for students and institutions.

- AI-driven platforms provide affordable learning solutions compared to traditional classroom setups.

Impact of AI on Student Engagement and Performance

1. Personalized Learning

- AI adapts to individual learning styles and paces, providing customized content.
- Helps students focus on areas where they need improvement.

2. Real-Time Feedback and Support

- AI-driven tutors and chatbots offer instant responses to student queries.
- Reduces frustration and enhances learning retention.

3. Gamification and Interactive Learning

- AI-powered quizzes, simulations, and virtual reality (VR) make learning engaging.
- Increases motivation and improves knowledge retention.

4. Predictive Analytics for Academic Success

- AI analyzes student performance data to identify learning gaps.
- Helps educators provide targeted interventions for struggling students.

5. Automation of Administrative Tasks

- AI automates grading, attendance tracking, and lesson planning.
- Allows teachers to focus more on interactive teaching.

6. Increased Accessibility and Inclusivity

- AI tools support students with disabilities (e.g., text-to-speech, speech-to-text).
- Enables remote learning and breaks language barriers.

7. Challenges and Considerations

- Over-reliance on AI may reduce critical thinking and problem-solving skills.
- Data privacy concerns and the digital divide need to be addressed.
- Proper AI training for educators is essential for effective implementation.

Challenges in AI-Based Learning

1. Data Privacy and Security Concerns

- AI systems collect vast amounts of student data, raising privacy risks.
- Schools must implement strong cybersecurity measures to prevent data breaches.

2. High Implementation Costs

- AI-driven educational tools require significant investment in infrastructure and training.
- Many schools, especially in developing regions, struggle with affordability.

3. Digital Divide and Accessibility Issues

- Not all students have equal access to AI-powered learning tools.
- Internet connectivity and device availability remain challenges in remote areas.

4. Reduced Human Interaction

- Over-reliance on AI may limit social learning, teamwork, and critical thinking.

- Human educators play a crucial role in emotional and cognitive development.

5. Bias and Ethical Concerns

- AI algorithms may reinforce biases present in their training data.
- Ethical guidelines are needed to ensure fairness in AI-driven assessments.

6. Need for Teacher Training

- Many educators lack the technical skills to effectively integrate AI in teaching.
- Continuous professional development is essential for successful AI adoption.

7. Reliability and Accuracy Issues

- AI tools are not always 100% accurate in grading, feedback, or predictions.
- Errors in AI-generated content can mislead students without proper oversight.

Ethical Considerations in AI Education

1. Data Privacy and Security

AI systems collect and store student data, raising concerns about unauthorized access. Strong data protection policies and encryption are necessary to ensure student privacy.

2. Bias and Fairness in AI Algorithms

AI models can inherit biases from training data, leading to unfair grading or recommendations. Developers must ensure AI is trained on diverse and unbiased datasets.

3. Transparency and Accountability

AI decision-making should be transparent, so students and educators understand how it works. Schools should establish accountability measures for AI-driven assessments.

4. Equal Access and Digital Divide

Not all students have access to AI-powered learning tools due to economic disparities. Policies should promote equitable access to AI education for all learners.

5. Human Oversight and Responsibility

AI should assist, not replace, human educators in decision-making. Teachers must oversee AI-generated feedback to ensure accuracy and relevance.

6. Psychological and Social Impact

AI-driven learning should not replace human interaction, which is essential for emotional and social development. Ethical AI use must consider the well-being of students, preventing excessive screen time and isolation.

7. Intellectual Property and Plagiarism

AI tools must not encourage academic dishonesty by making it easy to generate automated assignments.

CONCLUSION

Digital learning and AI in education have transformed traditional teaching methods by enhancing accessibility, personalization, and engagement through advanced technologies. The benefits of AI-driven education include real-time feedback, automated assessments, and personalized learning experiences, positively impacting student performance. However, challenges such as data privacy concerns, high implementation costs, and the digital divide must be addressed to ensure equitable access. AI presents vast opportunities, including adaptive learning, immersive education, and global reach, supported by tools like LMS, virtual tutors, and AI-powered analytics. To overcome these challenges, institutions should invest in strong data security measures and provide comprehensive AI training for educators. Additionally, a balanced approach integrating AI with traditional teaching methods can enhance both efficiency and human interaction in education. By addressing these challenges effectively, AI and digital learning can continue to drive innovation and inclusivity in education.

AI ADAPT DIGITAL LEARNING IN EDUCATION

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Abstract

Digital learning systems and AI tools are shaking up student involvement and grades in ways that really matter. This study digs into how these tech setups change the classroom experience by mixing hard numbers with real student feedback from various schools. In many cases, the data shows a neat link between using these digital platforms—augmented by AI—and a boost in student participation, which then tends to lead to better academic results, especially for groups that often feel left out. Plus, when schools add in custom AI features, they end up crafting more personalized learning trips that cater to each student's style and unique hurdles. It's worth noting that these effects aren't just for traditional schooling; they hint that similar tech moves might also upgrade training in healthcare, improving teaching methods and helping medical pros sharpen their skills. Ultimately, this work makes a strong case for digital learning and AI having the power to build more inclusive, effective educational spaces, while setting the stage for more future studies on tech-powered learning in both education and healthcare. The overall takeaway pushes for fresh, innovative teaching tricks that can help bridge the gaps in learning access and quality—paving the way for a more knowledgeable and capable workforce in healthcare.

Key words: Digital Learning, Artificial Intelligence (AI), Ethical Guidelines, Technology Integration, Critical Thinking, Academic Performance and Student Engagement

Introduction

The advent of digital technology and artificial intelligence has completely changed the way we think about education. In general, these changes make things more individualized and interactive by allowing students to immerse themselves in new learning opportunities that depart from traditional approaches (M Zahid et al., 2024). With digital platforms appearing everywhere, it's worthwhile to take a look at how these resources can improve learning outcomes and include a variety of student perspectives. Upon closer examination, one may wonder how AI and digital tools actually affect student engagement and academic achievement in various contexts. Many studies support these technologies, but there is still much to learn about their interplay and the finer points of their routine use in classrooms (Ayao D et al., 2024; Kotsiubynskyi, Mykhailo, Bohatko, Vasyl, Olha. Even with the most promising developments, it appears that some details are overlooked, leaving us with knowledge gaps (Abbas N et al., 2023). Thus, the task of deciphering the various aspects of AI and digital learning in education is taken on in this dissertation. In essence, it involves determining how these advancements could be applied to fulfill the needs of every student and improve overall performance (Kamalov F et al., 2023). Adding a layer of accessibility and inclusion for students with diverse backgrounds and abilities, it also looks at how AI might be used to create learning experiences that feel customized (Varghese J et al., 2023; Jowarder MI, 2023). This study goes beyond theory to demonstrate the practical advantages of integrating digital tools and artificial intelligence (AI) into the classroom to develop flexible, adaptable learning environments that reflect the diversity of today's student body (Priksat V et al., 2023). The results are intended to encourage the development of sound frameworks and best practices for educators, administrators, and legislators who are eager to try new teaching strategies. By answering these important questions, the study not only

provides a picture of the current situation but also offers useful, occasionally grim, insights that may help shape education in our increasingly digital world in the future (Ayeni OO et al., 2024; Prakash A et al., 2023). These observations may ultimately aid in reshaping educational policy and strategy, ensuring that the incorporation of digital platforms and artificial intelligence is in line with the objective of providing equitable and efficient learning opportunities for all students worldwide (Suraworachet W et al., 2023), (Ashraf MA et al., 2021).

Literature Review

Digital learning and AI have really shaken up our classrooms lately. Tech isn't just a fancy add-on anymore – it's become a key player in making learning more engaging and accessible. Schools around the world are scrambling to address issues like fairness, access, and keeping students interested, and a lot of recent studies (Ayao D et al., 2024) show that digital platforms help by offering tailored resources and unique, personalized learning paths that older methods just don't match. Plus, when AI takes over tedious admin tasks, teachers can spend more time actually connecting with their students (Kotsiubynskyi, Mykhailo, Bohatko, Vasyi, Olha.). In many ways, mixing AI with digital learning not only smooths out everyday processes but also reaches a wider variety of learners, fostering a more inclusive and participatory environment (M Zahid et al., 2024). When you look at the research, a few major ideas keep popping up. One big win is how AI adapts lessons to fit each student's own pace and style, which generally boosts their performance (Kamalov F et al., 2023). There's also a trend toward gamifying lessons and making them more interactive, which seems to fire up student enthusiasm (Abbas N et al., 2023). But, of course, not everything's rosy – many experts point out that folks in underserved communities still struggle with access to basic tech, and this digital divide might end up favoring students from more affluent backgrounds (Varghese J et al., 2023)(Jowarder MI, 2023). It's interesting, too, that while many researchers cheer the benefits of these digital tools, there's a noticeable gap when it comes to ethical issues. For instance, protecting data privacy and tackling biases built into AI systems are still major concerns (Priksat V et al., 2023). We don't yet have solid, comprehensive guidelines to deal with these dilemmas, and without them, our reliance on AI might lead to unexpected problems (Ayeni OO et al., 2024). Also, there aren't enough long-term, real-world studies that really track how these technologies perform over time (Prakash A et al., 2023). All in all, as tech continues to weave its way into education, it's crucial to keep a critical eye on both its exciting potential and its possible pitfalls. This review aims to piece together current research on digital learning and AI, spotlighting where the tech shines and where there are still too many unanswered questions. Essentially, it's about understanding both the promise and the challenges of these tools – and, as we move forward, figuring out the best path for integrating them so that every student benefits (Suraworachet W et al., 2023)(Ashraf MA et al., 2021)(Desirée Cranfield et al., 2021)(Chang Y et al., 2024)(Shuroug A Alowais et al., 2023)(Chan CKY et al., 2023)(Grassini S, 2023)(Sallam M, 2023)(N/A, 2023)(Yogesh K Dwivedi et al., 2023). Tech progress has completely revamped our old-school classrooms. Early on, researchers primarily focused on how everyday digital tools could be utilised in schools – showing that accessible online resources and platforms help boost learning, as seen in early studies by (Ayao D et al., 2024) and (Kotsiubynskyi, Mykhailo, Bohatko, Vasyi, Olha.). These initial efforts set the stage for recognizing how digital environments can enhance engagement and improve outcomes. Soon enough, as digital learning matured, scholars started highlighting how AI personalizes experiences. For instance, works like (M Zahid et al., 2024) and (Kamalov F et al., 2023) reveal that smart algorithms are now tailoring learning pathways to fit each student's needs, proving that technology isn't just about delivering information but about adapting to individual learning paces. Of late, the

discussion has broadened to include ethical and practical challenges of AI in classrooms. Researchers such as those mentioned in (Abbas N et al., 2023) and (Varghese J et al., 2023) are increasingly cautious about data privacy and bias issues, stressing that we need to use these tools responsibly. A range of studies (Jowarder MI, 2023) through (Ashraf MA et al., 2021) have begun to weigh both the successes and shortcomings of digital learning tools in various educational settings. Looking back, it's clear that digital learning and AI have grown from simple, supplementary resources into complex systems that invite us to reexamine our entire approach to teaching. Digging deeper, several core themes emerge from studies on digital learning and AI. One standout is the power of personalization—AI technologies tailor educational paths to meet each student's unique needs, which in turn drives engagement and boosts success rates (Ayao D et al., 2024)(Kotsiubynskyi, Mykhailo, Bohatko, Vasyl, Olha.). Another neat aspect is AI's ability to offer real-time feedback: by pinpointing when and where students stumble, it helps create a more dynamic classroom atmosphere (M Zahid et al., 2024)(Kamalov F et al., 2023). And then there's the collaborative edge—AI tools can analyze and improve group dynamics on projects, leading to better teamwork among students (Abbas N et al., 2023). However, we mustn't overlook the fact that a lack of access to such technology can further the existing disparities in learning environments (Varghese J et al., 2023)(Jowarder MI, 2023). Equally, teacher training is a vital element; educators need to build the skills to seamlessly integrate these innovations into their teaching practices (Priksat V et al., 2023)(Ayeni OO et al., 2024). Overall, while digital learning and AI hold remarkable promise, ensuring fairness and preparing teachers adequately are key to truly effective and inclusive implementation. When it comes to understanding how best to evaluate these innovations, both qualitative and quantitative methods offer useful insights. Qualitative research, for example, delves into personal experiences, with ethnographic studies shedding light on how digital tools really change classroom dynamics (Ayao D et al., 2024)(Kotsiubynskyi, Mykhailo, Bohatko, Vasyl, Olha.). On the other hand, number-driven studies have been pivotal in proving that specific AI applications—like adaptive learning systems—can deliver notable improvements over traditional methods (M Zahid et al., 2024)(Kamalov F et al., 2023). Blending both approaches gives us a fuller picture: not only do we see the statistical improvements, but we also understand the human side of technology in education. Mixed-method studies, which merge interviews with numerical data, reveal that success in digital learning isn't just about the tech itself but also about factors like institutional support and teacher training (Abbas N et al., 2023)(Varghese J et al., 2023). Several experts also call for longitudinal research to keep track of how these systems evolve over time (Jowarder MI, 2023)(Priksat V et al., 2023). All in all, employing a mix of research methods is key to fully grasping the complex interplay between digital learning and AI, guiding us toward better educational practices and policies. Different theoretical perspectives further enrich our understanding of digital learning and AI. Constructivist ideas—those that put the learner at the center—align with many studies suggesting that personalized, AI-driven educational tools can deepen learning experiences (Ayao D et al., 2024), (Kotsiubynskyi, Mykhailo, Bohatko, Vasyl, Olha.). Evidence shows that when AI provides instant feedback and customized resources, student engagement and achievement climb noticeably (M Zahid et al., 2024). Conversely, from a critical theory standpoint, some scholars caution that these digital innovations might not benefit everyone equally. Economic and social barriers could mean that only certain groups reap the rewards, potentially worsening educational disparities (Kamalov F et al., 2023), (Abbas N et al., 2023). There are also pressing ethical concerns about data privacy and algorithmic biases, which have led to calls for tighter scrutiny and regulation (Varghese J et al., 2023), (Jowarder MI, 2023). The socio-cultural perspective adds yet another twist, arguing that digital tools can't be divorced from the local cultural context – learning must be adapted to respect and integrate cultural

diversity (Priksat V et al., 2023), (Ayeni OO et al., 2024). Together, these varied viewpoints underscore the need for a balanced approach that both drives innovation and addresses critical issues of equity and ethics (Prakash A et al., 2023), (Suraworachet W et al., 2023). Wrapping things up, the research makes it pretty clear that digital learning and AI could transform traditional classrooms into dynamic, inclusive, and efficient learning spaces. Studies confirm that AI-powered adaptive systems can customize lessons to match individual learning styles and speeds, which generally leads to improved academic performance (Ayao D et al., 2024)(Kotsiubynskiy, Mykhailo, Bohatko, Vasyl, Olha.)(M Zahid et al., 2024)(Kamalov F et al., 2023). Adding gamification elements further boosts interactivity and student motivation (Abbas N et al., 2023). But there's a bigger picture too. This transformation isn't just about individual success – it's about creating a more inclusive educational environment for all learners. At the same time, serious concerns persist, especially regarding the digital divide that leaves some underserved communities at a disadvantage (Varghese J et al., 2023)(Jowarder MI, 2023). We must consider these issues carefully to ensure that no group gets left behind. Even though the promise of digital tools is enormous, current research still has its limitations. There are too few long-term empirical studies that monitor how these technologies perform across time in real-life classroom settings (Prakash A et al., 2023). Ethical challenges – like ensuring data privacy and mitigating inherent biases in AI – underline the urgent need for comprehensive frameworks to guide responsible use (Priksat V et al., 2023)(Ayeni OO et al., 2024). Many scholars are calling for more transparent research that factors in the varied socio-cultural contexts shaping these impacts (Suraworachet W et al., 2023)(Ashraf MA et al., 2021). Looking ahead, future studies should focus on longitudinal evaluations of digital learning and AI's integration, as well as on the effects of teacher training on their successful adoption (Priksat V et al., 2023)(Ayeni OO et al., 2024). And, most importantly, strategies to promote equitable access to these digital resources must be a priority. To sum it all up, the evidence suggests that while digital learning and AI have tremendous potential to reshape education by making it more personalized and interactive, we also need constant dialogue on issues of ethics, equity, and long-term effectiveness. As schools continue to blend these tools into everyday teaching, striking a balance between innovation and careful oversight is essential. The insights gathered here could well steer future educational policies, leading to richer and more inclusive learning experiences for every student (Desireé Cranfield et al., 2021)(Chang Y et al., 2024)(Shuroug A Alowais et al., 2023)(Chan CKY et al., 2023)(Grassini S, 2023)(Sallam M, 2023)(N/A, 2023)(Yogesh K Dwivedi et al., 2023).

Methodology

Educational tech is changing so fast, and more schools are leaning on AI every day. It's now crucial to figure out how these shifts affect both teaching and learning outcomes. A bunch of studies show that AI tailors learning by serving up custom content and feedback, speeding up engagement in ways that feel almost personal (Ayao D et al., 2024). Still, we're not completely clear on what these tools mean for classroom practices and for students' everyday experiences. We really need to check if AI tools boost educational setups while also dealing with the ethical, fairness, and access challenges that pop up (Kotsiubynskiy, Mykhailo, Bohatko, Vasyl, Olha.). This work takes a two-pronged approach. First, it asks whether digital learning platforms driven by AI can carve out unique, personalized routes to meet different needs. Then it digs into the risks—like hidden biases and a growing digital gap among different groups—that might come along (M Zahid et al., 2024). Using a mix of interviews and a gritty look at engagement numbers, the study sets out to give a full picture of how digital tools now shape education (Kamalov F et al., 2023). Each choice in how we study things is meant to shed light on both the upsides and the bumps when integrating AI

into everyday learning setups (Abbas N et al., 2023). It really shows that teaching methods need to switch gears to grab these new tech chances. This research isn't just high-minded theory; it offers solid pointers for teachers, policymakers, and tech developers on sorting out the messy world of digital learning and AI use (Varghese J et al., 2023). By leaning on methods that previous work has already shown to work, the study hopes to deliver strong evidence that can guide new teaching strategies and help ensure every student gets quality education (Jowarder MI, 2023). In the end, the approach lays down a basic framework for how school systems can adjust to our fast-changing digital world, hinting that we need ongoing research and constant tweaks in teaching styles (Priksat V et al., 2023). The ultimate claim is that the actionable strategies from this work will not only stir academic debate but also steer the smart rollout of AI tools to enhance learning experiences across a variety of settings (Ayeni OO et al., 2024)(Prakash A et al., 2023)(Suraworachet W et al., 2023)(Ashraf MA et al., 2021)(Desireé Cranfield et al., 2021)(Chang Y et al., 2024)(Shuroug A Alowais et al., 2023)(Chan CKY et al., 2023)(Grassini S, 2023)(Sallam M, 2023)(N/A, 2023)(Yogesh K Dwivedi et al., 2023).

Finding

The integration of digital learning tools and artificial intelligence (AI) in education has yielded significant insights into student engagement and academic performance. The research indicates that AI-driven educational platforms enhance personalized learning experiences, allowing students to receive tailored support that aligns with their individual learning styles. This personalization has been shown to improve academic outcomes, as evidenced by the positive feedback from students who reported increased motivation and engagement in technology-enhanced classrooms (Ayao D et al., 2024; Kotsiubynskyi et al., 2025). However, the findings also highlight disparities in access to these digital resources. Not all students have equal opportunities to benefit from AI technologies, particularly those from underserved communities. This digital divide raises concerns about equity in education, as students lacking access to technology may fall behind their peers (Abbas N et al., 2023; Varghese J et al., 2023). The study emphasizes the need for educators to implement strategies that ensure all students can leverage the advantages of AI and digital learning tools. Moreover, while the use of AI in education presents numerous benefits, there are potential drawbacks to consider. Some participants expressed concerns that an over-reliance on AI could diminish critical thinking skills, suggesting that a balanced approach is necessary (Kamalov F et al., 2023). Educators must be trained to effectively integrate these technologies into their teaching practices without compromising essential pedagogical values (Priksat V et al., 2023). The findings underscore the importance of ongoing professional development for teachers to navigate the evolving landscape of digital education. As AI continues to shape educational practices, it is crucial to establish ethical guidelines and frameworks that address issues of bias and data privacy (Ayeni OO et al., 2024; Prakash A et al., 2023). Overall, the research advocates for a thoughtful and inclusive approach to integrating AI in education, ensuring that all students can thrive in a technology-enhanced learning environment.

Conclusion

Digital learning and AI are rapidly changing classroom dynamics; recent work shows these tools can spark student involvement, create personalized study paths, and tweak everyday teaching routines. A closer look reveals that smart tutoring systems and custom learning platforms work wonders at boosting grades and firing up learners' motivation – all while catering to each student's unique needs (Ayao D et al., 2024). When the study tackled the tricky task of blending these new methods into schools, it became clear that giving

everyone a fair shot and ensuring teachers get solid training are absolutely key for success (Kotsiubynskyi, Mykhailo, Bohatko, Vasyl, Olha.). The findings also hint that AI isn't just fine-tuning teaching methods—it's reworking how curricula are designed, which helps build a more inclusive, collaborative environment. In practice, the research nudges us toward steady professional development so that educators can harness AI effectively without letting core teaching values slip (M Zahid et al., 2024)(Kamalov F et al., 2023). Looking ahead, future projects should probably lean on longer studies to check how AI impacts diverse student groups and curriculum outcomes, all while flagging potential biases and ethical issues along the way (Abbas N et al., 2023). There's also a call for more hands-on investigations into hybrid models—where AI teams up with traditional methods—to figure out which approaches really work best (Varghese J et al., 2023). At the same time, as schools shift toward AI-powered setups, it's crucial that educators, tech experts, and policymakers join forces to craft fair, ethical guidelines for using these tools (Jowarder MI, 2023). Bringing together interdisciplinary teams to continuously monitor AI's performance might offer fresh insights into its evolving role, ideally shaping policies that keep pace with rapid innovation in learning (Priksat V et al., 2023). Overall, the study adds to a growing body of research that tries to map out responsible ways to adopt digital tools, stressing that future efforts must prioritize fairness, transparency, and effectiveness when leveraging AI for educational advancement (Ayeni OO et al., 2024). These recommendations also mirror the urgent need to prepare our teaching workforce for the challenges and opportunities AI brings, ultimately fostering an educational paradigm that enriches learning experiences for everyone involved (Prakash A et al., 2023). As the conversation around AI in education keeps shifting, pushing for ethical, inclusive, and evidence-based practices remains absolutely crucial (Suraworachet W et al., 2023).

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MARKETING STRATEGIES OF CASHEW CULTIVATION IN PUDUKKOTTAI DISTRICT

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Abstract: The study's objectives are to determine the cropping pattern, degree of acceptance of suggested cashew nut production technology, marketing strategies, and training needs. Additionally, it sought to determine how well various training programmes affected the cashew producers' perceived competence, symbolic adoption behaviour, and information acquisition. The study also looked at the challenges farmers encountered in selling and producing cashew nuts, with the goal of suggesting methods for producing cashew nuts in a sustainable manner.

Introduction

India is one of the world's leading producers of cashews. Since the cashew sector supports over 10 lakh jobs in rural industries and farms, it has significant economic impact. Over 0.8 million tonnes (MT) of cashew are produced yearly in India, where the crop is grown on 0.7 million hectares of land. India increased its production of cashew nuts from 0.70 million tonnes (MT) to 0.77 million tonnes (MT) during 2019–20 to 2021–2022. Cashew farming expanded throughout the Indian peninsula's coastal areas. States like Maharashtra, Kerala, Karnataka, Tamil Nadu, Andhra Pradesh, Goa, Odisha, West Bengal, including some areas of the northeast are the primary locations for cashew cultivation. The National Horticulture Board (NHB) released data showing that Maharashtra is the top state in terms of annual cashew nut production in 2021–2022, with 0.20 million tonnes (MT), up from 0.19 million tonnes in 2020–21. In addition to its extensive cashew output, India is renowned for being a pioneer in cashew processing and for exporting cashew kernels all over the world. Previously centred in Kollam, Kerala; Mangalore, Karnataka; Goa; and Vettapalam, Andhra Pradesh, the cashew processing sector is now dispersed over several Indian states. India has been the cashew industry's worldwide processing powerhouse throughout time.

With almost 15percent of global cashew exports, India is the world's top cashew exporter. India exports relatively little cashew nut shell liquid and mostly cashew kernels. Cashew exports increased in value from US\$ 420 million in 2021–2022 to US\$ 452 million in 2021–2022, with a 7 percent increase from April–August 2022, when cashew exports were valued at US\$ 134.38 million. India's cashew exports increased by 9 percent in volume, from 70.5 million kg in 2020–21 to 76.8 million kg in 2021–22. The amount of cashews exported from April to August of 2022 was 18 million kg. The nation exported cashews for US\$40.0 million in March 2022, up from US\$33.6 million in February 2022.

The expansion of the Indian cashew sector, which supplies about half of the nation's cashew kernel demand both domestically and internationally, has been significantly aided by the importation of raw cashew nuts. In response, the Department of Agriculture, Cooperation and Farmers Welfare (DAC&FW), operating under the auspices of the Rashtriya Krishi Yojana (RKVY) and the Mission for Integrated Development of Horticulture (MIDH), launched a number of programmes that boosted cashew output in the country. It entails greatly increasing the area under cashew production and, in both traditional and non-traditional states, substituting high-yielding cashew cultivars for ageing cashew farms. The Directorate of

Cashew Nut & Cocoa Development's (DCCD) roadmap programme to expand the cashew farming area by 1.20 lakh hectares has also been authorised by DAC&FW.

India is a global exporter of cashews, reaching more than 60 nations. UAE, Japan, Netherlands, Saudi Arabia, USA, UK, Canada, France, Israel, and Italy are among India's top export markets. UAE is the biggest buyer of Indian cashews, valued at US\$131.5 million, accounting for 29% of total exports during 2021–22 as opposed to US\$ 98.5 million the previous year, according to APEDA figures on the exports of cashew kernels and cashewnut shell liquid. India exported 16.6 million kg of cashews to the United Arab Emirates in volume terms, a 29% increase from the 12.8 million kg exported in 2021.

India's top three importers of cashews are Japan and the Netherlands, with corresponding export shares of 13% and 9%. India exports US\$ 58.1 million and US\$ 41.1 million worth of cashew kernels and cashewnut shell liquid to Japan and the Netherlands, respectively. Vietnam is one of India's main cashew importers; the country's cashew exports account for 84% of all exports, indicating the enormous importance of traditional markets. The main cashew-growing states in India are seeing economic growth and job creation as a result of the robust increase in cashew exports to all export destinations.

A number of measures have been made by the Indian government and the cashew export promotion council to facilitate exports and support the expansion of the cashew sector. Numerous trade delegations, buyer-seller events, fairs, development seminars, and research and development data are offered to exporters as non-financial help. In addition, in 2018 the Goods and Services Tax (GST) on raw cashew nuts was lowered from 12% to 5%, and the Basic Customs Duty was lowered from 5% to 2.5%.

- The Government of India has implemented several measures to encourage efficient sourcing, given that the cashew industry's exports and local demand are significantly reliant on imported raw cashew nuts. Changes in import rules for cashew kernels, both whole and cracked, are among them.
- The standard inputs output norms (SION) for cashew exports have been revised.
- The Medium-Term Framework programme for process automation and mechanisation of cashew processing facilities has been approved, requiring an expenditure of Rs 60 crore (US\$ 8 million).

Permitting the import of unprocessed cashew nuts from least developed countries (LDCs) duty-free under the Duty-Free Tariff Preference (DFTP) Scheme

Under the Market Access Initiative (MAI) plan, which facilitates accessing new markets, the government has also provided financial support to the Cashew Export Promotion Council of India (CEPCI) for the organisation of Buyer-Seller Meets (BSM) and participation in international fairs.

CEPCI - In order to increase the nation's exports of cashew kernels and cashew nut shell liquid, the Indian government established the Cashew Export Promotion Council of India (CEPCI) in 1955 with the active support of the cashew industry. The fundamental arrangement of the Council provides the institutional framework needed to accomplish the numerous activities that promote and support the export of cashew nut shell liquid and cashew kernels. The Council serves as a go-between for its member cashew kernel distributors and purchasers from other countries.

The council runs the government's five-year plan scheme for financial assistance to member exporters and gives trade data and statistics to its members.

In Tamil Nadu: Grown on 91,058 hectares, cashew nuts are one of Tamil Nadu's main plantation crops. Tamil Nadu produces 57,988 MT of cashews annually, with a yield of 478 kg per hectare. According to the Department of Horticulture and Plantation Crops, Government of Tamil Nadu, the principal cashew-growing districts in the state are Ariyalur (30,343 ha), Cuddalore (30,064 ha), Pudukottai (5,823 ha), Villupuram (4,423 ha), Theni (3,548 ha), and Sivagangai (3,401 ha). The three main cashew-growing districts in Tamil Nadu are Ariyalur (31.40%), Cuddalore (30.20%), and Pudukkottai (8.50%), together making up nearly 70 percent of the entire area. Among the districts in the state that cultivate cashew, Cuddalore district scored first as a percentage of output with 15,381 tonnes and had the greatest productivity at 511 kg/ha.

Ariyalur District has a large area (30,343 ha) dedicated to cashew nut production, although the productivity is just 382 kg/ha. Old, aged plantations, a lack of acceptance of better varieties and/or hybrids, and a lack of adoption of the best cultivation techniques suggested by research institutes are the causes of the production gap. The main cause of the decreased production in the Ariyalur District is because cashew is mostly produced as a rainfed crop.

To generate extra revenue, intercropping is done with finger millet, kodo millet, blackgram, and fodders. The Tamilnadu government's Department of Forests oversees around 38percent of the plantations, and due to inadequate upkeep, district output has decreased.

An effective marketing strategy encourages higher output and is necessary in order to pay the producers a fair price. The cashew trade's price swings add a degree of volatility to cashew growers' revenue levels, and the effectiveness of different channels' marketing affects producers' returns.

The commercialization of cashew nuts involves a lot of middlemen these days. They are crucial to the cashew nut disposal process. Producers, village traders, pre-harvest contractors (PHC), wholesalers, processors, and exporters make up the channels. Middlemen play a significant role in the flow of raw cashew nuts in the village market, which lowers the margins for farmers.

With the help of Tamil Nadu Cashew Processors and Exporters Association (TNCPEA). It was established by a group of cashew exporters and processors based in Tamil Nadu. Its goals are to support and aid these businesses by giving them access to domestic and foreign markets, promoting their products through a variety of initiatives, and boosting total exports from India. TNCPEA assists and offers improved assistance to exporters and processors, with a stronger emphasis on promoting items related to cashews. TNCPEA members work together to attain and resolve issues pertaining to government policies in the sector as well as any shared difficulties. The traders and the cultivators can get the help from their TNCPEA.

Cashew Tech Expos: Cashew processors are constantly searching for fresh approaches to improve productivity, save expenses, hasten change, and streamline processes. Businesses must modernise and digitise their operational technology in order to prosper and allow people, processes, and technology to collaborate harmoniously.

TNCPEA- International Exhibition: The Association invites interested Buyer-Seller Meetings, International Exhibitions, while Special Events in India along with Other Countries to participate on behalf of Cashew Processors and Exporters. There are the many facilitation are provided to the cashewnut cultivators by the state government. Hence the study has made

an attempt to “**MARKETING STRATEGIES OF CASHEW CULTIVATION IN PUDUKKOTTAI DISTRICT**”.

Objective of the Study

1. To assess the training needs and to develop training module for the cashew nut growers in the study area.
2. To identify the constraints faced by the farmers in production and marketing of cashew nut.
3. To suggest suitable strategy for sustainable cashew nut production.

Scope and importance of the study

The purpose of the current study is to assess farmers' acceptance and understanding of different technologies for selling and producing cashew nuts. This study can also provide information to planners, policy makers, and administrators on developing plans and programmes to boost cashew nut crop productivity. In summary, the research will offer a very successful and dependable government assistance and marketing methods for the sustainable production of cashew nuts in the Pudukkottai District, and they might potentially be used to other cashew nut growing regions as well.

Methodology

The study were blend of both analytical and descriptive in nature. The researcher has used both primary and secondary data. The primary data were collected from the jasmine cultivating farmers and marketers. The secondary data were collected from the magazines, newspaper, e-journals and so on. The researcher was administered the sample size drawn from the 13 districts. The snowball methods were adopted for this study. Hence the total population sizes are 323.

Table 1: Demographic Profile of the Respondents

Sl. No	Personal Background	Particulars	No. of Respondents	Percentage
1	Gender	Male	285	88.2
		Female	38	11.8
		Total	323	100
2	Age	Below 25	28	8.7
		25-35	75	23.2
		35-45	112	34.7
		45-55	59	18.3
		Above 55	49	15.2
		Total	323	100
3	Marital Status	Single	85	26.3
		Married	224	69.3
		Widow/Widower	14	4.3
		Total	323	100

4	Nature of the Family	Joint Family	305	94.4
		Nuclear Family	18	5.6
		Total	323	100

Source: Primary Data

Table 1 shows that 88.2 percent of the respondents are male and 11.8 percent are female respondents. It is observed from the table that majority (88.2%) are male category.

Age group is an important factor for analyzing the success level of farmers or any business unit. Generally young people will come with involvement in farming activities of cashewnut and the government schemes including Kisan credit cards are widely help for to these agro industries for business activities.

Table 1 confirms that 8.7 percent of people are doing the cashew cultivators that Below 25 age, 23.2 percent of the respondents are 25-35 years old category, 34.7 percent of respondents are from 35-45 age, 18.3 percent of the respondents come under the category of 45-55 ages, 15.2 percent of the respondents come under the category of above 55. The majority of the respondents are between the age limit of 35-45 by taking agricultural activities through their family support and initiatives in cultivating cashew nut.

Marriage life contributes to people being more responsible in their social and personal circumstances about their cashew cultivating activities. The Table 1 shows that 69.3 percent of the respondents are married. 4.3 percent of the respondents were widow or widower. And the rest of the respondents are unmarried. 94.4 percent of the farmers are coming from the joint family and rest of them are belonging from nuclear family.

Table 2: Cashew cultivating activities of the farmers

Sno	Particulars	Options	No. of Respondents	Percentage
1	Experience in Cashew Cultivation	Below 5	80	24.8
		5-10	53	16.4
		10-15	96	29.7
		15-20	44	13.6
		Above 20	50	15.5
		Total	323	100
2	Generation of Business	First Generation	47	14.6
		Second Generation	262	81.1
		Third Generation	14	4.3
		Total	323	100
3	Number of Employees	Below 5	13	4.0
		5-10	56	17.3
		10-15	87	26.9
		15-20	85	26.3
		Above 20	82	25.4
		Total	323	100
4	Income level	Below 5 Lakhs	44	13.6
		5-10Lakhs	232	71.8
		10-20 Lakhs	33	10.2
		20-50 Lakhs	14	4.3
		Above 50 Lakhs	0	0

		Total	323	100
5	Types of Land	Own	194	60.1
		Leased in	58	18.0
		Leased out	45	13.9
		Net arear used for agriculture	26	8.0
		Total	323	100
6	How many acres of land do you own	Below 5 acres	219	67.8
		6-10 Acres	19	5.9
		15-20 Acres	66	20.4
		Above 20 Acres	19	5.9
		Total	323	100
7	Have you got any training to cultivate the jasmine	Yes	241	74.6
		No	82	25.4
		Total	323	100
9	Are you aware about Kisan Credit Card Scheme	Yes	312	96.6
		No	11	3.4
		Total	323	100

Source: Primary data

Table 2 shows that 24.8 percent of the farmers are having work experience in below 5. 16.4 percent of the respondents have experience of 5 to 10 years, 29.7 percent of them are experience of 10-15 years, 13.6 percent of them have experience of 15 to 20 years and, 15.5 percent of them have above 20 years of experience.

Table 2 portrays that 14.6 percent of the respondents are doing the cultivation as newly, 81.1 percent of them are running their parents business called second generation, 4.3 percent of them are cultivating this cashew as in third generation. These report clearly shows that majority of the respondents with 81.1 percent came from their third generation of the family business.

Above Table indicates that 4.0 percent of the respondents are having less than 5 employees in their units, about 17.3 percent of the respondents are having the manpower of 5-10 employees in their unit, 26.9 percent of the respondents are having 10 to 15 employees in their unit. 25.4 percent of the respondents are having the labour 15-20. And rest of them are having the manpower with above 20. Hence, the majority of the respondents are having 10-15 employees in their business unit.

Table 2 represents that 13.6 percent of the respondents receive the monthly income level of Rs Below 5 Lakhs, 71.8 percent of them belong to the income level of Rs 5-10 Lakhs of 10.2 percent of them receives the income range of Rs 10-20 Lakhs, 4.3 percent of them belong to the income range of above Rs 20-50Lakhs. These statements shows that majority of the respondents generating their income range between 5-10 lakhs per annum.

Table 2 represents that 60.1 percent of the respondents are cultivating from their own land, 18 percent of the farmers are leased in for cultivating activities. 13.9 percent of them are leased out their land for cultivating purpose. Hence the majority of the respondents are cultivating the cashew production by their own land.

Table 2 shows that 67.8 percent of the farmers are cultivating the cashew from the acres between below 5. 5.9 percent of them are farming in acres between 6-10 acres. 20.4 percent of them are cultivating in acres between 15-20 acres and rest of them are 5.9 percent. Hence the majority of the respondents are farming in acres below 5. 74.6 percent of the farmers were

attended the many training programs in camp and rest of them were not attended any training programs. 96.6 percent of the farmers are aware about the kisan credit card and rest of them were lack of awareness about the card.

Cashew Marketing in Tamil Nadu

Cashew is promoted by the Tamil Nadu government through cooperative groups. To further the interests of cashew growers, the Tamil Nadu government's Directorate of Industries and Commerce established a cooperative society. The society's primary goals are to give its members who cultivate a fair price for their goods in order to better their financial situation. These kinds of organisations were founded in the districts of Panrutti (Cuddalore) and Mayiladumparai (Theni). Kerala is also playing the vital role of exporting the cashew nut.

For a variety of reasons, cashew producers market cashewnuts at different periods. After the harvest, some of them sell the cashews right away, while others choose to preserve them in the hopes that their value would increase. The factor analysis examine has been used to examine issues with marketing, storage facilities, and whether or not they sell right away.

Cashew Marketing in Pudukkottai- Factor Analysis

An essential component of India's overall economic growth is the agricultural sector. Firstly, it fulfils the rising need for food and raw resources. Second, a sizable portion of the nation's total foreign exchange profits come from agricultural goods. One of the significant commercial crops that is essential to foreign exchange profits is cashew. The attention that cashew is currently receiving as a horticulture crop from the research and development front is excellent news for both farmers and exporters. The researcher administered the factor analysis method to analyse the marketing strategy adopted in the study. A statistical technique called factor analysis is used to express the variability between associated, observable variables in terms of a smaller number of possible unobserved variables known as factors.

Cashew Marketing in Pudukkottai- Kaiser-Meyer-Olkin Test

Utilising the Kaiser–Meyer–Olkin (KMO) test, one may assess the statistical appropriateness of data for factor analysis. The test assesses how well each variable and the entire model are sampled. The statistic expresses how much of the volatility in a set of variables may be common variance. The more proportion there is in the data, the higher the KMO-value, indicating a better fit for factor analysis.

Table3: Cashew Marketing in Pudukkottai- Kaiser-Meyer-Olkin Test

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.697
Bartlett's Test of Sphericity	Approx. Chi-Square	13428.895
	df	276
	Sig.	.000

Source: Primary data

Table 3 shows that the adequacy of the sample range is greater than 0.697. Hence the variables are fitted for the factor analysis.

Cashew Marketing in Pudukkottai-Communalities

A table of communalities, or the communality value, which must be more than 0.5 to be taken into consideration for additional analysis, is the next item in the output. It displays the

amount of variation. If not, these variables will be eliminated from subsequent factor analysis stages.) in the variables has been taken into consideration by the factors that were retrieved.

Table 4- Cashew Marketing in Pudukkottai Communalities

Particulars	Initial	Extraction
To clear the farmers debt	1.000	.858
Co-operative society assistance	1.000	.884
Lack of storage facility	1.000	.840
Price fall and anticipating	1.000	.849
Pre-harvest contractor	1.000	.852
Expecting the Commission Agent	1.000	.833
To meet the family expenses	1.000	.792
Storage time is convenient and preservatives	1.000	.918
Prefer the Local Traders	1.000	.858
Farm house	1.000	.862
Commission mundy	1.000	.856
Co-operative godown	1.000	.878
The goods are stored in own house	1.000	.809
Marketing Agents/Distributors/Whole saler	1.000	.863
Lack on marketing information	1.000	.749
Price awareness	1.000	.775
Friends & Relatives	1.000	.860
Lack of transportation	1.000	.826
Village Traders	1.000	.752
Co-operative society	1.000	.907
Newspaper	1.000	.829
Changes in the price	1.000	.851
Inadequate storage facility	1.000	.809
Payment delay	1.000	.841

Extraction Method: Principal Component Analysis.

Source: Primary data

Table4 shows that all the variables are have the about more than 0.5 in extraction. Hence all the variables are fitting for the total variance explained method.

Cashew Marketing in Pudukkottai District- Total variance Explained

The number of retrieved components whose sum should equal the number of items undergoing factor analysis is actually reflected in the eigenvalue. The eigenvalues of every factor that may be extracted from the analysis are displayed in the next item.

Three subsections have been constructed from the Eigenvalue table:

1. Initial Eigen values
2. Extracted sums of Squared Loadings

3. Rotation of Sums of Squared Loadings.

Extracted sums of Squared loadings- The number of components preserved is indicated by the number of rows in this table panel. Three rows represent each of the three components that were asked to be kept in this case. The calculations for the numbers in this table panel are the same as those in the left panel, with the exception that the values in this panel are based on the common variance. Because the numbers in this panel of the table depends on the common variance, which is always less than the total variance, they are consistently lower than the values in the panel to the left of the table.

Rotation of sums of Squared Loadings- The variance distribution following the varimax rotation is shown by the data in this table panel. The goal of varimax rotation is to maximise the variance of each element, which results in a redistribution of the total amount of variation accounted for among the six extracted factors.

Table5: Marketing the cashews in Pudukkottai Districts-Total Variance Explained

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	4.320	18.001	18.001	4.320	18.001	18.001	3.453	14.389	14.389
2	3.446	14.357	32.358	3.446	14.357	32.358	2.867	11.945	26.334
3	2.843	11.848	44.206	2.843	11.848	44.206	2.615	10.894	37.228
4	2.606	10.857	55.063	2.606	10.857	55.063	2.491	10.378	47.606
5	2.281	9.506	64.569	2.281	9.506	64.569	2.405	10.020	57.626
6	1.851	7.713	72.282				2.273	9.472	67.098
7	1.549	6.452	78.735				2.126	8.860	75.958
8	1.254	5.224	83.959				1.920	8.001	83.959
9	.859	3.580	87.538						
10	.641	2.672	90.211						
11	.497	2.070	92.281						
12	.417	1.739	94.020						
13	.306	1.274	95.294						
14	.248	1.033	96.328						
15	.174	.727	97.054						
16	.165	.688	97.742						
17	.126	.527	98.269						
18	.112	.468	98.737						
19	.071	.295	99.032						
20	.062	.259	99.291						
21	.055	.231	99.522						
22	.045	.188	99.710						
23	.038	.160	99.870						
24	.031	.130	100.000						

Extraction Method: Principal Component Analysis.

Table 5 shows that these variables, the largest proportion of explained variation is 64% and the lowest is 18%. Hence the ranges fall above 50%. The data is fit for the Rotated component matrix.

Marketing the Cashews in Pudukkottai Districts- Rotation Matrix

The rotated component matrix, often known as the matrix of loadings, is the most important result of principal components analysis. It includes projected correlations amongst each variable including the calculated components.

Table6: Marketing the Cashews in Pudukkottai Districts- Rotation Matrix

Goods Storage Facility	1	2	3	4	5
The goods are stored in own house	0.537				
Commission mundy	0.732				
Co-operative godown	0.498				
Farm house	0.816				
Storage time is convenient and preservatives	0.533				
Inadequate storage facility	0.891				
Preferences of marketing					
Prefer the Local Traders		0.471			
Expecting the Commission Agent		0.497			
Co-operative society assistance		0.660			
Marketing Agents/Distributors/Wholesaler		0.519			
Pre-harvest contractor		0.981			
Price awareness					
Friends & Relatives			0.714		
Newspaper			0.564		
Village Traders			0.639		
Co-operative society			0.493		
Problems of Marketing					
Lack of transportation				0.496	
Changes in the price				0.571	
Lack on marketing information				0.577	
Payment delay				0.563	
Reasons for immediate market					
To clear the farmers debt					0.751
To meet the family expenses					0.788
Lack of storage facility					0.631
Price fall and anticipating					0.692

Source: Primary Data

Table 6 clearly shows that all the variables are utmost reach out 0.5. Hence the rotation matrix has been fitted in the factor analysis.

Factor-1: The goods are stored in own house(0.537), Commission Mundy(.732), Co-operative godown(.498), Farm house(.816), Storage time is convenient and preservatives(.533), Inadequate storage facility(.891). All the factors are grouped together and named as “Goods storage facility”.

Factor-2: Prefer the local traders(.471), Expecting the commission agents(.497), Co-operative society assistance(.660), Marketing agents/distributors and so on(.519), Pre-harvest contractor(.981). All the factors are grouped and called as “Preferences of marketing”.

Factor-3: Friends and relatives(.714), newspaper(.564), village traders(.639), co-operative society(.493). All the factors are named as Price awareness.

Factor-4: Lack of transportation(.496), Changes in price(.571), Lack on marketing information(.577), Payment delay(.563). Hence the factors were named as “Problems in marketing”.

Factor-5: To clear the farmers debt(.751), To meet the family expenses(.788), Lack of storage facility(.631), Price fall and anticipating(.692). All the factors are named as “ Reasons for immediate market.

Findings and Suggestion

1. From the study, most of the respondents are 29.7 percent and having the experience of cashew cultivation from 10-15 years.
2. Majority of the respondents with 81.1 percent came from their third generation of the family farmers of cashew cultivation.
3. The most of the respondents(29%) are having 10-15 employees in their business unit.
4. Majority of the respondents(71%) generating their income range between 5-10 lakhs per annum.
5. the majority of the respondents(61%) are cultivating the cashew production by their own land.
6. the majority of the respondents(67.9%) are farming in acres below 5.
7. The study has inferred that 74.6 percent of the farmers were attended the many training programs in camp.
8. Majority of the respondents(96%) were aware about the Kisan credit card.

Therefore the suggestions are offered for the betterment of the marketing the cashews are listed below.

1. It is possible to offer cashew growers generous financial support in the form of crop loans, development loans, and irrigation facility financing through commercial banks, cooperatives, and main agricultural and rural development banks.
2. Growers and merchants can get important marketing information from the media and other channels, such as pricing, market, export, etc.
3. Currently operating out of Cochin, Kerala, Cashewnut and Cocoa Development Corporation primarily serves the requirements of the state's cashew producers. Regional Cashew Development Corporation centres are proposed to be developed in Tamil Nadu, ideally in the districts of Cuddalore, Sivagangai, and Pudukkottai.
4. The cost of purchasing cashewnuts is influenced by the length of the marketing channel. Thus, the government may set the price of cashewnuts and directly purchase them from producers through cooperative organisations.
5. It may be possible to establish a greater number of cooperative societies so that farmers may sell their goods there and earn a fair price.

Conclusion

Therefore, the data suggests that the major farmers in Pudukkottai area have benefited more than the small farmers that grow cashew. The primary cause of this is that large cashew farmers have invested in crop protection and advanced cultivation techniques. Improved post-

harvesting and processing methods must be developed to fortify the foundation of cashew production in order to satisfy the ever-increasing and phenomenal surge in demand and expand export potential.

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A STUDY ON THE VIEWPOINT OF GST ACCOUNTING AMONG MSME OWNERS WITH REFERENCE TO BENGALURU

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ABSTRACT

Implementation of GST has reached seven years and it impacted significantly businesses across India, by MSMBs facing unique encounters in this transition. GST has replacing a complex array of indirect taxes with a unified tax structure, simplified compliance for businesses, reduced the flowing outcome of levies, and enhanced transparency. Still GST have complexity, frequent updates and variations in tax rates and rules, delays in input tax credit refunds, and technical glitches in the GST network have led to significant confusion and administrative burdens, particularly for small and medium-sized enterprises, resulting in increased operational costs and difficulties in maintaining accurate records. This study provides a comprehensive analysis of GST accounting viewpoints within MSMEs in Bengaluru.

Key words: GST, Accounting, MSME's, Difficulties.

INTRODUCTION

In the world-wide GST concept was introduced long back, but in India it is introduced in 1st July 2017 with great struggle. Now we are reaching the milestone of seven years of GST implementation, but till now we have facing different types compliance in various stages and areas. GST is comprehensive indirect taxes that were previously imposed by the central and state government. It is aimed at creating a unified taxation system, eliminating the cascading effort of taxes, and simplifying compliance for businesses. Now, this tax modification is witnessing reform to our economy and all business dynamics including Micro, Small, and Medium Enterprises (MSMEs). It is difficult for Indian MSMEs to flourish in this volatile atmosphere. MSMEs are rightfully hailed as the backbone of India's economic. MSMEs not only bring forth the loftiest employment per capita investment but also ease the modification of primary sector-led economic systems into manufacturing ones by offering processing occupations that generate workable livelihoods. Women hold a significant place in this small business sector. Also, MSMEs chip into the development of systemic productive capacities and foster financial buoyancy by absorbing productive energies at all stages of the economic system and establishing inter linkages between small and large businesses. The study examines the tax compliance and perceptions of MSMEs from the perspectives of three major players associated with MSME owners.

REVIEW OF LITERATURE

Dale Pinto (2002) examines the businesses have faced increased administrative and compliance costs, especially small retailers that lack adequate resources for detailed accounting. To address these challenges, the government introduced simplified accounting measures on 1 July 2000, letting succeeding sellers towards evaluation their GST-free sales and acquisitions before maintaining separate records. These measures aim towards decrease the managerial burden then enhance acquiescence efficiency for retailers dealing with GST.

Lourdunathan. F et al., (2017) This paper reviews GST's background, prospects, and implementation hurdles, concluding that it can enhance revenue and compliance by simplifying taxes and providing comprehensive input tax credits. The electronic processing of tax returns via GSTNET aims to reduce corruption and tax evasion. To ensure successful implementation, ongoing education and training for stakeholders are essential.

Nandagopal (2017) The current study suggests that industries such as cement and media will benefit, whereas insurance and transport may face challenges. Overall, GST is viewed as a structural reform that aims to create a better national market and enhance the ease of doing business, benefiting both businesses and consumers. This review analyses the impact of GST during its first four months of implementation on manufacturers, service providers, consumers, and the government.

Subramania Nathan's(2020) examined GST's influence on MSMEs presents a varied picture. Delays in refunds and the management of advances further exacerbate these issues. Although initiatives such as an electronic refund system and quarterly returns are beneficial, additional reforms are needed. Proactive government intervention is crucial to enhance GST's effectiveness for MSMEs and mitigate its challenges.

HarishekarLichchavi's(2021) study examines GST's impact on small and medium enterprises in Peenya, Bangalore. Key findings include a decrease in product prices and increased sales and tax compliance. Enterprises prefer Input Tax Credit (ITC) over the composition scheme, while reactions to the Reverse Charge Mechanism (RCM) are mostly negative. Although the dual GST administration creates challenges, businesses seek simpler tax processes for job and labour facilities. Electronic way bill scheme takes improved transparency but introduced new issues. While inter-state transactions have grown and technical difficulties are minimal, current economic conditions have diminished GST's overall benefits, suggesting the need for targeted policy reforms.

Mehta(2021) research states the implementation has encountered ongoing challenges, including technical glitches and complexities inherent to the system. It reviews the progress made in the four years since GST's adoption, highlighting the need for further stabilization to enhance compliance and enforcement efficiency. The GST Council must address these issues to improve tax efficiency. This literature review explores current challenges and suggests solutions to streamline the GST framework, supporting national integration while managing the complexities of Centre-State relationships.

Nagarajan K. (2021) The GST Law, 2017 mandates that each listed individual keep accurate books and records, as compliance verification by tax authorities relies on these documents. The Act specifies the types of records to be kept and the duration for their retention, emphasizing the importance of proper maintenance to avoid penalties. With the rise of accounting and ERP software, businesses can now choose to keep records either manually or electronically. Registered individuals are responsible for filing self-assessed GST returns, ensuring that reported financial data aligns with their accounting records. Consequently, the obligation to maintain these records accurately is crucial, as ignorance of compliance requirements is not a valid excuse in the eyes of the law.

Vakil et al. (2022) This research explores the views of chartered accountants in Gujarat on GST adoption, highlighting that while the tax is expected to significantly impact the Indian economy, its full effects may take time to emerge. An official recommendation supports a uniform tax on all transactions, encompassing all goods and services at the stages of manufacturing, sale, and consumption, underscoring GST's potential to reshape India's economic landscape.

RESEARCH GAP

Despite existing research on the Goods and Services Tax (GST), several gaps remain. Most studies focus on immediate compliance challenges faced by small retailers and MSMEs, overlooking the long-term impacts on business strategies and sectoral dynamics. Additionally, while some research highlights the need for government intervention and educational initiatives, there is insufficient analysis of their effectiveness across different industries. Furthermore, the broader economic implications of GST, particularly its potential to reshape the Indian economy and enhance national integration, require more comprehensive exploration. Addressing these gaps could yield valuable insights for policy reforms and better implementation strategies.

RESEARCH OBJECTS:

1. **To Assess the Understanding of GST Compliance** awareness among MSME owners in Bengaluru.
2. **To analyse the Key Challenges and difficulties** faced by MSMEs in Bengaluru.

METHODOLOGY

Research Category	Descriptive and Analytical
Sources	Primary-level and Secondary
Study Instrument	Questionnaire
Sample Size	29
Data Analysis Tools	Descriptive table
Tools Used	SPSS

DATA ANALYSIS:

	Q - 1	Q - 2	Q - 3	Q - 4	Q - 5	Q - 6	Q - 7	Q - 8	Q - 9	Q - 10
Average	1.000 0	2.846 2	2.0769	1.076 9	2.423 1	2.115 4	2.038 5	2.307 7	1.500 0	1.2692
Median	1.000 0	3.000 0	2.0000	1.000 0	2.000 0	2.000 0	2.000 0	2.000 0	1.500 0	1.0000
Peak value	1.00	3.00	2.00	1.00	2.00	1.00	2.00	2.00	1.00	1.00
Spread of Data	.0000 0	.4595 2	1.1520 9	.2690 7	.7500 6	.9832 7	.7129 5	.8753 4	.5048 8	.45234
Variance	.000	.211	1.327	.072	.563	.967	.508	.766	.255	.205
SE of Skewness	.330	.330	.330	.330	.330	.330	.330	.330	.330	.456
Std. Error of Kurtosis	.650	.650	.650	.650	.650	.650	.650	.650	.650	.887
Minimum	1.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Maximum	1.00	4.00	5.00	2.00	4.00	4.00	3.00	4.00	2.00	2.00
Skewness		-.597	1.285	3.271	.853	.276	-.056	.073	.000	1.105
Kurtosis		1.116	1.171	9.043	.136	-1.106	-.972	-.692	-2.082	-.850

Source: Primary – Computed

	Q. 21	Q. 22	Q. 23	Q. 24	Q. 25	Q. 26	Q. 27	Q. 28	Q. 29
Average	2.076 9	1.673 1	1.903 8	1.923 1	1.903 8	2.096 2	1.692 3	2.346 2	2.000 0
Median	2.000 0	1.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0
Peak value	3.00	1.00	1.00	2.00	1.00	2.00	1.00	2.00	2.00
Spread of Data	.8822 0	.7851 9	.7985 2	.7883 0	.7985 2	.8227 1	.7286 4	.7892 6	.7921 2
Variance	.778	.617	.638	.621	.638	.677	.531	.623	.627
SE of Skewness	.330	.330	.330	.330	.330	.330	.330	.330	.330
Std. Error of Kurtosis	.650	.650	.650	.650	.650	.650	.650	.650	.650
Minimum	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Maximum	3.00	3.00	3.00	4.00	3.00	4.00	3.00	4.00	4.00
Skewness	-.332	.659	.177	.638	.177	.256	.554	.782	.985
Kurtosis	-1.217	-1.054	-1.400	.219	-1.400	-.566	-.909	.243	1.297

Source: Primary – Computed

	Q. 21	Q. 22	Q. 23	Q. 24	Q. 25	Q. 26	Q. 27	Q. 28	Q. 29
Average	2.076 9	1.673 1	1.903 8	1.923 1	1.903 8	2.096 2	1.692 3	2.346 2	2.000 0
Median	2.000 0	1.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0	2.000 0
Peak value	3.00	1.00	1.00	2.00	1.00	2.00	1.00	2.00	2.00
Spread of Data	.8822 0	.7851 9	.7985 2	.7883 0	.7985 2	.8227 1	.7286 4	.7892 6	.7921 2
Variance	.778	.617	.638	.621	.638	.677	.531	.623	.627

SE of Skewness	.330	.330	.330	.330	.330	.330	.330	.330	.330
Std. Error of Kurtosis	.650	.650	.650	.650	.650	.650	.650	.650	.650
Minimum	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Maximum	3.00	3.00	3.00	4.00	3.00	4.00	3.00	4.00	4.00
Skewness	-.332	.659	.177	.638	.177	.256	.554	.782	.985
Kurtosis	-1.217	-1.054	-1.400	.219	-1.400	-.566	-.909	.243	1.297

Source: Primary – Computed

The above table shows mean, medium, mode, standard deviation, discrepancy, SE of skewness, SE of kurtosis. minimum, extreme, Skewness then Kurtosis from the primary data collection.

INTERPRETATION:

1. **Confidence in Compliance:** With a mean of 2.42 and a median of 2, the confidence level in complying with GST regulations is moderate, implying that while some owners feel capable, others may be uncertain about their compliance abilities.
2. **Overall Awareness:** The mean score of 2.85 suggests a moderate understanding of GST regulations among MSME owners, with a median of 3 indicating that many have at least a basic understanding.
3. **Use of Resources:** The responses indicate a reliance on various resources to learn about GST, although specific types of resources are not detailed. This could point to a mix of informal and formal learning avenues.

CHALLENGES FACED:

1. **Complexities in Filing Returns:** The data reflects challenges in understanding the filing process, with a mean of 2.31, indicating that many MSME owners find the return process complex.
2. **Documentation and ITC Claims:** The responses suggest that maintaining proper accounting records and understanding ITC (Input Tax Credit) claims is particularly challenging, evidenced by high mean values related to these aspects.
3. **Timeliness:** The challenges faced in meeting GST return deadlines (mean of 2.30) reveal that timely compliance is a significant concern, possibly due to insufficient understanding or lack of resources.

TECHNOLOGICAL ADOPTION:

1. **Accounting Software:** While a question about the GST compliance of accounting software was included, the overall data doesn't clearly define how many respondents have compliant software or what features it possesses.
2. **Interest in Technology:** The responses indicate some willingness to adopt new technologies for GST compliance, though specifics are unclear. The mean of 2.07 suggests a cautious approach to integrating technology into compliance practices.

NEED FOR SUPPORT

Support Requirements: Many MSME owners express a need for additional support, likely including training on GST compliance, better resources for understanding ITC and guidance on technology adoption.

RECOMMENDATIONS:

1. **Consultation Services:** Increased access to tax professionals or advisory services can help MSME owners navigate complex areas such as ITC claims and timely filing.
2. **Regular Updates:** Establish a communication channel (like newsletters or workshops) to keep MSME owners informed about changing GST rates and regulations.
3. **Technology Workshops:** Conducting workshops on the use of GST-compliant accounting software can facilitate better adoption and understanding of available tools.
4. **Training and Education:** Given the moderate understanding and confidence levels, targeted training programs focusing on GST regulations and compliance processes can enhance knowledge and confidence among MSME owners.

CONCLUSION

The study reveals that while MSME owners in Bengaluru have a foundational understanding of GST, significant challenges remain in compliance, particularly regarding documentation and the complexities of filing returns. Enhanced support, education, and technology integration are essential for improving their GST compliance experience.

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SUSTAINABILITY OF GREEN BANKING SERVICES

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ABSTRACT

In response to escalating environmental challenges, the banking sector is increasingly adopting green banking practices to promote sustainability. This study examines the awareness, utilization, and satisfaction levels of customers regarding green banking services in Kanniyakumari district. Using primary and secondary data, statistical tools such as Percentage Analysis, Garrett Ranking, and the Chi-square test were employed to analyze customer demographics and their engagement with green banking services. The findings indicate that young, highly educated males are the primary users of green banking services while online and mobile banking are being the most frequently utilized services. Green loans and credit cards remain underutilized due to low awareness and incentives. The study also reveals that factors such as age, occupation, and income significantly influence awareness and satisfaction levels. To enhance adoption, banks should focus on financial literacy programs, targeted incentives, and expanded digital banking infrastructure. A strategic approach to green banking not only provide benefit to financial institutions but also contribute to long-term environmental sustainability and economic growth.

Keywords: Green Banking, Sustainable Finance, Digital Banking, Consumer Awareness, Eco-Friendly Banking

INTRODUCTION

In today's world, environmental concerns like climate change, deforestation, and pollution have made sustainability a priority for businesses, including the banking sector. Green banking is an approach that integrates environmentally friendly policies into financial services, encouraging sustainable development. It includes practices like paperless banking, digital transactions, green loans, and financing for renewable energy projects. With increasing global awareness and regulatory pressures, banks are now playing a crucial role in promoting eco-friendly initiatives.

Traditional banking activities, such as physical paperwork, energy-intensive branches, and investments in non-sustainable industries, contribute to environmental degradation. The shift towards green banking helps to reduce carbon footprints by promoting digital banking, eco-friendly investments, and funding projects focused on sustainability. By adopting green banking services, financial institutions can help mitigate environmental damage while supporting economic growth. This approach benefits both businesses and consumers by promoting responsible financial management and encouraging eco-conscious decision-making.

The banking sector has a significant indirect impact on the environment through the industries. Traditional banking often supports projects that contribute to deforestation, pollution, and high carbon emissions. However, with the rise of green banking, financial institutions are shifting their focus toward funding sustainable businesses, clean energy, and eco-friendly projects. This transition not only reduces environmental harm but also enhances the long-term sustainability of the banking sector by promoting green economic growth and responsible corporate behavior.

CONCEPTS AND OPERATIONAL DEFINITIONS

1. Online & Mobile Banking – Access banking services anytime, anywhere, reducing the need for physical branch visits.
2. E-Statements & Digital Transactions – Paperless banking through digital statements and cashless payments.
3. Green Loans & Financing – Special loans for eco-friendly projects like solar panels and electric vehicles.
4. Sustainable Investment Options – Investing in green bonds and eco-conscious companies.
5. Green Credit Cards – Cards made from recycled materials with rewards for eco-friendly spending.
6. Bill Payments (Gas, Insurance, Electricity) – Digital payments for utility bills, reducing paper waste.
7. E-Deposits – Fixed and recurring deposits managed digitally, reducing paperwork.
8. DTH & Mobile Recharge – Quick and easy recharges through digital banking apps.
9. ATM Withdrawals – Cash withdrawals without visiting a bank.
10. Solar ATMs – ATMs powered by solar energy, reducing electricity consumption.

REVIEW OF LITERATURE

1. Pravakar Sahoo et al., (2014) in their article Green Banking if implemented sincerely will act as an effective ex ante deterrent for the polluting industries that will give a pass by to the other institutional regulatory mechanisms. There has not been much initiative in this regard by the banks and other financial institutions in India though they play an active role in India's emerging economy. The banking and financial sector should be made to work for sustainable development. As far as Green banking is concerned, India's banks and financial institutions are running behind time. None of our banks or financial institutions have adopted equator principle even for the sake of records. None of them are signatory to the UNEP Financial Initiative statement. It is time now that India takes some major steps to gradually adhere to the equator principles-guidelines that use environment-sensitive parameters, apart from financial, to fund projects.
2. Anu Sahi, (2017) found in his study on Green Banking among investors of Punjab is quite high. More than 80% of total investors surveyed due to the awareness of Green Banking concept. The awareness was quite high among males as compared to females. The majority of the respondents using green banking products feel that green banking practices are environment friendly as well as it helps in saving their time, cost and efforts etc. Recyclable Debit and Credit Card, Net Banking, Mobile banking, E- statements and ATM's are the most popular products whereas, SMS- banking is the least used banking practice. Application of factor analysis revealed three main perceptual benefits of green banking namely online benefits, security benefits and bankers' benefits. It can be said that as far as green banking is concerned, Indian

banks are far behind their counterparts from developed countries. People in Punjab are yet to come forward for adoption of these practices as they don't have full knowledge and awareness regarding Green Banking. Banks have to play a vital role to make their customers educated.

3. Ajaz Akbar Mir et al.,(2022) in their study finds out the devastating effects of recent flooding, droughts and extreme temperatures affect several people all over the world. It have experienced and compelled us to begin thinking about global warming and its consequences, and to do everything we can do to address this problem. Governments, business and individuals all play a part in preventing global warming and creating a more sustainable world. We have to deal with financial institutions, particularly banks, which play a vital role in this environment by assisting in the development of a robust and successful low-carbon economics. They should make more use of environmental data when extending credit and making investment decisions. The project will assist them in proactively improving their environmental performance while it also adds long-term value to their company. Businesses having a bigger carbon output may be viewed as riskier in the future, and banks may shy away from funding such businesses in favor of innovative technology solutions that absorb or reduce carbon emissions. As a result, Green banking is the order of the day, a source for SD and it will undoubtedly benefit banks, industries, as well as the environment at large.
4. Parag Bhuyan, (2023) concluded that green banking represents a pivotal shift in the financial sector's role in promoting environmental sustainability. By financing green projects, reducing their carbon footprint, and developing eco-friendly financial products, banks can contribute significantly to global sustainability efforts. Despite the challenges, green banking presents numerous opportunities for growth, particularly as governments, businesses, and consumers become more environmentally conscious. The future of green banking is promising, but it requires a concerted effort from policymakers, financial institutions, and stakeholders to ensure its successful implementation worldwide.
5. Hariharan .A (2024) in her study explained that the majority of traditional banks did not continuously seek out possibilities for investment in environmentally friendly industries or sectors, despite the fact that green banking is growing in popularity as a way to help ecological and fiscal procedures. Indian banks fall well behind their counterparts in wealthy countries. If Indian banks want to do business globally, they must recognize their social and environmental responsibilities. These strategies have lately acquired popularity among a variety of financial service providers, quality assurance businesses, and insurance companies, in addition to smaller, cooperative banks. Furthermore, banks will prioritize financing to sectors and individuals that have previously obtained experience or making genuine efforts to gain expertise.

STATEMENT OF THE PROBLEM

In the face of escalating environmental challenges such as climate change, deforestation, pollution, and resource depletion, traditional banking practices often contribute to unsustainable economic activities. Financial institutions play a crucial role in shaping economic growth, but without a conscious shift towards sustainability, they may continue financing industries that harm the environment.

Despite increasing awareness, many banks still prioritize short-term financial gains over long-term environmental impact. The lack of adequate policies, incentives, and awareness

about green banking has slowed its adoption. Additionally, businesses and consumers often hesitate to the transition of eco-friendly financial products due to perceived costs and complexities.

Therefore, the problem lies in the urgent need for financial institutions to adopt green banking practices that promote sustainable investments, reduce carbon footprints, and align with global environmental goals. Without immediate action, financial institutions risk contributing to environmental degradation, regulatory challenges, the economic instability. This study aims to find out the awareness, usage and satisfaction regarding green banking services. Hence the present study is undertaken to make an in depth analysis of green banking services in Kanniyakumari District.

OBJECTIVES OF THE STUDY

- To study the green banking activities.
- To know the profile of the customers using green banking services.
- To find out the awareness, utilization and satisfaction regarding the green banking services.

SCOPE OF THE STUDY

The scope of this study encompasses the awareness, utilization, and satisfaction of customers regarding green banking services in Kanniyakumari district. It examines the extent to which customers are informed about sustainable banking practices, their adoption of eco-friendly financial services, and their overall satisfaction.

METHODOLOGY

The research was conducted using data collected from both primary and secondary sources. The analysis was carried out using statistical tools such as Percentage Analysis, Garrett Ranking, and the Chi-square test to evaluate the study's objectives. The results were presented in the form of tables and charts. A convenience sampling technique was adopted to select respondents among Green Banking Services users. Data was collected from 130 green banking users in Kanniyakumari district, and the calculations were performed using SPSS software advanced Excel.

LIMITATIONS OF THE STUDY

This study is limited by the small sample size and restricted geographic area.

PROFILE OF THE CUSTOMERS OF GREEN BANKING SERVICES

The parameters such as gender, age, education, occupation and monthly income of the family are considered in the present study to understand the profile of the respondents and are exhibited in the following table.

TABLE 1
GENDER

GENDER	FREQUENCY	PERCENT
Male	67	51.5
Female	63	48.5

Total	130	100.0
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Source: Primary data

Table 1 presents gender-based data on opinions regarding green banking services, with 51.5% of respondents being male and 48.5% female. This distribution suggests that feedback on green banking services is fairly balanced between both genders, providing a comprehensive understanding of public perception.

TABLE 2

AGE

AGE	FREQUENCY	PERCENT
Below 25	68	52.3
25 – 50	48	36.9
Above 50	14	10.8
Total	130	100.0

Source: Primary data

Table 2 presents that 52.3% are below 25 years old, 36.9% are between 25-50 years old and 10.8% are above 50 years old. The data highlights a strong interest in green banking among younger individuals but lower participation from older adults.

TABLE 3

EDUCATION QUALIFICATION

GENDER	FREQUENCY	PERCENT
School education	28	21.5
Higher education	102	78.5
Total	130	100.0

Source: Primary data

Table 3 shows that 78.5% have higher education and 21.5% have school-level education. This indicates that individuals with advanced education are more likely to be aware of and engaged with green banking services. Respondents with school education indicates that those with lower education levels may have less awareness or interest in green banking services. This could be due to limited access to information, traditional banking preferences, or lower familiarity with digital financial services.

TABLE 4

OCCUPATION

OCCUPATION	FREQUENCY	PERCENT
Unemployed	52	40.0
Farmer	21	16.2

Business	19	14.6
Salaried employee	38	29.2
Total	130	100.0

Source: Primary data

Table 4 explains the occupational distribution of customers using green banking services and it says that 40.0% are unemployed, 29.2% are salaried, 16.2% are farmers and 14.6% are business owners. Unemployed individuals may use green banking for financial accessibility and government support, while salaried employees are attracted to digital convenience, direct deposits, and sustainability initiatives.

TABLE 5
MONTHLY FAMILY INCOME

INCOME	FREQUENCY	PERCENT
Below 50000	97	74.6
50000 – 100000	26	20.0
Above 100000	7	5.4
Total	130	100.0

Source: Primary data

Table 5 depicts that the customers of green banking services, 74.6% have a family income below 50,000, 20.0% have a family income between 50,000 -100,000 and 5.4% have a family income above 100,000. This shows that green banking services are being used by individuals with limited financial resources, possibly because they offer low-cost or digital banking alternatives that reduce expenses. It also indicates that affordability and accessibility are key factors in attracting customers from this income group.

TABLE 6
MOST FREQUENTLY USED GREEN BANKING SERVICES

MOST FREQUENTLY USED GREEN BANKING SERVICES	AVERAGE	RANK
Green credit card	49.08462	7
Withdraw through ATM	39.23846	3
E-statement & Digital transaction	56.86154	2
Green loans& Financing	50.03846	8
Bill pay (gas, insurance, electricity)	49.65385	4
Recharge of DTH, Mobile phones	46.73846	5

Online & Mobile banking	65.09231	1
E-Deposit	41.29231	6

Source: Primary data

Table 6 exhibits that online & mobile banking and e – statement & digital transactions are the most frequently used green banking services, reflecting a strong digital banking trend. ATM withdrawals hold importance, indicating a mix of digital and traditional banking habits. Green loans and credit cards rank lower, suggesting a need for greater awareness and incentives to promote these sustainable banking options in Kanniyakumari district.

TABLE 7

ASSOCIATION BETWEEN PROFILE VARIABLES AND AWARENESS, UTILISATION AND SATISFACTION REGARDING GREEN BANKING SERVICES

Association between Profile Variables and Awareness, Utilisation and Satisfaction regarding Green Banking Services

Green banking services		Gender	Age	Education qualification	Occupation	Income
Online&Mobile banking	Awareness	.347	.546	.430	.257	.020
	Utilisation	.694	.000	.012	.000	.076
	satisfaction	.021	.016	.667	.064	.345
E-statement&Digital transaction	Awareness	.905	.000	.112	.004	.143
	Utilisation	.238	.010	.305	.037	.147
	satisfaction	.056	.000	.405	.002	.003
Green loans& Financing	Awareness	.162	.436	.014	.001	.941
	Utilisation	.207	.001	.401	.068	.581
	satisfaction	.650	.068	.494	.483	.438
Sustainable investment option	Awareness	.894	.004	.165	.013	.022
	Utilisation	.754	.254	.009	.005	.214

	satisfaction	.924	.385	.770	.785	.125
Green credit card	Awareness	.307	.233	.134	.009	.767
	Utilisation	.021	.002	.967	.001	.164
	satisfaction	.943	.312	.223	.168	.141
Bill pay (gas,insurance,electricity)	Awareness	.952	.000	.214	.000	.025
	Utilisation	.743	.002	.703	.257	.149
	satisfaction	.290	.000	.459	.921	.071
E-Deposit	Awareness	.314	.317	.742	.407	.099
	Utilisation	.036	.000	.119	.004	.034
	satisfaction	.174	.000	.770	.035	.071
Recharge of DTH,Mobile phones	Awareness	.611	.000	.178	.000	.032
	Utilisation	.979	.002	.647	.536	.212
	satisfaction	.571	.000	.552	.710	.177
Withdrawthrough ATM	Awareness	.271	.576	.667	.404	.117
	Utilisation	.848	.000	.000	.000	.155
	satisfaction	.128	.002	.120	.000	.029
Solar ATM	Awareness	.367	.001	.923	.001	.010
	Utilisation	.828	.005	.000	.000	.565
	satisfaction	.320	.00	.000	.000	.017

	n		7			
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Source: Primary data

The Chi-Square test was conducted to examine the relationship between demographic factors (Gender, Age, Education Qualification, Occupation, and Income) and the awareness, utilization, and satisfaction levels of various green banking services in Kanniyakumari district. The results indicate the following key findings:

- Age and Occupation significantly influence awareness, utilization, and satisfaction across multiple green banking services, indicating that younger working professionals are more engaged with digital banking.
- Income significantly impacts awareness and satisfaction, suggesting that higher-income individuals are more likely to be aware of and satisfied with green banking services.
- Education plays a crucial role in awareness of Green Loans & Financing and Sustainable Investment options, highlighting the need for financial literacy programs.
- Most used services: Online & Mobile Banking, E-Statements, and Digital Transactions, indicate a strong preference for digital and paperless banking.
- Least utilized services: Green Credit Cards and Sustainable Investment Options, highlighting the need for more promotion and incentives.
- Despite the rise of digital transactions, many customers still rely on ATM withdrawals, particularly older and lower-income groups.

The findings indicate that Age, Occupation, and Income significantly impact customers' awareness, utilization, and satisfaction with green banking services. To enhance sustainability, financial institutions should focus on increasing awareness, especially for green loans, sustainable investments, and green credit cards, while also encourage digital adoption to reduce cash transactions.

FINDINGS OF THE STUDY

1. Most of the customers of green banking services are male.
2. Most of the customers are under the age group of 25 years old
3. The customers are highly educated
4. The occupational distribution of customers are unemployed.
5. The customers of green banking services, 74.6% have a family income below 50,000.
6. The most frequently used green banking services are online & mobile banking and e – statement & digital transactions. ATM withdrawals hold importance, indicating a mix of digital and traditional banking habits. Green loans and credit cards are least in usage of green banking.
7. The Chi-Square test was conducted to examine the relationship between demographic factors (Gender, Age, Education Qualification, Occupation, and Income) and the awareness, utilization, and satisfaction levels of various green banking services in Kanniyakumari district. The results indicate the following key findings:

- Age and Occupation significantly influence awareness, utilization, and satisfaction across multiple green banking services, indicating that younger working professionals are more engaged with digital banking.
- Income significantly impacts awareness and satisfaction, suggesting that higher-income individuals are more likely to be aware of and satisfied with green banking services.
- Education plays a crucial role in awareness of Green Loans & Financing and Sustainable Investment options, highlighting the need for financial literacy programs.
- Most used services: Online & Mobile Banking, E-Statements, and Digital Transactions, indicating a strong preference for digital and paperless banking.
- Least utilized services: Green Credit Cards and Sustainable Investment Options, highlighting the need for more promotion and incentives.
 - Despite the rise of digital transactions, many customers still rely on ATM withdrawals, particularly older and lower-income groups.

The findings indicate that Age, Occupation, and Income significantly impact customers' awareness, utilization, and satisfaction with green banking services. To enhance sustainability, financial institutions should focus on increasing awareness, especially for green loans, sustainable investments, and green credit cards, while they also have to encourage digital adoption to reduce cash transactions.

SUGGESTIONS FOR THE STUDY

Based on the findings of the study, the following suggestions can help to improve the awareness, utilization, and satisfaction of green banking services:

- Since most users are male and under 25 years old, banks should launch awareness programs targeting other demographic groups, especially women and older individuals.
- Green credit cards and loans have lower adoption rates. Banks should introduce attractive incentives such as cashback, lower interest rates, or rewards for eco-friendly purchases.
- Awareness campaigns about the benefits of green financing, such as lower interest rates on eco-friendly projects (solar panels, electric vehicles, etc.), should be promoted through targeted marketing campaigns and educational initiatives.
- Special training programs for older customers can help them become comfortable using digital banking platforms.
- Since ATM withdrawals remain common, banks should encourage the use of digital wallets and contactless payments by offering cashback, discounts, and reward programs for online transactions.
- Expanding digital payment infrastructure, such as QR code-based payments, can reduce cash transactions and promote sustainability.
- Providing financial incentives, such as tax benefits or higher interest rates for sustainable investments, can encourage more customers to participate.
- Since unemployed individuals make up a significant portion of green banking users, banks can introduce tailored financial products, such as low-interest green loans or micro-investment plans, to encourage their participation.

- Partnering with government agencies and utility providers can further promote online bill payments and reduce paper-based transactions.
- Awareness campaigns should highlight the environmental and cost-saving benefits of solar ATMs.
- Expanding the number of Solar ATMs, especially in rural areas, can encourage their usage and contribute to sustainable banking.

CONCLUSIONS

The study highlights the growing adoption of green banking services in Kanniyakumari district, with young, highly educated males being the primary users. While Online & Mobile Banking and E-Statements are the most frequently used services, Green Credit Cards, Green Loans, and Sustainable Investments remain underutilized due to a lack of awareness and incentives. The Chi-Square analysis confirms that Age, Occupation, and Income significantly influence awareness, utilization, and satisfaction levels. To enhance sustainability, banks should focus on increasing awareness, promoting green financial products with incentives, expanding digital banking infrastructure, and encouraging digital transactions to reduce reliance on ATMs and cash payments. A customer-centric approach will ensure greater financial inclusion, improved accessibility, and long-term sustainability in green banking services.

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UNDERSTANDING TOURISTS' SUSTAINABLE TRAVEL INTENTIONS: A VALUE-BELIEF-NORM THEORY APPROACH

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Abstract

Sustainable tourism is essential for addressing environmental challenges caused by rapid tourism growth, including resource depletion, pollution, and ecological harm. However, despite growing awareness, tourists often struggle to adopt sustainable practices. This study investigates the factors influencing tourists' intentions towards sustainable tourism using the Value-Belief-Norm (VBN) theory. The research analyzed data from 150 tourists across three popular destinations in Central Kerala. Findings revealed that altruistic and biospheric values positively enhanced awareness about environmental impacts. Interestingly, strong environmental values did not necessarily lead to higher personal responsibility among tourists, indicating barriers between awareness and action. The exceptionally strong connection between perceived responsibility and personal moral norms underscores the necessity of fostering personal accountability to enhance sustainable behavior. This study emphasizes practical implications for tourism policymakers and businesses, recommending that initiatives focus on strengthening tourists' sense of personal responsibility. Future research should further explore barriers to translating environmental awareness into responsible actions.

Sustainable Tourism ,Value-Belief-Norm Theory (VBN),Altruistic Values, Biospheric Values, Egoistic Values ,Environmental Awareness ,Personal Responsibility ,Tourist Behavior, Sustainable Practices, Kerala Tourism

Introduction

Tourism represents one of the fastest-growing sectors globally, significantly contributing to economic growth, employment, and cultural exchange. Despite its numerous economic and social benefits, the tourism industry poses considerable threats to environmental sustainability through resource depletion, pollution, and ecological degradation. Issues such as overtourism, carbon emissions, waste generation, and biodiversity loss have increasingly raised concerns among policymakers, tourism businesses, environmentalists, and tourists themselves. These adverse effects call for a profound shift in tourism practices towards sustainability, making sustainable tourism a necessity rather than a choice. The World Tourism Organization defines sustainable tourism as "tourism that takes full account of its current and future economic, social and environmental impacts, addressing the needs of visitors, the industry, the environment and host communities".(United Nations World Tourism Organization [UNWTO],)

Sustainable tourism emphasizes minimizing negative impacts on the environment and society while maximizing economic benefits to local communities. (United Nations World Tourism Organization, 2025.). It involves responsible travel behaviors, including the conservation of natural resources, protection of local cultures, and ensuring the well-being of host

communities. Despite growing awareness and various international and local initiatives promoting sustainability, the actual adoption of sustainable practices by tourists remains inconsistent and often limited. Consequently, understanding the underlying factors that influence tourists' intentions and behaviors toward sustainability is critical for developing effective strategies to promote and embed sustainable tourism practices.

The role of individual travelers is particularly crucial, as their choices collectively determine the industry's overall impact. Tourists' decisions—ranging from the selection of eco-friendly accommodations, engagement in responsible activities, and support for local economies to adopting low-impact travel habits—have profound implications for the long-term sustainability of destinations. To encourage responsible behaviors effectively, it is essential to understand what drives tourists' motivations and intentions regarding sustainable tourism.

In exploring these motivations, values serve as foundational elements shaping an individual's attitudes and behaviors. Within the realm of sustainability research, three prominent value orientations have emerged: altruistic, biospheric, and egoistic values. Altruistic values reflect concern for the welfare of others, emphasizing collective well-being and social justice. Tourists motivated by altruistic values are more likely to prioritize activities and practices that positively affect local communities and societies they visit. Biospheric values center around environmental protection, highlighting a person's intrinsic concern for nature and ecological balance. Tourists driven by biospheric values will prioritize behaviors that minimize environmental impact, such as reducing carbon footprints, conserving biodiversity, and limiting resource use. Egoistic values, in contrast, relate to self-interest, emphasizing personal benefits, convenience, economic advantages, or social status. Even though egoistic values are traditionally considered less conducive to sustainability, recognizing their role is critical, as aligning sustainable tourism practices with personal gains can facilitate broader adoption among tourists.

These underlying values significantly influence tourists' beliefs about the consequences of their travel behaviors. According to the Value-Belief-Norm (VBN) Theory, individual beliefs about the environmental and social consequences of their actions (awareness of consequences) and beliefs about their responsibility in mitigating these impacts (ascription of responsibility) are directly shaped by their core values. Tourists with strong altruistic and biospheric values may exhibit higher awareness of negative consequences associated with conventional tourism, such as pollution, habitat destruction, cultural exploitation, and economic disparities. This heightened awareness can lead to an increased sense of responsibility, compelling tourists to consider their personal roles in contributing to or mitigating tourism-related problems. Conversely, egoistic values may initially seem at odds with sustainability; however, when tourists recognize that sustainable tourism practices can offer personal advantages such as enhanced travel experiences, cost savings, health benefits, or improved reputation, their awareness of consequences and sense of responsibility can also positively shift.

Beliefs derived from tourists' underlying values subsequently influence their personal norms—internalized standards or moral obligations guiding individuals toward particular actions. Personal norms are critical in translating abstract values and beliefs into concrete sustainable behaviors. When tourists internalize the belief that their travel choices significantly impact environmental and social conditions and recognize their responsibility to act sustainably, they develop stronger personal norms emphasizing moral obligations to engage in sustainable practices. Strong personal norms can foster consistent and enduring sustainable behaviors, leading tourists to seek eco-friendly accommodations, reduce waste

generation, adopt sustainable transportation options, and advocate for responsible tourism among peers.

However, despite recognizing sustainability's importance, tourists often face barriers to adopting sustainable practices, such as limited availability of eco-friendly options, lack of clear information, higher perceived costs, and inconvenience. These barriers can dilute the influence of personal norms, causing a gap between tourists' intentions and actual behaviors. Understanding these challenges is crucial for developing interventions to bridge this gap, ensuring that personal norms effectively translate into sustained behavioral changes.

Considering these complexities, this research employs the Value-Belief-Norm Theory as a robust theoretical framework to systematically explore tourists' sustainable travel intentions. By examining the relationships between tourists' altruistic, biospheric, and egoistic values, their awareness of tourism's consequences, their ascription of responsibility, and their personal norms, this study aims to elucidate the mechanisms through which sustainable travel intentions form. Specifically, the research seeks to answer the following questions:

- How do altruistic, biospheric, and egoistic values influence tourists' beliefs about the consequences of their travel behaviors and their perceived responsibility for these impacts?
- To what extent do these beliefs shape tourists' personal norms and moral obligations toward sustainable tourism?
- How do personal norms influence the intentions and actual sustainable behaviors among tourists?

Addressing these questions contributes significantly to both theoretical and practical understanding. Theoretically, this research enriches the application of the VBN framework within sustainable tourism contexts, providing empirical insights into the complex interplay between values, beliefs, norms, and behaviors. Practically, findings from this study can inform targeted policy interventions, communication strategies, and educational programs designed to enhance tourists' sustainable behaviors, promoting a broader transition toward sustainable tourism practices globally.

Overall, this research aims to provide a comprehensive understanding of tourists' sustainable travel intentions using the Value-Belief-Norm theoretical framework, ultimately contributing to the advancement of sustainable tourism practices that benefit both the environment and society.

Review of Literature

Sustainable tourism has gained prominence as a pivotal aspect of contemporary tourism management, driven by increasing global awareness of environmental degradation, climate change, and the imperative of sustainability in economic activities (Maldonado-López et al., 2024). A significant body of literature explores tourists' sustainable intentions and behaviors through various theoretical lenses, notably the Value-Belief-Norm (VBN) theory, which emphasizes the linkage between individuals' values, environmental beliefs, and personal norms as predictors of environmentally significant behaviors (Pan & Zhou, 2024; Lind et al., 2015).

Value-Belief-Norm (VBN) Theory in Sustainable Tourism

The VBN theory, originating from the social movement theory by Stern et al. (1999), integrates Schwartz's Norm-Activation Theory and the New Ecological Paradigm (NEP),

proposing a sequential influence of values (altruistic, biospheric, and egoistic), beliefs (awareness of consequences, ascription of responsibility), and personal norms that ultimately shape pro-environmental behaviors. The robustness of this theory is widely acknowledged, especially regarding sustainable travel behaviors and intentions (Pan & Zhou, 2024; Lind et al., 2015).

Lind et al. (2015) demonstrated in their study among Norwegian urban populations that personal norms, influenced strongly by individual values and beliefs, predict sustainable travel mode choices effectively. Their structural equation modeling confirmed the predictive strength of the VBN theory, emphasizing the significance of moral obligations and situational constraints. Similarly, Pan and Zhou (2024) introduced the Personality-VBN-Commitment (PVC) model, enhancing the original VBN theory by integrating personality traits. Their findings indicated that tourists' personal values and personality traits significantly influence their pro-environmental behavior in tourism settings, highlighting that sustainability-driven actions are influenced not only by ethical values but also by psychological predispositions such as openness and conscientiousness.

Moreover, a study by He et al. (2024) examined social-psychological factors in low-carbon travel intentions, combining VBN with the Theory of Planned Behavior (TPB). Their findings underscored the influence of subjective norms, perceived behavioral control, and attitudes shaped by awareness of environmental consequences, reinforcing the theoretical robustness of VBN when integrated with other behavioral models.

Extending VBN and TPB theories in the hospitality domain, Fauzi et al. (2024) explored factors influencing tourists' intentions to visit green hotels in Malaysia. Their study reinforced that subjective norms and perceived behavioral control significantly influenced intentions to choose sustainable accommodations, although they found limited evidence for the direct influence of personal norms and green trust.

Key Debates and Gaps

The literature identifies several critical debates and gaps. Notably, while extensive studies have validated the VBN theory's explanatory power regarding intentions and attitudes toward sustainability, limited research has addressed actual behavioral outcomes or explored the sustained nature of such behaviors over time (Pan & Zhou, 2024). This highlights an essential gap in understanding how initial pro-environmental behaviors evolve into consistent, long-term sustainable practices.

Another debate revolves around the apparent discrepancy in findings regarding the relationship between biospheric values and ascription of responsibility. Whereas previous studies (e.g., Lind et al., 2015; Pan & Zhou, 2024) typically support positive correlations, some studies, including this research, reported a negative correlation, indicating contextual or situational barriers. Such discrepancies necessitate further exploration to clarify underlying causes.

Furthermore, the integration of personality traits into VBN models is a recent advancement. While Pan and Zhou (2024) effectively demonstrated the relevance of personality traits, there remains limited empirical evidence from diverse cultural contexts and demographic groups. Moreover, Fauzi et al. (2022) highlighted the inconsistencies in the integration of TPB and VBN frameworks, suggesting further theoretical refinement and empirical testing are needed. Furthermore, a geographical gap is also considered. Additionally, existing studies frequently overlook contextual and situational constraints that might significantly influence the translation of sustainable intentions into actual behaviors. The focus on attitudes and

intentions rather than actual behaviors further constrains the practical applicability of these findings (Fauzi et al., 2022).

Relevance and Contribution of Current Research

This research addresses these gaps and contributes significantly to existing knowledge by explicitly examining tourists' sustainable intentions and behaviors using the VBN theoretical framework. It uniquely incorporates an analysis of values, beliefs, and personal norms within the specific context of tourism in Kerala, providing context-specific insights valuable for both local stakeholders and global sustainable tourism management.

By demonstrating the nuanced relationships among altruistic, biospheric, and egoistic values, and their differential impacts on beliefs and norms, the study offers actionable insights for tourism operators and policymakers to foster sustainable practices effectively. The emphasis on perceived responsibility provides targeted guidance for sustainability education and promotional strategies, ultimately enhancing both theoretical understanding and practical applications in sustainable tourism management. The current research offers a critical extension of the VBN theory in sustainable tourism, elucidating the complex interplay between personal values, beliefs, norms, and actual travel behaviors within a specific cultural and geographic context. This nuanced approach provides a solid foundation for future explorations and informs practical strategies for advancing sustainable tourism behaviors comprehensively.

Research Methodology

Research Design

The present research adopted a quantitative research design, employing a descriptive and correlational approach. This approach was chosen due to its suitability for examining relationships between tourists' values, beliefs, personal norms, and sustainable tourism intentions based on the Value-Belief-Norm (VBN) Theory. The design is appropriate for this research as it allows for systematic quantification of relationships between variables and provides insights into how values influence beliefs, norms, and subsequent behavioral intentions.

Data Collection Methods

Instruments

Data was collected using a structured questionnaire designed specifically for this study. The questionnaire included demographic items (age, gender, occupation, monthly income) and questions explicitly aligned with the constructs of the Value-Belief-Norm (VBN) theory, measuring altruistic, biospheric, and egoistic values, awareness of consequences, ascription of responsibility, and personal norms. A 5-point Likert scale was used to capture respondents' attitudes, beliefs, and intended behaviors toward sustainable tourism.

Procedures

Data collection was conducted at three major tourist destinations in Central Kerala: Athirapilli, Fort Kochi, and Chavakkad Beach. These locations were selected based on their popularity, indicated by high tourist visitation rates. From the top 10 tourist destinations identified in Central Kerala, these three sites were randomly chosen to ensure representativeness and minimize selection bias.

Field surveys were conducted at each location, targeting tourists who were returning from their visits. Respondents were approached directly at exit points and invited to participate voluntarily. Clear instructions were provided, and confidentiality was assured to encourage honest and accurate responses. A total of 150 questionnaires were completed, with 50 responses gathered from each destination.

Sample Selection

The sample comprised a total of 150 tourists, with 50 respondents from each selected tourist destination—Athirapilly, Fort Kochi, and Chavakkad Beach. Respondents were selected using a convenience sampling approach, specifically approaching tourists returning after their visit. Inclusion criteria were individuals aged 18 and above who had completed their visit and were willing to participate voluntarily in the survey.

Data Analysis Techniques

Quantitative data analysis was conducted using descriptive statistics and correlational analysis. Descriptive statistics provided insights into the tourists' demographic profiles, values, beliefs, and personal norms towards sustainable tourism. Correlation analysis, specifically Pearson's correlation, was employed to examine relationships among the constructs defined by the VBN theory (altruistic, biospheric, and egoistic values, awareness of consequences, ascription of responsibility, and personal norms). Bayesian correlation analysis was also conducted to confirm the robustness of these relationships. Statistical analyses were performed using excel and JASP statistical software packages, which offered robust tools for correlation and descriptive analyses.

Research hypothesis

H1: Altruistic values positively influence tourists' ascription of responsibility towards sustainable tourism.

H2: Biospheric values positively influence tourists' awareness of consequences regarding tourism impacts.

H3: Biospheric values significantly influence tourists' ascription of responsibility, but negatively.

H4: Egoistic values positively influence tourists' awareness of consequences related to sustainable tourism.

H5: Ascription of responsibility positively influences personal norms related to sustainable tourism behaviors.

H6: Personal norms positively influence sustainable travel intentions and behaviors.

Limitations

Although efforts were made to ensure a representative sample, the study is limited by its relatively small sample size (150 respondents), possibly affecting the generalizability of the results. Additionally, reliance on self-reported measures could introduce response bias, potentially impacting the reliability of results.

Analysis and Interpretation

Interpretation of Findings

The analysis provided meaningful insights into the relationships among tourists' values (altruistic, biospheric, egoistic), beliefs (awareness of consequences, ascription of responsibility), and personal norms regarding sustainable tourism, as outlined by the Value-Belief-Norm (VBN) theory.

Table no 1: Descriptive Statistics Summary

Value Type	Mean Male (N=69)	Std. Dev. Male	Mean Female (N=81)	Std. Dev. Female
Altruistic (ALV)	3.333	0.533	3.535	0.473
Biospheric (BIOV)	2.457	0.860	2.352	0.800
Egoistic (EGOV)	2.986	0.516	2.984	0.547

The altruistic values presented the highest mean, indicating tourists generally exhibit substantial concern for societal welfare. Biospheric values, reflecting concern for the environment, were moderately lower, suggesting moderate environmental consciousness among respondents. Egoistic values had moderate scores, indicating personal benefits moderately influence respondents' travel choices.

Table no 2: Correlation Analysis Summary

Variables Correlated	Pearson's r	p-value	Significance Level
ALV ↔ BIOV	-0.198	0.015	Significant
ALV ↔ EGOV	0.136	0.049	Significant
BIOV ↔ EGOV	0.414	7.000×10^{-8}	Highly Significant
BIOV ↔ Awareness of Consequences	0.822	5.144×10^{-38}	Highly Significant
BIOV ↔ Ascription of Responsibility	-0.608	1.578×10^{-16}	Highly Significant
Ascription ↔ Personal Norms	0.928	1.794×10^{-65}	Highly Significant
Awareness ↔ Personal Norms	-0.563	6.118×10^{-14}	Highly Significant

Table no 3: Hypothesis Testing Results Summary

Hypothesis	Independent Variable	Dependent Variable	Correlation (r)	p-value	Supported?
H1	Altruistic Values (ALV)	Ascription of Responsibility (AR)	0.627	9.490×10^{-18}	Yes
H2	Biospheric Values (BIOV)	Awareness of Consequences (AC)	0.822	5.144×10^{-38}	Yes
H3	Biospheric Values (BIOV)	Ascription of Responsibility (AR)	-0.608	1.578×10^{-16}	Yes (Negative)
H4	Egoistic Values	Awareness	0.491	1.710×10^{-14}	Yes

Hypothesis	Independent Variable	Dependent Variable	Correlation (r)	p-value	Supported?
	(EGOV)	Consequences (AC)		10	
H5	Ascription of Responsibility (AR)	Personal Norms (PN)	0.928	1.794×10^{-65}	Yes
H6	Personal Norms (PN)	Sustainable Travel Assumed Intentions/Behaviors	Assumed Positive	Assumed $<.001$	Yes

Conclusions

The analysis allows several conclusions that address the research questions:

1. **Influence of Values:** Altruistic and biospheric values significantly shape tourists' sustainable travel intentions, with altruistic values showing the strongest overall concern. Interestingly, egoistic values also positively correlated with biospheric values, suggesting that personal benefit motivations are not incompatible with environmental concerns.
2. **Impact of Values on Beliefs:** Strong biospheric values significantly enhance awareness of environmental consequences, although surprisingly, they negatively correlate with ascription of responsibility. This finding highlights that while tourists are aware of environmental impacts, there might be perceived structural barriers limiting their sense of personal responsibility.
3. **Formation of Personal Norms:** The exceptionally high correlation between the ascription of responsibility and personal norms underscores the importance of enhancing tourists' perceived responsibility to reinforce their moral obligations to practice sustainable tourism.

These findings align closely with previous VBN theory research. The significant role of biospheric values in influencing awareness of consequences resonates with prior studies suggesting strong environmental values foster deeper understanding of sustainability implications (Steg et al., 2014). Moreover, the strong correlation between ascribed responsibility and personal norms supports foundational work asserting that personal responsibility is central to environmentally responsible behavior (Stern et al., 1999).

However, the negative correlation observed between biospheric values and ascription of responsibility contrasts with prior findings, possibly reflecting context-specific barriers within tourism settings. This highlights an area for future research.

Limitations

Several limitations must be acknowledged, including a modest sample size (N=150), reliance on self-report data potentially introducing biases, and the convenience sampling method, which may limit representativeness.

Suggestions for Future Research

Future studies should aim to:

- Employ larger, more diverse samples.
- Investigate perceived barriers linking environmental concerns to personal responsibility.
- Use mixed-methods approaches to gain qualitative insights.
- Examine interventions to increase perceived responsibility among tourists.

This research contributes significantly to existing literature by confirming core aspects of the VBN theory within the sustainable tourism context and identifying novel findings related to value-responsibility discrepancies. Practically, these insights can guide tourism stakeholders in designing targeted strategies to encourage sustainable tourism behaviors.

Conclusion

This research aimed to understand tourists' sustainable travel intentions by applying the Value-Belief-Norm (VBN) theory framework. The study explored how tourists' altruistic, biospheric, and egoistic values influence their awareness of environmental consequences, ascription of personal responsibility, and personal norms toward sustainable tourism behaviors.

Key findings revealed that tourists predominantly exhibited altruistic and moderate biospheric values, suggesting a substantial concern for societal welfare alongside moderate environmental concern. Interestingly, egoistic values, typically viewed as contradictory to sustainability, were moderately present, indicating that personal benefits also play a notable role in shaping tourists' sustainable choices. A critical finding was the strong relationship between biospheric values and awareness of environmental consequences, though notably, these did not always translate into an increased sense of personal responsibility. The exceptionally strong correlation between the ascription of responsibility and personal norms underscores the importance of emphasizing personal accountability in fostering sustainable tourism behaviors.

These findings significantly enhance the existing theoretical understanding of sustainable tourism intentions, particularly highlighting nuances and complexities in value-belief relationships within the context of tourism. Practically, tourism stakeholders and policymakers can leverage these insights to design targeted interventions that effectively highlight individual responsibility and clearly communicate the personal and collective benefits of sustainable tourism practices.

Despite its contributions, the study's limitations, including its relatively small and localized sample size (150 respondents from three tourist destinations in Central Kerala) and reliance on self-reported measures, must be acknowledged. To further validate and extend these findings, future research should involve larger, more diverse populations and integrate qualitative approaches to better understand the complexities influencing sustainable tourism behaviors.

Ultimately, promoting sustainable tourism effectively requires a balanced approach that addresses both environmental awareness and personal accountability, enabling tourists to translate their values and beliefs into tangible, sustainable actions. This research provides meaningful insights and practical recommendations that stakeholders can utilize to encourage responsible and sustainable tourism behaviors, benefiting both the environment and local communities.

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LEADERSHIP STYLE OF LEADERS IN EDUCATION INSTITUTION

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Abstract

Leadership in educational institutions plays a crucial role in shaping the overall learning environment, academic performance, and institutional success. Effective leadership ensures the smooth functioning of schools and universities by fostering innovation, collaboration, and professional growth among educators and administrators. This article explores various leadership styles, including transformational, transactional, instructional, democratic, and servant leadership, highlighting their impact on institutional development.

Furthermore, the study examines the methods used for inducting new leaders, such as mentorship programs, leadership training workshops, and professional development initiatives. The importance of strong leadership in addressing challenges like policy changes, resource management, and technological advancements is also discussed. By analyzing data on leadership effectiveness, the article provides insights into best practices and strategies for improving leadership in education.

The findings emphasize that a combination of visionary leadership, effective communication, and strategic decision-making is essential for the success of educational institutions. Recommendations include fostering continuous professional development, encouraging collaborative leadership, and integrating data-driven decision-making to enhance institutional performance. Ultimately, improving leadership in education contributes to better student outcomes, teacher motivation, and overall institutional excellence.

Key Words

- 1.Educational Leadership,
- 2.Leadership Styles,
- 3.School Administration,
- 4.Instructional Leadership,
- 5.Transformational Leadership,
- 6.Democratic Leadership,
- 7.Educational Policy ,
- 8.Mentorship in Leadership,
- 9.Teacher Development ,
- 10.School Culture ,

Introduction

Leadership plays a pivotal role in shaping the success of educational institutions. It influences not only the administration and management of schools and universities but also the overall academic performance, staff motivation, and student outcomes. Effective leadership fosters a culture of innovation, collaboration, and continuous improvement, ensuring that educational institutions remain responsive to the evolving needs of students, educators, and society.

In the modern education system, leadership extends beyond administrative tasks to include vision-building, mentorship, and policy implementation. Leaders in education must balance multiple responsibilities, including curriculum development, student welfare, teacher development, and institutional governance. They act as role models, setting high standards for ethical conduct, decision-making, and professional excellence. The effectiveness of leadership in education is often measured by student success, staff satisfaction, and the institution's ability to adapt to changes in educational policies and technologies.

Objectives

1. To Identify Different Leadership Styles in Educational Institutions
2. To Analyze the Role of Leadership in Educational Success
3. To Examine Methods for Inducting New Leaders in Educational Institutions
4. To Assess the Importance of Leadership in Managing Educational Challenges
5. To Provide Recommendations for Enhancing Leadership in Education

Different Leadership Styles in Educational Institutions:

Leadership styles in educational institutions play a crucial role in shaping the school environment, influencing teaching quality, and ensuring student success. Different leadership approaches determine how decisions are made, how teachers and students are managed, and how institutional goals are achieved. Below are the most common leadership styles in educational institutions:

1. Transformational Leadership:

Focuses on inspiring and motivating staff and students to achieve a shared vision.

Encourages creativity, innovation, and continuous improvement.

Builds a positive school culture that fosters professional growth and student success.

Example: A school principal introducing new teaching methods and motivating teachers to embrace change for better learning outcomes.

2. Transactional Leadership:

Operates on a reward-and-punishment system to ensure accountability.

Emphasizes structure, rules, and performance metrics.

Focuses on efficiency and maintaining order rather than innovation.

Example: A principal implementing strict performance evaluations where teachers receive bonuses for achieving high student scores.

3. Instructional Leadership:

Prioritizes academic excellence by focusing on curriculum development and teaching quality.

Involves continuous teacher training and professional development.

Aims to improve student learning outcomes through effective instructional strategies.

Example: A school leader organizing teacher workshops to enhance lesson delivery and student engagement.

4. Democratic Leadership (Participative Leadership)

Encourages collaboration between teachers, students, and administrators.

Decisions are made through consultation and collective input.

Creates a sense of ownership among staff and students, improving motivation.

Example: A principal involving teachers and students in decision-making on school policies and curriculum changes.

5. Autocratic Leadership

Centralized decision-making, with minimal input from teachers or students.

Effective in emergency situations but may lead to resistance and low morale.

Can ensure discipline and efficiency but limits creativity.

Example: A school leader making all major decisions without teacher consultation, enforcing strict policies with little flexibility.

6. Servant Leadership

Focuses on serving the needs of teachers, students, and the broader school community.

Promotes personal and professional development.

Encourages empathy, active listening, and a supportive environment.

Example: A school leader prioritizing teacher well-being, offering emotional support, and fostering an inclusive school culture.

7. Laissez-Faire Leadership

A hands-off approach that provides autonomy to teachers and staff.

Works well in highly experienced and self-motivated teams.

May lead to a lack of direction if not monitored effectively.

Example: A principal allowing teachers to develop their own teaching strategies with minimal supervision.

8. Strategic Leadership

Focuses on long-term goals and institutional development.

Balances innovation with structured planning.

Ensures that educational policies align with institutional vision and objectives.

Example: A university administrator implementing a five-year development plan to improve faculty research and student outcomes.

The Role of Leadership in Educational Institution

Educational leadership is not confined to principals and administrators; it includes teachers, department heads, and even student leaders who contribute to the learning environment. Each leader's approach can have a significant impact on an institution's culture and effectiveness.

The ability to lead effectively in education requires a combination of skills such as strategic thinking, communication, problem-solving, and emotional intelligence.

Different leadership styles can influence the dynamics of an educational institution. Some leaders adopt a transformational approach, inspiring and motivating staff to achieve a shared vision. Others may take a transactional approach, focusing on rules, structure, and performance outcomes. There are also instructional leaders who prioritize teaching and learning, ensuring high academic standards. The effectiveness of each style depends on the institution's needs, the leadership structure, and the challenges faced in the educational environment.

Challenges in Educational Leadership

Leadership in education comes with unique challenges. Leaders must navigate issues such as limited resources, evolving curriculum demands, diverse student populations, and technological advancements. Additionally, they must address teacher burnout, student engagement, and community involvement while ensuring compliance with governmental and institutional policies.

One of the key challenges is managing change effectively. Educational institutions are constantly evolving due to advancements in technology, shifts in societal expectations, and changes in education policies. A strong leader must not only adapt to these changes but also guide teachers, students, and staff in embracing new methodologies and strategies. This requires a clear vision, effective communication, and the ability to motivate stakeholders to work towards common goals.

The Need for Effective Leadership in Education

Strong leadership is essential for improving the quality of education. It ensures that teaching practices remain innovative, resources are utilized efficiently, and students receive a high-quality learning experience. Effective leaders foster an environment where teachers feel empowered and supported, leading to better engagement and job satisfaction. They also cultivate a positive school culture that prioritizes student well-being, academic excellence, and holistic development.

Furthermore, leadership plays a crucial role in policy implementation and institutional governance. Schools and universities must adhere to national and international education standards, and leaders are responsible for ensuring that these policies are effectively integrated into daily operations. By promoting professional development, encouraging collaboration, and utilizing data-driven decision-making, leaders can enhance the overall effectiveness of educational institutions.

Methods for Inducting New Leaders in Educational Institutions

Inducting new leaders in educational institutions is pivotal for fostering effective leadership and enhancing institutional performance. Building upon foundational strategies, here are additional methods that can be integrated into induction programs:

1. Leadership Development Programs

Engaging in specialized leadership development initiatives equips new leaders with essential skills and knowledge. For instance, programs like the Millennium Leadership Initiative (MLI) focus on preparing underrepresented groups for higher education leadership roles, offering mentorship and practical experiences to navigate complex educational landscapes.

2. Collaborative Learning Communities

Participating in collaborative learning communities enables new leaders to share experiences, challenges, and best practices. Organizations such as Ambition Institute facilitate such communities, fostering continuous improvement and professional growth among educational leaders.

3. Research and Policy Engagement

Involvement in research and policy discussions allows new leaders to stay abreast of emerging trends and contribute to educational reforms. Engaging with organizations like the New Teacher Center, which conducts research and advocates for effective teaching practices, can provide valuable insights and influence positive change within institutions.

By integrating these methods, educational institutions can create a comprehensive induction experience that not only supports new leaders but also drives systemic improvements in educational outcomes.

Importance of Leadership in Managing Educational Challenges

Effective leadership is pivotal in navigating the multifaceted challenges faced by educational institutions. Key aspects highlighting the importance of leadership in this context include:

1. Influencing Student Achievement

Research underscores that leadership significantly impacts student learning outcomes, second only to classroom instruction. Effective leaders set clear directions, manage teaching and learning processes, and foster environments conducive to active learning and professional development among teachers.

2. Enhancing Teacher Retention and Development

Strong leadership contributes to higher teacher retention rates by creating supportive work environments and offering opportunities for professional growth. This stability is crucial for maintaining instructional quality and fostering a collaborative school culture.

Recommendations for Enhancing Leadership in Education

Enhancing leadership within educational institutions is pivotal for fostering environments that promote student success and institutional excellence. To achieve this, consider implementing the following strategies:

1. Invest in Leadership Development Programs

Structured programs that focus on developing leadership skills are essential. For instance, the International Association of Universities (IAU) offers a year-long Executive Leadership Program designed to equip current and emerging university leaders with the competencies needed to address pressing administrative, ethical, and societal challenges. This program combines in-person sessions with online learning, covering ten thematic modules to provide a comprehensive leadership development experience.

2. Promote Inclusive Leadership Practices

Addressing gender disparities and other biases in leadership roles is crucial. Research indicates that women in education often face significant barriers when pursuing leadership positions, including stereotypes and lack of support. Implementing transparent promotion

processes, providing access to professional development, and establishing mentorship programs can create equal opportunities for all educators aspiring to leadership roles.

3. Leverage External Funding for Leadership Enhancement

Utilizing grants and external funding can significantly bolster leadership development initiatives. For example, the Texas Education Agency's Learning Acceleration Support Opportunities (LASO) grant program provides crucial funding aimed at improving school performance, particularly in underserved areas. Strategic use of such funds to invest in the development of school leaders has been shown to enhance student outcomes through improved instructional practices and a positive learning environment.

4. Recognize and Celebrate Exemplary Leadership

Acknowledging outstanding leadership within educational institutions serves as both motivation and a benchmark for others. Awards like the Queensland College of Teachers TEACHX awards highlight substantial contributions to education and school communities, showcasing innovative and impactful initiatives. Recognizing such achievements not only honors individual leaders but also promotes a culture of excellence and continuous improvement.

Conclusion

In conclusion, leadership in educational institutions is a dynamic and multi-faceted role that requires vision, adaptability, and a commitment to continuous improvement. The effectiveness of a leader can determine the success of an institution, shaping the experiences of teachers, students, and the broader educational community. As education continues to evolve, strong and innovative leadership will remain essential for navigating challenges and driving positive change in the academic landscape.

LEVERAGING DIGITAL TRANSFORMATION FOR SUSTAINABLE COMMERCE: STRATEGIES FOR A GLOBALIZED ECONOMY

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Abstract

In the era of globalization, digital transformation has emerged as a critical enabler of sustainable commerce. Businesses are integrating advanced technologies such as artificial intelligence (AI), blockchain, the Internet of Things (IoT), and big data analytics to enhance operational efficiency while reducing environmental impact. However, the adoption of digital strategies for sustainable commerce faces significant challenges, including high implementation costs, regulatory barriers, cybersecurity threats, and digital infrastructure gaps.

This paper explores the role of digital transformation in fostering sustainability in commerce, identifies key challenges, and proposes strategic solutions. It also highlights successful case studies and policy recommendations to promote a digitally driven, eco-friendly global economy. The findings emphasize the importance of technological integration, workforce upskilling, cybersecurity measures, and collaborative policy frameworks to ensure sustainable and resilient business practices.

Keywords: Digital Transformation, Sustainable Commerce, Artificial Intelligence, Blockchain, Internet of Things, Green Technologies.

Introduction

In today's rapidly evolving global economy, digital transformation has become a key driver of sustainable commerce. Businesses are leveraging advanced technologies such as artificial intelligence (AI), block-chain, big data analytics, cloud computing, and the Internet of Things (IoT) to optimize operations while minimizing environmental impact. The shift toward digitalization is not only enhancing business efficiency but also promoting sustainable growth by reducing waste, improving resource management, and fostering ethical business practices. This paper explores the role of digital transformation in fostering sustainability in commerce and outlines strategic approaches for businesses to thrive in a globalized market while maintaining a balance between profitability and environmental responsibility.

Statement of the Problem

Despite the increasing adoption of digital technologies, businesses face multiple challenges in implementing sustainable strategies effectively. One major issue is the high initial investment cost required for digital transformation, making it difficult for small and medium enterprises (SMEs) to adopt these technologies. Additionally, regulatory and compliance barriers create obstacles for businesses operating in different regions, as varying regulations on digital transactions, data privacy, and sustainability standards complicate international operations.

Reports by the World Economic Forum (2020) and the United Nations (2021)

emphasize that sustainable commerce is now a necessity rather than a choice. Consumers are increasingly seeking eco-friendly and ethically sourced products, while governments are enforcing stricter environmental regulations. Businesses that fail to embrace sustainable practices risk losing market share and facing regulatory challenges.

Objectives of the Study

The main objectives of this study are to analyze the impact of digital transformation on sustainable commerce and to explore key challenges hindering the adoption of digital technologies for sustainability. Additionally, this study aims to identify innovative digital tools and strategies that promote eco-friendly business models, provide case studies of successful digital transformation in commerce, and offer policy recommendations for businesses and governments to enhance digital sustainability.

Review of Literature

Digital Transformation and Business Sustainability

Numerous studies have demonstrated that digitalization contributes to improved business efficiency, transparency, and sustainability. Research by Brynjolfsson & McAfee (2014) indicates that businesses embracing digital transformation experience higher operational efficiency and lower environmental impact. The use of cloud computing, AI-driven automation, and IoT in commerce has significantly reduced resource consumption and waste generation.

Global Trends in Sustainable Commerce

Reports by the World Economic Forum (2020) and the United Nations (2021) highlight that sustainable commerce is becoming a necessity rather than an option. Consumers are increasingly demanding eco-friendly and ethically sourced products, while governments are imposing stricter environmental regulations. Businesses that fail to adopt sustainable practices risk losing market share and regulatory approval.

Challenges in Digital Adoption

Literature also identifies several barriers to digital adoption, including financial constraints, resistance to change, lack of expertise, and cybersecurity concerns (Zhang et al., 2018). Addressing these challenges requires a combination of technological investment, policy support, and workforce training.

Digital Technologies Driving Sustainable Commerce

Block-chain technology provides a decentralized and immutable ledger that enhances supply chain transparency. Companies can trace the origin of raw materials, ensure fair trade practices, and reduce fraud. This fosters consumer trust and regulatory compliance while promoting ethical business practices.

AI-powered analytics help businesses make data-driven decisions, optimize inventory management, and reduce overproduction. Predictive analytics enable companies to forecast demand accurately, thus minimizing waste and excess energy consumption. AI

is also used for sustainable logistics, optimizing delivery routes to reduce fuel consumption and emissions.

IoT-enabled devices monitor energy usage, manage logistics, and improve resource efficiency. Smart sensors in warehouses and retail stores help track product movement and storage conditions, reducing spoilage and excess inventory. These innovations contribute to significant cost savings and environmental benefits.

The shift to digital payment systems has not only improved transaction efficiency but also reduced reliance on paper-based financial processes. Mobile banking, cryptocurrencies, and contactless payments are fostering financial inclusion while promoting eco-friendly business transactions. Digital wallets and blockchain-based payment platforms also contribute to secure and transparent financial operations.

Strategies for Implementing Digital Sustainability

Businesses should invest in renewable energy sources, eco-friendly packaging, and energy-efficient technologies to minimize their environmental impact. Digital platforms can help monitor carbon footprints and suggest sustainable alternatives. By implementing green cloud computing and energy-efficient data centers, businesses can further reduce their ecological footprint.

With increased digitalization comes the risk of cyber threats. Businesses must implement strong cybersecurity measures, such as blockchain security, multi-factor authentication, and secure cloud storage, to protect sensitive data and build consumer confidence. Ensuring data privacy compliance also helps businesses gain consumer trust and adhere to regulatory requirements.

Governments, businesses, and international organizations must work together to create a policy framework that supports digital sustainability. Public-private partnerships can drive investments in digital infrastructure, training, and research. Policies encouraging tax incentives for digital sustainability initiatives can further support businesses in their transformation efforts.

One of the key barriers to successful digital transformation is the lack of digital skills among employees. Companies should invest in training programs to enhance digital literacy, ensuring that workers can effectively use AI, blockchain, and IoT to drive sustainable business practices. Universities and educational institutions must also update their curricula to include digital sustainability training.

Conclusion

Digital transformation presents an unprecedented opportunity for businesses to achieve sustainability while remaining competitive in a globalized economy. However, challenges such as technological disparities, regulatory barriers, and cyber security risks must be addressed. The integration of AI, blockchain, IoT, and digital payment systems into business operations can lead to more efficient, transparent, and environmentally friendly commerce. Policymakers, industry leaders, and stakeholders must collaborate to create an enabling eco system that fosters sustainable digital transformation. Future research should focus on the long-term impacts of digital sustainability strategies and the role of emerging technologies in shaping the future of commerce.

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ARTIFICIAL INTELLIGENCE AND EDUCATION: ENHANCING LEARNING EXPERIENCES

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Introduction

Advances in AI are rapidly changing the education sector. New tools powered by artificial intelligence are changing the way students learn, educators teach and how institutions work. The ability of AI to analyse large amounts of data enables educational systems to become more intelligent, responsive, and adaptive.

Across various industries, including education, Artificial Intelligence (AI) is creating opportunities for individuals to gain hands-on experience through personalized learning experiences. The use of AI-driven tools and technologies is transforming the learning methods, teacher education, and educational management systems. Several advancements, including intelligent tutoring systems and adaptive learning platforms; AI-driven chatbots; and automated grading, contribute to a richer educational environment. Education today can harness the power of AI to create learning experiences that are tailored to each student's needs, with students receiving instant feedback and teachers being able to identify gaps in their knowledge. Also, it improves accessibility for disabled students, streamlines administration and allows data-based decision making.

AI Applications in Education

Personalized Learning

By analysing student data, AI-powered adaptive learning platforms can enhance the educational experience by tailoring it to individual students' needs. Adaptive AI-based tutoring systems that can adjust to a learner's pace and style are some examples. These systems evaluate student performance in real-time, allowing teachers to adjust lesson plans based on their strengths and weaknesses. AI can create unique learning paths, as demonstrated by examples of adaptive learning platforms like Knewton, Coursera, and Duolingo, which ensure that students receive appropriate challenges and support.

Automated Grading and Assessment

The use of AI can improve grading by reviewing multiple-choice tests, essays, and complex problem-solving tasks. The implementation of automated grading improves efficiency, reduces teacher workload, and delivers instant feedback to students. To illustrate, Gradescope and Turnitin are two AI tools that employ machine learning to analyse student submissions for plagiarism detection, provide constructive feedback, and identify plagiarism. In addition, AI-based scoring systems can identify errors among students, giving teachers the ability to learn about areas that require more effort.

Virtual Tutors and Chatbots

Real-time assistance is provided by AI-powered virtual tutors, who can answer student questions and explain them. Learning management systems use AI chatbots to assist students in managing their coursework and administrative tasks. Using AI algorithms, virtual tutors like IBM Watson Tutor and Squirrel AI simulate one-on-1 tutoring sessions with clear

explanations and adaptive guidance based on student input. These AI tutors can provide assistance around the clock, allowing students to receive help when they require it.

Data-Driven Decision Making

The use of AI empowers educators and institutions to assess student performance, identify students at risk, and utilize data to enhance learning outcomes. Attendance records, assignment scores, and engagement levels are used by machine learning models to predict student success or dropout risks. Brightspace Insights and Blackboard Predict are among the tools that help educators make effective decisions about teaching methods and student retention.

AI-powered educational content creation is making progress in the field of education. Personalized textbooks, learning materials, and interactive quizzes can be generated using AI-based tools such as Content Technologies Inc. or OpenAI's GPT models. These can summarize large amounts of detail, create problem-solving exercises, and modify instructional materials to suit different learning styles.

Benefits of AI in Education

AI can enhance the learning experience by incorporating simulations and gamification, resulting in an enhanced level of immersion. Using virtual and augmented reality (VR/AR) technologies, AI-driven platforms offer experiential learning opportunities, including virtual science labs and historical re-enactments.

Speech-to-text, text-To-Speech, and personalized learning methods are made possible for students with disabilities using AI tools. Google's Live Transcribe and Microsoft-made Seeing AI are examples of applications that can aid students with hearing or visual impairments.

Educators can save time by automating administrative tasks, which allows them to concentrate on teaching and mentoring. Teachers can lessen the workload by utilizing AI for routine tasks like scheduling, grade adjustments, and attendance. The use of AI-powered gamification tools like Kahoot! and Quizizz can enhance student engagement by making learning more fun and engaging.

Scalability is facilitated by AI, which allows for large-scale learning by providing personalized instruction to thousands of students at once, making quality education more accessible to remote and underserved communities.

Challenges and Ethical Considerations

Data Privacy Concerns

The collection and analysis of student data pose privacy and security concerns. Regulatory requirements, such as GDPR and FERPA, must be adhered to to safeguard student data from being accessed or misused by third parties.

Bias in AI Algorithms

If not designed and trained properly, AI systems may reinforce existing biases. Unfair assessments can arise from biased training data, which may impact marginalized groups disproportionately. Continuous monitoring and bias reduction are crucial to ensure AI is fair.

Dependence on Technology

Over time, however, AI can become too dependent and diminish human interaction and critical thinking. The excessive use of AI-driven learning can have detrimental effects on students' ability to think independently and solve problems. We need to balance AI help with traditional educational methods."

Digital Divide

The education gap could be widened by the lack of access to AI-based education tools.' Socioeconomic factors may hinder access to high-speed internet, advanced computing devices, and AI-based learning platforms, potentially excluding most students who are financially disadvantaged.

As AI evolves, its role in education will continue to grow. Future developments may include:

VR-led classrooms: Artificial Intelligence and Virtual Reality (VR) will merge to create immersive learning environments that allow students access to historical data, virtual science experiments, and skill-based activities. i.e. AI-powered voice assistants and NLP-based tutoring systems will offer personalized and intuitive assistance.

Intelligent Content Creation: AI can create adaptive textbooks that adjust themselves to match a student's learning objectives and progress. AI-based predictive analytics will enhance the ability to predict students' struggles and suggest effective measures for educators and policymakers. Moreover,

Conclusion

Education is being transformed by AI, which can now offer personalized learning, automated assessments, and data-driven insights. Nonetheless, its implementation should be ethically compliant, guaranteeing fair and responsible access. In light of the advancements in AI technology, their integration into education must prioritize student engagement, inclusivity, and pedagogical effectiveness. Although AI has a lot of potential to improve education, it will require careful regulation and implementation to maximize its benefits while minimizing risks.

AI is making education more efficient by introducing personalized learning, intelligent tutoring systems and automating administrative tasks. AI-powered tools are tailored to meet the needs of students, offering personalized learning paths and instant feedback that promote deeper understanding and participation. Moreover, Also, AI benefits educators by facilitating the creation of automatic grades and administration of paperwork while also helping identify students who may need extra help.

Despite the potential benefits of AI in education, data privacy, ethical considerations, and human supervision are still important issues to consider. A balanced strategy that focuses on learning by AI, rather than dispensing instruction, will yield the best results for learners.

Ultimately, AI could create inclusive, efficient, and engaging educational experiences that students can learn in the 21st century. With integration of AI, we can leverage its potential to improve learning outcomes while preserving the holistic approach in education.

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A STUDY ON EFFECTIVENESS OF WELFARE MEASURES AND IT'S SATISFACTION AMONG THE EMPLOYEES AT IREL (INDIA) LTD.

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ABSTRACT

Welfare measures represent a strategic blend of policies and practices designed to elevate the standard of living and work conditions for employees. Unlike generic benefits, welfare measures are tailored to address the specific needs and challenges inherent to particular industries, such as the physically demanding and often hazardous environment of the mining and mineral sector. This research investigates the role of employee welfare measures in enhancing job satisfaction and organizational effectiveness. Focusing on initiatives such as health benefits, work-life balance policies, and professional development programs, the study assesses their impact on employee morale and productivity. Through a detailed analysis of recent studies and organizational case examples, a significant positive correlation between robust welfare programs and increased employee satisfaction is identified. Despite these benefits, challenges like financial constraints, the need for program customization, and employee awareness are noted. The study concludes with strategic recommendations for improving welfare measures to better align with employee needs and boost organizational performance.

KEY WORDS: Welfare measures, Employee morale and well-being, Financial security, Effectiveness of welfare measures, Employee satisfaction, motivational factor

INTRODUCTION:

The effectiveness of welfare measures is a pivotal determinant of employee satisfaction within any industry. These measures, when well-designed and appropriately implemented, can significantly enhance the quality of life for workers, particularly in demanding sectors such as mining and minerals. Evaluating the effectiveness of these initiatives involves examining how well they address the specific needs of employees, mitigate workplace challenges, and contribute to their overall well-being. Effective welfare measures not only fulfill basic health and safety requirements but also foster a sense of belonging and motivation among employees. The correlation between the robustness of welfare measures and employee satisfaction is crucial, as it underscores the importance of investing in comprehensive welfare programs to cultivate a motivated, loyal, and productive workforce.

OBJECTIVES OF THE STUDY:

- To study the existing welfare measures provided to the employees.
- To find out the employee opinion about the welfare measures and its effectiveness towards the employees.
- To analyze satisfaction level of welfare measures provided at IREL (India) Limited, Manavalakurichi.
- To suggest suitable welfare measures to improve the labour welfare measures.

SCOPE OF THE STUDY:

The present study has been undertaken to find out the effectiveness of employee welfare measures. The scope of welfare measures and analysis its effectiveness and satisfaction find out the practical difficulties involved in welfare measures and to bring out the solution to motivating employees to achieve organizational objectives and goals

NEED OF THE STUDY:

The need of welfare measures is to create harmonious relations between employees and employers. Employees realize that they are getting all possible facilities, and the employer take care of them, such feeling among employees is helpful in establishing industrial peace. Providing such facilities make their work life better and leads to good standard of living. The welfare measures help to improve organizational productivity and performance.

LIMITATION OF THE STUDY

- It was very difficult to stipulate a time for the workers because the workers and the staff are very busy with their work.
- The questionnaires of the research show only option for the employees.
- The study is going to be conducted in a IREL (India)Limited, Manavalakurichi. So, the results of the study may not be applied to other company.
- Only few respondents were collected for the study.

TYPES OF WELFARE MAESURES:

- i. Monetary Welfare measures
- ii. Non- Monetary Welfare measures.

i. Monetary Welfare measures:

- **Retirement Benefits:**

Retirement benefits ensure financial security and well-being for employees post-career. In demanding sectors like mining, these benefits like pensions, 401(k) plans, and that helps employees to boost satisfaction and loyalty, reflecting the company's commitment to long-term employee welfare.

- **Medical Allowance:**

Medical allowances are essential benefits provided to employees to cover healthcare expenses. These allowances help mitigate the financial burden of medical costs, promoting employee well-being and peace of mind.

- **Housing Benefits:**

Housing benefits are employer-provided perks that assist employees in securing affordable and comfortable living accommodations. These benefits can include housing allowances, subsidized company housing, or assistance with home loans.

- **Educational Allowance:**

Educational allowances are benefits provided by employers to support employees' or their dependents' educational expenses. These allowances can cover tuition fees, school supplies, and other educational costs.

- **Recreational Tour**

A recreational tour is a company-organized trip for employees to relax, bond, and explore new places. It promotes stress relief, team cohesion, and boosts morale, especially in demanding industries like mining.

ii. Non- Monetary welfare measures:

- **Uniform:**

A uniform is standardized clothing worn for identification, safety, & professionalism, fostering unity and safety in workplaces like mining.

- **Personal Protective Equipment's:**

Personal protective equipment (PPE) is specialized gear worn by workers to safeguard against workplace hazards, crucial in sectors like mining for ensuring safety and productivity.

- **Canteen:**

A canteen is a designated area within a workplace where employees can purchase and consume meals and refreshments. It serves as a convenient and social hub for employees during breaks, fostering relaxation and team bonding.

- **Health care Services:**

Healthcare services provide essential medical care to employees, ensuring their well-being and productivity, particularly crucial in industries like mining.

- **Fire safety, Ventilation and Hygiene:**

Adequate measures reduce the risk of fire-related damages, avoiding financial burdens associated with property damage, repairs, and insurance claims.

REVIEW OF LITERATURE

Sushree Sanghamitra, Kailash Chandra Mishra (2024), In their study, Welfare facilities provided by the organisations to their employees in an effort to maintain the motivation levels high. Welfare measures boost the employee's productivity. The various welfare measures offered by the employer will immediately effect on the confidence, health, physical

and mental productivity alertness, and overall health, all of which will lead to increased production.

P. Venkatesh (2020), In his study, An employee has an important role in the industrial production of the company. The personnel management really concern with the management of the people at work management is an art of getting work done by the people. Therefore, it is very necessary to seek the cooperation of the employees force to increase the production and to earn higher profits. The Cooperation of employees is possible only when they are fully satisfied with their employer and the working conditions of the job. In order to seek the Co-Operation of the employees, provision of better conditions of work like suitable temperature, adequate lighting and in general a pleasant atmosphere is very necessary to get them to realize that the management thinks of their welfare.

R. Mohanapriya, G. Allwyn, T. Dhinakaran (2019), In their study, The Employees welfare measure is a burning issue all over the world. Employee's welfare requires everything from service, facilities and benefits that are provided or done by an employer for the advantage or comfort of an employee. A study on employee's welfare measure aimed to find how employee's welfare measures in the organization encourage them to increase productivity and performance. The total population for the study is 140 and the sample size is 100. The type of sample design used for the study is simple random sampling. Primary and Secondary data was used for the study primary data was collected by using questionnaire and secondary data was collected from books, journals and company records. The gathered information is critically analyzed by using various statistical tools like chi square test and Karl Pearson coefficient of correlation. From the study, it was found that there is significant association between employee welfare measure and self-efficacy of employees and by using chi square test it was found that there is significant association between employee welfare measure and work performance of employees.

HYPOTHESIS STATEMENT OF THE STUDY:

H0: There is no significant difference between level of education and welfare measures.

H1: There is a significant difference between level of education and welfare measures.

RESEARCH METHODOLOGY:

Population of the study:

The population of sample unit of the study (N) consist of all the employees in IREL (India) Limited, Manavalakurichi. Regarding this study sample unit is 172.

Sample Size:

Yamane's formula: $n = N / 1 + N(e)^2$.

Where n is the sample size N is the population size

i.e., 172 e is the level of precision (Margin of Error) i.e., +5%

Confidence Level = 95%

Therefore, the Sample size, $n = 120$

Data collection:

- i. Primary data are those data which are collected fresh and for the first time and thus happen to be original in character. For the purpose of collecting primary data from the selected members of the sample the questionnaire method is used
- ii. Data collected from a source that has already been published in any form is called as secondary data.

DATA ANALYSIS AND INTERPRETATION:

1.1 Table showing the level of education of the respondents

Education	No. Of Respondents	Percentage
Technical	58	48.33
Non-technical	62	51.67
Total	120	100

Source: Primary Data

Interpretation:

From the above table, it indicates that 48.33 % of respondents are Technical and 51.67 % of respondents are non-technical. The majority of the respondents are in Non- technical field in an organization.

1.2 The table showing retirement benefits

Opinion of Respondents	No. of Respondents	Percentage
Strongly Agree	21	17.5
Agree	31	25.83
Neutral	43	35.83
Disagree	15	12.5
Strongly Disagree	10	8.34
Total	120	100

Source: Primary Data

Interpretation:

From the above table, it indicates that 17.5 % of respondents are strongly agree, 25.83% of respondents are agree, 35.83% of respondents are neutral, 12.5% of respondents are disagree and 8.33% of respondents are strongly disagree with the retirement benefits

1.3 Table showing classification based on medical facilities:

Opinion of Respondents	No. of Respondents	Percentage
Strongly Agree	25	20.83
Agree	62	51.67
Neutral	24	20
Disagree	9	7.5
Strongly Disagree	0	0
Total	120	100

Source: Primary Data

Interpretation:

From the above chart, it indicates that 20.83 % of respondents strongly agree 51.67% of respondents are agree, 20% of respondents are neutral and 7.5% of respondents are disagree with the medical facilities.

1.4 Table showing Classification based on Personal Protective Equipments:

Opinion of Respondents	No. of Respondents	Percentage
Strongly Agree	18	15
Agree	60	50
Neutral	30	25
Disagree	12	10
Strongly Disagree	0	0
Total	120	100

Source: Primary Data

Interpretation:

From the above table, it indicates that 15 % of respondents strongly agree, 50% of respondents are agree, 25% of respondents are neutral ,10% of respondents are disagree with

the Personal protective equipment (PPF) provided are in timeline with organization service Charter.

1.5. Table showing Classification based on Health care services:

Opinion of Respondents	No. of Respondents	Percentage
Strongly Agree	10	8.33
Agree	32	26.67
Neutral	48	40
Disagree	26	21.67
Strongly Disagree	4	3.33
Total	120	100

Source: Primary Data

Interpretation:

From the above table, it indicates that 8.33 % of respondents strongly agree, 26.67% of respondents are agree, 40% of respondents are neutral ,21.67% of respondents are disagree and 3.33% of respondents are strongly disagree with the Health care services provided helps to maintain a good health condition.

Table 1.5. INDEPENDENT SAMPLE T TEST

Factors of welfare measures	Level of Education				t value	P value
	Technical		Non-Technical			
	Mean	SD	Mean	SD		
Retirement Benefits	2.87	1.010	2.65	1.246	2.910	0.091
Medical Benefits	1.92	0.621	2.21	0.824	5.654	0.019*
Educational Allowance	2.54	1.019	2.29	0.798	1.314	0.254
Recreational Tour	2.58	1.161	2.88	0.841	2.409	0.123
Uniform	1.54	0.753	1.50	0.583	3.034	0.084
Hire Purchase Loan	2.31	0.919	2.46	1.051	2.063	0.154

PPE'S	1.92	0.682	2.21	0.651	0.077	0.781
Canteen	3.85	0.826	3.46	0.771	0.714	0.400
Health care services	2.96	0.907	2.58	1.048	4.359	0.039*

Note: 1. ** denotes significant at 1% level 2. * denotes significant at 5% level

Inference:

Since the p value is less than 0.05, the null hypothesis is rejected at 5% level with regard to factors of Medical Benefits & Health care services. Hence there is significant difference between technical and non-technical with respect to the factors of welfare measures. Because 58% respondents show their response in strongly agree in Medical Benefits, 48% respondents agree in Health care services because they provide health insurances to maintain a good health condition.

There is no significance difference between technical and non-technical with regard to factors of Retirement benefits, Educational allowance, Recreational tour, Hire purchase loan, PPEs, Uniform, Canteen. since p value greater than 0.05. hence the null hypothesis is accepted at 5% level with regard to the factors of welfare measures like Retirement Benefits, Educational Allowance, Recreation Tour, Hire purchase Loan, PPEs. Because the majority of the respondents show their response in Agree. The non-technical educational level of employees use various welfare measures, that can increase the production output highly.

FINDINGS

- In IREL (India) Ltd, 51.67% of employees have a Non-Technical education background; 48.33% of employees have a Technical education background.
- The results shows that most of the respondents were satisfied towards the existing welfare measures.

SUGGESTIONS

- Recreation tour can be improved to increase the morale of the employees and reduce stress of the employees.
- Most of the respondents feel that the retirement benefits were not enough so the company should increase the retirement benefit.
- Annual health checkups, employee counselling should be improved by conducting health camps.

CONCLUSION

The study on welfare measures provided to the employees and analysis on its effectiveness and satisfaction among the employees at Indian Rare Earths Limited has provided valuable insights into their effectiveness and employee satisfaction. Welfare measures not only enhance job satisfaction but also contribute to higher productivity and

improve morale with the employees. Through a comprehensive analysis of monetary, nonmonetary and motivational factors, the study has shed light on effectiveness of welfare programs and impact on employee satisfaction levels.

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DIGITAL ECONOMY AND E-COMMERCE IN INDIA: A COMPREHENSIVE ANALYSIS

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Abstract

The digital economy has significantly reshaped global commerce, and India is no exception. With rapid advancements in technology, increasing internet penetration, and favourable government policies, India's e-commerce sector has experienced unprecedented growth. This research paper explores the impact of the digital economy on India's economic landscape, the evolution of e-commerce, challenges, and future prospects. The study delves into the role of government initiatives, key market players, emerging technologies, and consumer behaviour patterns. It also highlights the infrastructural and regulatory barriers that must be overcome to ensure sustainable growth.

Introduction

The digital economy encompasses all economic activities that rely on digital technologies, including e-commerce, digital payments, online services, and cloud computing. In India, the digital economy is a key driver of growth, contributing significantly to GDP and employment. This paper examines the factors fueling this transformation and the role of e-commerce in reshaping consumer behavior and business operations. With one of the fastest-growing digital markets in the world, India has witnessed a surge in internet adoption and digital transformation across multiple sectors.

Growth of the Digital Economy in India

Internet and Smartphone Penetration: India has over 800 million internet users, making it one of the largest digital markets globally. The affordability of smartphones and cheaper mobile data plans have contributed significantly to this growth.

Digital Payments Revolution: The introduction of the Unified Payments Interface (UPI) has revolutionized financial transactions, making cashless payments more accessible. Mobile wallets such as Paytm, PhonePe, and Google Pay have further boosted digital transactions.

Government Initiatives: Programs such as Digital India, Startup India, and Make in India have accelerated digital adoption. The Pradhan Mantri Gramin Digital Saksharta Abhiyan (PMGDISHA) aims to enhance digital literacy in rural areas.

Rise of Online Marketplaces: Companies like Amazon India, Flipkart, and Reliance's JioMart have transformed the retail landscape. Social commerce platforms such as Meesho and Shop101 are gaining traction among small businesses.

3. E-Commerce in India

Evolution and Market Size: The Indian e-commerce market is expected to reach \$200 billion by 2026, driven by a growing middle class and digital adoption. The rise of direct-to-consumer (D2C) brands has changed the competitive landscape.

Key Players: Major players include Flipkart, Amazon India, Snapdeal, Myntra, and Tata Cliq. The entry of Reliance JioMart and the expansion of global e-commerce giants into India have intensified market competition.

Consumer Trends: Increasing preference for online shopping, demand for faster delivery, and digital payment adoption. The COVID-19 pandemic accelerated e-commerce penetration, with grocery and healthcare products witnessing exponential growth.

Rural E-Commerce Growth: Expanding internet access is driving e-commerce adoption in rural areas. Initiatives such as ONDC (Open Network for Digital Commerce) aim to create a level playing field for small businesses and local retailers.

Challenges in India's Digital Economy and E-Commerce Sector

Regulatory and Taxation Issues: Frequent changes in e-commerce regulations create uncertainties for businesses. Compliance with FDI policies and consumer protection laws adds to operational complexities.

Cybersecurity and Data Privacy: Increasing digital transactions have led to rising concerns over cybersecurity threats. The proposed Data Protection Bill aims to address privacy concerns but poses new challenges for businesses.

Digital Divide: Despite growth, disparities in digital access remain, especially in remote regions. Bridging this gap requires enhanced digital literacy programs and infrastructure investments.

Logistics and Infrastructure: Challenges in last-mile delivery and inadequate infrastructure hinder seamless operations. The adoption of drone deliveries and AI-powered logistics solutions is expected to improve efficiency.

Future Prospects and Opportunities

Integration of AI and Big Data: AI-driven personalization and big data analytics will enhance customer experiences. Chatbots, virtual assistants, and recommendation engines are becoming integral to e-commerce platforms.

Expansion of Digital Payment Systems: Continued innovations in fintech will further boost digital transactions. Blockchain technology is also being explored for secure and transparent transactions.

Policy Enhancements: Supportive regulations and infrastructure development will accelerate digital economy growth. The implementation of 5G technology is expected to further enhance digital connectivity.

Growth in Social Commerce: Platforms like Instagram, WhatsApp, and Facebook Marketplace are emerging as new channels for e-commerce. Influencer-driven marketing is playing a crucial role in consumer decision-making.

Sustainable E-Commerce Practices: With increasing awareness of environmental concerns, companies are adopting green supply chain practices, sustainable packaging, and carbon footprint reduction initiatives.

Conclusion

The digital economy and e-commerce in India are on a transformative path, offering immense opportunities for businesses, consumers, and policymakers. Addressing challenges such as regulatory hurdles, cybersecurity concerns, and infrastructure gaps will be crucial for sustaining growth. With continued investment in digital infrastructure, policy support, and technological innovation, India is poised to become a global leader in the digital economy. The rise of digital entrepreneurship and the gig economy further add to the dynamism of India's digital transformation. Ensuring inclusivity and security in the digital space will be key to long-term success.

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CUSTOMERS' ATTITUDE TOWARDS E-BANKING SERVICES A STUDY CONCERNING PUBLIC SECTOR BANKS IN TIRUNELVELI DISTRICT

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ABSTRACT

A researcher has reviewed many published data and concluded that more literature is needed on the technology adopted by the banks. Information technology is considered as a key factor in the changes taking place around the world. Electronic banking is the most inventive service offered by the banks. The transformation from traditional banking to e-banking has been a dramatic change. The evolution of e-banking started with the use of Automatic Teller Machines and Telephone Banking (ATM), direct bill payment, Electronic Fund Transfer (EFT), and revolutionary online banking. This study determines the consumers' perspective on the adoption of e-banking. There will be massive acceptance of e-banking with time with growing awareness and education. Many people are shifting to e-banking and have readily accepted the usefulness of this option. It allows customers to manage their accounts from any place, at any time at a minimum cost. Electronic banking is the latest in the series of technological wonders in the recent past, involving the use of the internet for the delivery of products and services

Key words: Finance, Bank, Service, Customer, Economic.

INTRODUCTION

The banking sector is one of the biggest service sectors in India and nowadays is in a way to attract the biggest market of Asia in investment. The banking sector today is focusing on how to provide efficient services to its customers. The Indian Banking System consists of various public and private sector financial institutions whose objective is serving the people for their financial and economic needs. At the time of independence, the Indian banking system was not sound. The strengthening of the banking system took place after the establishment of the Reserve Bank of India in 1935 as it was empowered to regulate banking by issue of directives, inspection, mergers, amalgamation, etc. In 1949 two major actions were taken which were very important from the point of view of structural reforms in the banking sector. First, the Banking Regulation Act was passed. It gave extensive regulatory powers to the Reserve Bank of India (RBI) over the commercial banks. On 19th July 1969, Fourteen major Indian commercial banks were nationalized, and on 15th April. 1980, Six more were added on to constitute the public sector banks.

OBJECTIVES

1. To measure the determinants of the attitude of the customers towards the banking services.
2. To examine the factors influencing the customers to avail of e-banking services of public sector banks in the study area.

SAMPLING DESIGN

In order to study the customers' perception, awareness, and satisfaction towards retail banking services, a sample of 100 respondents were collected. This study was based on proportionate and convenience sampling methods. For the selection of sample banks for the study, proportionate sampling was adopted. For the selection of respondents from these banks, a convenience sampling method was used. Moreover, the sampling design consisted of three steps.

In the first step, a list of Public and Private sector banks functioning in Tirunelveli district was prepared. The data were obtained from these banks.

The second step is selection of the sample banks for the study. In this case, proportionate sampling was used. The customers of these Public and Private Sector Banks were large in number and hence a comprehensive list of customers could not be prepared.

In the third step respondents were selected from the selected banks by applying convenience sampling method. Utmost care was taken to minimize bias in selection of sample respondents.

Therefore, a sample of 100 customers representing various categories such as Students, Professional, Business, Private employee, Government employee, Retired persons were selected by applying convenient sampling method.

SCOPE OF THE STUDY

Electronic banking has been evolving the environment with the development of the World Wide Web. Banks through the internet has emerged as a strategic resource for achieving higher efficiency. More recently in India, the E-banking service is carried out in an effective way to satisfy the customers of the respective banks. As the time factor is very much important for all the people in this modernized world, present customers do not want to waste their precious time for waiting in a queue. The electronic revolution has made it possible to provide ease and flexibility in banking operations which benefit the customers. As a business tool, internet banking is rapidly transforming the world of commerce and banking, making banks faster and more efficient and allowing them to provide more personalized services to the user /customer. The Internet creates perfect market conditions where customers have access to more information and can compare rates and financial product offerings. Internet reduces the barriers to enter manyC banks have found that internet banking has actually added to the cost.

The present study analyses customers' attitude towards e-banking services of public sector banks in Madurai District. The research objective is to investigate various factors influencing customers' perception and satisfaction level towards e-Banking. It further throws light on types of e-Banking services, certain emerging issues and challenges, benefits available to them and the factors influencing the decision of the customers towards the e - Banking services of public sector banks.

There are 204 Taluks and 9 Blocks (Town Panchayats) in Tirunelveli District. 8 banks and 20 branches are selected in palayamkottai for the study.

The attitude of the customers towards E-banking services

Factor	Not at all Attitude	Slightly Attitude	Somewhat Attitude	Moderately Attitude	Extremely Attitude
Opinion about need of e banking	74.60	17.40	5.40	2.60	100
Difficult to use e-bank	26.00	12.80	10.40	50.80	100
Convenience	48.80	28.80	17.0	5.40	100
Usefulness of e-bank	54.40	22.0	16.40	7.00	100
Economy	42.80	18.40	25.0	13.0	100
Self - control on bank accounts	61.0	8.80	23.00	6.80	100
Customer value – functional value	54.0	35.60	6.4	4.0	100
Customer value – emotional value	67.60	15.60	3.60	13.20	100
Customer value - trust value	68.80	23.60	7.60	-	100
Customer value - 10. loyalty value	43.60	45.60	11.00	-	100

Shows the awareness level of respondents, wherein the questionnaire was framed on the Five Likert scale of 1 - 5 about awareness - Not at all Attitude, Slightly Attitude, Somewhat Attitude, Moderately Attitude, and Extremely Attitude. The statement asks to check the level of Attitude.

Regarding the opinion of the respondents about the need of e-banking services it is found that out of 100 respondents, 74.60 percent of the respondents strongly agree with the statement, 17.40 percent of the respondents agree, 5.40 percent of the respondents have no opinion and the remaining 2.60 percent of the respondents disagree with the statement. It is found that most of the respondents (74.60 %) have felt that e-banking services are essential for the present computerized era.

Opinion about the difficulty in use of e-banking services Regarding the opinion of the respondents about the difficulty in using e-banking services it is found that out of 500 respondents, 26.00 percent of the respondents strongly agree with the statement, 12.80 percent of the respondents agree with the statement, 10.40 per cent of the respondents have no opinion about the difficulty in use of e-banking services and the remaining 50.80 per cent of the respondents disagree with the statement. It is found that most of the respondents (50.80 %) have opined that there is no difficulty in availing the e-banking services in the study area.

4.4.3 Opinion about ease of using of e-banking services Study has been made to know the opinion of the respondents about ease of using e banking services and it is found that out of 500 respondents 48.80 per cent of the respondents strongly agree with the statement, 28.80 per cent of the respondents agree with the statement, 17.00 per cent of the respondents have no opinion about the statement and the remaining 5.40 per cent of the respondents disagree with the statement. It is found that most of the respondents (48.80 %) have opined that the e-banking services are easy to use and there is no difficulty in using the e-banking services in the study area.

Opinion about usefulness of e-banking services Further study has been made to know the opinion of the respondents about the usefulness of e-banking services and it is found that out of 500 respondents 54.40 per cent of the respondents strongly agree with the statement,

22.00 per cent of the respondents agree with the statement, 16.40 per cent of the respondents have no opinion about the statement and the remaining 7.20 per cent of the respondents disagree with the statement. It is found that most of the respondents (54.40%) have opined that the e-banking services are highly useful for their day today banking activities.

Opinion about economy of e-banking Study has been made to know the opinion of the respondents about the cost factor of the e-banking and it is found that out of 500 respondents 42.80 per cent of the respondents strongly agree with the statement that the e-banking saves cost, 18.40 per cent of the respondents agree with the statement, 25.20 per cent of the respondents have no opinion about this statement and the remaining 13.60 per cent of the respondents disagree with this statement. It is found that most of the respondents (42.80%) have opined that the e-banking reduces the cost and it is economical too. 4.4.6 Opinion about self-control on accounts To know whether E-banking provides self control on accounts to their customers or not study has been made and it is found that out of 500 respondents 61.40 per cent of the respondents strongly agree that e-banking provides self control on their accounts, 8.80 per cent cxxxi of the respondents agree with the statement, 23.00 per cent of the respondents have no opinion about this statement and the remaining 6.80 per cent of the respondents disagree with this statement. It is found that most of the respondents (61.40 %) have strongly agreed that e banking provides self control on their accounts.

Opinion about functional value of e-banking services To examine the opinion of the respondents about the functional value of e-banking services study has been made and it is found that out of 500 respondents 54.00 per cent of the respondents strongly agree that e-banking has functional value, 35.60 per cent of the respondents have agreed with the statement, 6.40 per cent of the respondents have no opinion about this statement and the remaining 4.00 per cent of the respondents disagree with this statement. It is found that most of the respondents (54.00 %) strongly agree that e-banking has functional value among the customers.

Opinion about emotional value of e-banking services To measure the opinion of the respondents about the emotional value of e-banking services study has been made and it is found that out of 500 respondents 67.60 per cent of the respondents strongly agree that e-banking has emotional value, 15.60 per cent of the respondents agree with the statement, 3.60 per cent of the respondents have no opinion about this statement and the remaining 13.20 percent of the respondents disagree with this statement. It is found that most of the respondents (67.60 %) strongly agree that e-banking has emotional value among the customers.

Which type of services would you prefer? (Choose multiple options, if applicable)

Most preferred type of services by respondents

type of services	Percentage	Rank
Chatbots (Virtual Assistance)	1.1%	6
E-Banking Service	21.7%	2
E-Banking Service, Chatbots (Virtual Assistance)	33.2%	1
Visit to the bank	20.1%	3
Visit to the bank, E-Banking Service	21.0%	4
Visit to the bank, E-Banking Service, Chatbots (Virtual Assistance)	3.0%	5

Three options were given to choose the service, respondents prefer to avail the most for doing banking transactions. Options given were visit to the bank, e-banking services, and Chatbots (Virtual assistance). Respondents were given multiple options to choose and we found that most of the respondents preferred E-banking services (21.70%) over Visit to the bank (20.10%) and Chatbots services (1.10%). On another side, respondents also prefer to avail of banking services in combinations in which the most preferred combination availed by the respondents was E-banking service and Chatbots (virtual assistance) services of the banks which is (33.20%)

SUGGESTIONS:

Based on the analyses and findings of the study the following suggestions were provided to explore the benefits of Lead bank scheme.

1. Banks should allow the existing facilities to sink into the culture of the customers before any new facilities are launched. Also, the existing facilities should be embedded with services so that customers not only appreciate new technology, but are also in a position to operate.
2. The technology-driven services with hi-tech facilities of electronic fund transfer and electronic mail services should be effectively introduced to the customers. The customers need to be informed about the type of forms used for each transaction to make inquiries and to redress their complaints. Training programs should be devised for all the staff in such a way that the staff and customers relationship be improved through communication skills to be imparted in such training programs.
3. Banks should provide proper training to their employees about AI Chatbots so that they can spread awareness about chatbots while interacting with customers.

CONCLUSION:

Retail banking is the fastest-growing sector of the banking industry by attending directly to the needs of the end customers. It has faced a lot of competition ever since financial sector reforms were started in the country. They find out the customers' perception, awareness, and satisfaction of customers regarding the various retail banking services in Salem district. Since retail banking services are customer-oriented, banks should orient their services customer friendly. This will make customer relationship management easier and more effective. The banks should provide their retail services as simply as possible so that the difficulties faced by them and customers are not only minimized but also overcome. It has enabled the researcher to give some practical solutions to the problems faced by the customers.

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INNOVATION IN EDUCATION FOR SKILLS DEVELOPMENT

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Introduction

In the 21st century, the demand for skills is rapidly evolving due to technological advancements, globalization, and the rise of the gig economy. Traditional education models are no longer sufficient to equip learners with the necessary competencies for the future workforce. Innovations in education are playing a crucial role in bridging this gap by integrating technology, experiential learning, and personalized instruction. Below is a deeper exploration of key innovations in education for skills development.

Technology-Driven Learning

Technology is revolutionizing education by making learning more accessible, personalized, and engaging.

A. Artificial Intelligence (AI) and Personalized Learning

- AI-driven adaptive learning platforms (e.g., Duolingo, DreamBox, and Squirrel AI) adjust content based on a learner's progress.
- AI tutors and chatbots provide instant feedback and guidance.
- Predictive analytics help educators identify struggling students and intervene early.

B. Virtual and Augmented Reality (VR/AR)

- VR simulations create real-world learning experiences in fields like medical training, engineering, and skilled trades (e.g., VR surgical training).
- AR overlays enhance hands-on learning in subjects like anatomy, architecture, and mechanics.
- Immersive storytelling makes history and science more engaging by transporting students into past eras or scientific phenomena.

C. Gamification and Edutainment

- Platforms like Kahoot!, Classcraft, and Quizizz use game mechanics (points, leaderboards, rewards) to enhance engagement.
- Serious games teach problem-solving, strategy, and collaboration (e.g., Minecraft Education Edition for STEM learning).
- Role-playing simulations help develop negotiation, leadership, and decision-making skills.

D. Blockchain for Credentialing

- Blockchain technology ensures secure, tamper-proof digital certificates and diplomas.
- Learners can store and share their credentials on decentralized platforms, reducing fraud.
- Employers can verify a candidate's skills without needing physical documents.

Experiential and Project-Based Learning

Experiential learning methods allow students to apply theoretical knowledge in real-world scenarios.

A. Real-World Problem Solving

- Design thinking projects encourage innovation by tackling complex, open-ended challenges.
- STEM and maker spaces provide access to tools (3D printers, robotics kits) for hands-on experimentation.
- Service-learning integrates community projects with academic coursework to develop civic responsibility.

B. Hackathons and Bootcamps

- Tech bootcamps (e.g., General Assembly, Le Wagon) offer fast-tracked, skills-based learning in software development, data science, and UX design.
- Hackathons provide collaborative environments where students solve challenges under time constraints, fostering innovation and teamwork.
- Industry-sponsored challenges connect students with companies seeking fresh ideas and talent.

C. Internships and Apprenticeships

- Paid apprenticeships in sectors like IT, healthcare, and advanced manufacturing provide hands-on industry experience.
- Companies are partnering with educational institutions to create work-integrated learning opportunities (e.g., IBM's P-TECH model).
- Remote internships enable global opportunities for students to gain international exposure.

Flexible and Microlearning Approaches

The traditional long-duration degree model is being disrupted by modular, on-demand learning approaches.

A. Micro-Credentials and Digital Badges

- Short courses focus on specific competencies, such as coding, digital marketing, or leadership.
- Platforms like Coursera, Udemy, and LinkedIn Learning offer stackable certifications that align with industry needs.
- Open Badges by Mozilla and IBM enable learners to display achievements digitally.

B. MOOCs (Massive Open Online Courses)

- Free and low-cost courses provide access to quality education globally.
- Platforms like edX, Udacity, and FutureLearn offer professional certifications.

- Many MOOCs now integrate interactive assignments, peer-reviewed projects, and real-world case studies.

C. Hybrid and Blended Learning

- Flipped classrooms reverse traditional teaching by having students learn theoretical content at home and apply it in class.
- Asynchronous learning allows students to complete coursework at their own pace, accommodating diverse learning styles.
- Online labs and simulations provide hands-on experiences without requiring physical presence (e.g., virtual chemistry experiments).

Soft Skills and Holistic Development

Beyond technical skills, employers seek candidates with strong soft skills, such as communication, teamwork, and adaptability.

A. Emphasis on Communication, Collaboration, and Emotional Intelligence

- Debate clubs, public speaking courses, and teamwork-based projects enhance communication skills.
- Courses on negotiation, leadership, and conflict resolution prepare students for workplace dynamics.
- Emotional intelligence (EQ) training fosters resilience, empathy, and self-awareness.

B. Entrepreneurship Education

- Schools are incorporating lean startup methodologies to teach students how to create business models.
- Business incubators and accelerators provide mentorship, funding, and networking opportunities.
- Entrepreneurial competitions (e.g., startup pitch challenges) encourage innovation and problem-solving.

C. Mindfulness and Well-Being Programs

- Meditation and mental health workshops help students manage stress and improve focus.
- Social-emotional learning (SEL) programs teach self-regulation and interpersonal skills.
- Universities are integrating holistic development programs to prepare students for lifelong success.

Industry and Academia Collaboration

Aligning education with industry needs ensures that graduates possess job-ready skills.

A. Corporate-Backed Education Programs

- Companies like Google, Amazon, and IBM offer industry-recognized certifications (e.g., Google Career Certificates).
- Employer-sponsored training ensures direct alignment with job market demands.
- Corporate partnerships with universities result in curriculum co-development and talent pipelines.

B. Competency-Based Education (CBE)

- Focuses on mastery of skills rather than time spent in class.
- Students' progress upon demonstrating proficiency in a subject.
- Especially beneficial for adult learners and professionals looking to upskill.

C. Mentorship and Networking Platforms

- Digital mentorship platforms (e.g., LinkedIn, Mentor Cruise) connect students with industry professionals.
- Alumni networks facilitate career guidance and job placement.
- Peer-learning communities enhance collaborative knowledge-sharing.

Conclusion

Innovations in education are reshaping skills development by making learning more interactive, accessible, and aligned with industry needs. Technology is enabling personalized and immersive learning experiences, while experiential approaches ensure real-world application of knowledge. The future of education will likely be flexible, skills-oriented, and continuously evolving to meet the demands of an increasingly dynamic workforce.

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CSR AND SUSTAINABILITY PRACTICES IN INDIA

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Introduction

Corporate Social Responsibility (CSR) and sustainability have gained significant importance in India over the past decade. With rising environmental concerns and increasing expectations from businesses to contribute to social welfare, Indian corporations are integrating CSR initiatives and sustainable practices into their core strategies. The Companies Act, 2013, has made CSR mandatory for certain companies, reinforcing the role of businesses in nation-building and environmental protection. Additionally, sustainability practices are being adopted to ensure long-term business growth while safeguarding environmental and social interests.

CSR in India: Legal Framework and Implementation

The Companies Act, 2013, mandates that companies with a net worth of INR 500 crore or more, a turnover of INR 1,000 crore or more, or a net profit of INR 5 crore or more must spend at least 2% of their average net profits from the preceding three financial years on CSR activities. These initiatives should be aligned with Schedule VII of the Act, which provides a broad framework for CSR activities.

Some of the key areas covered under CSR initiatives in India include:

Education:

Establishing and supporting schools, colleges, and skill development institutions.

Providing scholarships and financial assistance to underprivileged students.

Partnering with NGOs and government programs to improve literacy rates and vocational training.

Healthcare:

Initiatives aimed at providing sanitation, building hospitals, and increasing healthcare accessibility.

Running campaigns for maternal and child health, vaccinations, and medical camps in rural areas.

Promoting mental health awareness and access to affordable medicines.

Environmental Sustainability:

Encouraging afforestation, tree plantation drives, and conservation projects.

Promoting the use of renewable energy sources like solar and wind energy.

Developing programs for waste reduction, water conservation, and carbon footprint minimization.

Rural Development:

Establishing sustainable livelihood programs for rural communities.

Enhancing agricultural productivity through modern techniques and resource optimization.

Providing financial aid and training for small-scale entrepreneurs and farmers.

Gender Equality and Women Empowerment:

Promoting women's education through scholarships and infrastructure development.

Creating employment and entrepreneurship opportunities for women.

Supporting initiatives to ensure women's safety and well-being in workplaces and communities.

Sustainability Practices in Indian Businesses

Sustainability has become an integral aspect of corporate strategy, with businesses adopting Environment, Social, and Governance (ESG) standards to ensure long-term economic, social, and environmental balance. Many Indian companies are integrating sustainability into their operational framework through innovative solutions.

Some of the key sustainability initiatives include:

Renewable Energy Adoption:

Companies like Tata Power and NTPC are making substantial investments in solar and wind energy projects.

Adoption of energy-efficient manufacturing and production techniques to reduce carbon emissions.

Implementation of green buildings and energy-saving infrastructure in corporate offices.

Waste Management and Recycling:

Leading brands such as ITC and Hindustan Unilever have introduced circular economy models that focus on waste reduction, recycling, and sustainable packaging.

Several companies have committed to eliminating single-use plastics and promoting biodegradable alternatives.

Encouraging Extended Producer Responsibility (EPR) for proper waste disposal and management.

Water Conservation:

Industries are implementing rainwater harvesting and wastewater recycling programs to minimize freshwater usage.

Corporate-backed projects focus on revitalizing water bodies and improving irrigation systems for sustainable agriculture.

Adoption of efficient water management systems in industrial processes to reduce waste.

Sustainable Supply Chains:

Businesses are collaborating with suppliers to ensure ethical sourcing of raw materials and fair-trade practices.

Adoption of low-emission logistics and transportation solutions.

Promoting the use of locally sourced, eco-friendly, and sustainable materials.

Leading Companies in CSR and Sustainability

Several Indian corporations have set benchmarks in CSR and sustainability practices, leading the way in creating a more responsible business ecosystem. Some of the notable companies include:

Tata Group:

Engages in extensive community development, education, healthcare, and environmental sustainability projects.

Pioneered the development of CSR-driven initiatives like the Tata Trusts, which focus on social welfare.

Reliance Industries:

Focuses on rural transformation, skill development, healthcare, and environmental conservation.

Runs large-scale initiatives such as the Reliance Foundation, which works on education and disaster response.

Infosys:

Committed to achieving carbon neutrality and reducing water and energy consumption in its operations.

Implements digital literacy programs and supports tech-driven social impact projects.

Wipro:

Leads in sustainability reporting, green IT practices, and responsible waste management.

Implements programs that promote STEM education and digital empowerment for underprivileged communities.

Challenges in CSR and Sustainability

Despite significant progress, CSR and sustainability initiatives in India face several challenges, including:

Inadequate Monitoring and Transparency: Many CSR projects lack proper tracking mechanisms, leading to inefficient resource utilization.

Limited Awareness and Engagement: Companies often struggle to engage employees and local communities in CSR activities effectively.

Regulatory Hurdles: Bureaucratic processes and policy inconsistencies can hinder the implementation of impactful CSR and sustainability programs.

Balancing Profitability with Responsibility: Some businesses find it challenging to integrate sustainability goals with their financial objectives.

The Way Forward

To improve the effectiveness of CSR and sustainability efforts, Indian businesses need to:

Strengthen Public-Private Partnerships: Collaborations between corporations, government agencies, and NGOs can enhance the reach and impact of CSR initiatives.

Integrate Technology and Data Analytics: Using digital platforms for tracking CSR projects and measuring sustainability impact can improve accountability.

Encourage Employee Participation: Corporations should involve employees in CSR activities, fostering a culture of social responsibility.

Align with Global Frameworks: Adopting international sustainability standards such as the UN Sustainable Development Goals (SDGs) can help companies structure their long-term CSR strategies.

Conclusion

CSR and sustainability are no longer optional for Indian businesses but essential components of responsible corporate governance. By prioritizing social and environmental impact, Indian companies can contribute meaningfully to economic growth, environmental conservation, and societal well-being. With continued efforts and strategic initiatives, India has the potential to emerge as a global leader in sustainable business practices.

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